

Question ID 7dea2114

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 7dea2114

Immanuel purchased a certain rare coin on January 1. The function $f(x) = 65(1.03)^x$, where $0 \leq x \leq 10$, gives the predicted value, in dollars, of the rare coin x years after Immanuel purchased it. What is the best interpretation of the statement “ $f(8)$ is approximately equal to **82**” in this context?

- A. When the rare coin's predicted value is approximately **82** dollars, it is **8%** greater than the predicted value, in dollars, on January 1 of the previous year.
- B. When the rare coin’s predicted value is approximately **82** dollars, it is **8** times the predicted value, in dollars, on January 1 of the previous year.
- C. From the day Immanuel purchased the rare coin to **8** years after Immanuel purchased the coin, its predicted value increased by a total of approximately **82** dollars.
- D. **8** years after Immanuel purchased the rare coin, its predicted value is approximately **82** dollars.

ID: 7dea2114 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the function $f(x) = 65(1.03)^x$ gives the predicted value, in dollars, of a certain rare coin x years after Immanuel purchased it. It follows that $f(x)$ represents the predicted value, in dollars, of the coin x years after Immanuel purchased it. Since the value of $f(8)$ is the value of $f(x)$ when $x = 8$, it follows that “ $f(8)$ is approximately equal to **82**” means that $f(x)$ is approximately equal to **82** when $x = 8$. Therefore, the best interpretation of the statement “ $f(8)$ is approximately equal to **82**” in this context is **8** years after Immanuel purchased the rare coin, its predicted value is approximately **82** dollars.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Hard

Question ID 4690d45d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 4690d45d

$$P(t) = 1,800(1.02)^t$$

The function P gives the estimated number of marine mammals in a certain area, where t is the number of years since a study began. What is the best interpretation of $P(0) = 1,800$ in this context?

- A. The estimated number of marine mammals in the area was **102** when the study began.
- B. The estimated number of marine mammals in the area was **1,800** when the study began.
- C. The estimated number of marine mammals in the area increased by **102** each year during the study.
- D. The estimated number of marine mammals in the area increased by **1,800** each year during the study.

ID: 4690d45d Answer

Correct Answer: B

Rationale

Choice B is correct. The function P gives the estimated number of marine mammals in a certain area, where t is the number of years since a study began. Since the value of $P(0)$ is the value of $P(t)$ when $t = 0$, it follows that $P(0) = 1,800$ means that the value of $P(t)$ is **1,800** when $t = 0$. Since t is the number of years since the study began, it follows that $t = 0$ is **0** years since the study began, or when the study began. Therefore, the best interpretation of $P(0) = 1,800$ in this context is the estimated number of marine mammals in the area was **1,800** when the study began.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 877bd007

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 877bd007

$f(x) = 4x^2 - 50x + 126$

The given equation defines the function f . For what value of x does $f(x)$ reach its minimum?

ID: 877bd007 Answer

Correct Answer: 25/4, 6.25

Rationale

The correct answer is $\frac{25}{4}$. The given equation can be rewritten in the form $f(x) = a(x - h)^2 + k$, where a , h , and k are constants. When $a > 0$, h is the value of x for which $f(x)$ reaches its minimum. The given equation can be rewritten as $f(x) = 4(x^2 - \frac{50}{4}x) + 126$, which is equivalent to $f(x) = 4(x^2 - \frac{50}{4}x + (\frac{50}{8})^2 - (\frac{50}{8})^2) + 126$. This equation can be rewritten as $f(x) = 4((x - \frac{50}{8})^2 - (\frac{50}{8})^2) + 126$, or $f(x) = 4(x - \frac{50}{8})^2 - 4(\frac{50}{8})^2 + 126$, which is equivalent to $f(x) = 4(x - \frac{25}{4})^2 - \frac{121}{4}$. Therefore, $h = \frac{25}{4}$, so the value of x for which $f(x)$ reaches its minimum is $\frac{25}{4}$. Note that 25/4 and 6.25 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 50dfcb6a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 50dfcb6a

$$x - 29 = (x - a)(x - 29)$$

Which of the following are solutions to the given equation, where a is a constant and $a > 30$?

- I. a
- II. $a + 1$
- III. 29
- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II, and III

ID: 50dfcb6a Answer

Correct Answer: C

Rationale

Choice C is correct. Subtracting the expression $(x - 29)$ from both sides of the given equation yields $0 = (x - a)(x - 29) - (x - 29)$, which can be rewritten as $0 = (x - a)(x - 29) + (-1)(x - 29)$. Since the two terms on the right-hand side of this equation have a common factor of $(x - 29)$, it can be rewritten as $0 = (x - 29)(x - a + (-1))$, or $0 = (x - 29)(x - a - 1)$. Since $x - a - 1$ is equivalent to $x - (a + 1)$, the equation $0 = (x - 29)(x - a - 1)$ can be rewritten as $0 = (x - 29)(x - (a + 1))$. By the zero product property, it follows that $x - 29 = 0$ or $x - (a + 1) = 0$. Adding 29 to both sides of the equation $x - 29 = 0$ yields $x = 29$. Adding $a + 1$ to both sides of the equation $x - (a + 1) = 0$ yields $x = a + 1$. Therefore, the two solutions to the given equation are 29 and $a + 1$. Thus, only $a + 1$ and 29 , not a , are solutions to the given equation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 1ee78af2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 1ee78af2

The function f is defined by $f(x) = a^x + b$, where a and b are constants and $a > 0$. In the xy -plane, the graph of $y = f(x)$ has a y -intercept at $(0, -25)$ and passes through the point $(2, 23)$. What is the value of $a + b$?

ID: 1ee78af2 Answer

Correct Answer: -19

Rationale

The correct answer is -19 . It's given that function f is defined by $f(x) = a^x + b$, where a and b are constants and $a > 0$. It's also given that the graph of $y = f(x)$ in the xy -plane has a y -intercept at $(0, -25)$ and passes through the point $(2, 23)$. Since the graph has a y -intercept at $(0, -25)$, $f(0) = -25$. Substituting 0 for x in the given equation yields $f(0) = a^0 + b$, or $f(0) = 1 + b$, and substituting -25 for $f(0)$ in this equation yields $-25 = 1 + b$. Subtracting 1 from each side of this equation yields $-26 = b$. Substituting -26 for b in the equation $f(x) = a^x + b$ yields $f(x) = a^x - 26$. Since the graph also passes through the point $(2, 23)$, $f(2) = 23$. Substituting 2 for x in the equation $f(x) = a^x - 26$ yields $f(2) = a^2 - 26$, and substituting 23 for $f(2)$ yields $23 = a^2 - 26$. Adding 26 to each side of this equation yields $49 = a^2$. Taking the square root of both sides of this equation yields $\pm 7 = a$. Since it's given that $a > 0$, the value of a is 7 . It follows that the value of $a + b$ is $7 - 26$, or -19 .

Question Difficulty: Hard

Question ID e1708eb0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: e1708eb0

x	$h(x)$
0	1.23
2	1.54
4	1.94

The table shows the exponential relationship between the number of years, x , since Hana started training in pole vault, and the estimated height $h(x)$, in meters, of her best pole vault for that year. Which of the following functions best represents this relationship, where $x \leq 4$?

- A. $h(x) = 1.12(0.23)^x$
- B. $h(x) = 1.12(1.23)^x$
- C. $h(x) = 1.23(0.12)^x$
- D. $h(x) = 1.23(1.12)^x$

ID: e1708eb0 Answer

Correct Answer: D

Rationale

Choice D is correct. The table shows an increasing exponential relationship between the number of years, x , since Hana started training in pole vault and the estimated height $h(x)$, in meters, of her best pole vault for that year. The relationship can be written as $h(x) = Ca^x$, where C and a are positive constants. It's given that when $x = 0$, $h(x) = 1.23$. Substituting 0 for x and 1.23 for $h(x)$ in $h(x) = Ca^x$ yields $1.23 = Ca^0$, or $1.23 = C$. Substituting 1.23 for C in $h(x) = Ca^x$ yields $h(x) = 1.23a^x$. It's also given that when $x = 2$, $h(x) = 1.54$. Substituting 2 for x and 1.54 for $h(x)$ in $h(x) = 1.23a^x$ yields $1.54 = 1.23a^2$. Dividing each side of this equation by 1.23 yields $\frac{1.54}{1.23} = \frac{1.23a^2}{1.23}$, or a^2 is approximately equal to 1.252. Since a is positive, a is approximately equal to $\sqrt{1.252}$, or 1.12. Substituting 1.12 for a in $h(x) = 1.23a^x$ yields $h(x) = 1.23(1.12)^x$.

Choice A is incorrect. When $x = 0$, the value of $h(x)$ in this function is equal to 1.12 rather than 1.23, and it is decreasing rather than increasing.

Choice B is incorrect. When $x = 0$, the value of $h(x)$ in this function is equal to 1.12 rather than 1.23.

Choice C is incorrect. This function is decreasing rather than increasing.

Question Difficulty: Hard

Question ID b3f03f6a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: b3f03f6a

A rectangle has a length that is **15** times its width. The function $y = (15w)(w)$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. Which of the following is the best interpretation of **15w** in this context?

- A. The length of the rectangle, in feet
- B. The area of the rectangle, in square feet
- C. The difference between the length and the width of the rectangle, in feet
- D. The width of the rectangle, in feet

ID: b3f03f6a Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that a rectangle has a length that is **15** times its width. It's also given that the function $y = (15w)(w)$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. The area of a rectangle can be calculated by multiplying the rectangle's length by its width. Since the rectangle has a length that is **15** times its width, it follows that w represents the width of the rectangle, in feet, and **15w** represents the length of the rectangle, in feet. Therefore, the best interpretation of **15w** in this context is that it's the length of the rectangle, in feet.

Choice B is incorrect. This is the best interpretation of y , not **15w**, in the given function.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. This is the best interpretation of w , not **15w**, in the given function.

Question Difficulty: Medium

Question ID 08efd6e7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 08efd6e7

An investment account was opened with an initial value of \$890. The value of the account doubled every 10 years. Which equation represents the value of the account $M(t)$, in dollars, t years after the account was opened?

- A. $M(t) = 890\left(\frac{1}{2}\right)^{\frac{t}{10}}$
- B. $M(t) = 890\left(\frac{1}{10}\right)^{\frac{t}{2}}$
- C. $M(t) = 890(2)^{\frac{t}{10}}$
- D. $M(t) = 890(10)^{\frac{t}{2}}$

ID: 08efd6e7 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that t represents the number of years since the account was opened. Therefore, $\frac{t}{10}$ represents the number of 10-year periods since the account was opened. Since the value of the account doubles during each of these 10-year periods, the value of the account can be found by multiplying the initial value by $\frac{t}{10}$ factors of 2. This is equivalent to $2^{\frac{t}{10}}$. It's given that the initial value of the account is \$890. Therefore, the value of the account $M(t)$, in dollars, t years after the account was opened can be represented by $M(t) = 890(2)^{\frac{t}{10}}$.

Choice A is incorrect. This equation represents the value of an account if the value of the account halves, not doubles, every 10 years.

Choice B is incorrect. This equation represents the value of an account if the value of the account decreases by 90%, not doubles, every 2, not 10, years.

Choice D is incorrect. This equation represents the value of an account if the value of the account increases by a factor of 10, not doubles, every 2, not 10, years.

Question Difficulty: Medium

Question ID 7046a0da

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 7046a0da

$$\begin{aligned}x^2 + y + 10 &= 10 \\ 8x + 16 - y &= 0\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

- A. -16
- B. -4
- C. 2
- D. 8

ID: 7046a0da Answer

Correct Answer: B

Rationale

Choice B is correct. Adding y to each side of the second equation in the given system of equations yields $8x + 16 = y$. Substituting $8x + 16$ for y in the first equation yields $x^2 + 8x + 16 + 10 = 10$. Subtracting 10 from each side of this equation yields $x^2 + 8x + 16 = 0$. This equation can be rewritten as $(x + 4)^2 = 0$. Taking the square root of each side of this equation yields $x + 4 = 0$. Subtracting 4 from each side of this equation yields $x = -4$. Therefore, the value of x is -4 .

Choice A is incorrect. This is the value of y , not x .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID b28007f8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: b28007f8

The function f is defined by $f(x) = |x - 4x|$. What value of a satisfies $f(5) - f(a) = -15$?

- A. -20
- B. 5
- C. 10
- D. 45

ID: b28007f8 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the function f is defined by $f(x) = |x - 4x|$. It's also given that $f(5) - f(a) = -15$. Substituting 5 for x in the function $f(x) = |x - 4x|$ yields $f(5) = |5 - 4(5)|$ and substituting a for x in the function $f(x) = |x - 4x|$ yields $f(a) = |a - 4a|$. Therefore, $f(5) = 15$ and $f(a) = |-3a|$. Substituting 15 for $f(5)$ and $|-3a|$ for $f(a)$ in the equation $f(5) - f(a) = -15$ yields $15 - |-3a| = -15$. Subtracting 15 from both sides of this equation yields $-|-3a| = -30$. Dividing both sides of this equation by -1 yields $|-3a| = 30$. By the definition of absolute value, if $|-3a| = 30$, then $-3a = 30$ or $-3a = -30$. Dividing both sides of each of these equations by -3 yields $a = -10$ or $a = 10$, respectively. Thus, of the given choices, a value of a that satisfies $f(5) - f(a) = -15$ is 10 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 763c1f25

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 763c1f25

$$\begin{aligned}y + k &= x + 26 \\ y - k &= x^2 - 5x\end{aligned}$$

In the given system of equations, k is a constant. The system has exactly one distinct real solution. What is the value of k ?

ID: 763c1f25 Answer

Correct Answer: 17.5, 35/2

Rationale

The correct answer is $\frac{35}{2}$. Subtracting the second equation from the first equation yields $(y + k) - (y - k) = x + 26 - (x^2 - 5x)$, or $2k = -x^2 + 6x + 26$. This is equivalent to $x^2 - 6x + (2k - 26) = 0$. It's given that the system has exactly one distinct real solution; therefore, this equation has exactly one distinct real solution. An equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has exactly one distinct real solution when the discriminant, $b^2 - 4ac$, is equal to 0. The equation $x^2 - 6x + (2k - 26) = 0$ is of this form, where $a = 1$, $b = -6$, and $c = 2k - 26$. Substituting these values into the discriminant, $b^2 - 4ac$, yields $(-6)^2 - 4(1)(2k - 26)$. Setting the discriminant equal to 0 yields $(-6)^2 - 4(1)(2k - 26) = 0$, or $-8k + 140 = 0$. Subtracting 140 from both sides of this equation yields $-8k = -140$. Dividing both sides of this equation by -8 yields $k = \frac{35}{2}$. Note that 35/2 and 17.5 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 2584d355

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 2584d355

Which of the following functions has(have) a minimum value at -3 ?

I. $f(x) = -6(3)^x - 3$

II. $g(x) = -3(6)^x$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

ID: 2584d355 Answer

Correct Answer: D

Rationale

Choice D is correct. A function of the form $f(x) = a(b)^x + c$, where $a < 0$ and $b > 1$, is a decreasing function. Both of the given functions are of this form; therefore, both are decreasing functions. If a function f is decreasing as the value of x increases, the corresponding value of $f(x)$ decreases; therefore, the function doesn't have a minimum value. Thus, neither of the given functions has a minimum value.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 0ccd0652

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 0ccd0652

$$\begin{aligned}8x + y &= -11 \\ 2x^2 &= y + 341\end{aligned}$$

The graphs of the equations in the given system of equations intersect at the point (x, y) in the xy -plane. What is a possible value of x ?

- A. -15
- B. -11
- C. 2
- D. 8

ID: 0ccd0652 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the graphs of the equations in the given system of equations intersect at the point (x, y) . Therefore, this intersection point is a solution to the given system. The solution can be found by isolating y in each equation. The given equation $8x + y = -11$ can be rewritten to isolate y by subtracting $8x$ from both sides of the equation, which gives $y = -8x - 11$. The given equation $2x^2 = y + 341$ can be rewritten to isolate y by subtracting 341 from both sides of the equation, which gives $2x^2 - 341 = y$. With each equation solved for y , the value of y from one equation can be substituted into the other, which gives $2x^2 - 341 = -8x - 11$. Adding $8x$ and 11 to both sides of this equation results in $2x^2 + 8x - 330 = 0$. Dividing both sides of this equation by 2 results in $x^2 + 4x - 165 = 0$. This equation can be rewritten by factoring the left-hand side, which yields $(x + 15)(x - 11) = 0$. By the zero-product property, if $(x + 15)(x - 11) = 0$, then $(x + 15) = 0$, or $(x - 11) = 0$. It follows that $x = -15$, or $x = 11$. Since only -15 is given as a choice, a possible value of x is -15 .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 62ba1727

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 62ba1727

$$y = x^2 - 14x + 22$$

The given equation relates the variables x and y . For what value of x does the value of y reach its minimum?

ID: 62ba1727 Answer

Correct Answer: 7

Rationale

The correct answer is **7**. When an equation is of the form $y = ax^2 + bx + c$, where a , b , and c are constants, the value of y reaches its minimum when $x = -\frac{b}{2a}$. Since the given equation is of the form $y = ax^2 + bx + c$, it follows that $a = 1$, $b = -14$, and $c = 22$. Therefore, the value of y reaches its minimum when $x = -\frac{(-14)}{2(1)}$, or $x = 7$.

Question Difficulty: Hard

Question ID 155d1adf

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 155d1adf

A rectangle has a length of x units and a width of $(x - 15)$ units. If the rectangle has an area of **76** square units, what is the value of x ?

- A. **4**
- B. **19**
- C. **23**
- D. **76**

ID: 155d1adf Answer

Correct Answer: B

Rationale

Choice B is correct. The area of a rectangle is equal to its length multiplied by its width. Multiplying the given length, x units, by the given width, $(x - 15)$ units, yields $x(x - 15)$ square units. If the rectangle has an area of **76** square units, it follows that $x(x - 15) = 76$, or $x^2 - 15x = 76$. Subtracting **76** from both sides of this equation yields $x^2 - 15x - 76 = 0$. Factoring the left-hand side of this equation yields $(x - 19)(x + 4) = 0$. Applying the zero product property to this equation yields two solutions: $x = 19$ and $x = -4$. Since x is the rectangle's length, in units, which must be positive, the value of x is **19**.

Choice A is incorrect. This is the width, in units, of the rectangle, not the value of x .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the area, in square units, of the rectangle, not the value of x .

Question Difficulty: Medium

Question ID 288c81be

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 288c81be

The function f is defined by $f(x) = x^3 + 15$. What is the value of $f(2)$?

- A. 20
- B. 21
- C. 23
- D. 24

ID: 288c81be Answer

Correct Answer: C

Rationale

Choice C is correct. The value of $f(2)$ is the value of $f(x)$ when $x = 2$. Substituting 2 for x in the given function yields $f(2) = (2)^3 + 15$, or $f(2) = 8 + 15$, which is equivalent to $f(2) = 23$. Therefore, the value of $f(2)$ is 23.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $f(2)$ when $f(x) = x(3) + 15$, rather than $f(x) = x^3 + 15$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID a714b25a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: a714b25a

$|x - 2| = 9$

What is one possible solution to the given equation?

ID: a714b25a Answer

Correct Answer: 11, -7

Rationale

The correct answer is **11** or **−7**. By the definition of absolute value, if $|x - 2| = 9$, then $x - 2 = 9$ or $x - 2 = -9$. Adding **2** to both sides of the equation $x - 2 = 9$ yields $x = 11$. Adding **2** to both sides of the equation $x - 2 = -9$ yields $x = -7$. Thus, the given equation, $|x - 2| = 9$, has two possible solutions, **11** and **−7**. Note that 11 and -7 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID 5b205ed2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 5b205ed2

$(x - 1)^2 = -4$

How many distinct real solutions does the given equation have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

ID: 5b205ed2 Answer

Correct Answer: D

Rationale

Choice D is correct. Any quantity that is positive or negative in value has a positive value when squared. Therefore, the left-hand side of the given equation is either positive or zero for any value of x . Since the right-hand side of the given equation is negative, there is no value of x for which the given equation is true. Thus, the number of distinct real solutions for the given equation is zero.

Choices A, B, and C are incorrect and may result from conceptual errors.

Question Difficulty: Hard

Question ID 09032b95

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 09032b95

$|4x - 3| = -9$

How many solutions does the given equation have?

- A. Zero
- B. One
- C. Two
- D. Infinitely many

ID: 09032b95 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $|4x - 3| = -9$. By the definition of absolute value, the left-hand side of this equation, $|4x - 3|$, must be nonnegative. Since the right-hand side of this equation, -9 , is negative, it follows that no value of x can make this equation true; therefore, the given equation has zero solutions.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 62821aef

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 62821aef

$f(x) = (x - 14)(x + 19)$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

- A. -266
- B. -19
- C. $-\frac{33}{2}$
- D. $-\frac{5}{2}$

ID: 62821aef Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that $f(x) = (x - 14)(x + 19)$, which can be rewritten as $f(x) = x^2 + 5x - 266$. Since the coefficient of the x^2 -term is positive, the graph of $y = f(x)$ in the xy -plane opens upward and reaches its minimum value at its vertex. The x -coordinate of the vertex is the value of x such that $f(x)$ reaches its minimum. For an equation in the form $f(x) = ax^2 + bx + c$, where a , b , and c are constants, the x -coordinate of the vertex is $-\frac{b}{2a}$. For the equation $f(x) = x^2 + 5x - 266$, $a = 1$, $b = 5$, and $c = -266$. It follows that the x -coordinate of the vertex is $-\frac{5}{2(1)}$, or $-\frac{5}{2}$. Therefore, $f(x)$ reaches its minimum when the value of x is $-\frac{5}{2}$.

Alternate approach: The value of x for the vertex of a parabola is the x -value of the midpoint between the two x -intercepts of the parabola. Since it's given that $f(x) = (x - 14)(x + 19)$, it follows that the two x -intercepts of the graph of $y = f(x)$ in the xy -plane occur when $x = 14$ and $x = -19$, or at the points $(14, 0)$ and $(-19, 0)$. The midpoint between two points, (x_1, y_1) and (x_2, y_2) , is $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$. Therefore, the midpoint between $(14, 0)$ and $(-19, 0)$ is $(\frac{14+(-19)}{2}, \frac{0+0}{2})$, or $(-\frac{5}{2}, 0)$. It follows that $f(x)$ reaches its minimum when the value of x is $-\frac{5}{2}$.

Choice A is incorrect. This is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane.

Choice B is incorrect. This is one of the x -coordinates of the x -intercepts of the graph of $y = f(x)$ in the xy -plane.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 2b8f11e6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 2b8f11e6

The function f is defined by $f(x) = 7x^3$. In the xy -plane, the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ down 2 units. Which equation defines function g ?

- A. $g(x) = \frac{7}{2}x^3$
- B. $g(x) = 7x^{\frac{3}{2}}$
- C. $g(x) = 7x^3 + 2$
- D. $g(x) = 7x^3 - 2$

ID: 2b8f11e6 Answer

Correct Answer: D

Rationale

Choice D is correct. If the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ down k units in the xy -plane, the function g can be defined by an equation of the form $g(x) = f(x) - k$. It's given that $f(x) = 7x^3$ and the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ down 2 units. Substituting $7x^3$ for $f(x)$ and 2 for k in the equation $g(x) = f(x) - k$ yields $g(x) = 7x^3 - 2$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

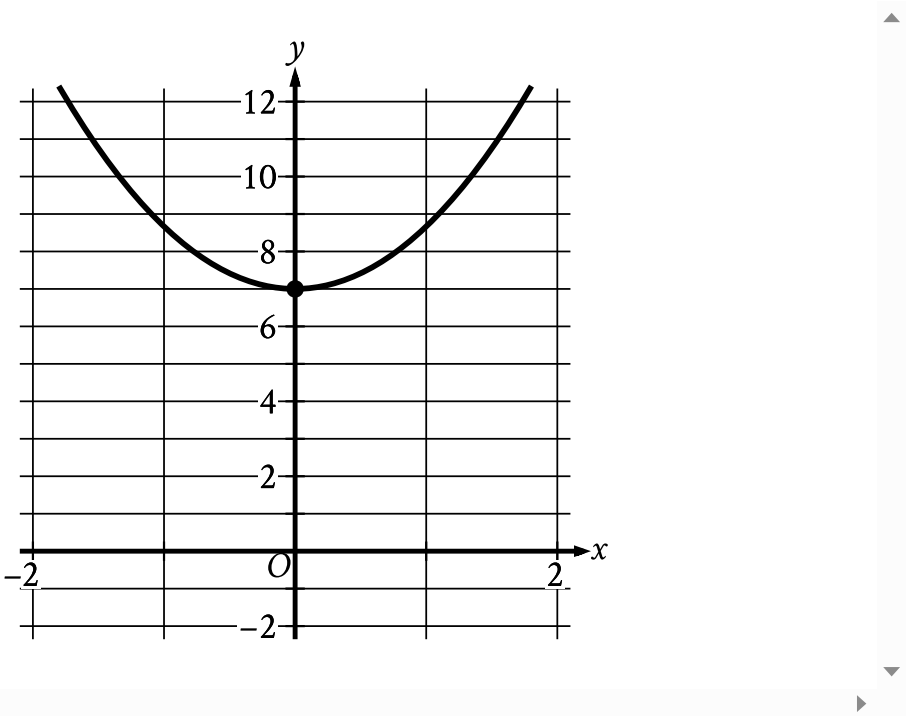
Choice C is incorrect. This equation defines a function g for which the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ up, not down, 2 units.

Question Difficulty: Easy

Question ID 3cc0acf5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 3cc0acf5



The parabola shown intersects the y -axis at the point (x, y) . What is the value of y ?

ID: 3cc0acf5 Answer

Correct Answer: 7

Rationale

The correct answer is **7**. It's given that the parabola intersects the y -axis at the point (x, y) . The graph shows that the parabola intersects the y -axis at the point $(0, 7)$. Therefore, the value of y is **7**.

Question Difficulty: Easy

Question ID 5a7fef39

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 5a7fef39

According to a model, the estimated volume of a certain tree in a forest increases by **13.9%** for every increase of **100** centimeters in the tree's height. The equation $y = ab^{\frac{x}{100}}$ represents this model, where y represents the estimated volume, in cubic centimeters, of this tree in the forest, x represents the tree's height, in centimeters, and a and b are constants. In this equation, what is the value of b ?

- A. **0.139**
- B. **1**
- C. **1.139**
- D. **13.9**

ID: 5a7fef39 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that equation $y = ab^{\frac{x}{100}}$ represents the model, where y is the estimated volume, in cubic centimeters, of a certain tree in a forest and x is the tree's height, in centimeters. For an exponential equation of this form to represent a model where the value of y increases as the value of x increases, it must follow that the values of a and b are positive. It's also given that the tree's estimated volume increases by **13.9%** for every increase of **100** centimeters in the tree's height. Therefore, for every **100**-centimeter increase in the tree's height, the estimated volume of the tree is **1.139** times larger. In other words, $ab^{\frac{x+100}{100}} = 1.139ab^{\frac{x}{100}}$, which is equivalent to $ab^{\frac{x}{100}}b^{\frac{100}{100}} = 1.139ab^{\frac{x}{100}}$. Since a and b are positive, dividing both sides of this equation by $ab^{\frac{x}{100}}$ yields $b^{\frac{100}{100}} = 1.139$, or $b = 1.139$. Therefore, the value of b is **1.139**.

Choice A is incorrect. This is the given percent increase in decimal form, not the value of b .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the given percent increase, not the value of b .

Question Difficulty: Hard

Question ID 6c30cbbe

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 6c30cbbe

$$\frac{(x+9)(x-9)}{x+9} = 7$$

What is the solution to the given equation?

- A. 7
- B. 9
- C. 16
- D. 63

ID: 6c30cbbe Answer

Correct Answer: C

Rationale

Choice C is correct. Since the left-hand side of the given equation has a factor of $x + 9$ in both the numerator and the denominator, the solution to the given equation can be found by solving the equation $x - 9 = 7$. Adding 9 to both sides of this equation yields $x = 16$. Substituting 16 for x in the given equation yields $\frac{(16+9)(16-9)}{16+9} = 7$, or $7 = 7$. Therefore, the solution to the given equation is 16.

Choice A is incorrect. Substituting 7 for x in the given equation yields $\frac{(7+9)(7-9)}{7+9} = 7$, or $-2 = 7$, which is false.

Choice B is incorrect. Substituting 9 for x in the given equation yields $\frac{(9+9)(9-9)}{9+9} = 7$, or $0 = 7$, which is false.

Choice D is incorrect. Substituting 63 for x in the given equation yields $\frac{(63+9)(63-9)}{63+9} = 7$, or $54 = 7$, which is false.

Question Difficulty: Medium

Question ID 423265ec

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 423265ec

$x^2 - 2x - 9 = 0$

One solution to the given equation can be written as $1 + \sqrt{k}$, where k is a constant. What is the value of k ?

- A. 8
- B. 10
- C. 20
- D. 40

ID: 423265ec Answer

Correct Answer: B

Rationale

Choice B is correct. Adding 9 to each side of the given equation yields $x^2 - 2x = 9$. To complete the square, adding 1 to each side of this equation yields $x^2 - 2x + 1 = 9 + 1$, or $(x - 1)^2 = 10$. Taking the square root of each side of this equation yields $x - 1 = \pm\sqrt{10}$. Adding 1 to each side of this equation yields $x = 1 \pm \sqrt{10}$. Since it's given that one of the solutions to the equation can be written as $1 + \sqrt{k}$, the value of k must be 10.

Alternate approach: It's given that $1 + \sqrt{k}$ is a solution to the given equation. It follows that $x = 1 + \sqrt{k}$. Substituting $1 + \sqrt{k}$ for x in the given equation yields $(1 + \sqrt{k})^2 - 2(1 + \sqrt{k}) - 9 = 0$, or $(1 + \sqrt{k})(1 + \sqrt{k}) - 2(1 + \sqrt{k}) - 9 = 0$. Expanding the products on the left-hand side of this equation yields $1 + 2\sqrt{k} + k - 2 - 2\sqrt{k} - 9 = 0$, or $k - 10 = 0$. Adding 10 to each side of this equation yields $k = 10$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID d91f65a7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: d91f65a7

$$7m = 5(n + p)$$

The given equation relates the positive numbers m , n , and p . Which equation correctly gives n in terms of m and p ?

- A. $n = \frac{5p}{7m}$
- B. $n = \frac{7m}{5} - p$
- C. $n = 5(7m) + p$
- D. $n = 7m - 5 - p$

ID: d91f65a7 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the equation $7m = 5(n + p)$ relates the positive numbers m , n , and p . Dividing both sides of the given equation by 5 yields $\frac{7m}{5} = n + p$. Subtracting p from both sides of this equation yields $\frac{7m}{5} - p = n$, or $n = \frac{7m}{5} - p$. It follows that the equation $n = \frac{7m}{5} - p$ correctly gives n in terms of m and p .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 9c14cee3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 9c14cee3

$$x = 8a(b + 9)$$

The given equation relates the positive numbers a , b , and x . Which equation correctly expresses a in terms of b and x ?

- A. $a = \frac{x}{8} - (b + 9)$
- B. $a = \frac{x}{8(b+9)}$
- C. $a = \frac{8(b+9)}{x}$
- D. $a = 8x(b + 9)$

ID: 9c14cee3 Answer

Correct Answer: B

Rationale

Choice B is correct. To express a in terms of b and x , the given equation can be rewritten such that a is isolated on one side of the equation. Since it's given that b is a positive number, $b + 9$ is not equal to zero. Therefore, dividing both sides of the given equation by $8(b + 9)$ yields the equivalent equation $\frac{x}{8(b+9)} = a$, or $a = \frac{x}{8(b+9)}$.

Choice A is incorrect. This equation is equivalent to $x = 8(a + (b + 9))$.

Choice C is incorrect. This equation is equivalent to $x = \frac{8(b+9)}{a}$.

Choice D is incorrect. This equation is equivalent to $x = \frac{a}{8(b+9)}$.

Question Difficulty: Hard

Question ID ccc723a5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: ccc723a5

Which expression is equivalent to $23x^3 + 2x^2 + 9x$?

- A. $23x(x^2 + 2x + 9)$
- B. $9x(23x^3 + 2x^2 + 1)$
- C. $x(23x^2 + 2x + 9)$
- D. $34(x^3 + x^2 + x)$

ID: ccc723a5 Answer

Correct Answer: C

Rationale

Choice C is correct. Since x is a common factor of each term in the given expression, the given expression can be rewritten as $x(23x^2 + 2x + 9)$.

Choice A is incorrect. This expression is equivalent to $23x^3 + 46x^2 + 207x$.

Choice B is incorrect. This expression is equivalent to $207x^4 + 18x^3 + 9x$.

Choice D is incorrect. This expression is equivalent to $34x^3 + 34x^2 + 34x$.

Question Difficulty: Medium

Question ID d7706113

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: d7706113

$$y = x + 9$$
$$y = x^2 + 16x + 63$$

A solution to the given system of equations is (x, y) . What is the greatest possible value of x ?

- A. **−6**
- B. 7
- C. 9
- D. 63

ID: d7706113 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $y = x + 9$ and $y = x^2 + 16x + 63$; therefore, it follows that $x + 9 = x^2 + 16x + 63$. This equation can be rewritten as $x + 9 = (x + 9)(x + 7)$. Subtracting $(x + 9)$ from both sides of this equation yields $0 = (x + 9)(x + 7) - (x + 9)$. This equation can be rewritten as $0 = (x + 9)((x + 7) - 1)$, or $0 = (x + 9)(x + 6)$. By the zero product property, $x + 9 = 0$ or $x + 6 = 0$. Subtracting 9 from both sides of the equation $x + 9 = 0$ yields $x = -9$. Subtracting 6 from both sides of the equation $x + 6 = 0$ yields $x = -6$. Therefore, the given system of equations has solutions, (x, y) , that occur when $x = -9$ and $x = -6$. Since -6 is greater than -9 , the greatest possible value of x is -6 .

Choice B is incorrect. This is the negative of the greatest possible value of x when $y = 0$ for the second equation in the given system of equations.

Choice C is incorrect. This is the value of y when $x = 0$ for the first equation in the given system of equations.

Choice D is incorrect. This is the value of y when $x = 0$ for the second equation in the given system of equations.

Question Difficulty: Hard

Question ID 2272fa21

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 2272fa21

Which expression is equivalent to $9x + 6x + 2y + 3y$?

- A. $3x + 5y$
- B. $6x + 8y$
- C. $12x + 8y$
- D. $15x + 5y$

ID: 2272fa21 Answer

Correct Answer: D

Rationale

Choice D is correct. Combining like terms in the given expression yields $(9x + 6x) + (2y + 3y)$, or $15x + 5y$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID dfb44d1d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: dfb44d1d

$$q - 29r = s$$

The given equation relates the positive numbers q , r , and s . Which equation correctly expresses q in terms of r and s ?

- A. $q = s - 29r$
- B. $q = s + 29r$
- C. $q = 29rs$
- D. $q = -\frac{s}{29r}$

ID: dfb44d1d Answer

Correct Answer: B

Rationale

Choice B is correct. Adding $29r$ to each side of the given equation yields $q = s + 29r$. Therefore, the equation $q = s + 29r$ correctly expresses q in terms of r and s .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID b3d52f66

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: b3d52f66

$$\begin{aligned}x + 5 &= 14 \\ y &= 4x^2 + 4\end{aligned}$$

At what point (x, y) do the graphs of the equations in the given system intersect?

- A. $(9, 324)$
- B. $(9, 328)$
- C. $(14, 4)$
- D. $(14, 788)$

ID: b3d52f66 Answer

Correct Answer: B

Rationale

Choice B is correct. The point (x, y) at which the graphs of the given equations intersect is the solution to the system of equations. Subtracting 5 from both sides of the equation $x + 5 = 14$ yields $x = 9$. Substituting 9 for x in the equation $y = 4x^2 + 4$ yields $y = 4(9)^2 + 4$, or $y = 328$. Therefore, the graphs of the equations in the given system intersect at the point $(9, 328)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 11066939

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 11066939

The function f is defined by $f(x) = 5x^2$. What is the value of $f(8)$?

- A. 40
- B. 50
- C. 80
- D. 320

ID: 11066939 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the function f is defined by $f(x) = 5x^2$. Substituting 8 for x in $f(x) = 5x^2$ yields $f(8) = 5(8)^2$, which is equivalent to $f(8) = 5(64)$, or $f(8) = 320$. Therefore, the value of $f(8)$ is 320.

Choice A is incorrect. This is the value of $f(8)$ if $f(x) = 5x$.

Choice B is incorrect. This is the value of $f(8)$ if $f(x) = 5(x + 2)$.

Choice C is incorrect. This is the value of $f(8)$ if $f(x) = (5x)(2)$.

Question Difficulty: Medium

Question ID b8be51aa

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: b8be51aa

Which expression is equivalent to $5x^2 - 50xy^2$?

- A. $5x(x - 10y^2)$
- B. $5x(x - 50y^2)$
- C. $5x^2(10xy^2)$
- D. $5x^2(50xy^2)$

ID: b8be51aa Answer

Correct Answer: A

Rationale

Choice A is correct. Since each term of the given expression has a factor of $5x$, it can be rewritten as $5x(x) - 5x(10y^2)$, or $5x(x - 10y^2)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID e8e66998

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: e8e66998

$$\frac{12}{n} - \frac{2}{t} = -\frac{2}{w}$$

The given equation relates the variables n , t , and w , where $n > 0$, $t > 0$, and $w > t$. Which expression is equivalent to n ?

- A. $12tw$
- B. $6(t - w)$
- C. $\frac{w-t}{6tw}$
- D. $\frac{6tw}{w-t}$

ID: e8e66998 Answer

Correct Answer: D

Rationale

Choice D is correct. Adding $\frac{2}{t}$ to each side of the given equation yields $\frac{12}{n} = -\frac{2}{w} + \frac{2}{t}$. The fractions on the right side of this equation have a common denominator of tw ; therefore, the equation can be written as $\frac{12}{n} = \frac{2w}{tw} - \frac{2t}{tw}$, or $\frac{12}{n} = \frac{2w-2t}{tw}$, which is equivalent to $\frac{12}{n} = \frac{2(w-t)}{tw}$. Dividing each side of this equation by 2 yields $\frac{6}{n} = \frac{w-t}{tw}$. Since n , t , w , and $w - t$ are all positive quantities, taking the reciprocal of each side of the equation $\frac{6}{n} = \frac{w-t}{tw}$ yields an equivalent equation: $\frac{n}{6} = \frac{tw}{w-t}$. Multiplying each side of this equation by 6 yields $n = \frac{6tw}{w-t}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is equivalent to $\frac{1}{n}$ rather than n .

Question Difficulty: Hard

Question ID 6a30c880

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 6a30c880

Which expression is equivalent to $16x^3y^2 + 14xy$?

- A. $2xy(8xy + 7)$
- B. $2xy(8x^2y + 7)$
- C. $14xy(2x^2y + 1)$
- D. $14xy(8x^2y + 1)$

ID: 6a30c880 Answer

Correct Answer: B

Rationale

Choice B is correct. Since $2xy$ is a common factor of each term in the given expression, the expression can be rewritten as $2xy(8x^2y + 7)$.

Choice A is incorrect. This expression is equivalent to $16x^2y^2 + 14xy$.

Choice C is incorrect. This expression is equivalent to $28x^3y^2 + 14xy$.

Choice D is incorrect. This expression is equivalent to $112x^3y^2 + 14xy$.

Question Difficulty: Easy

Question ID b35afde7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: b35afde7

The area A , in square centimeters, of a rectangular cutting board can be represented by the expression $w(w + 9)$, where w is the width, in centimeters, of the cutting board. Which expression represents the length, in centimeters, of the cutting board?

- A. $w(w + 9)$
- B. w
- C. 9
- D. $(w + 9)$

ID: b35afde7 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the expression $w(w + 9)$ represents the area, in square centimeters, of a rectangular cutting board, where w is the width, in centimeters, of the cutting board. The area of a rectangle can be calculated by multiplying its length by its width. It follows that the length, in centimeters, of the cutting board is represented by the expression $(w + 9)$.

Choice A is incorrect. This expression represents the area, in square centimeters, of the cutting board, not its length, in centimeters.

Choice B is incorrect. This expression represents the width, in centimeters, of the cutting board, not its length.

Choice C is incorrect. This is the difference between the length, in centimeters, and the width, in centimeters, of the cutting board, not its length, in centimeters.

Question Difficulty: Medium

Question ID ce145b5d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: ce145b5d

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

- A. $3x^2 + 7x + 14b$
- B. $3x^2 + 28x + 14b$
- C. $3x^2 + 42x + 14b$
- D. $3x^2 + 49x + 14b$

ID: ce145b5d Answer

Correct Answer: D

Rationale

Choice D is correct. Since each choice has a term of $3x^2$, which can be written as $(3x)(x)$, and each choice has a term of $14b$, which can be written as $(7)(2b)$, the expression that has a factor of $x + 2b$, where b is a positive integer constant, can be represented as $(3x + 7)(x + 2b)$. Using the distributive property of multiplication, this expression is equivalent to $3x(x + 2b) + 7(x + 2b)$, or $3x^2 + 6xb + 7x + 14b$. Combining the x-terms in this expression yields $3x^2 + (7 + 6b)x + 14b$. It follows that the coefficient of the x-term is equal to $7 + 6b$. Thus, from the given choices, $7 + 6b$ must be equal to $7, 28, 42$, or 49 . Therefore, $6b$ must be equal to $0, 21, 35$, or 42 , respectively, and b must be equal to $\frac{0}{6}, \frac{21}{6}, \frac{35}{6}$, or $\frac{42}{6}$, respectively. Of these four values of b , only $\frac{42}{6}$, or 7 , is a positive integer. It follows that $7 + 6b$ must be equal to 49 because this is the only choice for which the value of b is a positive integer constant. Therefore, the expression that has a factor of $x + 2b$ is $3x^2 + 49x + 14b$.

Choice A is incorrect. If this expression has a factor of $x + 2b$, then the value of b is 0 , which isn't positive.

Choice B is incorrect. If this expression has a factor of $x + 2b$, then the value of b is $\frac{21}{6}$, which isn't an integer.

Choice C is incorrect. If this expression has a factor of $x + 2b$, then the value of b is $\frac{35}{6}$, which isn't an integer.

Question Difficulty: Hard

Question ID 5628913b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 5628913b

At the time that an article was first featured on the home page of a news website, there were **40** comments on the article. An exponential model estimates that at the end of each hour after the article was first featured on the home page, the number of comments on the article had increased by **190%** of the number of comments on the article at the end of the previous hour. Which of the following equations best represents this model, where C is the estimated number of comments on the article t hours after the article was first featured on the home page and $t \leq 4$?

- A. $C = 40(1.19)^t$
- B. $C = 40(1.9)^t$
- C. $C = 40(19)^t$
- D. $C = 40(2.9)^t$

ID: 5628913b Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that an exponential model estimates that the number of comments on an article increased by a fixed percentage at the end of each hour. Therefore, the model can be represented by an exponential equation of the form $C = Ka^t$, where C is the estimated number of comments on the article t hours after the article was first featured on the home page and K and a are constants. It's also given that when the article was first featured on the home page of the news website, there were **40** comments on the article. This means that when $t = 0$, $C = 40$. Substituting **0** for t and **40** for C in the equation $C = Ka^t$ yields $40 = Ka^0$, or $40 = K$. It's also given that the number of comments on the article at the end of an hour had increased by **190%** of the number of comments on the article at the end of the previous hour. Multiplying the percent increase by the number of comments on the article at the end of the previous hour yields the number of estimated additional comments the article has on its home page: $(40)(\frac{190}{100})$, or **76** comments. Thus, the estimated number of comments for the following hour is the sum of the comments from the end of the previous hour and the number of additional comments, which is **40 + 76**, or **116**. This means that when $t = 1$, $C = 116$. Substituting **1** for t , **116** for C , and **40** for K in the equation $C = Ka^t$ yields $116 = 40a^1$, or $116 = 40a$. Dividing both sides of this equation by **40** yields $2.9 = a$. Substituting **40** for K and **2.9** for a in the equation $C = Ka^t$ yields $C = 40(2.9)^t$. Thus, the equation that best represents this model is $C = 40(2.9)^t$.

Choice A is incorrect. This model represents a situation where the number of comments at the end of each hour increased by **19%** of the number of comments at the end of the previous hour, rather than **190%**.

Choice B is incorrect. This model represents a situation where the number of comments at the end of each hour increased by **90%** of the number of comments at the end of the previous hour, rather than **190%**.

Choice C is incorrect. This model represents a situation where the number of comments at the end of each hour was **19** times the number of comments at the end of the previous hour, rather than increasing by **190%** of the number of comments at the end of the previous hour.

Question Difficulty: Hard

Question ID 99b73a30

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 99b73a30

For the exponential function f , the value of $f(0)$ is c , where c is a constant. Of the following equations that define the function f , which equation shows the value of c as the coefficient or the base?

- A. $f(x) = 22(1.5)^{x+1}$
- B. $f(x) = 33(1.5)^x$
- C. $f(x) = 49.5(1.5)^{x-1}$
- D. $f(x) = 74.25(1.5)^{x-2}$

ID: 99b73a30 Answer

Correct Answer: B

Rationale

Choice B is correct. Each of the given choices is an equation of the form $f(x) = a(b)^{x-k}$, where a , b , and k are constants. For an equation of this form, the coefficient, a , is equal to the value of the function when the exponent is equal to 0, or when $x = k$. It follows that in the equation $f(x) = 33(1.5)^x$, the coefficient, **33**, is equal to the value of $f(0)$. Substituting 0 for x in this equation yields $f(0) = 33(1.5)^0$, which is equivalent to $f(0) = 33(1)$, or $f(0) = 33$. Thus, the value of c is **33** and the equation $f(x) = 33(1.5)^x$ shows the value of c as the coefficient.

Choice A is incorrect. This equation shows the value of $f(-1)$, not $f(0)$, as the coefficient.

Choice C is incorrect. This equation shows the value of $f(1)$, not $f(0)$, as the coefficient.

Choice D is incorrect. This equation shows the value of $f(2)$, not $f(0)$, as the coefficient.

Question Difficulty: Medium

Question ID ec396a63

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: ec396a63

Which expression is equivalent to $(8x^3 + 8) - (x^3 - 2)$?

- A. $8x^3 + 6$
- B. $7x^3 + 10$
- C. $8x^3 + 10$
- D. $7x^3 + 6$

ID: ec396a63 Answer

Correct Answer: B

Rationale

Choice B is correct. The given expression is equivalent to $8x^3 + 8 - x^3 - (-2)$, or $8x^3 + 8 - x^3 + 2$. Combining like terms in this expression yields $7x^3 + 10$.

Choice A is incorrect. This expression is equivalent to $(8x^3 + 8) - 2$, not $(8x^3 + 8) - (x^3 - 2)$.

Choice C is incorrect. This expression is equivalent to $(8x^3 + 8) - (-2)$, not $(8x^3 + 8) - (x^3 - 2)$.

Choice D is incorrect. This expression is equivalent to $(8x^3 + 8) - (x^3 + 2)$, not $(8x^3 + 8) - (x^3 - 2)$.

Question Difficulty: Medium

Question ID d684dd1a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: d684dd1a

The value of a certain fishing boat was \$24,000 when it was purchased. Each year, the value of the fishing boat decreased by 20% of the previous year’s value. Which equation represents this situation, where v is the value of the fishing boat, in dollars, and y is the number of years since the fishing boat was purchased?

- A. $v = 0.8(24,000)^y$
- B. $v = 1.2(24,000)^y$
- C. $v = 24,000(0.8)^y$
- D. $v = 24,000(1.2)^y$

ID: d684dd1a Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that the value of a certain fishing boat was \$24,000 when it was purchased. It’s also given that each year, the value of the fishing boat decreased by 20% of the previous year’s value. It follows that this situation can be represented by the equation $v = a(1 - \frac{r}{100})^y$, where v is the value of the boat, in dollars, y years since it was purchased, a is the value of the boat, in dollars, when it was purchased, and r is the percent by which the value of the boat decreases each year from the previous year’s value. Substituting 24,000 for a and 20 for r in this equation yields $v = 24,000(1 - \frac{20}{100})^y$, or $v = 24,000(0.8)^y$. Therefore, $v = 24,000(0.8)^y$ represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This equation represents a situation where each year the value of the boat increased, not decreased, by 20% of the previous year’s value.

Question Difficulty: Medium

Question ID 40d75cea

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 40d75cea

$$f(t) = 8,000(0.65)^t$$

The given function f models the number of coupons a company sent to their customers at the end of each year, where t represents the number of years since the end of **1998**, and $0 \leq t \leq 5$. If $y = f(t)$ is graphed in the ty -plane, which of the following is the best interpretation of the y -intercept of the graph in this context?

- A. The minimum estimated number of coupons the company sent to their customers during the **5** years was **1,428**.
- B. The minimum estimated number of coupons the company sent to their customers during the **5** years was **8,000**.
- C. The estimated number of coupons the company sent to their customers at the end of **1998** was **1,428**.
- D. The estimated number of coupons the company sent to their customers at the end of **1998** was **8,000**.

ID: 40d75cea Answer

Correct Answer: D

Rationale

Choice D is correct. The y -intercept of a graph in the ty -plane is the point where $t = 0$. For the given function f , the y -intercept of the graph of $y = f(t)$ in the ty -plane can be found by substituting **0** for t in the equation $y = 8,000(0.65)^t$, which gives $y = 8,000(0.65)^0$. This is equivalent to $y = 8,000(1)$, or $y = 8,000$. Therefore, the y -intercept of the graph of $y = f(t)$ is **(0, 8,000)**. It's given that the function f models the number of coupons a company sent to their customers at the end of each year. Therefore, $f(t)$ represents the estimated number of coupons the company sent to their customers at the end of each year. It's also given that t represents the number of years since the end of **1998**. Therefore, $t = 0$ represents **0** years since the end of **1998**, or the end of **1998**. Thus, the best interpretation of the y -intercept of the graph of $y = f(t)$ is that the estimated number of coupons the company sent to their customers at the end of **1998** was **8,000**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID b8fa47e9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: b8fa47e9

$k^2 - 53 = 91$

What is the positive solution to the given equation?

- A. 144
- B. 72
- C. 38
- D. 12

ID: b8fa47e9 Answer

Correct Answer: D

Rationale

Choice D is correct. Adding **53** to each side of the given equation yields $k^2 = 144$. Taking the square root of each side of this equation yields $k = \pm 12$. Therefore, the positive solution to the given equation is **12**.

Choice A is incorrect. This is the positive solution to the equation $k^2 - 53 = 20,683$, not $k^2 - 53 = 91$.

Choice B is incorrect. This is the positive solution to the equation $k^2 - 53 = 5,131$, not $k^2 - 53 = 91$.

Choice C is incorrect. This is the positive solution to the equation $k^2 - 53 = 1,391$, not $k^2 - 53 = 91$.

Question Difficulty: Medium

Question ID eae04f9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: eae04f9

The function f is defined by $f(x) = a\sqrt{x + b}$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ passes through the point $(-24, 0)$, and $f(24) < 0$. Which of the following must be true?

- A. $f(0) = 24$
- B. $f(0) = -24$
- C. $a > b$
- D. $a < b$

ID: eae04f9 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that $f(24) < 0$. Substituting 24 for $f(x)$ in the equation $f(x) = a\sqrt{x + b}$ yields $f(24) = a\sqrt{24 + b}$. Therefore, $a\sqrt{24 + b} < 0$. Since $\sqrt{24 + b}$ can't be negative, it follows that $a < 0$. It's also given that the graph of $y = f(x)$ passes through the point $(-24, 0)$. It follows that when $x = -24$, $f(x) = 0$. Substituting -24 for x and 0 for $f(x)$ in the equation $f(x) = a\sqrt{x + b}$ yields $0 = a\sqrt{-24 + b}$. By the zero product property, either $a = 0$ or $\sqrt{-24 + b} = 0$. Since $a < 0$, it follows that $\sqrt{-24 + b} = 0$. Squaring both sides of this equation yields $-24 + b = 0$. Adding 24 to both sides of this equation yields $b = 24$. Since $a < 0$ and b is 24 , it follows that $a < b$ must be true.

Choice A is incorrect. The value of $f(0)$ is $a\sqrt{b}$, which must be negative.

Choice B is incorrect. The value of $f(0)$ is $a\sqrt{b}$, which could be -24 , but doesn't have to be.

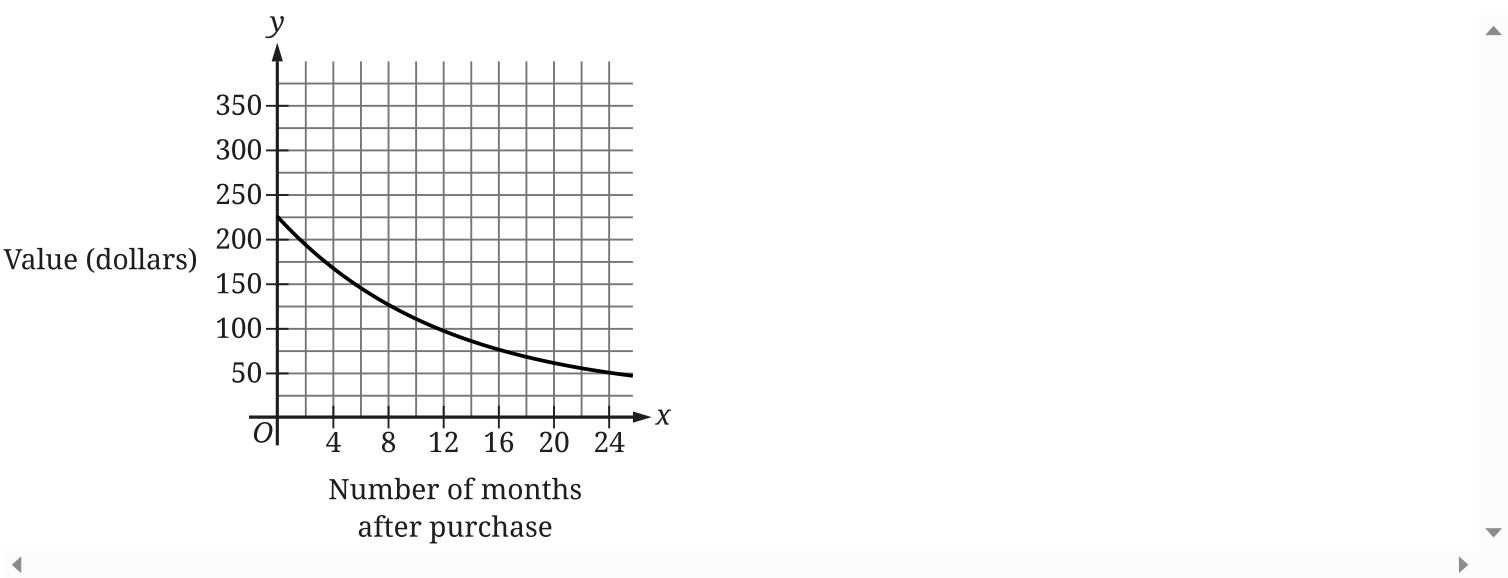
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 80850e93

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 80850e93



The graph shown gives the estimated value, in dollars, of a tablet as a function of the number of months since it was purchased. What is the best interpretation of the y -intercept of the graph in this context?

- A. The estimated value of the tablet was \$225 when it was purchased.
- B. The estimated value of the tablet 24 months after it was purchased was \$225.
- C. The estimated value of the tablet had decreased by \$225 in the 24 months after it was purchased.
- D. The estimated value of the tablet decreased by approximately 2.25% each year after it was purchased.

ID: 80850e93 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the graph shown gives the estimated value y , in dollars, of a tablet as a function of the number of months since it was purchased, x . The y -intercept of a graph is the point at which the graph intersects the y -axis, or when x is 0. The graph shown intersects the y -axis at the point (0, 225). It follows that 0 months after the tablet was purchased, or when the tablet was purchased, the estimated value of the tablet was 225 dollars. Therefore, the best interpretation of the y -intercept is that the estimated value of the tablet was \$225 when it was purchased.

Choice B is incorrect. The estimated value of the tablet 24 months after it was purchased was \$50, not \$225.

Choice C is incorrect. The estimated value of the tablet had decreased by \$225 – \$50, or \$175, not \$225, in the 24 months after it was purchased.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 3d8c1891

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 3d8c1891

$$-16x^2 - 8x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

ID: 3d8c1891 Answer

Correct Answer: -1

Rationale

The correct answer is -1 . A quadratic equation in the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has exactly one solution when its discriminant, $b^2 - 4ac$, is equal to 0 . In the given equation, $-16x^2 - 8x + c = 0$, $a = -16$ and $b = -8$. Substituting -16 for a and -8 for b in $b^2 - 4ac$ yields $(-8)^2 - 4(-16)(c)$, or $64 + 64c$. Since the given equation has exactly one solution, $64 + 64c = 0$. Subtracting 64 from both sides of this equation yields $64c = -64$. Dividing both sides of this equation by 64 yields $c = -1$. Therefore, the value of c is -1 .

Question Difficulty: Hard

Question ID b589168d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: b589168d

The exponential function g is defined by $g(x) = 19 \cdot a^x$, where a is a positive constant. If $g(3) = 2,375$, what is the value of $g(4)$?

ID: b589168d Answer

Correct Answer: 11875

Rationale

The correct answer is **11,875**. It's given that the exponential function g is defined by $g(x) = 19 \cdot a^x$, where a is a positive constant, and $g(3) = 2,375$. It follows that when $x = 3$, $g(x) = 2,375$. Substituting **3** for x and **2,375** for $g(x)$ in the given equation yields $2,375 = 19 \cdot a^3$. Dividing each side of this equation by **19** yields $125 = a^3$. Taking the cube root of both sides of this equation gives $a = 5$. Substituting **4** for x and **5** for a in the equation $g(x) = 19 \cdot a^x$ yields $g(4) = 19 \cdot 5^4$, or $g(4) = 11,875$. Therefore, the value of $g(4)$ is **11,875**.

Question Difficulty: Hard

Question ID 91570fae

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 91570fae

The expression $(3x - 23)(19x + 6)$ is equivalent to the expression $ax^2 + bx + c$, where a , b , and c are constants. What is the value of b ?

ID: 91570fae Answer

Correct Answer: -419

Rationale

The correct answer is -419 . It's given that the expression $(3x - 23)(19x + 6)$ is equivalent to the expression $ax^2 + bx + c$, where a , b , and c are constants. Applying the distributive property to the given expression, $(3x - 23)(19x + 6)$, yields $(3x)(19x) + (3x)(6) - (23)(19x) - (23)(6)$, which can be rewritten as $57x^2 + 18x - 437x - 138$. Combining like terms yields $57x^2 - 419x - 138$. Since this expression is equivalent to $ax^2 + bx + c$, it follows that the value of b is -419 .

Question Difficulty: Hard

Question ID 6f8f1208

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 6f8f1208

Which expression is equivalent to $13x^2 - 7x^2$?

- A. $-91x^2$
- B. $6x^2$
- C. $20x^2$
- D. $40x^2$

ID: 6f8f1208 Answer

Correct Answer: B

Rationale

Choice B is correct. Since each term in the given expression has a common factor of x^2 , it can be rewritten as $(13 - 7)x^2$, or $6x^2$. Therefore, the expression $6x^2$ is equivalent to $13x^2 - 7x^2$.

Alternate approach: Since the two terms of the given expression are both constant multiples of x^2 , they are like terms and can, therefore, be combined through subtraction. Subtracting like terms in the expression $13x^2 - 7x^2$ yields $6x^2$.

Choice A is incorrect. This expression is equivalent to $(13x)(-7x)$, not $13x^2 - 7x^2$.

Choice C is incorrect. This expression is equivalent to $13x^2 + 7x^2$, not $13x^2 - 7x^2$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID db59f2eb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: db59f2eb

Which expression is equivalent to $12x^3 - 5x^3$?

- A. $7x^6$
- B. $17x^3$
- C. $7x^3$
- D. $17x^6$

ID: db59f2eb Answer

Correct Answer: C

Rationale

Choice C is correct. The given expression shows subtraction of two like terms. The two terms can be subtracted as follows: $12x^3 - 5x^3 = (12 - 5)x^3$, or $7x^3$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the result of adding, not subtracting, the two like terms.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 38bdfb69

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 38bdfb69

$$y = 0.25x^2 - 7.5x + 90.25$$

The equation gives the estimated stock price y , in dollars, for a certain company x days after a new product launched, where $0 \leq x \leq 20$. Which statement is the best interpretation of $(x, y) = (1, 83)$ in this context?

- A. The company's estimated stock price increased **\$83** every day after the new product launched.
- B. The company's estimated stock price increased **\$1** every **83** days after the new product launched.
- C. **1** day after the new product launched, the company's estimated stock price is **\$83**.
- D. **83** days after the new product launched, the company's estimated stock price is **\$1**.

ID: 38bdfb69 Answer

Correct Answer: C

Rationale

Choice C is correct. In the given equation, x represents the number of days after a new product launched, where $0 \leq x \leq 20$, and y represents the estimated stock price, in dollars, for a certain company. Therefore, the best interpretation of $(x, y) = (1, 83)$ in this context is that **1** day after the new product launched, the company's estimated stock price is **\$83**.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 3f9633f7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 3f9633f7

If $a = 4k + 5r$ and $b = 7k - 12r + 3$, which expression is equivalent to $a - b$?

- A. $-3k + 17r + 3$
- B. $-3k + 17r - 3$
- C. $-3k - 7r - 3$
- D. $-3k - 7r + 3$

ID: 3f9633f7 Answer

Correct Answer: B

Rationale

Choice B is correct. Substituting $4k + 5r$ for a and $7k - 12r + 3$ for b in the expression $a - b$ yields $(4k + 5r) - (7k - 12r + 3)$. This expression can be rewritten as $(4k + 5r) + (-1)(7k - 12r + 3)$. By applying the distributive property, this expression can be rewritten as $4k + 5r - 7k + 12r - 3$, which is equivalent to $(4k - 7k) + (5r + 12r) - 3$, or $-3k + 17r - 3$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

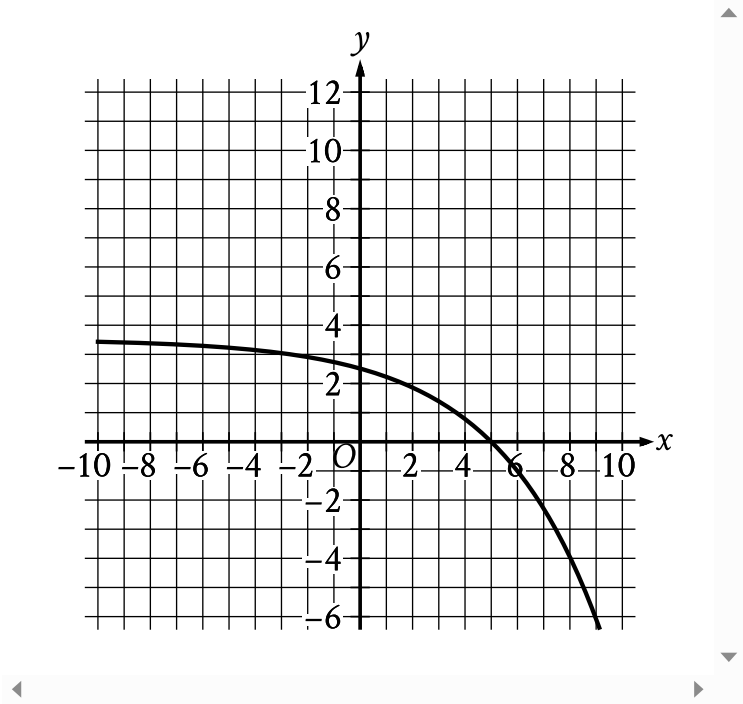
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 77b35575

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 77b35575



What is the x-intercept of the graph shown?

- A. $(-5, 0)$
- B. $(5, 0)$
- C. $(-2, 0)$
- D. $(2, 0)$

ID: 77b35575 Answer

Correct Answer: B

Rationale

Choice B is correct. An x -intercept of a graph in the xy -plane is a point at which the graph crosses the x -axis. The graph shown crosses the x -axis at the point $(5, 0)$. Therefore, the x -intercept of the graph shown is $(5, 0)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID b9413f0f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: b9413f0f

Which expression is equivalent to $2\left(x - \frac{9}{2}\right)(x + 8)$?

- A. $2x^2 - 72$
- B. $2x^2 + 7x - 72$
- C. $2x^2 - x - 36$
- D. $x^2 + 7x - 36$

ID: b9413f0f Answer

Correct Answer: B

Rationale

Choice B is correct. In the given expression, $2\left(x - \frac{9}{2}\right)$ can be rewritten as $2x - 2\left(\frac{9}{2}\right)$, or $2x - 9$. Applying the distributive property to the expression $(2x - 9)(x + 8)$ yields $2x(x + 8) - 9(x + 8)$, or $2x^2 + 16x - 9x - 72$. Combining like terms in this expression yields $2x^2 + 7x - 72$. Thus, $2\left(x - \frac{9}{2}\right)(x + 8)$ is equivalent to $2x^2 + 7x - 72$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

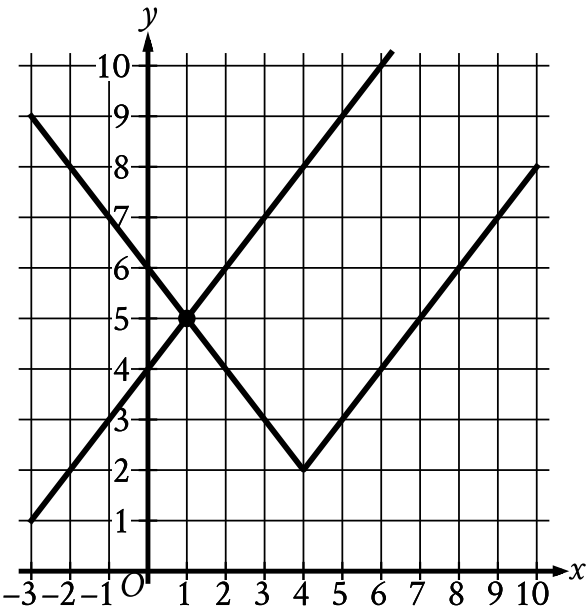
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID af691219

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: af691219



The graph of a system of an absolute value function and a linear function is shown. What is the solution (x, y) to this system of two equations?

- A. $(-1, 5)$
- B. $(0, 4)$
- C. $(1, 5)$
- D. $(4, 2)$

ID: af691219 Answer

Correct Answer: C

Rationale

Choice C is correct. The solution to the system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear function and the absolute value function shown intersect at the point $(1, 5)$. Thus, the solution to the system is $(1, 5)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the y-intercept of the graph of the linear function.

Choice D is incorrect. This is the vertex of the graph of the absolute value function.

Question ID f7474219

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: f7474219

Which expression is equivalent to $4(x^2 + 6)$?

- A. $4x^2 + 24$
- B. $4x^2 + 10$
- C. $4x^2 + 6$
- D. $4x^2 - 2$

ID: f7474219 Answer

Correct Answer: A

Rationale

Choice A is correct. The expression $4(x^2 + 6)$ can be rewritten as $4(x^2) + 4(6)$, which is equivalent to $4x^2 + 24$.

Choice B is incorrect. This expression is equivalent to $4(x^2 + \frac{5}{2})$, not $4(x^2 + 6)$.

Choice C is incorrect. This expression is equivalent to $4(x^2 + \frac{3}{2})$, not $4(x^2 + 6)$.

Choice D is incorrect. This expression is equivalent to $4(x^2 - \frac{1}{2})$, not $4(x^2 + 6)$.

Question Difficulty: Easy

Question ID 70cdc6d2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 70cdc6d2

A sample of a certain isotope takes **29** years to decay to half its original mass. The function $s(t) = 184(0.5)^{\frac{t}{29}}$ gives the approximate mass of this isotope, in grams, that remains t years after a **184**-gram sample starts to decay. Which statement is the best interpretation of $s(87) = 23$ in this context?

- A. Approximately **23** grams of the sample remains **87** years after the sample starts to decay.
- B. The mass of the sample has decreased by approximately **23** grams **87** years after the sample starts to decay.
- C. The mass of the sample has decreased by approximately **87** grams **23** years after the sample starts to decay.
- D. Approximately **87** grams of the sample remains **23** years after the sample starts to decay.

ID: 70cdc6d2 Answer

Correct Answer: A

Rationale

Choice A is correct. In the given function, $s(t)$ represents the approximate mass, in grams, of the sample that remains t years after the sample starts to decay. It follows that the best interpretation of $s(87) = 23$ is that approximately **23** grams of the sample remains **87** years after the sample starts to decay.

Choice B is incorrect. The mass of the sample has decreased by approximately $184 - 23$, or **161**, grams, not **23** grams, **87** years after the sample starts to decay.

Choice C is incorrect. The mass of the sample has decreased by approximately **78** grams, not **87** grams, **23** years after the sample starts to decay.

Choice D is incorrect. This would be the best interpretation of $s(23) = 87$, not $s(87) = 23$.

Question Difficulty: Medium

Question ID 4bb4cfa1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 4bb4cfa1

The given expression can be rewritten as $\frac{a}{6}x^2 + \frac{b}{6}x + \frac{c}{6}$, where a , b , and c are constants. What is the value of $a + b + c$?

- A. -612
- B. -306
- C. -102
- D. -17

ID: 4bb4cfa1 Answer

Correct Answer: A

Rationale

Choice A is correct. Expanding each squared binomial in the given expression yields $-14(25x^2 - 20x + 4) + 6(25x^2 - 30x + 9)$. Applying the distributive property to both terms of this expression yields $(-350x^2 + 280x - 56) + (150x^2 - 180x + 54)$. Combining like terms in this expression yields $-200x^2 + 100x - 2$. Since the given expression can be rewritten as $\frac{a}{6}x^2 + \frac{b}{6}x + \frac{c}{6}$, it follows that $\frac{a}{6} = -200$, $\frac{b}{6} = 100$, and $\frac{c}{6} = -2$. Multiplying both sides of each of these equations by 6 yields $a = -1,200$, $b = 600$, and $c = -12$, respectively. Substituting $-1,200$ for a , 600 for b , and -12 for c in the expression $a + b + c$ yields $-1,200 + 600 - 12$, or -612 .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $\frac{a}{6} + \frac{b}{6} + \frac{c}{6}$, not $a + b + c$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID c95fef9c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: c95fef9c

The function f is defined by $f(x) = x^3 + 15$. What is the value of $f(2)$?

ID: c95fef9c Answer

Correct Answer: 23

Rationale

The correct answer is **23**. The value of $f(2)$ is the value of $f(x)$ when $x = 2$. The function f is defined by $f(x) = x^3 + 15$. Substituting **2** for x in this equation yields $f(2) = 2^3 + 15$. This equation can be rewritten as $f(2) = 8 + 15$, or $f(2) = 23$. Therefore, the value of $f(2)$ is **23**.

Question Difficulty: Medium

Question ID 5d4df2fd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 5d4df2fd

$$y = 9\left(\frac{a}{7}\right)^{x+c} - b$$

How many times does the graph of the given equation in the xy -plane cross the x -axis, where a , b , and c are positive constants such that $a > 7$ and $b > c$?

- A. Zero
- B. One
- C. Two
- D. Three

ID: 5d4df2fd Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that $y = 9\left(\frac{a}{7}\right)^{x+c} - b$, where a , b , and c are positive constants such that $a > 7$ and $b > c$. The graph of any equation in the xy -plane crosses the x -axis when there is a point on the graph for which the value of y is equal to 0, and the value of y changes sign when the graph passes through that point. Since $a > 7$, the base $\frac{a}{7}$ is greater than 1, so the expression $9\left(\frac{a}{7}\right)^{x+c} - b$ increases as x increases. Since b is positive, there is a point on the graph of $y = 9\left(\frac{a}{7}\right)^{x+c} - b$ for which the value of y is equal to 0. When the graph passes through this point, the value of y changes sign from negative to positive. Therefore, the graph of the given equation crosses the x -axis exactly one time.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 0ac483b4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 0ac483b4

$$P = N(19 - C)$$

The given equation relates the positive numbers P , N , and C . Which equation correctly expresses C in terms of P and N ?

- A. $C = \frac{19 + P}{N}$
- B. $C = \frac{19 - P}{N}$
- C. $C = 19 + \frac{P}{N}$
- D. $C = 19 - \frac{P}{N}$

ID: 0ac483b4 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the values of P , N , and C are positive. Therefore, dividing each side of the given equation by N yields $\frac{P}{N} = 19 - C$. Subtracting 19 from each side of this equation yields $\frac{P}{N} - 19 = -C$. Dividing each side of this equation by -1 yields $19 - \frac{P}{N} = C$, or $C = 19 - \frac{P}{N}$.

Choice A is incorrect. This equation is equivalent to $P = NC - 19$, not $P = N(19 - C)$.

Choice B is incorrect. This equation is equivalent to $P = 19 - NC$, not $P = N(19 - C)$.

Choice C is incorrect. This equation is equivalent to $P = N(C - 19)$, not $P = N(19 - C)$.

Question Difficulty: Hard

Question ID 077b2830

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 077b2830

A scientist initially measures **12,000** bacteria in a growth medium. **4** hours later, the scientist measures **24,000** bacteria. Assuming exponential growth, the formula $P = C(2)^{rt}$ gives the number of bacteria in the growth medium, where r and C are constants and P is the number of bacteria t hours after the initial measurement. What is the value of r ?

- A. $\frac{1}{12,000}$
- B. $\frac{1}{4}$
- C. **4**
- D. **12,000**

ID: 077b2830 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that the formula $P = C(2)^{rt}$ gives the number of bacteria in a growth medium, where r and C are constants and P is the number of bacteria t hours after the initial measurement. It’s also given that a scientist initially measures **12,000** bacteria in the growth medium. Since the initial measurement is **0** hours after the initial measurement, it follows that when $t = 0$, $P = 12,000$. Substituting **0** for t and **12,000** for P in the given equation yields $12,000 = C(2)^{r(0)}$, or $12,000 = C(2)^0$, which is equivalent to $12,000 = C$. It’s given that **4** hours later, the scientist measures **24,000** bacteria, or when $t = 4$, $P = 24,000$. Substituting **4** for t , **24,000** for P , and **12,000** for C in the given equation yields $24,000 = 12,000(2)^{4r}$. Dividing each side of this equation by **12,000** yields $2 = 2^{4r}$, or $2^1 = 2^{4r}$, which is equivalent to $1 = 4r$. Dividing both sides of this equation by **4** yields $\frac{1}{4} = r$. Therefore, the value of r is $\frac{1}{4}$.

Choice A is incorrect. This is the value of the reciprocal of C .

Choice C is incorrect. This is the value of the reciprocal of r .

Choice D is incorrect. This is the value of C .

Question Difficulty: Hard

Question ID db9e8c54

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: db9e8c54

A ball is dropped from an initial height of **22** feet and bounces off the ground repeatedly. The function h estimates that the maximum height reached after each time the ball hits the ground is **85%** of the maximum height reached after the previous time the ball hit the ground. Which equation defines h , where $h(n)$ is the estimated maximum height of the ball after it has hit the ground n times and n is a whole number greater than **1** and less than **10**?

- A. $h(n) = 22(0.22)^n$
- B. $h(n) = 22(0.85)^n$
- C. $h(n) = 85(0.22)^n$
- D. $h(n) = 85(0.85)^n$

ID: db9e8c54 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that for the function h , $h(n)$ is the estimated maximum height, in feet, of the ball after it has hit the ground n times. It's also given that the function h estimates that the maximum height reached after each time the ball hits the ground is **85%** of the maximum height reached after the previous time the ball hit the ground. It follows that h is a decreasing exponential function that can be written in the form $h(n) = a(\frac{p}{100})^n$, where a is the initial height, in feet, the ball was dropped from and the function estimates that the maximum height reached after each time the ball hits the ground is $p\%$ of the maximum height reached after the previous time the ball hit the ground. It's given that the ball is dropped from an initial height of **22** feet. Therefore, $a = 22$. Since the function h estimates that the maximum height reached after each time the ball hits the ground is **85%** of the maximum height reached after the previous time the ball hit the ground, $p = 85$. Substituting **22** for a and **85** for p in the equation $h(n) = a(\frac{p}{100})^n$ yields $h(n) = 22(\frac{85}{100})^n$, or $h(n) = 22(0.85)^n$.

Choice A is incorrect. This function estimates that the maximum height reached after each time the ball hits the ground is **22%**, not **85%**, of the maximum height reached after the previous time the ball hit the ground.

Choice C is incorrect. This function estimates that the ball is dropped from an initial height of **85** feet, not **22** feet, and that the maximum height reached after each time the ball hits the ground is **22%**, not **85%**, of the maximum height reached after the previous time the ball hit the ground.

Choice D is incorrect. This function estimates that the ball is dropped from an initial height of **85** feet, not **22** feet.

Question Difficulty: Medium

Question ID 9f39eced

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 9f39eced

Which expression is equivalent to $6x^8y^2 + 12x^2y^2$?

- A. $6x^2y^2(2x^6)$
- B. $6x^2y^2(x^4)$
- C. $6x^2y^2(x^6 + 2)$
- D. $6x^2y^2(x^4 + 2)$

ID: 9f39eced Answer

Correct Answer: C

Rationale

Choice C is correct. Since each term of the given expression has a common factor of $6x^2y^2$, it may be rewritten as $6x^2y^2(x^6) + 6x^2y^2(2)$, or $6x^2y^2(x^6 + 2)$.

Choice A is incorrect. This expression is equivalent to $12x^8y^2$, not $6x^8y^2 + 12x^2y^2$.

Choice B is incorrect. This expression is equivalent to $6x^6y^2$, not $6x^8y^2 + 12x^2y^2$.

Choice D is incorrect. This expression is equivalent to $6x^6y^2 + 12x^2y^2$, not $6x^8y^2 + 12x^2y^2$.

Question Difficulty: Hard

Question ID a6b17375

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: a6b17375

$f(t) = 55t - 2t^2$

The function f is defined by the given equation. The function g is defined by $g(t) = f(t) + 3$. Which expression represents the maximum value of $g(t)$?

- A. $3 + \left(\frac{55}{2}\right)^2$
- B. $3 + 2\left(\frac{55}{4}\right)^2$
- C. $3 - 2\left(\frac{55}{4}\right)^2$
- D. $3 - \left(\frac{55}{2}\right)^2$

ID: a6b17375 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that function g is defined by $g(t) = f(t) + 3$ and that $f(t) = 55t - 2t^2$. Substituting $55t - 2t^2$ for $f(t)$ in $g(t) = f(t) + 3$ yields $g(t) = 55t - 2t^2 + 3$, or $g(t) = -2t^2 + 55t + 3$. The maximum value of $g(t)$ can be found by completing the square to rewrite the equation defining g in the form $g(t) = a(t - h)^2 + k$, where the maximum value of the function is k , which occurs when $t = h$, and a is a negative constant. The equation $g(t) = -2t^2 + 55t + 3$ is equivalent to $g(t) = -2\left(t^2 - \frac{55}{2}t\right) + 3$, which can be rewritten as $g(t) = -2\left(t^2 - \frac{55}{2}t + \left(\frac{55}{4}\right)^2\right) + 3 + 2\left(\frac{55}{4}\right)^2$, or $g(t) = -2\left(t - \frac{55}{4}\right)^2 + 3 + 2\left(\frac{55}{4}\right)^2$. This equation is in the form $g(t) = a(t - h)^2 + k$, where $a = -2$, $h = \frac{55}{4}$, and $k = 3 + 2\left(\frac{55}{4}\right)^2$. Thus, the maximum value of $g(t)$ is $3 + 2\left(\frac{55}{4}\right)^2$.

Alternate approach: Since the function f is a quadratic function, the maximum value of $f(t)$ occurs at the value of t that's halfway between the two zeros of the function. The zeros of function f can be found by substituting 0 for $f(t)$ in the equation defining f , which yields $0 = 55t - 2t^2$. This equation can be rewritten as $0 = t(55 - 2t)$. By the zero product property, it follows that $t = 0$ or $55 - 2t = 0$. Subtracting 55 from each side of the equation $55 - 2t = 0$ yields $-2t = -55$. Dividing each side of this equation by -2 yields $t = \frac{55}{2}$. Therefore, the zeros of function f are 0 and $\frac{55}{2}$. The value that's halfway between 0 and $\frac{55}{2}$ can be found by calculating the average of 0 and $\frac{55}{2}$, which is $\frac{0 + \frac{55}{2}}{2}$, or $\frac{55}{4}$. It follows that the maximum of function f occurs when $t = \frac{55}{4}$. Substituting $\frac{55}{4}$ for t in the equation defining function f yields $f\left(\frac{55}{4}\right) = 55\left(\frac{55}{4}\right) - 2\left(\frac{55}{4}\right)^2$, which is equivalent to $f\left(\frac{55}{4}\right) = \frac{55^2}{4} - 2\left(\frac{55^2}{4^2}\right)$. Multiplying $\frac{55^2}{4}$ by $\frac{4}{4}$ in this equation to get a common denominator yields $f\left(\frac{55}{4}\right) = 4\left(\frac{55^2}{4^2}\right) - 2\left(\frac{55^2}{4^2}\right)$, or $f\left(\frac{55}{4}\right) = 2\left(\frac{55^2}{4^2}\right)$, which is equivalent to $f\left(\frac{55}{4}\right) = 2\left(\frac{55}{4}\right)^2$. Thus, the maximum value of $f(t)$ is $2\left(\frac{55}{4}\right)^2$. Since the equation defining $g(t)$ is $g(t) = f(t) + 3$, the maximum value of $g(t)$ is 3 greater than the maximum value of $f(t)$. It follows that the maximum value of $g(t)$ is $3 + 2\left(\frac{55}{4}\right)^2$.

- Choice A is incorrect and may result from conceptual or calculation errors.
- Choice C is incorrect and may result from conceptual or calculation errors.
- Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 64a7a257

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 64a7a257

$$\sqrt{k - x} = 58 - x$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

ID: 64a7a257 Answer

Correct Answer: 231

Rationale

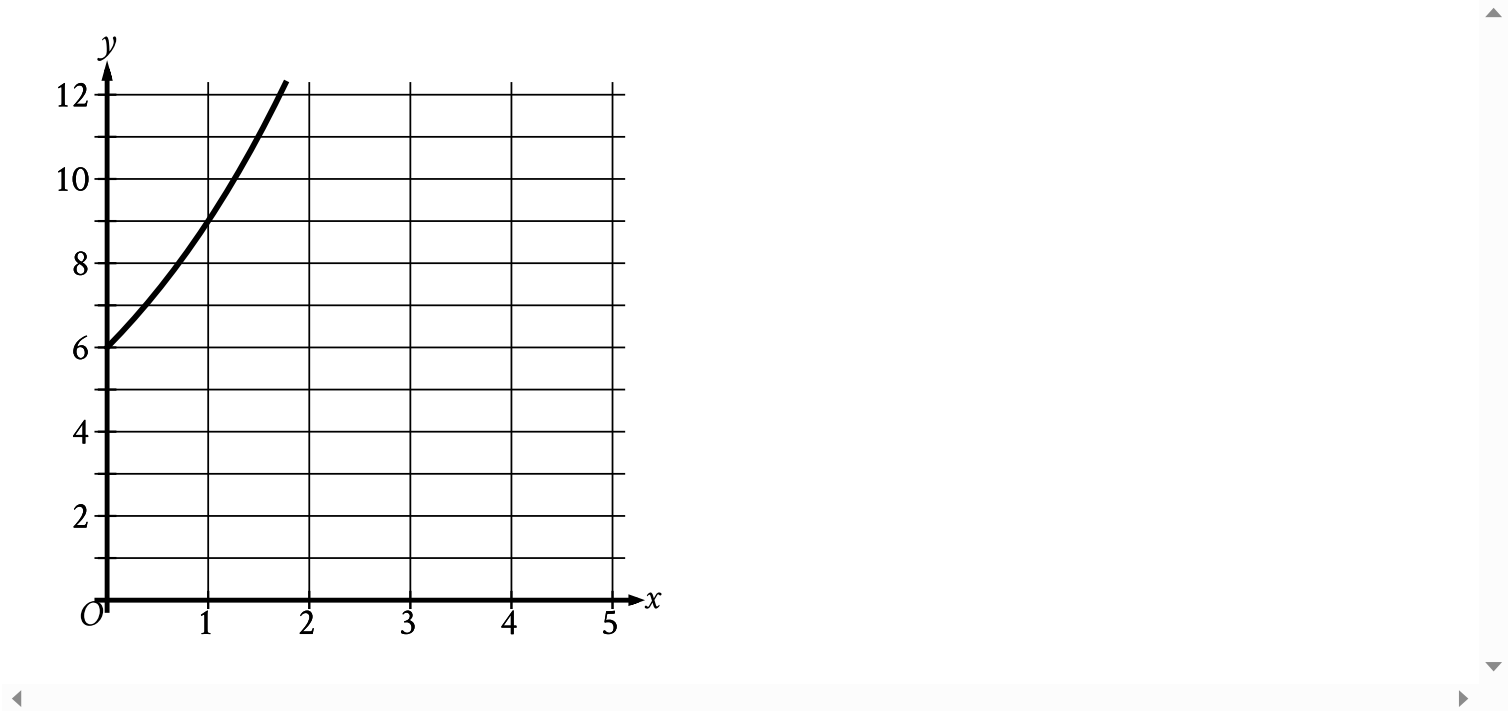
The correct answer is **231**. It's given that $\sqrt{k - x} = 58 - x$. Squaring both sides of this equation yields $k - x = (58 - x)^2$, which is equivalent to the given equation if $58 - x > 0$. It follows that if a solution to the equation $k - x = (58 - x)^2$ satisfies $58 - x > 0$, then it's also a solution to the given equation; if not, it's extraneous. The equation $k - x = (58 - x)^2$ can be rewritten as $k - x = 3,364 - 116x + x^2$. Adding x to both sides of this equation yields $k = x^2 - 115x + 3,364$. Subtracting k from both sides of this equation yields $0 = x^2 - 115x + (3,364 - k)$. The number of solutions to a quadratic equation in the form $0 = ax^2 + bx + c$, where a , b , and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. Substituting -115 for b , 1 for a , and $3,364 - k$ for c in $b^2 - 4ac$ yields $(-115)^2 - 4(1)(3,364 - k)$, or $4k - 231$. The equation $0 = x^2 - 115x + (3,364 - k)$ has exactly one real solution if the discriminant is equal to zero, or $4k - 231 = 0$. Subtracting **231** from both sides of this equation yields $4k = 231$. Dividing both sides of this equation by **4** yields $k = 57.75$. Therefore, if $k = 57.75$, then the equation $0 = x^2 - 115x + (3,364 - k)$ has exactly one real solution. Substituting **57.75** for k in this equation yields $0 = x^2 - 115x + (3,364 - 57.75)$, or $0 = x^2 - 115x + 3,306.25$, which is equivalent to $0 = (x - 57.5)^2$. Taking the square root of both sides of this equation yields $0 = x - 57.5$. Adding **57.5** to both sides of this equation yields **57.5** = x . To check whether this solution satisfies $58 - x > 0$, the solution, **57.5**, can be substituted for x in $58 - x > 0$, which yields $58 - 57.5 > 0$, or $0.5 > 0$. Since **0.5** is greater than **0**, it follows that if $k = 57.75$, or $4k = 231$, then the given equation has exactly one real solution. If $4k < 231$, then the discriminant, $4k - 231$, is negative and the given equation has no solutions. Therefore, the minimum possible value of $4k$ is **231**.

Question Difficulty: Hard

Question ID 2bd489b5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 2bd489b5



The graph gives the estimated population y , in thousands, of a town x years since 2003, where $0 \leq x \leq 5$. Which of the following best describes the increase in the estimated population from $x = 0$ to $x = 1$?

- A. The estimated population at $x = 1$ is 0.5 times the estimated population at $x = 0$.
- B. The estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.
- C. The estimated population at $x = 1$ is 2.5 times the estimated population at $x = 0$.
- D. The estimated population at $x = 1$ is 3.5 times the estimated population at $x = 0$.

ID: 2bd489b5 Answer

Correct Answer: B

Rationale

Choice B is correct. On the graph shown, the y -axis represents estimated population, in thousands. The graph shows that when $x = 0$, the y -coordinate is 6. Therefore, the estimated population at $x = 0$ is 6 thousand. The graph also shows that when $x = 1$, the y -coordinate is 9. Therefore, the estimated population at $x = 1$ is 9 thousand. Dividing 9 thousand by 6 thousand yields 1.5; therefore, 9 thousand is 1.5 times 6 thousand. It follows that the estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID 1efd2fad

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 1efd2fad

The number of cells of a certain bacteria doubles every **25** minutes during its growth period when placed in a medium of milk in a controlled laboratory environment. If a sample of **830** cells of this bacteria is placed in milk in a controlled laboratory environment, which function f best represents the number of cells of the bacteria x minutes after the growth period begins, where x is less than the number of minutes in the growth period?

- A. $f(x) = 830\left(\frac{1}{2}\right)^{\frac{x}{25}}$
- B. $f(x) = 830(2)^{\frac{x}{25}}$
- C. $f(x) = 830\left(\frac{1}{3}\right)^{\frac{x}{25}}$
- D. $f(x) = 830(3)^{\frac{x}{25}}$

ID: 1efd2fad Answer

Correct Answer: B

Rationale

Choice B is correct. For a bacteria with an initial population of P cells and a doubling time of m minutes, the number of cells, $f(x)$, of the bacteria x minutes after the growth period begins can be given by the equation $f(x) = P(2)^{\frac{x}{m}}$. It's given that there are **830** cells of the bacteria when the growth period begins. Therefore, $P = 830$. It's also given that the number of cells of the bacteria doubles every **25** minutes during the growth period. Therefore, $m = 25$. Substituting **830** for P and **25** for m in the equation $f(x) = P(2)^{\frac{x}{m}}$ yields $f(x) = 830(2)^{\frac{x}{25}}$.

Choice A is incorrect. This function represents a situation where the number of cells decreases by half, rather than doubles, every **25** minutes.

Choice C is incorrect. This function represents a situation where the number of cells decreases by two-thirds, rather than doubles, every **25** minutes.

Choice D is incorrect. This function represents a situation where the number of cells triples, rather than doubles, every **25** minutes.

Question Difficulty: Medium

Question ID 44c23cf1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 44c23cf1

The function $f(w) = 6w^2$ gives the area of a rectangle, **in square feet (ft^2)**, if its width is w **ft** and its length is 6 times its width. Which of the following is the best interpretation of $f(14) = 1,176$?

- A. If the width of the rectangle is **14 ft**, then the area of the rectangle is **1,176 ft^2** .
- B. If the width of the rectangle is **14 ft**, then the length of the rectangle is **1,176 ft**.
- C. If the width of the rectangle is **1,176 ft**, then the length of the rectangle is **14 ft**.
- D. If the width of the rectangle is **1,176 ft**, then the area of the rectangle is **14 ft^2** .

ID: 44c23cf1 Answer

Correct Answer: A

Rationale

Choice A is correct. The function f gives the area of the rectangle, in ft^2 , if its width is w **ft**. Since the value of $f(14)$ is the value of $f(w)$ if $w = 14$, it follows that $f(14) = 1,176$ means that $f(w)$ is 1,176 if $w = 14$. In the given context, this means that if the width of the rectangle is **14 ft**, then the area of the rectangle is **1,176 ft^2** .

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from interpreting $f(w)$ as the width, in **ft**, of the rectangle if its area is w ft^2 , rather than as the area, in ft^2 , of the rectangle if its width is w **ft**.

Question Difficulty: Medium

Question ID 1d367f3e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 1d367f3e

$$\frac{55}{x+6} = x$$

What is the positive solution to the given equation?

ID: 1d367f3e Answer

Correct Answer: 5

Rationale

The correct answer is **5**. Multiplying both sides of the given equation by $x + 6$ results in $55 = x(x + 6)$. Applying the distributive property of multiplication to the right-hand side of this equation results in $55 = x^2 + 6x$. Subtracting **55** from both sides of this equation results in $0 = x^2 + 6x - 55$. The right-hand side of this equation can be rewritten by factoring. The two values that multiply to -55 and add to **6** are **11** and -5 . It follows that the equation $0 = x^2 + 6x - 55$ can be rewritten as $0 = (x + 11)(x - 5)$. Setting each factor equal to **0** yields two equations: $x + 11 = 0$ and $x - 5 = 0$. Subtracting **11** from both sides of the equation $x + 11 = 0$ results in $x = -11$. Adding **5** to both sides of the equation $x - 5 = 0$ results in $x = 5$. Therefore, the positive solution to the given equation is **5**.

Question Difficulty: Medium

Question ID 99a754e3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 99a754e3

$$f(x) = (x - 1)(x + 3)(x - 2)$$

In the xy -plane, when the graph of the function f , where $y = f(x)$, is shifted up 6 units, the resulting graph is defined by the function g . If the graph of $y = g(x)$ crosses through the point $(4, b)$, where b is a constant, what is the value of b ?

ID: 99a754e3 Answer

Correct Answer: 48

Rationale

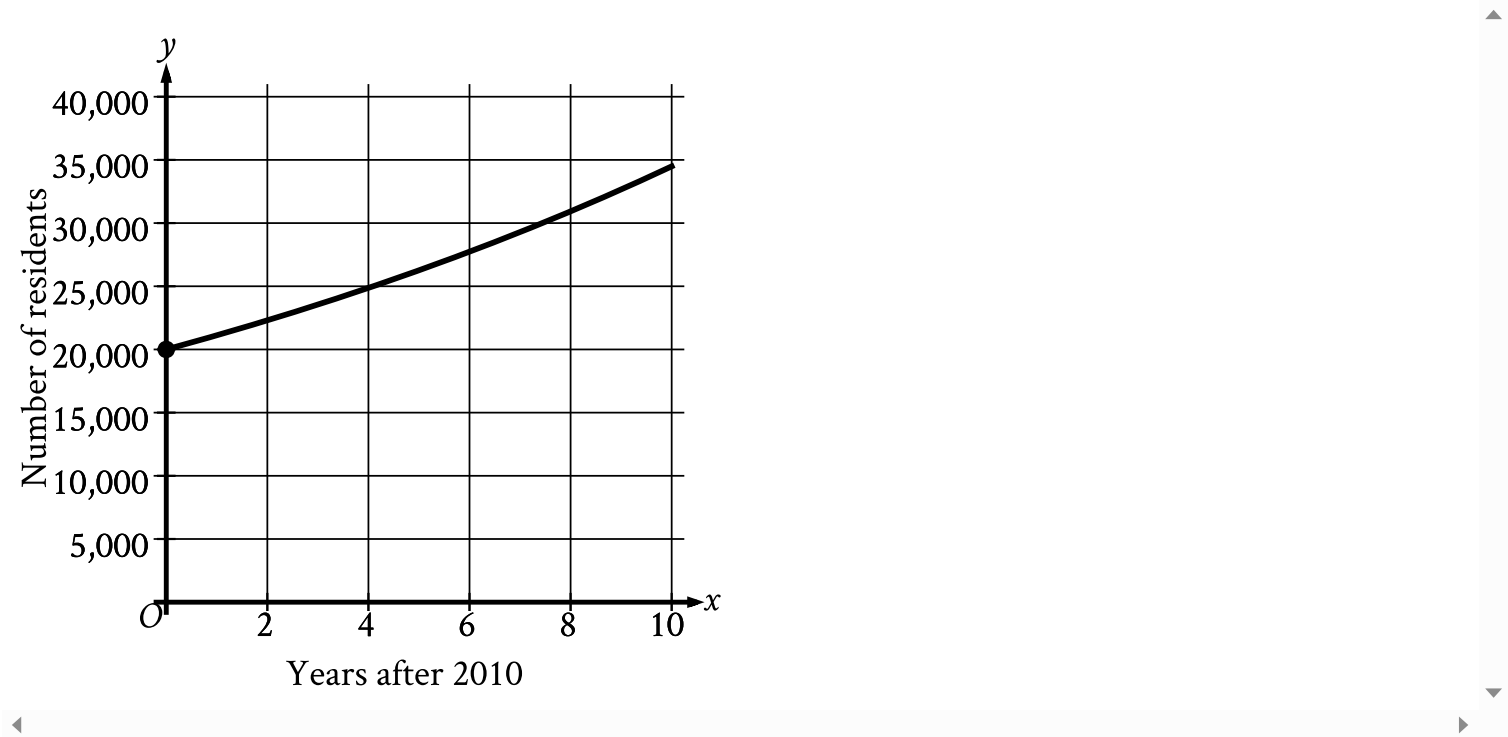
The correct answer is **48**. It's given that in the xy -plane, when the graph of the function f , where $y = f(x)$, is shifted up 6 units, the resulting graph is defined by the function g . Therefore, function g can be defined by the equation $g(x) = f(x) + 6$. It's given that $f(x) = (x - 1)(x + 3)(x - 2)$. Substituting $(x - 1)(x + 3)(x - 2)$ for $f(x)$ in the equation $g(x) = f(x) + 6$ yields $g(x) = (x - 1)(x + 3)(x - 2) + 6$. For the point $(4, b)$, the value of x is 4 . Substituting 4 for x in the equation $g(x) = (x - 1)(x + 3)(x - 2) + 6$ yields $g(4) = (4 - 1)(4 + 3)(4 - 2) + 6$, or $g(4) = 48$. It follows that the graph of $y = g(x)$ crosses through the point $(4, 48)$. Therefore, the value of b is **48**.

Question Difficulty: Hard

Question ID ecf4620a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: ecf4620a



The graph shown models the number of residents of a certain city x years after **2010**. How many residents does this model estimate the city had in **2010**?

- A. 0
- B. 2,000
- C. 20,000
- D. 25,000

ID: ecf4620a Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that x represents years after **2010**. Therefore, **2010** is represented by $x = 0$. On the model shown, the point with an x -coordinate of **0** has a y -coordinate of **20,000**. Thus, the model estimates that in **2010**, the city had **20,000** residents.

Choice A is incorrect. This is the value of x that represents the year **2010**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is approximately the number of residents the model estimates the city had in **2014**, not **2010**.

Question Difficulty: Easy

Question ID 8f7cb906

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 8f7cb906

The expression $90y^5 - 54y^4$ is equivalent to $ry^4(15y - 9)$, where r is a constant. What is the value of r ?

ID: 8f7cb906 Answer

Correct Answer: 6

Rationale

The correct answer is **6**. Applying the distributive property to the expression $ry^4(15y - 9)$ yields $15ry^5 - 9ry^4$. Since $90y^5 - 54y^4$ is equivalent to $ry^4(15y - 9)$, it follows that $90y^5 - 54y^4$ is also equivalent to $15ry^5 - 9ry^4$. Since these expressions are equivalent, it follows that corresponding coefficients are equivalent. Therefore, $90 = 15r$ and $-54 = -9r$. Solving either of these equations for r will yield the value of r . Dividing both sides of $90 = 15r$ by 15 yields $6 = r$. Therefore, the value of r is **6**.

Question Difficulty: Medium

Question ID 2553455c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 2553455c

$$y = 3,600(a)^x$$

The given equation, where a is a positive constant, gives the predicted number of bacteria, y , in a growth medium x hours after the number of bacteria was initially measured. According to the equation, what was the predicted number of bacteria initially measured in the growth medium?

ID: 2553455c Answer

Correct Answer: 3600

Rationale

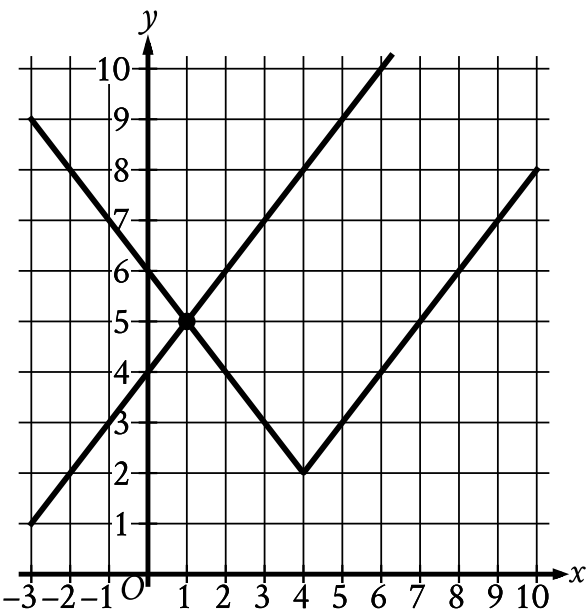
The correct answer is **3,600**. The predicted number of bacteria initially measured in the growth medium is the value of y when $x = 0$. Substituting **0** for x in the given equation yields $y = 3,600(a)^0$. Since a is positive, this equation is equivalent to $y = 3,600(1)$. Therefore, when $x = 0$, the value of y is **3,600**. Thus, the predicted number of bacteria initially measured in the growth medium is **3,600**.

Question Difficulty: Medium

Question ID af691219

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: af691219



The graph of a system of an absolute value function and a linear function is shown. What is the solution (x, y) to this system of two equations?

- A. $(-1, 5)$
- B. $(0, 4)$
- C. $(1, 5)$
- D. $(4, 2)$

ID: af691219 Answer

Correct Answer: C

Rationale

Choice C is correct. The solution to the system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear function and the absolute value function shown intersect at the point $(1, 5)$. Thus, the solution to the system is $(1, 5)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the y-intercept of the graph of the linear function.

Choice D is incorrect. This is the vertex of the graph of the absolute value function.

Question ID 3189516f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 3189516f

The function g is defined by $g(x) = x^2 + 9$. For which value of x is $g(x) = 25$?

- A. 4
- B. 5
- C. 9
- D. 13

ID: 3189516f Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $g(x) = x^2 + 9$. Substituting **25** for $g(x)$ in this equation yields $25 = x^2 + 9$. Subtracting **9** from both sides of this equation yields $16 = x^2$. Taking the square root of each side of this equation yields $x = \pm 4$. It follows that $g(x) = 25$ when the value of x is **4** or **−4**. Only **4** is listed among the choices.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID dc7115cf

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: dc7115cf

Which expression is equivalent to $9x^2 + 5x$?

- A. $x(9x + 5)$
- B. $5x(9x + 1)$
- C. $9x(x + 5)$
- D. $x^2(9x + 5)$

ID: dc7115cf Answer

Correct Answer: A

Rationale

Choice A is correct. Since x is a factor of each term in the given expression, the expression is equivalent to $x(9x) + x(5)$, or $x(9x + 5)$.

Choice B is incorrect. This expression is equivalent to $45x^2 + 5x$, not $9x^2 + 5x$.

Choice C is incorrect. This expression is equivalent to $9x^2 + 45x$, not $9x^2 + 5x$.

Choice D is incorrect. This expression is equivalent to $9x^3 + 5x^2$, not $9x^2 + 5x$.

Question Difficulty: Easy

Question ID 993dd564

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 993dd564

Which expression is equivalent to $(d - 6)(8d^2 - 3)$?

- A. $8d^3 - 14d^2 - 3d + 18$
- B. $8d^3 - 17d^2 + 48$
- C. $8d^3 - 48d^2 - 3d + 18$
- D. $8d^3 - 51d^2 + 48$

ID: 993dd564 Answer

Correct Answer: C

Rationale

Choice C is correct. Applying the distributive property to the given expression yields $d(8d^2 - 3) - 6(8d^2 - 3)$. Applying the distributive property once again to this expression yields $(d)(8d^2) + (d)(-3) + (-6)(8d^2) + (-6)(-3)$, or $8d^3 + (-3d) + (-48d^2) + 18$. This expression can be rewritten as $8d^3 - 48d^2 - 3d + 18$. Thus, $(d - 6)(8d^2 - 3)$ is equivalent to $8d^3 - 48d^2 - 3d + 18$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID fb8113d1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: fb8113d1

$$(9x^4 + 8x^3) - (7x^3 + 5x^2) - (3x^2 + x)$$

The given expression can be written in the form $9x^4 + x^3 + ax^2 - x$, where a is a constant. What is the value of a ?

- A. -8
- B. -2
- C. 2
- D. 8

ID: fb8113d1 Answer

Correct Answer: A

Rationale

Choice A is correct. The given expression is equivalent to $9x^4 + 8x^3 - 7x^3 - 5x^2 - 3x^2 - x$. By combining like terms, this can be written as $9x^4 + x^3 - 8x^2 - x$. This expression is in the form $9x^4 + x^3 + ax^2 - x$, where a is a constant. Therefore, the value of a is -8 .

Choice B is incorrect. This may result from adding $-5x^2$ and $3x^2$ rather than $-5x^2$ and $-3x^2$.

Choice C is incorrect. This may result from adding $5x^2$ and $-3x^2$ rather than $-5x^2$ and $-3x^2$.

Choice D is incorrect. This may result from adding $5x^2$ and $3x^2$ rather than $-5x^2$ and $-3x^2$.

Question Difficulty: Hard

Question ID a0228fe7

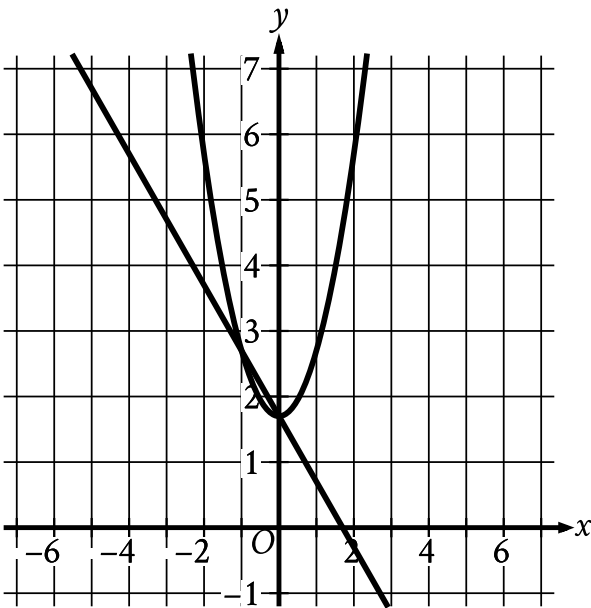
Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: a0228fe7

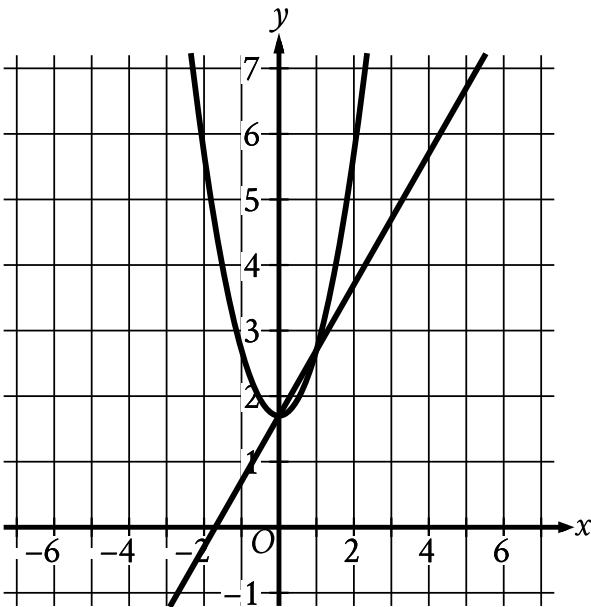
$y = x^2 + 1.7$
 $y = 1.7 - x$

Which graph represents the given system of equations?

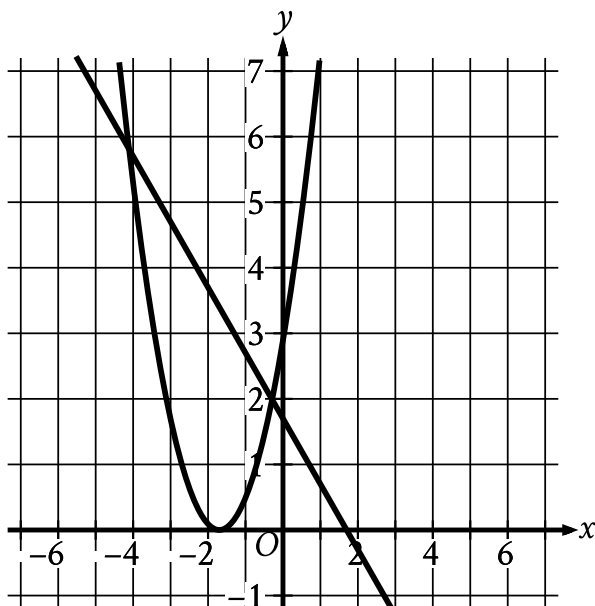
A.



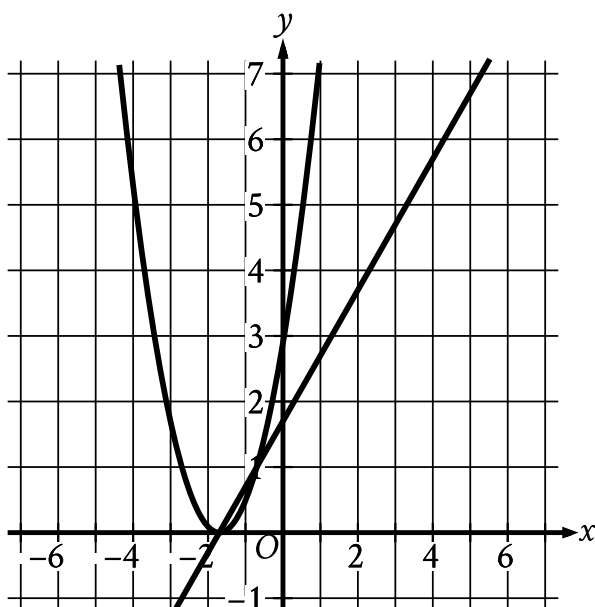
B.



C.



D.



ID: a0228fe7 Answer

Correct Answer: A

Rationale

Choice A is correct. The graph of a quadratic equation in the form $y = x^2 + c$ has its vertex at $(0, c)$. The first equation in the given system of equations is $y = x^2 + 1.7$, so the graph of this quadratic equation has its vertex at $(0, 1.7)$. The graph of a linear equation of the form $y = b - x$ has a slope of -1 and a y-intercept at $(0, b)$. The second equation in the given system of equations is $y = 1.7 - x$, so the graph of this linear equation has a slope of -1 and a y-intercept at $(0, 1.7)$. Of the choices, only choice A has the graph of a quadratic equation with its vertex at $(0, 1.7)$ and the graph of a linear equation with a slope of -1 and a y-intercept at $(0, 1.7)$.

Choice B is incorrect. This graph represents a system in which the second equation is $y = 1.7 + x$, not $y = 1.7 - x$.

Choice C is incorrect. This graph represents a system in which the first equation is $y = (x + 1.7)^2$, not $y = x^2 + 1.7$.

Choice D is incorrect. This graph represents a system in which the first equation is $y = (x + 1.7)^2$, not $y = x^2 + 1.7$, and the second equation is $y = 1.7 + x$, not $y = 1.7 - x$.

Question ID cfee492e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: cfee492e

$y = 576^{(2x+2)}$

The graph of the given equation in the xy -plane has a y -intercept of (r, s) . Which of the following equivalent equations displays the value of s as a constant, a coefficient, or the base?

- A. $y = 331,776^{(x+1)}$
- B. $y = 24^{(4x+4)}$
- C. $y = \frac{1}{24}(24)^{(4x+5)}$
- D. $y = \frac{1}{576}(576)^{(2x+3)}$

ID: cfee492e Answer

Correct Answer: A

Rationale

Choice A is correct. The y -intercept of a graph in the xy -plane is the point where $x = 0$. Substituting 0 for x in the given equation, $y = 576^{(2x+2)}$, yields $y = 576^{(2(0)+2)}$, which is equivalent to $y = 576^2$, or $y = 331,776$. Therefore, the graph of the given equation in the xy -plane has a y -intercept of $(0, 331,776)$. It follows that $r = 0$ and $s = 331,776$. Thus, the equivalent equation $y = 331,776^{(x+1)}$ displays the value of s as the base.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

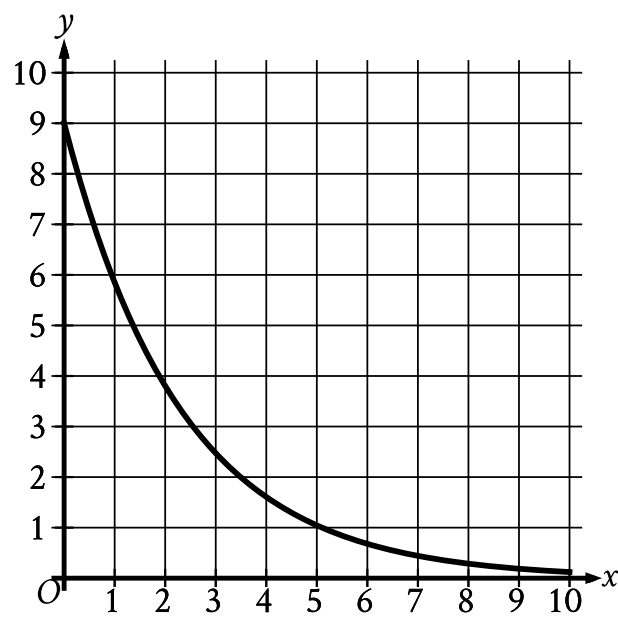
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 92280623

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 92280623



The graph gives the estimated number of catalogs y , in thousands, a company sent to its customers at the end of each year, where x represents the number of years since the end of **1992**, where $0 \leq x \leq 10$. Which statement is the best interpretation of the y -intercept in this context?

- A. The estimated total number of catalogs the company sent to its customers during the first **10** years was **9,000**.
- B. The estimated total number of catalogs the company sent to its customers from the end of **1992** to the end of **2002** was **90**.
- C. The estimated number of catalogs the company sent to its customers at the end of **1992** was **9**.
- D. The estimated number of catalogs the company sent to its customers at the end of **1992** was **9,000**.

ID: 92280623 Answer

Correct Answer: D

Rationale

Choice D is correct. The y -intercept of the graph is the point at which the graph crosses the y -axis, or the point for which the value of x is **0**. Therefore, the y -intercept of the given graph is the point **(0, 9)**. It's given that x represents the number of years since the end of **1992**. Therefore, $x = 0$ represents **0** years since the end of **1992**, which is the same as the end of **1992**. It's also given that y represents the estimated number of catalogs, in thousands, that the company sent to its customers at the end of the year. Therefore, $y = 9$ represents **9,000** catalogs. It follows that the y -intercept **(0, 9)** means that the estimated number of catalogs the company sent to its customers at the end of **1992** was **9,000**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

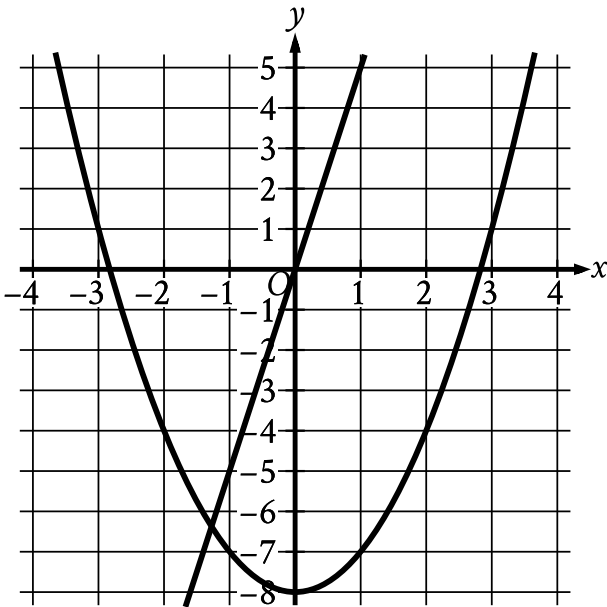
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 8c2b81ae

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 8c2b81ae



The graph of a system of a linear and a quadratic equation is shown. Which system of equations is represented by the graph?

- A. $y = -5x$
 $y = x^2 + 8$
- B. $y = -5x$
 $y = x^2 - 8$
- C. $y = 5x$
 $y = x^2 + 8$
- D. $y = 5x$
 $y = x^2 - 8$

ID: 8c2b81ae Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the graph shows a system consisting of a linear and a quadratic equation in the xy -plane. In each of the given choices, the linear equation is in the form $y = mx$, where m is a constant, and the quadratic equation is in the form $y = x^2 + c$, where c is a constant. The graph of a linear equation in the form $y = mx$ has a slope of m and a y -intercept at $(0, 0)$. Given two points on the line, (x_1, y_1) and (x_2, y_2) , the slope can be calculated using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The graph of the linear equation in this system has a y -intercept at $(0, 0)$ and passes through the point $(1, 5)$. Substituting the points $(0, 0)$ and $(1, 5)$ for (x_1, y_1) and (x_2, y_2) in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{5 - 0}{1 - 0}$, which is equivalent to $m = \frac{5}{1}$, or $m = 5$. It

follows that the linear equation is $y = 5x$. The graph of a quadratic equation in the form $y = x^2 + c$ has a vertex at $(0, c)$. The graph of the quadratic equation in this system has a vertex at $(0, -8)$. It follows that the quadratic equation is $y = x^2 + (-8)$, or $y = x^2 - 8$. Thus, the system of equations represented by the graph is $y = 5x$ and $y = x^2 - 8$.

Choice A is incorrect. The graph of this system of equations has a line with a slope of -5 , not 5 , and a parabola with a vertex at $(0, 8)$, not $(0, -8)$.

Choice B is incorrect. The graph of this system of equations has a line with a slope of -5 , not 5 .

Choice C is incorrect. The graph of this system of equations has a parabola with a vertex at $(0, 8)$, not $(0, -8)$.

Question Difficulty: Medium

Question ID ce925185

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: ce925185

$$\begin{aligned}y &= 4x \\ y &= x^2 - 12\end{aligned}$$

A solution to the given system of equations is (x, y) , where $x > 0$. What is the value of x ?

ID: ce925185 Answer

Correct Answer: 6

Rationale

The correct answer is **6**. It’s given that $y = 4x$ and $y = x^2 - 12$. Since $y = 4x$, substituting $4x$ for y in the second equation of the given system yields $4x = x^2 - 12$. Subtracting $4x$ from both sides of this equation yields $0 = x^2 - 4x - 12$. This equation can be rewritten as $0 = (x - 6)(x + 2)$. By the zero product property, $x - 6 = 0$ or $x + 2 = 0$. Adding **6** to both sides of the equation $x - 6 = 0$ yields $x = 6$. Subtracting **2** from both sides of the equation $x + 2 = 0$ yields $x = -2$. Therefore, solutions to the given system of equations occur when $x = 6$ and when $x = -2$. It’s given that a solution to the given system of equations is (x, y) , where $x > 0$. Since **6** is greater than **0**, it follows that the value of x is **6**.

Question Difficulty: Hard

Question ID 35fed7ec

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 35fed7ec

$x^2 = -841$

How many distinct real solutions does the given equation have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

ID: 35fed7ec Answer

Correct Answer: D

Rationale

Choice D is correct. Since the square of a real number is never negative, the given equation isn't true for any real value of x . Therefore, the given equation has zero distinct real solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 94c258fa

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 94c258fa

The function $f(x) = 240,000(1.22)^x$ gives a company’s predicted annual revenue, in dollars, x years after the company started selling jewelry online, where $0 < x \leq 10$. What is the best interpretation of the statement “ $f(5)$ is approximately equal to 648,650” in this context?

- A. 5 years after the company started selling jewelry online, its predicted annual revenue is approximately 648,650 dollars.
- B. 5 years after the company started selling jewelry online, its predicted annual revenue will have increased by a total of approximately 648,650 dollars.
- C. When the company’s predicted annual revenue is approximately 648,650 dollars, it is 5 times the predicted annual revenue for the previous year.
- D. When the company’s predicted annual revenue is approximately 648,650 dollars, it is 5% greater than the predicted annual revenue for the previous year.

ID: 94c258fa Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that the function f gives a company’s predicted annual revenue, in dollars, x years after the company started selling jewelry online. Since the value of $f(5)$ is the value of $f(x)$ when $x = 5$, it follows that “ $f(5)$ is approximately equal to 648,650” means that $f(x)$ is approximately equal to 648,650 when $x = 5$. Therefore, the best interpretation of the given statement is that 5 years after the company started selling jewelry online, its predicted annual revenue is approximately 648,650 dollars.

Choice B is incorrect. The function f gives the predicted annual revenue, not the predicted increase in annual revenue.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. In the given function, x represents the number of years after the company started selling jewelry online, not the percent increase in revenue from the previous year.

Question Difficulty: Medium

Question ID 6955a2ac

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 6955a2ac

$$(5x^3 - 3) - (-4x^3 + 8)$$

The given expression is equivalent to $bx^3 - 11$, where b is a constant. What is the value of b ?

ID: 6955a2ac Answer

Correct Answer: 9

Rationale

The correct answer is **9**. The given expression can be rewritten as $(5x^3 - 3) + (-1)(-4x^3 + 8)$. By applying the distributive property, this expression can be rewritten as $5x^3 - 3 + 4x^3 + (-8)$, which is equivalent to $(5x^3 + 4x^3) + (-3 + (-8))$. Adding like terms in this expression yields $9x^3 - 11$. Since it's given that $(5x^3 - 3) - (-4x^3 + 8)$ is equivalent to $bx^3 - 11$, it follows that $9x^3 - 11$ is equivalent to $bx^3 - 11$. Therefore, the coefficients of x^3 in these two expressions must be equivalent, and the value of b must be **9**.

Question Difficulty: Hard

Question ID 1a4b397c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 1a4b397c

$$64x^2 - (16a + 4b)x + ab = 0$$

In the given equation, a and b are positive constants. The sum of the solutions to the given equation is $k(4a + b)$, where k is a constant. What is the value of k ?

ID: 1a4b397c Answer

Correct Answer: .0625, 1/16

Rationale

The correct answer is $\frac{1}{16}$. Let p and q represent the solutions to the given equation. Then, the given equation can be rewritten as $64(x - p)(x - q) = 0$, or $64x^2 - 64(p + q)x + 64pq = 0$. Since this equation is equivalent to the given equation, it follows that $-(16a + 4b) = -64(p + q)$. Dividing both sides of this equation by -64 yields $\frac{16a + 4b}{64} = p + q$, or $\frac{1}{16}(4a + b) = p + q$. Therefore, the sum of the solutions to the given equation, $p + q$, is equal to $\frac{1}{16}(4a + b)$. Since it's given that the sum of the solutions to the given equation is $k(4a + b)$, where k is a constant, it follows that $k = \frac{1}{16}$. Note that 1/16, .0625, 0.062, and 0.063 are examples of ways to enter a correct answer.

Alternate approach: The given equation can be rewritten as $64x^2 - 4(4a + b)x + ab = 0$, where a and b are positive constants. Dividing both sides of this equation by 4 yields $16x^2 - (4a + b)x + \frac{ab}{4} = 0$. The solutions for a quadratic equation in the form $Ax^2 + Bx + C = 0$, where A , B , and C are constants, can be calculated using the quadratic formula, $x = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$ and $x = \frac{-B - \sqrt{B^2 - 4AC}}{2A}$. It follows that the sum of the solutions to a quadratic equation in the form $Ax^2 + Bx + C = 0$ is $\frac{-B + \sqrt{B^2 - 4AC}}{2A} + \frac{-B - \sqrt{B^2 - 4AC}}{2A}$, which can be rewritten as $\frac{-B + -B + \sqrt{B^2 - 4AC} - \sqrt{B^2 - 4AC}}{2A}$, which is equivalent to $\frac{-2B}{2A}$, or $-\frac{B}{A}$. In the equation $16x^2 - (4a + b)x + \frac{ab}{4} = 0$, $A = 16$, $B = -(4a + b)$, and $C = \frac{ab}{4}$. Substituting 16 for A and $-(4a + b)$ for B in $-\frac{B}{A}$ yields $-\frac{-(4a + b)}{16}$, which can be rewritten as $\frac{1}{16}(4a + b)$. Thus, the sum of the solutions to the given equation is $\frac{1}{16}(4a + b)$. Since it's given that the sum of the solutions to the given equation is $k(4a + b)$, where k is a constant, it follows that $k = \frac{1}{16}$.

Question Difficulty: Hard

Question ID 30ceb7c8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 30ceb7c8

$$f(x) = (x - 2)(x + 15)$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

ID: 30ceb7c8 Answer

Correct Answer: -13/2, -6.5

Rationale

The correct answer is $-\frac{13}{2}$. The value of x for which $f(x)$ reaches its minimum can be found by rewriting the given equation in the form $f(x) = (x - h)^2 + k$, where $f(x)$ reaches its minimum, k , when the value of x is h . The given equation, $f(x) = (x - 2)(x + 15)$, can be rewritten as $f(x) = x^2 + 13x - 30$. By completing the square, this equation can be rewritten as $f(x) = \left(x^2 + 13x + \left(\frac{13}{2}\right)^2\right) - 30 - \left(\frac{13}{2}\right)^2$, which is equivalent to $f(x) = \left(x + \frac{13}{2}\right)^2 - \frac{289}{4}$, or $f(x) = \left(x - \left(-\frac{13}{2}\right)\right)^2 - \frac{289}{4}$. Therefore, $f(x)$ reaches its minimum when the value of x is $-\frac{13}{2}$. Note that -13/2 and -6.5 are examples of ways to enter a correct answer.

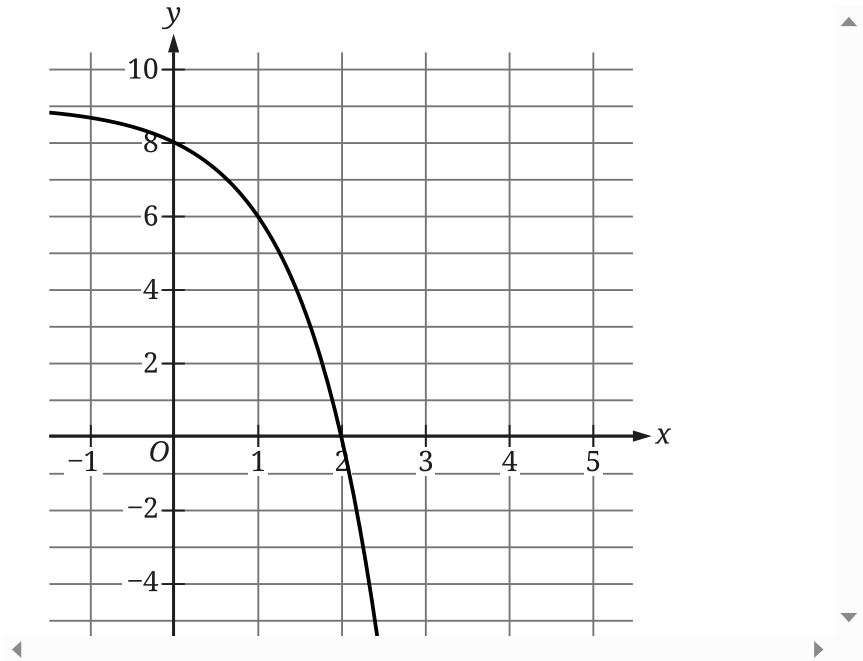
Alternate approach: The graph of $y = f(x)$ in the xy -plane is a parabola. The value of x for the vertex of a parabola is the x -value of the midpoint between the two x -intercepts of the parabola. Since it's given that $f(x) = (x - 2)(x + 15)$, it follows that the two x -intercepts of the graph of $y = f(x)$ in the xy -plane occur when $x = 2$ and $x = -15$, or at the points $(2, 0)$ and $(-15, 0)$. The midpoint between two points, (x_1, y_1) and (x_2, y_2) , is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$. Therefore, the midpoint between $(2, 0)$ and $(-15, 0)$ is $\left(\frac{2 + (-15)}{2}, \frac{0 + 0}{2}\right)$, or $\left(-\frac{13}{2}, 0\right)$. It follows that $f(x)$ reaches its minimum when the value of x is $-\frac{13}{2}$. Note that -13/2 and -6.5 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID c26d46ff

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: c26d46ff



The graph of $y = f(x) + 4$ is shown. Which equation defines function f ?

- A. $f(x) = -3^x + 1$
- B. $f(x) = -3^x + 5$
- C. $f(x) = -3^x + 8$
- D. $f(x) = -3^x + 9$

ID: c26d46ff Answer

Correct Answer: B

Rationale

Choice B is correct. The graph of $y = f(x) + 4$ is shown, and the graph passes through the point $(0, 8)$. Substituting 0 for x and 8 for y in this equation yields $8 = f(0) + 4$. Subtracting 4 from both sides of this equation yields $4 = f(0)$. Therefore, when $x = 0$, the value of $f(x)$ is 4 . Each of the given choices is in the form $f(x) = -3^x + k$, where k is a constant. Substituting 0 for x and 4 for $f(x)$ in this equation yields $4 = -3^0 + k$, or $4 = -1 + k$. Adding 1 to both sides of this equation yields $5 = k$. Substituting 5 for k in the equation $f(x) = -3^x + k$ yields $f(x) = -3^x + 5$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID bee2bb22

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: bee2bb22

Which of the following expressions is(are) a factor of $3x^2 + 20x - 63$?

- I. $x - 9$

II. $3x - 7$
- A. I only

B. II only

C. I and II

D. Neither I nor II

ID: bee2bb22 Answer

Correct Answer: B

Rationale

Choice B is correct. The given expression can be factored by first finding two values whose sum is **20** and whose product is **$3(-63)$** , or **-189** . Those two values are **27** and **-7** . It follows that the given expression can be rewritten as **$3x^2 + 27x - 7x - 63$** . Since the first two terms of this expression have a common factor of **$3x$** and the last two terms of this expression have a common factor of **-7** , this expression can be rewritten as **$3x(x + 9) - 7(x + 9)$** . Since the two terms of this expression have a common factor of **$(x + 9)$** , it can be rewritten as **$(3x - 7)(x + 9)$** . Therefore, expression II, **$3x - 7$** , is a factor of **$3x^2 + 20x - 63$** , but expression I, **$x - 9$** , is not a factor of **$3x^2 + 20x - 63$** .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

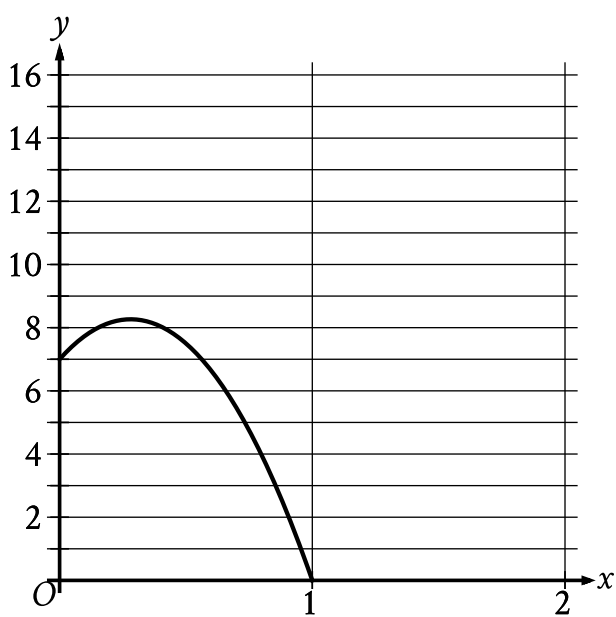
Question Difficulty: Hard

Question ID 4fe2a7ee

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

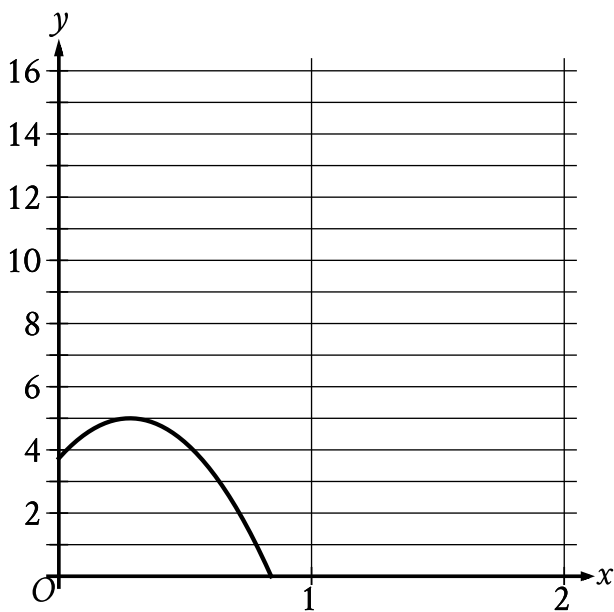
ID: 4fe2a7ee

During the first part of an experiment, a ball was launched from a **7-foot**-tall platform. The graph shows the height y , in feet, of the ball x seconds after it was launched during the first part of the experiment.

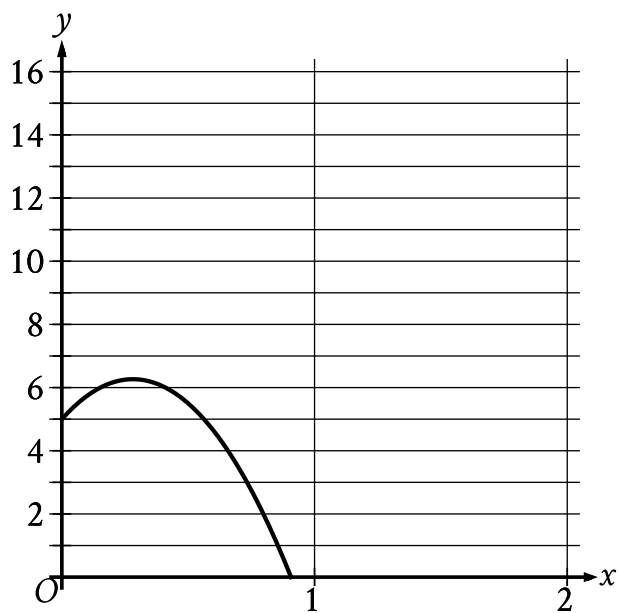


During the second part of the experiment, the ball was launched the same way, but from a platform that is **2** feet shorter than the first platform. Which of the following graphs could represent the height y , in feet, of the ball x seconds after it was launched during the second part of the experiment?

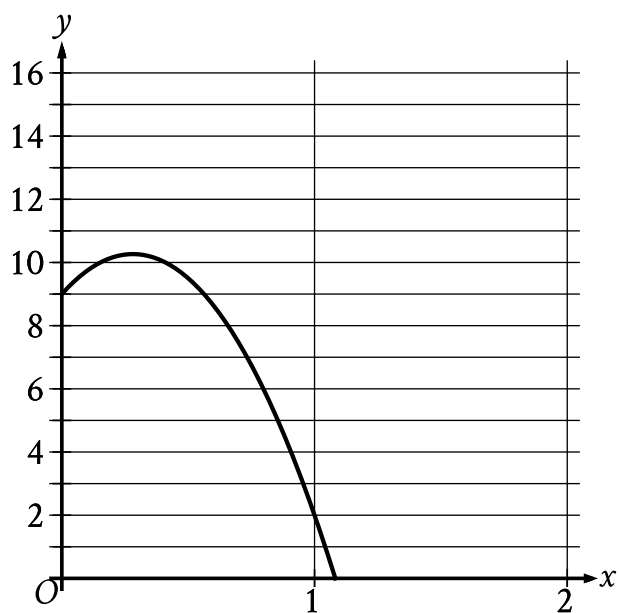
A.



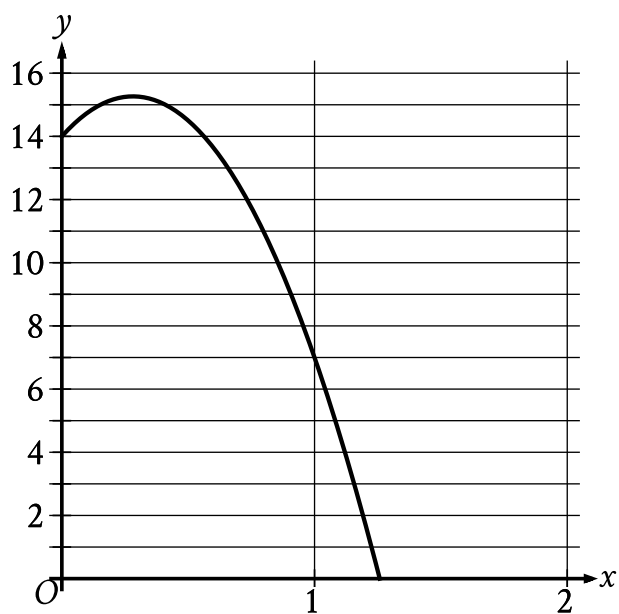
B.



C.



D.



Correct Answer: B

Rationale

Choice B is correct. It's given that y represents the height, in feet, of the ball x seconds after it was launched. It's also given that during the first part of an experiment, a ball was launched from a **7**-foot-tall platform. Therefore, the y -coordinate of the y -intercept of the given graph, **7**, represents the platform height, in feet. During the second part of the experiment, the platform the ball was launched from was **2** feet shorter than the platform in the first part of the experiment. It follows that the height of the platform in the second part of the experiment was $7 - 2$ feet, or **5** feet. Therefore, the y -coordinate of the y -intercept of the graph representing the second part of the experiment must be **5**. Only choice B satisfies this condition.

Choice A is incorrect. This could represent the graph if the ball were launched from a platform that was about **3** feet shorter rather than **2** feet shorter.

Choice C is incorrect. This could represent the graph if the ball were launched from a platform that was **2** feet taller rather than **2** feet shorter.

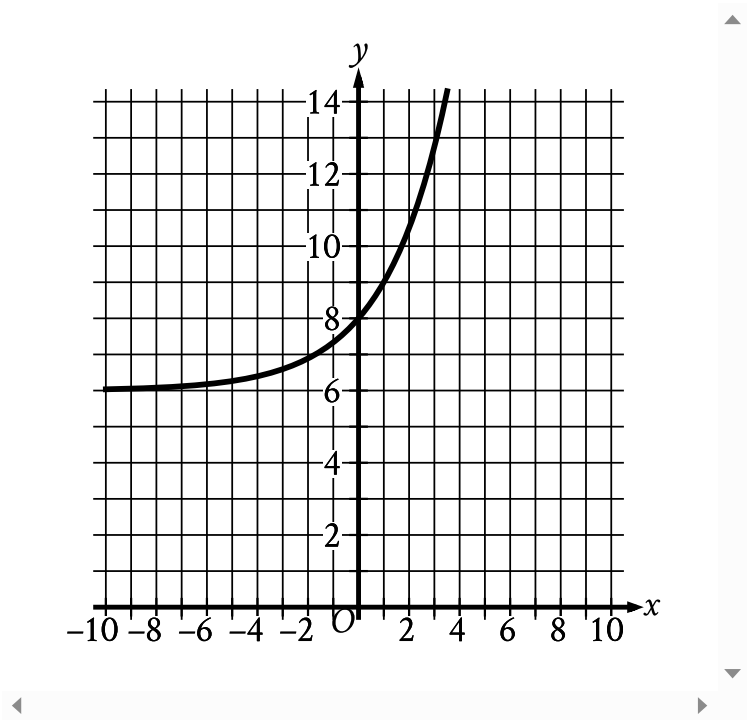
Choice D is incorrect. This could represent the graph if the ball were launched from a platform that was twice as tall rather than **2** feet shorter.

Question Difficulty: Easy

Question ID a2166248

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: a2166248



What is the y-intercept of the graph shown?

- A. $(-8, 0)$
- B. $(-6, 0)$
- C. $(0, 6)$
- D. $(0, 8)$

ID: a2166248 Answer

Correct Answer: D

Rationale

Choice D is correct. The y-intercept of a graph in the xy-plane is the point at which the graph crosses the y-axis. The graph shown crosses the y-axis at the point $(0, 8)$. Therefore, the y-intercept of the graph shown is $(0, 8)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 46e37d4b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 46e37d4b

$$y - 57 = px$$

The given equation relates the positive numbers p , x , and y . Which equation correctly expresses y in terms of p and x ?

- A. $y = 57x + p$
- B. $y = px + 57$
- C. $y = 57px$
- D. $y = \frac{px}{57}$

ID: 46e37d4b Answer

Correct Answer: B

Rationale

Choice B is correct. Adding 57 to each side of the given equation yields $y = px + 57$. Therefore, the equation $y = px + 57$ correctly expresses y in terms of p and x .

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID ad607d67

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: ad607d67

x	y
1	11
2	19
3	a

The table shows three values of x and their corresponding values of y for the equation $y = 4(2)^x + 3$. In the table, a is a constant. What is the value of a ?

- A. 67
- B. 35
- C. 32
- D. 27

ID: ad607d67 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the table shows three values of x and their corresponding values of y for the equation $y = 4(2)^x + 3$. It's also given that when $x = 3$ the corresponding value of y is a , and a is a constant. Substituting 3 for x and a for y in the given equation yields $a = 4(2)^3 + 3$, or $a = 35$. Therefore, the value of a is 35.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID ba5a0193

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: ba5a0193

A certain bowl was sold online for prices varying between \$7 and \$13. The number of bowls sold can be modeled by a quadratic function based on the price, where 5 bowls were sold for \$12 each and the maximum of 9 bowls were sold for \$10 each. According to the model, what is the number of bowls that were sold for \$8 each?

ID: ba5a0193 Answer

Correct Answer: 5

Rationale

The correct answer is 5. It's given that the number of bowls sold can be modeled by a quadratic function based on the price, where a maximum of 9 bowls were sold for \$10 each and 5 bowls were sold for \$12 each. This quadratic function can be defined by an equation of the form $N(p) = a(p - h)^2 + k$, where $N(p)$ is the number of bowls sold when the price is \$ p each, and a , h , and k are constants such that the function reaches its maximum value, k , when $p = h$. Since the model indicates the maximum number of bowls sold is 9 when the price is \$10 each, $N(p)$ reaches its maximum value, 9, when $p = 10$. Therefore, $k = 9$ and $h = 10$. Substituting 10 for h and 9 for k in the function $N(p) = a(p - h)^2 + k$ yields $N(p) = a(p - 10)^2 + 9$. Since the model indicates the number of bowls sold is 5 when the price is \$12 each, the value of $N(p)$ is 5 when $p = 12$. Substituting 12 for p and 5 for $N(p)$ in the equation $N(p) = a(p - 10)^2 + 9$ yields $5 = a(12 - 10)^2 + 9$, or $5 = 4a + 9$. Subtracting 9 from both sides of this equation yields $-4 = 4a$. Dividing both sides of this equation by 4 yields $-1 = a$. Therefore, the model can be represented by the equation $N(p) = -(p - 10)^2 + 9$. Substituting 8 for p in this equation yields $N(8) = -(8 - 10)^2 + 9$, or $N(8) = -4 + 9$, or $N(8) = 5$. Therefore, according to the model, the number of bowls sold for \$8 each was 5.

Question Difficulty: Hard

Question ID e9eb0862

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: e9eb0862

x	24	30	32
$f(x)$	-8	-8	8

For the quadratic function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f ?

- A. $f(x) = (x - 24)(x - 30) + 4$
- B. $f(x) = (x - 24)(x - 30) - 8$
- C. $f(x) = (x - 8)(x - 32) + 32$
- D. $f(x) = (x - 8)(x - 32) - 32$

ID: e9eb0862 Answer

Correct Answer: B

Rationale

Choice B is correct. According to the table, $f(24) = -8$, $f(30) = -8$, and $f(32) = 8$. Let the quadratic function g be defined by $g(x) = f(x) + 8$. Therefore, $g(24) = f(24) + 8$, $g(30) = f(30) + 8$, and $g(32) = f(32) + 8$. Substituting -8 for $f(24)$ in the equation $g(24) = f(24) + 8$ yields $g(24) = -8 + 8$, or $g(24) = 0$. Substituting -8 for $f(30)$ in the equation $g(30) = f(30) + 8$ yields $g(30) = -8 + 8$, or $g(30) = 0$. Substituting 8 for $f(32)$ in the equation $g(32) = f(32) + 8$ yields $g(32) = 8 + 8$, or $g(32) = 16$. By the factor theorem, if $g(24) = 0$ and $g(30) = 0$, then $g(x) = k(x - 24)(x - 30)$, where k is a constant. Since $g(32) = 16$, substituting 32 for x and 16 for $g(x)$ in the equation $g(x) = k(x - 24)(x - 30)$ yields $16 = k(32 - 24)(32 - 30)$, or $16 = k(8)(2)$, which is equivalent to $16 = 16k$. Dividing both sides of this equation by 16 yields $1 = k$. Substituting 1 for k in the equation $g(x) = k(x - 24)(x - 30)$ yields $g(x) = (x - 24)(x - 30)$. Since $g(x) = f(x) + 8$, it follows that $g(x) - 8 = f(x)$. Subtracting 8 from both sides of the equation $g(x) = (x - 24)(x - 30)$ yields $g(x) - 8 = (x - 24)(x - 30) - 8$. Substituting $f(x)$ for $g(x) - 8$ in this equation yields $f(x) = (x - 24)(x - 30) - 8$.

Choice A is incorrect. If this equation defined f , then when $x = 24$, the value of $f(x)$ would be 4 , not -8 .

Choice C is incorrect. If this equation defined f , then when $x = 32$, the value of $f(x)$ would be 32 , not 8 .

Choice D is incorrect. If this equation defined f , then when $x = 32$, the value of $f(x)$ would be -32 , not 8 .

Question Difficulty: Hard

Question ID 7a7ed411

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 7a7ed411

$$\frac{-54}{w} = 6$$

What is the solution to the given equation?

ID: 7a7ed411 Answer

Correct Answer: -9

Rationale

The correct answer is -9 . Since w is in the denominator of a fraction in the given equation, w can't be equal to 0 . Since w isn't equal to 0 , multiplying both sides of the given equation by w yields an equivalent equation, $-54 = 6w$. Dividing both sides of this equation by 6 yields $-9 = w$. Therefore, -9 is the solution to the given equation.

Question Difficulty: Medium

Question ID c2ff4625

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: c2ff4625

The quadratic function g models the depth, in meters, below the surface of the water of a seal t minutes after the seal entered the water during a dive. The function estimates that the seal reached its maximum depth of **302.4** meters **6** minutes after it entered the water and then reached the surface of the water **12** minutes after it entered the water. Based on the function, what was the estimated depth, to the nearest meter, of the seal **10** minutes after it entered the water?

ID: c2ff4625 Answer

Correct Answer: 168

Rationale

The correct answer is **168**. The quadratic function g gives the estimated depth of the seal, $g(t)$, in meters, t minutes after the seal enters the water. It's given that function g estimates that the seal reached its maximum depth of **302.4** meters **6** minutes after it entered the water. Therefore, function g can be expressed in vertex form as $g(t) = a(t - 6)^2 + 302.4$, where a is a constant. Since it's also given that the seal reached the surface of the water after **12** minutes, $g(12) = 0$. Substituting **12** for t and **0** for $g(t)$ in $g(t) = a(t - 6)^2 + 302.4$ yields $0 = a(12 - 6)^2 + 302.4$, or $36a = -302.4$. Dividing both sides of this equation by **36** gives $a = -8.4$. Substituting -8.4 for a in $g(t) = a(t - 6)^2 + 302.4$ gives $g(t) = -8.4(t - 6)^2 + 302.4$. Substituting **10** for t in $g(t)$ gives $g(10) = -8.4(10 - 6)^2 + 302.4$, which is equivalent to $g(10) = -8.4(4)^2 + 302.4$, or $g(10) = 168$. Therefore, the estimated depth, to the nearest meter, of the seal **10** minutes after it entered the water was **168** meters.

Question Difficulty: Hard

Question ID a5dc10ab

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: a5dc10ab

$$p(x) + 57 = x^2$$

The given equation relates the value of x and its corresponding value of $p(x)$ for the function p . What is the minimum value of the function p ?

- A. $-3,249$
- B. -57
- C. 57
- D. $3,249$

ID: a5dc10ab Answer

Correct Answer: B

Rationale

Choice B is correct. For a quadratic function defined by an equation of the form $p(x) = a(x - h)^2 + k$, where a , h , and k are constants and $a > 0$, the minimum value of the function is k . Subtracting 57 from both sides of the given equation yields $p(x) = x^2 - 57$. This function is in the form $p(x) = a(x - h)^2 + k$, where $a = 1$, $h = 0$, and $k = -57$. Therefore, the minimum value of the function p is -57 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID dbb15274

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: dbb15274

$-9x^2 + 30x + c = 0$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

- A. 3
- B. 0
- C. -25
- D. -53

ID: dbb15274 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the equation $-9x^2 + 30x + c = 0$ has exactly one solution. A quadratic equation of the form $ax^2 + bx + c = 0$ has exactly one solution if and only if its discriminant, $-4ac + b^2$, is equal to zero. It follows that for the given equation, $a = -9$ and $b = 30$. Substituting -9 for a and 30 for b into $b^2 - 4ac$ yields $30^2 - 4(-9)(c)$, or $900 + 36c$. Since the discriminant must equal zero, $900 + 36c = 0$. Subtracting $36c$ from both sides of this equation yields $900 = -36c$. Dividing each side of this equation by -36 yields $-25 = c$. Therefore, the value of c is -25 .

Choice A is incorrect. If the value of c is 3 , this would yield a discriminant that is greater than zero. Therefore, the given equation would have two solutions, rather than exactly one solution.

Choice B is incorrect. If the value of c is 0 , this would yield a discriminant that is greater than zero. Therefore, the given equation would have two solutions, rather than exactly one solution.

Choice D is incorrect. If the value of c is -53 , this would yield a discriminant that is less than zero. Therefore, the given equation would have no real solutions, rather than exactly one solution.

Question Difficulty: Hard

Question ID 7a65a4aa

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 7a65a4aa

$$\begin{aligned}x &= 8 \\ y &= x^2 + 8\end{aligned}$$

The graphs of the equations in the given system of equations intersect at the point (x, y) in the xy -plane. What is the value of y ?

- A. 8
- B. 24
- C. 64
- D. 72

ID: 7a65a4aa Answer

Correct Answer: D

Rationale

Choice D is correct. Since the graphs of the equations in the given system intersect at the point (x, y) , the point (x, y) represents a solution to the given system of equations. The first equation of the given system of equations states that $x = 8$. Substituting 8 for x in the second equation of the given system of equations yields $y = 8^2 + 8$, or $y = 72$. Therefore, the value of y is 72.

Choice A is incorrect. This is the value of x , not y .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID b30bb27f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: b30bb27f

$$p = \frac{k}{4j+9}$$

The given equation relates the distinct positive numbers p , k , and j . Which equation correctly expresses $4j + 9$ in terms of p and k ?

- A. $4j + 9 = \frac{k}{p}$
- B. $4j + 9 = kp$
- C. $4j + 9 = k - p$
- D. $4j + 9 = \frac{p}{k}$

ID: b30bb27f Answer

Correct Answer: A

Rationale

Choice A is correct. To express $4j + 9$ in terms of p and k , the given equation must be solved for $4j + 9$. Since it's given that j is a positive number, $4j + 9$ is not equal to zero. Therefore, multiplying both sides of the given equation by $4j + 9$ yields the equivalent equation $p(4j + 9) = k$. Since it's given that p is a positive number, p is not equal to zero. Therefore, dividing each side of the equation $p(4j + 9) = k$ by p yields the equivalent equation $4j + 9 = \frac{k}{p}$.

Choice B is incorrect. This equation is equivalent to $p = \frac{4j+9}{k}$.

Choice C is incorrect. This equation is equivalent to $p = k - 4j - 9$.

Choice D is incorrect. This equation is equivalent to $p = k(4j + 9)$.

Question Difficulty: Hard

Question ID 7e7255de

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 7e7255de

The function p is defined by $p(n) = 7n^3$. What is the value of n when $p(n)$ is equal to 56?

- A. 2
- B. $\frac{8}{3}$
- C. 7
- D. 8

ID: 7e7255de Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $p(n) = 7n^3$. Substituting 56 for $p(n)$ in this equation yields $56 = 7n^3$. Dividing each side of this equation by 7 yields $8 = n^3$. Taking the cube root of each side of this equation yields $2 = n$. Therefore, when $p(n)$ is equal to 56, the value of n is 2.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 996ee9fc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 996ee9fc

Which expression is a factor of $2x^2 + 38x + 10$?

- A. 2
- B. $5x$
- C. $38x$
- D. $2x^2$

ID: 996ee9fc Answer

Correct Answer: A

Rationale

Choice A is correct. Since 2 is a common factor of each of the terms in the given expression, the expression can be rewritten as $2(x^2 + 19x + 5)$. Therefore, the factors of the given expression are 2 and $x^2 + 19x + 5$. Of these two factors, only 2 is listed as a choice.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is a term of the given expression, not a factor of the given expression.

Choice D is incorrect. This is a term of the given expression, not a factor of the given expression.

Question Difficulty: Medium

Question ID 7d86a6ad

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 7d86a6ad

$f(x) = (x - 10)(x + 13)$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

- A. -130
- B. -13
- C. $-\frac{23}{2}$
- D. $-\frac{3}{2}$

ID: 7d86a6ad Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that $f(x) = (x - 10)(x + 13)$, which can be rewritten as $f(x) = x^2 + 3x - 130$. Since the coefficient of the x^2 -term is positive, the graph of $y = f(x)$ in the xy -plane opens upward and reaches its minimum value at its vertex. The x -coordinate of the vertex is the value of x such that $f(x)$ reaches its minimum. For an equation in the form $f(x) = ax^2 + bx + c$, where a , b , and c are constants, the x -coordinate of the vertex is $-\frac{b}{2a}$. For the equation $f(x) = x^2 + 3x - 130$, $a = 1$, $b = 3$, and $c = -130$. It follows that the x -coordinate of the vertex is $-\frac{3}{2(1)}$, or $-\frac{3}{2}$. Therefore, $f(x)$ reaches its minimum when the value of x is $-\frac{3}{2}$.

Alternate approach: The value of x for the vertex of a parabola is the x -value of the midpoint between the two x -intercepts of the parabola. Since it's given that $f(x) = (x - 10)(x + 13)$, it follows that the two x -intercepts of the graph of $y = f(x)$ in the xy -plane occur when $x = 10$ and $x = -13$, or at the points $(10, 0)$ and $(-13, 0)$. The midpoint between two points, (x_1, y_1) and (x_2, y_2) , is $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$. Therefore, the midpoint between $(10, 0)$ and $(-13, 0)$ is $(\frac{10+(-13)}{2}, \frac{0+0}{2})$, or $(-\frac{3}{2}, 0)$. It follows that $f(x)$ reaches its minimum when the value of x is $-\frac{3}{2}$.

Choice A is incorrect. This is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane.

Choice B is incorrect. This is one of the x -coordinates of the x -intercepts of the graph of $y = f(x)$ in the xy -plane.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 9514e764

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 9514e764

$g(x) = x^2 + 55$

What is the minimum value of the given function?

- A. 0
- B. 55
- C. 110
- D. 3,025

ID: 9514e764 Answer

Correct Answer: B

Rationale

Choice B is correct. For a quadratic function defined by an equation of the form $g(x) = a(x - h)^2 + k$, where a , h , and k are constants and $a > 0$, the minimum value of the function is k . In the given function, $a = 1$, $h = 0$, and $k = 55$. Therefore, the minimum value of the given function is **55**.

Choice A is incorrect. This is the value of x for which the given function reaches its minimum value, not the minimum value of the function.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 67f9ce96

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 67f9ce96

At the time of posting a video, a social media channel had **53** subscribers. Each day for five days after the video was posted, the number of subscribers doubled from the number the previous day. Which equation gives the total number of subscribers, n , to the channel d days after the video was posted?

- A. $n = (53)^d$
- B. $n = 53(2)^d$
- C. $n = 53\left(\frac{1}{2}\right)^d$
- D. $n = (53)^2 + d$

ID: 67f9ce96 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that each day for five days after a social media channel posted a video, the number of subscribers doubled from the number the previous day. Since the number of subscribers doubled each day, the relationship between n and d can be represented by an exponential equation of the form $n = ab^d$, where a is the number of subscribers at the time of posting the video and the number of subscribers to the channel increases by a factor of b each day. It's given that at the time of posting the video, the channel had **53** subscribers. Therefore, $a = 53$. It's also given that the number of subscribers doubled, or increased by a factor of **2**, from the number the previous day. Therefore, $b = 2$. Substituting **53** for a and **2** for b in the equation $n = ab^d$ yields $n = 53(2)^d$.

Choice A is incorrect. This equation gives the total number of subscribers to a channel for which the initial number of subscribers was **1** and the number increased each day by **53** times the number from the previous day.

Choice C is incorrect. This equation gives the total number of subscribers to a channel for which the number of subscribers each day was half the number from the previous day, rather than double the number.

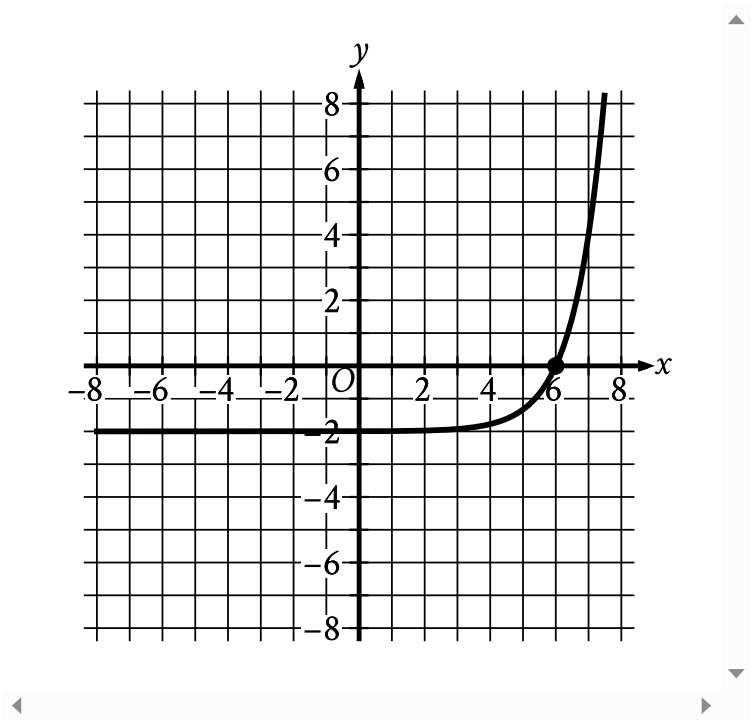
Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 5b574c25

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 5b574c25



What is the x-coordinate of the x-intercept of the graph shown?

ID: 5b574c25 Answer

Correct Answer: 6

Rationale

The correct answer is **6**. An x-intercept of a graph is a point on the graph where it intersects the x-axis, or where the value of **y** is **0**. The graph shown intersects the x-axis at the point **(6, 0)**. Therefore, the x-coordinate of the x-intercept of the graph shown is **6**.

Question Difficulty: Easy

Question ID 1845987d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 1845987d

$$f(x) = (x + 7)^2 + 4$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

ID: 1845987d Answer

Correct Answer: -7

Rationale

The correct answer is -7 . For a quadratic function defined by an equation of the form $f(x) = a(x - h)^2 + k$, where a , h , and k are constants and $a > 0$, the function reaches its minimum when $x = h$. In the given function, $a = 1$, $h = -7$, and $k = 4$. Therefore, the value of x for which $f(x)$ reaches its minimum is -7 .

Question Difficulty: Hard

Question ID 77c8bd4a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 77c8bd4a

For the function f , $f(0) = 86$, and for each increase in x by 1, the value of $f(x)$ decreases by 80%. What is the value of $f(2)$?

ID: 77c8bd4a Answer

Correct Answer: 3.44, 86/25

Rationale

The correct answer is **3.44**. It's given that $f(0) = 86$ and that for each increase in x by 1, the value of $f(x)$ decreases by 80%. Because the output of the function decreases by a constant percentage for each 1-unit increase in the value of x , this relationship can be represented by an exponential function of the form $f(x) = a(b)^x$, where a represents the initial value of the function and b represents the rate of decay, expressed as a decimal. Because $f(0) = 86$, the value of a must be 86. Because the value of $f(x)$ decreases by 80% for each 1-unit increase in x , the value of b must be $(1 - 0.80)$, or 0.2. Therefore, the function f can be defined by $f(x) = 86(0.2)^x$. Substituting 2 for x in this function yields $f(2) = 86(0.2)^2$, which is equivalent to $f(2) = 86(0.04)$, or $f(2) = 3.44$. Either **3.44** or **86/25** may be entered as the correct answer.

Alternate approach: It's given that $f(0) = 86$ and that for each increase in x by 1, the value of $f(x)$ decreases by 80%. Therefore, when $x = 1$, the value of $f(x)$ is $(100 - 80)\%$, or 20%, of 86, which can be expressed as $(0.20)(86)$. Since $(0.20)(86) = 17.2$, the value of $f(1)$ is 17.2. Similarly, when $x = 2$, the value of $f(x)$ is 20% of 17.2, which can be expressed as $(0.20)(17.2)$. Since $(0.20)(17.2) = 3.44$, the value of $f(2)$ is 3.44. Either **3.44** or **86/25** may be entered as the correct answer.

Question Difficulty: Hard

Question ID 1d8b3958

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 1d8b3958

$3x^2 - 15x + 18 = 0$

How many distinct real solutions are there to the given equation?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

ID: 1d8b3958 Answer

Correct Answer: B

Rationale

Choice B is correct. The number of solutions to a quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. If the value of the discriminant is positive, then the quadratic equation has exactly two distinct real solutions. If the value of the discriminant is equal to zero, then the quadratic equation has exactly one real solution. If the value of the discriminant is negative, then the quadratic equation has zero real solutions. In the given equation, $3x^2 - 15x + 18 = 0$, $a = 3$, $b = -15$, and $c = 18$. Substituting 3 for a , -15 for b , and 18 for c in $b^2 - 4ac$ yields $(-15)^2 - 4(3)(18)$, or 9 . Since the value of the discriminant is positive, the given equation has exactly two distinct real solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

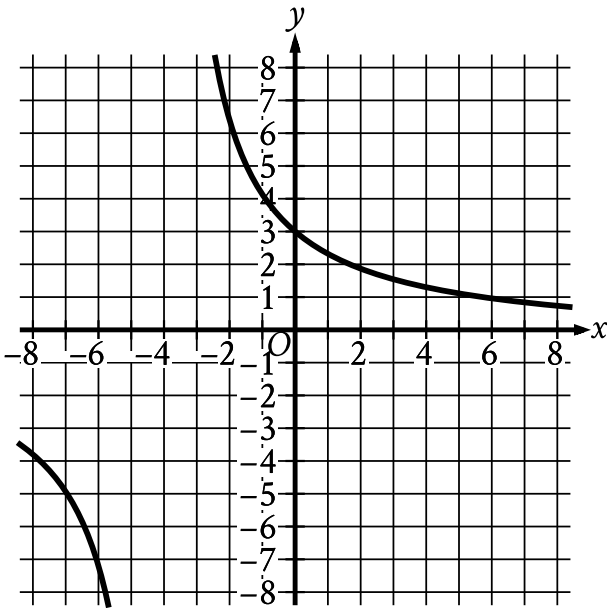
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 65aef65f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 65aef65f



The graph of $y = f(x)$ is shown in the xy -plane. The value of $f(0)$ is an integer. What is the value of $f(0)$?

ID: 65aef65f Answer

Correct Answer: 3

Rationale

The correct answer is **3**. The value of $f(0)$ is the value of y on the graph of $y = f(x)$ in the xy -plane that corresponds with $x = 0$. It's given that the value of $f(0)$ is an integer. For the graph of $y = f(x)$ shown, when $x = 0$, the corresponding integer value of y is **3**. Therefore, the value of $f(0)$ is **3**.

Question Difficulty: Medium

Question ID f52b072c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: f52b072c

The function k is defined by $k(x) = x^3 + 110$. What is the value of $k(x)$ when $x = 2$?

- A. 118
- B. 116
- C. 115
- D. 110

ID: f52b072c Answer

Correct Answer: A

Rationale

Choice A is correct. Substituting 2 for x in the given equation yields $k(2) = 2^3 + 110$, which is equivalent to $k(2) = 8 + 110$, or $k(2) = 118$. Therefore, the value of $k(x)$ when $x = 2$ is 118 .

Choice B is incorrect. This is the value of $k(x)$ when $x = 2$ if $k(x) = 3x + 110$, not $k(x) = x^3 + 110$.

Choice C is incorrect. This is the value of $k(x)$ when $x = 2$ if $k(x) = x + 3 + 110$, not $k(x) = x^3 + 110$.

Choice D is incorrect. This is the value of $k(x)$ when $x = 0$, not $x = 2$.

Question Difficulty: Easy

Question ID 9888ca87

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 9888ca87

The function f is defined by $f(x) = \frac{16}{x}$. What is the value of $f(x)$ when $x = 17$?

- A. $\frac{16}{17}$
- B. $\frac{17}{16}$
- C. 16
- D. 17

ID: 9888ca87 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $f(x) = \frac{16}{x}$. Substituting 17 for x in this function yields $f(17) = \frac{16}{17}$. Therefore, when $x = 17$, the value of $f(x)$ is $\frac{16}{17}$.

Choice B is incorrect. This is the value of the reciprocal of $f(x)$ when $x = 17$.

Choice C is incorrect. This is the value of $f(x)$ when $x = 1$.

Choice D is incorrect. This is the value of x when $x = 17$.

Question Difficulty: Easy

Question ID 707185e0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 707185e0

The solutions to $x^2 + 6x + 7 = 0$ are r and s , where $r < s$. The solutions to $x^2 + 8x + 8 = 0$ are t and u , where $t < u$. The solutions to $x^2 + 14x + c = 0$, where c is a constant, are $r + t$ and $s + u$. What is the value of c ?

ID: 707185e0 Answer

Correct Answer: 31

Rationale

The correct answer is **31**. Subtracting **7** from both sides of the equation $x^2 + 6x + 7 = 0$ yields $x^2 + 6x = -7$. To complete the square, adding $\left(\frac{6}{2}\right)^2$, or 3^2 , to both sides of this equation yields $x^2 + 6x + 3^2 = -7 + 3^2$, or $(x + 3)^2 = 2$. Taking the square root of both sides of this equation yields $x + 3 = \pm\sqrt{2}$. Subtracting **3** from both sides of this equation yields $x = -3 \pm \sqrt{2}$. Therefore, the solutions r and s to the equation $x^2 + 6x + 7 = 0$ are $-3 - \sqrt{2}$ and $-3 + \sqrt{2}$. Since $r < s$, it follows that $r = -3 - \sqrt{2}$ and $s = -3 + \sqrt{2}$. Subtracting **8** from both sides of the equation $x^2 + 8x + 8 = 0$ yields $x^2 + 8x = -8$. To complete the square, adding $\left(\frac{8}{2}\right)^2$, or 4^2 , to both sides of this equation yields $x^2 + 8x + 4^2 = -8 + 4^2$, or $(x + 4)^2 = 8$. Taking the square root of both sides of this equation yields $x + 4 = \pm\sqrt{8}$, or $x + 4 = \pm2\sqrt{2}$. Subtracting **4** from both sides of this equation yields $x = -4 \pm 2\sqrt{2}$. Therefore, the solutions t and u to the equation $x^2 + 8x + 8 = 0$ are $-4 - 2\sqrt{2}$ and $-4 + 2\sqrt{2}$. Since $t < u$, it follows that $t = -4 - 2\sqrt{2}$ and $u = -4 + 2\sqrt{2}$. It's given that the solutions to $x^2 + 14x + c = 0$, where c is a constant, are $r + t$ and $s + u$. It follows that this equation can be written as $(x - (r + t))(x - (s + u)) = 0$, which is equivalent to $x^2 - (r + t + s + u)x + (r + t)(s + u) = 0$. Therefore, the value of c is $(r + t)(s + u)$. Substituting $-3 - \sqrt{2}$ for r , $-4 - 2\sqrt{2}$ for t , $-3 + \sqrt{2}$ for s , and $-4 + 2\sqrt{2}$ for u in this equation yields $\left((-3 - \sqrt{2}) + (-4 - 2\sqrt{2})\right)\left((-3 + \sqrt{2}) + (-4 + 2\sqrt{2})\right)$, which is equivalent to $(-7 - 3\sqrt{2})(-7 + 3\sqrt{2})$, or $(-7)(-7) - (3\sqrt{2})(3\sqrt{2})$, which is equivalent to $49 - 18$, or **31**. Therefore, the value of c is **31**.

Question Difficulty: Hard

Question ID 9f0e721d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 9f0e721d

$$8j = k + 15m$$

The given equation relates the distinct positive numbers j , k , and m . Which equation correctly expresses j in terms of k and m ?

- A. $j = \frac{k}{8} + 15m$
- B. $j = k + \frac{15m}{8}$
- C. $j = 8(k + 15m)$
- D. $j = \frac{k+15m}{8}$

ID: 9f0e721d Answer

Correct Answer: D

Rationale

Choice D is correct. To express j in terms of k and m , the given equation must be solved for j . Dividing each side of the given equation by 8 yields $j = \frac{k+15m}{8}$.

Choice A is incorrect. This is equivalent to $8j = k + 120m$.

Choice B is incorrect. This is equivalent to $8j = 8k + 15m$.

Choice C is incorrect. This is equivalent to $\frac{j}{8} = k + 15m$.

Question Difficulty: Medium

Question ID d699d0b2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: d699d0b2

If $4^{8c} = \sqrt[3]{4^7}$, what is the value of c ?

ID: d699d0b2 Answer

Correct Answer: .2916, .2917, 7/24

Rationale

The correct answer is $\frac{7}{24}$. An expression of the form $\sqrt[n]{a^m}$, where m and n are integers greater than 1 and $a \geq 0$, is equivalent to $a^{\frac{m}{n}}$. Therefore, the expression on the right-hand side of the given equation, $\sqrt[3]{4^7}$, is equivalent to $4^{\frac{7}{3}}$. Thus, $4^{8c} = 4^{\frac{7}{3}}$. It follows that $8c = \frac{7}{3}$. Dividing both sides of this equation by 8 yields $c = \frac{7}{24}$. Note that 7/24, .2916, .2917, 0.219, and 0.292 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 194aa933

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 194aa933

$$P(t) = 24.8(1.036)^t$$

The function P gives the predicted population, in millions, of a certain country for the period from 1984 to 2018, where t is the number of years after 1984. According to the model, what is the best interpretation of the statement " $P(8)$ is approximately equal to 32.91"?

- A. In 1984, the predicted population of this country was approximately 8 million.
- B. In 1984, the predicted population of this country was approximately 32.91 million.
- C. 8 years after 1984, the predicted population of this country was approximately 32.91 million.
- D. 32.91 years after 1984, the predicted population of this country was approximately 8 million.

ID: 194aa933 Answer

Correct Answer: C

Rationale

Choice C is correct. The function P gives the predicted population, in millions, of a certain country for the period from 1984 to 2018, where t is the number of years after 1984. Since the value of $P(8)$ is the value of $P(t)$ when $t = 8$, it follows that " $P(8)$ is approximately equal to 32.91" means that the value of $P(t)$ is approximately equal to 32.91 when $t = 8$. Therefore, the best interpretation of the statement " $P(8)$ is approximately equal to 32.91" is that 8 years after 1984, the predicted population of this country was approximately 32.91 million.

Choice A is incorrect. In 1984, the predicted population of this country was 24.8 million, not approximately 8 million.

Choice B is incorrect. In 1984, the predicted population of this country was 24.8 million, not approximately 32.91 million.

Choice D is incorrect. 32.91 years after 1984, the predicted population of this country was $24.8(1.036)^{32.91}$ million, or approximately 79.42 million, not approximately 8 million.

Question Difficulty: Medium

Question ID 9a614311

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 9a614311

$$y = 2x^2 - 21x + 64$$
$$y = 3x + a$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. What is the value of x ?

- A. -8
- B. -6
- C. 6
- D. 8

ID: 9a614311 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. Therefore, (x, y) is the only solution to the given system of equations. The given system of equations can be solved by subtracting the second equation, $y = 3x + a$, from the first equation, $y = 2x^2 - 21x + 64$. This yields $y - y = (2x^2 - 21x + 64) - (3x + a)$, or $0 = 2x^2 - 24x + 64 - a$. Since the given system has only one solution, this equation has only one solution. A quadratic equation in the form $rx^2 + sx + t = 0$, where r , s , and t are constants, has one solution if and only if the discriminant, $s^2 - 4rt$, is equal to zero. Substituting 2 for r , -24 for s , and $-a + 64$ for t in the expression $s^2 - 4rt$ yields $(-24)^2 - (4)(2)(64 - a)$. Setting this expression equal to zero yields $(-24)^2 - (4)(2)(64 - a) = 0$, or $8a + 64 = 0$. Subtracting 64 from both sides of this equation yields $8a = -64$. Dividing both sides of this equation by 8 yields $a = -8$. Substituting -8 for a in the equation $0 = 2x^2 - 24x + 64 - a$ yields $0 = 2x^2 - 24x + 64 + 8$, or $0 = 2x^2 - 24x + 72$. Factoring 2 from the right-hand side of this equation yields $0 = 2(x^2 - 12x + 36)$. Dividing both sides of this equation by 2 yields $0 = x^2 - 12x + 36$, which is equivalent to $0 = (x - 6)(x - 6)$, or $0 = (x - 6)^2$. Taking the square root of both sides of this equation yields $0 = x - 6$. Adding 6 to both sides of this equation yields $x = 6$.

Choice A is incorrect. This is the value of a , not x .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 21be5f04

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 21be5f04

The area of a triangle is **270** square centimeters. The length of the base of the triangle is **12** centimeters greater than the height of the triangle. What is the height, in centimeters, of the triangle?

- A. 15
- B. 18
- C. 30
- D. 36

ID: 21be5f04 Answer

Correct Answer: B

Rationale

Choice B is correct. The area, A , of a triangle is given by the formula $A = \frac{1}{2}bh$, where b represents the length of the base of the triangle and h represents its height. It's given that the area of a triangle is **270** square centimeters and that the length of the base of this triangle is **12** centimeters greater than the height of the triangle. Let x represent the height, in centimeters, of the triangle. It follows that the length of the base of the triangle can be expressed as $x + 12$. Substituting **270** for A , x for h , and $x + 12$ for b in the formula $A = \frac{1}{2}bh$ yields $270 = \frac{1}{2}(x + 12)(x)$, or $270 = \frac{1}{2}x(x + 12)$. Multiplying both sides of this equation by **2** yields $540 = x(x + 12)$. Applying the distributive property on the right-hand side of this equation yields $540 = x^2 + 12x$. Subtracting **540** from both sides of this equation yields $0 = x^2 + 12x - 540$. In factored form, this equation is equivalent to $(x + 30)(x - 18) = 0$. Applying the zero product property, it follows that $x + 30 = 0$ or $x - 18 = 0$. Subtracting **30** from both sides of the equation $x + 30 = 0$ yields $x = -30$. Adding **18** to both sides of the equation $x - 18 = 0$ yields $x = 18$. Since x represents the height of the triangle, it must be positive. Therefore, the height, in centimeters, of the triangle is **18**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

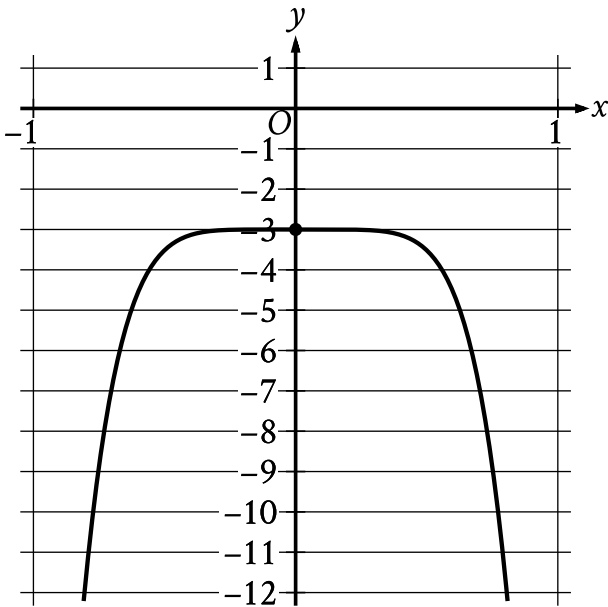
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID baf439a1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: baf439a1



The graph of the polynomial function f , where $y = f(x)$, is shown. The y -intercept of the graph is $(0, y)$. What is the value of y ?

ID: baf439a1 Answer

Correct Answer: -3

Rationale

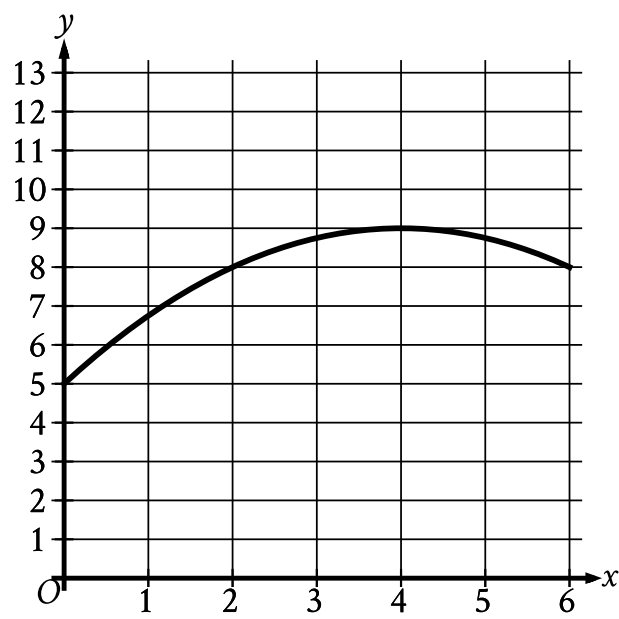
The correct answer is -3 . The y -intercept of the graph of a function in the xy -plane is the point where the graph crosses the y -axis. The graph of the polynomial function shown crosses the y -axis at the point $(0, -3)$. It's given that the y -intercept of the graph is $(0, y)$. Thus, the value of y is -3 .

Question Difficulty: Medium

Question ID fb72f2ed

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: fb72f2ed



The graph models the number of active projects a company was working on x months after the end of November **2012**, where $0 \leq x \leq 6$. According to the model, what is the predicted number of active projects the company was working on at the end of November **2012**?

- A. 0
- B. 5
- C. 8
- D. 9

ID: fb72f2ed Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the graph models the number of active projects a company was working on x months after the end of November **2012**. Therefore, the value of x that corresponds to the end of November **2012** is **0**. The point at which $x = 0$ is the y-intercept of the graph. It follows that the y-intercept of the graph shown is the point **(0, 5)**. Therefore, according to the model, the predicted number of active projects the company was working on at the end of November **2012** is **5**.

Choice A is incorrect. This is the value of x that corresponds to the end of November **2012**, not the predicted number of active projects the company was working on at the end of November **2012**.

Choice C is incorrect. This is the predicted number of active projects the company was working on **2** months after the end of November **2012**.

Choice D is incorrect. This is the predicted number of active projects the company was working on **4** months after the end of November **2012**.

Question Difficulty: Hard

Question ID e7d33124

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: e7d33124

Which expression is equivalent to $12x + 27$?

- A. $12(9x + 1)$
- B. $27(12x + 1)$
- C. $3(4x + 9)$
- D. $3(9x + 24)$

ID: e7d33124 Answer

Correct Answer: C

Rationale

Choice C is correct. Each term in the given expression, $12x + 27$, has a common factor of 3 . Therefore, the expression can be rewritten as $3(4x) + 3(9)$, or $3(4x + 9)$. Thus, the expression $3(4x + 9)$ is equivalent to the given expression.

Choice A is incorrect. This expression is equivalent to $108x + 12$, not $12x + 27$.

Choice B is incorrect. This expression is equivalent to $324x + 27$, not $12x + 27$.

Choice D is incorrect. This expression is equivalent to $27x + 72$, not $12x + 27$.

Question Difficulty: Easy

Question ID dfbf8890

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: dfbf8890

A quadratic function models a projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. The model estimates that the projectile was launched from an initial height of **7** meters above the ground and reached a maximum height of **51.1** meters above the ground **3** seconds after the launch. How many seconds after the launch does the model estimate that the projectile will return to a height of **7** meters?

- A. **3**
- B. **6**
- C. **7**
- D. **9**

ID: dfbf8890 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that a quadratic function models the projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. It follows that an equation representing the model can be written in the form $f(x) = a(x - h)^2 + k$, where $f(x)$ is the projectile's estimated height above the ground, in meters, x seconds after the launch, a is a constant, and k is the maximum height above the ground, in meters, the model estimates the projectile reached h seconds after the launch. It's given that the model estimates the projectile reached a maximum height of **51.1** meters above the ground **3** seconds after the launch. Therefore, $k = 51.1$ and $h = 3$. Substituting **51.1** for k and **3** for h in the equation $f(x) = a(x - h)^2 + k$ yields $f(x) = a(x - 3)^2 + 51.1$. It's also given that the model estimates that the projectile was launched from an initial height of **7** meters above the ground. Therefore, when $x = 0$, $f(x) = 7$. Substituting **0** for x and **7** for $f(x)$ in the equation $f(x) = a(x - 3)^2 + 51.1$ yields $7 = a(0 - 3)^2 + 51.1$, or $7 = 9a + 51.1$. Subtracting **51.1** from both sides of this equation yields $-44.1 = 9a$. Dividing both sides of this equation by **9** yields $-4.9 = a$. Substituting -4.9 for a in the equation $f(x) = a(x - 3)^2 + 51.1$ yields $f(x) = -4.9(x - 3)^2 + 51.1$. Therefore, the equation $f(x) = -4.9(x - 3)^2 + 51.1$ models the projectile's height, in meters, above the ground x seconds after it was launched. The number of seconds after the launch that the model estimates that the projectile will return to a height of **7** meters is the value of x when $f(x) = 7$. Substituting **7** for $f(x)$ in $f(x) = -4.9(x - 3)^2 + 51.1$ yields $7 = -4.9(x - 3)^2 + 51.1$. Subtracting **51.1** from both sides of this equation yields $-44.1 = -4.9(x - 3)^2$. Dividing both sides of this equation by -4.9 yields $9 = (x - 3)^2$. Taking the square root of both sides of this equation yields two equations: $3 = x - 3$ and $-3 = x - 3$. Adding **3** to both sides of the equation $3 = x - 3$ yields $6 = x$. Adding **3** to both sides of the equation $-3 = x - 3$ yields $0 = x$. Since **0** seconds after the launch represents the time at which the projectile was launched, **6** must be the number of seconds the model estimates that the projectile will return to a height of **7** meters.

Alternate approach: It's given that a quadratic function models the projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. It's also given that the model estimates that the projectile was launched from an initial height of **7** meters above the ground and reached a maximum height of **51.1** meters above the ground **3** seconds after the launch. Since the model is quadratic, and quadratic functions are symmetric, the model estimates that for any given height less than the maximum height, the time the projectile takes to travel from the given height to the maximum height is the same as the time the projectile takes to travel from the maximum height back to the given height. Thus, since the model estimates the projectile took **3** seconds to travel from **7** meters above the ground to its maximum height of **51.1** meters above the ground, the model also estimates the projectile will take **3** more seconds to travel from its maximum height of **51.1** meters above the ground

back to **7** meters above the ground. Thus, the model estimates that the projectile will return to a height of **7** meters **3** seconds after it reaches its maximum height, which is **6** seconds after the launch.

Choice A is incorrect. The model estimates that **3** seconds after the launch, the projectile reached a height of **51.1** meters, not **7** meters.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 5108645e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 5108645e

$5x^2 + 10x + 16 = 0$

How many distinct real solutions does the given equation have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

ID: 5108645e Answer

Correct Answer: D

Rationale

Choice D is correct. The number of solutions of a quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. If the value of the discriminant is positive, then the quadratic equation has exactly two distinct real solutions. If the value of the discriminant is equal to zero, then the quadratic equation has exactly one real solution. If the value of the discriminant is negative, then the quadratic equation has zero real solutions. In the given equation, $5x^2 + 10x + 16 = 0$, $a = 5$, $b = 10$, and $c = 16$. Substituting these values for a , b , and c in $b^2 - 4ac$ yields $(10)^2 - 4(5)(16)$, or -220 . Since the value of its discriminant is negative, the given equation has zero real solutions. Therefore, the number of distinct real solutions the given equation has is zero.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID cdc398ca

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: cdc398ca

$$p(t) = 90,000(1.06)^t$$

The given function p models the population of Lowell t years after a census. Which of the following functions best models the population of Lowell m months after the census?

- A. $r(m) = \frac{90,000}{12}(1.06)^m$
- B. $r(m) = 90,000\left(\frac{1.06}{12}\right)^m$
- C. $r(m) = 90,000\left(\frac{1.06}{12}\right)^{\frac{m}{12}}$
- D. $r(m) = 90,000(1.06)^{\frac{m}{12}}$

ID: cdc398ca Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the function p models the population of Lowell t years after a census. Since there are **12** months in a year, m months after the census is equivalent to $\frac{m}{12}$ years after the census. Substituting $\frac{m}{12}$ for t in the equation $p(t) = 90,000(1.06)^t$ yields $p\left(\frac{m}{12}\right) = 90,000(1.06)^{\frac{m}{12}}$. Therefore, the function r that best models the population of Lowell m months after the census is $r(m) = 90,000(1.06)^{\frac{m}{12}}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 583767ba

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 583767ba

Function f is defined by $f(x) = -a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x) - 15$ has a y -intercept at $(0, -\frac{99}{7})$. The product of a and b is $\frac{65}{7}$. What is the value of a ?

ID: 583767ba Answer

Correct Answer: 5

Rationale

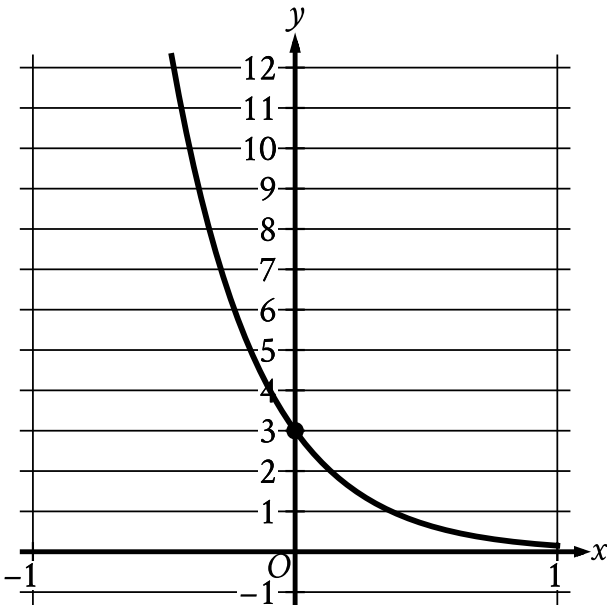
The correct answer is 5. It’s given that $f(x) = -a^x + b$. Substituting $-a^x + b$ for $f(x)$ in the equation $y = f(x) - 15$ yields $y = -a^x + b - 15$. It’s given that the y -intercept of the graph of $y = f(x) - 15$ is $(0, -\frac{99}{7})$. Substituting 0 for x and $-\frac{99}{7}$ for y in the equation $y = -a^x + b - 15$ yields $-\frac{99}{7} = -a^0 + b - 15$, which is equivalent to $-\frac{99}{7} = -1 + b - 15$, or $-\frac{99}{7} = b - 16$. Adding 16 to both sides of this equation yields $\frac{13}{7} = b$. It’s given that the product of a and b is $\frac{65}{7}$, or $ab = \frac{65}{7}$. Substituting $\frac{13}{7}$ for b in this equation yields $(a)(\frac{13}{7}) = \frac{65}{7}$. Dividing both sides of this equation by $\frac{13}{7}$ yields $a = 5$.

Question Difficulty: Hard

Question ID 4a05c831

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 4a05c831



The graph of the exponential function f is shown, where $y = f(x)$. The y -intercept of the graph is $(0, y)$. What is the value of y ?

ID: 4a05c831 Answer

Correct Answer: 3

Rationale

The correct answer is **3**. It's given that the y -intercept of the graph shown is $(0, y)$. The graph passes through the point $(0, 3)$. Therefore, the value of y is **3**.

Question Difficulty: Easy

Question ID de3cc97b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: de3cc97b

The first term of a sequence is **9**. Each term after the first is **4** times the preceding term. If *w* represents the *n*th term of the sequence, which equation gives *w* in terms of *n*?

- A. $w = 4(9^n)$
- B. $w = 4(9^{n-1})$
- C. $w = 9(4^n)$
- D. $w = 9(4^{n-1})$

ID: de3cc97b Answer

Correct Answer: D

Rationale

Choice D is correct. Since *w* represents the *n*th term of the sequence and **9** is the first term of the sequence, the value of *w* is **9** when the value of *n* is **1**. Since each term after the first is **4** times the preceding term, the value of *w* is **9(4)** when the value of *n* is **2**. Therefore, the value of *w* is **9(4)(4)**, or **9(4)²**, when the value of *n* is **3**. More generally, the value of *w* is **9(4ⁿ⁻¹)** for a given value of *n*. Therefore, the equation $w = 9(4^{n-1})$ gives *w* in terms of *n*.

Choice A is incorrect. This equation describes a sequence for which the first term is **36**, rather than **9**, and each term after the first is **9**, rather than **4**, times the preceding term.

Choice B is incorrect. This equation describes a sequence for which the first term is **4**, rather than **9**, and each term after the first is **9**, rather than **4**, times the preceding term.

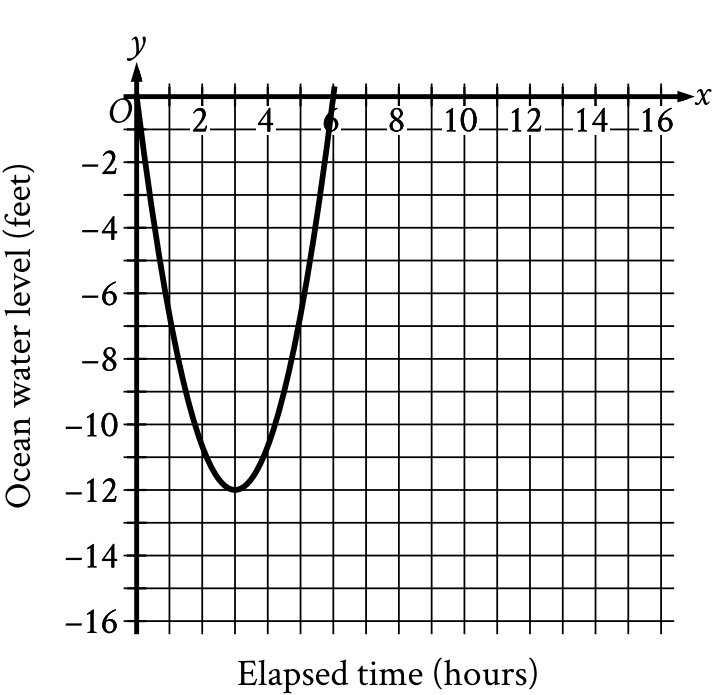
Choice C is incorrect. This equation describes a sequence for which the first term is **36**, rather than **9**.

Question Difficulty: Hard

Question ID d84cbe50

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: d84cbe50



Scientists recorded data about the ocean water levels at a certain location over a period of **6** hours. The graph shown models the data, where **$y = 0$** represents sea level. Which table gives values of **x** and their corresponding values of **y** based on the model?

A.

x	y
0	-12
0	3
3	6

B.

x	y
0	0
3	12
0	-6

C.

x	y
0	0
3	-12
6	0

D.

x	y
0	0
12	3
-6	0

ID: d84cbe50 Answer

Correct Answer: C

Rationale

Choice C is correct. Each point (x, y) on the graph represents an elapsed time x , in hours, and the corresponding ocean water level y , in feet, at a certain location based on the model. The graph shown passes through the points $(0, 0)$, $(3, -12)$, and $(6, 0)$. Thus, the table in choice C gives the values of x and their corresponding values of y based on the model.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID fb5f1cf7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: fb5f1cf7

If $4\sqrt{2x} = 16$, what is the value of $6x$?

- A. 24
- B. 48
- C. 72
- D. 96

ID: fb5f1cf7 Answer

Correct Answer: B

Rationale

Choice B is correct. Dividing each side of the given equation by 4 yields $\sqrt{2x} = 4$. Squaring both sides of this equation yields $2x = 16$. Multiplying each side of this equation by 3 yields $6x = 48$. Therefore, the value of $6x$ is 48.

Choice A is incorrect. This is the value of $3x$, not $6x$.

Choice C is incorrect. This is the value of $9x$, not $6x$.

Choice D is incorrect. This is the value of $12x$, not $6x$.

Question Difficulty: Medium

Question ID 42045837

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 42045837

A right rectangular prism has a height of **9** inches. The length of the prism's base is x inches, which is **7** inches more than the width of the prism's base. Which function V gives the volume of the prism, in cubic inches, in terms of the length of the prism's base?

- A. $V(x) = x(x + 9)(x + 7)$
- B. $V(x) = x(x + 9)(x - 7)$
- C. $V(x) = 9x(x + 7)$
- D. $V(x) = 9x(x - 7)$

ID: 42045837 Answer

Correct Answer: D

Rationale

Choice D is correct. The volume of a right rectangular prism can be represented by a function V that gives the volume of the prism, in cubic inches, in terms of the length of the prism's base. The volume of a right rectangular prism is equal to the area of its base times its height. It's given that the length of the prism's base is x inches, which is **7** inches more than the width of the prism's base. This means that the width of the prism's base is $x - 7$ inches. It follows that the area of the prism's base, in square inches, is $x(x - 7)$ and the volume, in cubic inches, of the prism is $x(x - 7)(9)$. Thus, the function V that gives the volume of this right rectangular prism, in cubic inches, in terms of the length of the prism's base, x , is $V(x) = 9x(x - 7)$.

Choice A is incorrect. This function would give the volume of the prism if the height were **9** inches more than the length of its base and the width of the base were **7** inches more than its length.

Choice B is incorrect. This function would give the volume of the prism if the height were **9** inches more than the length of its base.

Choice C is incorrect. This function would give the volume of the prism if the width of the base were **7** inches more than its length, rather than the length of the base being **7** inches more than its width.

Question Difficulty: Hard

Question ID 27711b27

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 27711b27

A company has a newsletter. In January 2018, there were 1,300 customers subscribed to the newsletter. For the next 24 months after January 2018, the total number of customers subscribed to the newsletter each month was 7% greater than the total number subscribed the previous month. Which equation gives the total number of customers, c , subscribed to the company's newsletter m months after January 2018, where $m \leq 24$?

- A. $c = 1,300(0.07)^m$
- B. $c = 1,300(1.07)^m$
- C. $c = 1,300(1.7)^m$
- D. $c = 1,300(7)^m$

ID: 27711b27 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that in January 2018, there were 1,300 customers subscribed to a company's newsletter and for the next 24 months after January 2018, the total number of customers subscribed to the newsletter each month was 7% greater than the total number subscribed the previous month. It follows that this situation can be represented by the equation $c = a(1 + \frac{r}{100})^m$, where c is the total number of customers subscribed to the company's newsletter m months after January 2018, a is the number of customers subscribed to the newsletter in January 2018, and the total number of customers subscribed to the newsletter each month was $r\%$ greater than the total number subscribed the previous month. Substituting 1,300 for a and 7 for r in this equation yields $c = 1,300(1 + \frac{7}{100})^m$, or $c = 1,300(1.07)^m$.

Choice A is incorrect. This equation represents a situation where the total number of customers subscribed each month was 93% less, not 7% greater, than the total number subscribed the previous month.

Choice C is incorrect. This equation represents a situation where the total number of customers subscribed each month was 70%, not 7%, greater than the total number subscribed the previous month.

Choice D is incorrect. This equation represents a situation where the total number of customers subscribed each month was 600%, not 7%, greater than the total number subscribed the previous month.

Question Difficulty: Medium

Question ID ef091e05

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: ef091e05

$f(x) = \frac{a-19}{x} + 5$

In the given function f , a is a constant. The graph of function f in the xy -plane, where $y = f(x)$, is translated **3** units down and **4** units to the right to produce the graph of $y = g(x)$. Which equation defines function g ?

- A. $g(x) = \frac{a-19}{x+4} + 2$
- B. $g(x) = \frac{a-19}{x-4} + 2$
- C. $g(x) = \frac{a-22}{x+4} + 5$
- D. $g(x) = \frac{a-22}{x-4} + 5$

ID: ef091e05 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the graph of $y = g(x)$ is produced by translating the graph of $y = f(x)$ **3** units down and **4** units to the right in the xy -plane. Therefore, function g can be defined by an equation in the form $g(x) = f(x - 4) - 3$. Function f is defined by the equation $f(x) = \frac{a-19}{x} + 5$, where a is a constant. Substituting $x - 4$ for x in the equation $f(x) = \frac{a-19}{x} + 5$ yields $f(x - 4) = \frac{a-19}{x-4} + 5$. Substituting $\frac{a-19}{x-4} + 5$ for $f(x - 4)$ in the equation $g(x) = f(x - 4) - 3$ yields $g(x) = \frac{a-19}{x-4} + 5 - 3$, or $g(x) = \frac{a-19}{x-4} + 2$. Therefore, the equation that defines function g is $g(x) = \frac{a-19}{x-4} + 2$.

Choice A is incorrect. This equation defines a function whose graph is produced by translating the graph of $y = f(x)$ **3** units down and **4** units to the left, not **3** units down and **4** units to the right.

Choice C is incorrect. This equation defines a function whose graph is produced by translating the graph of $y = f(x)$ **4** units to the left, not **3** units down and **4** units to the right.

Choice D is incorrect. This equation defines a function whose graph is produced by translating the graph of $y = f(x)$ **4** units to the right, not **3** units down and **4** units to the right.

Question Difficulty: Hard

Question ID 8e83d9b7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 8e83d9b7

A model predicts that the population of Bergen was **15,000** in **2005**. The model also predicts that each year for the next **5** years, the population p increased by **4%** of the previous year's population. Which equation best represents this model, where x is the number of years after **2005**, for $x \leq 5$?

- A. $p = 0.96(15,000)^x$
- B. $p = 1.04(15,000)^x$
- C. $p = 15,000(0.96)^x$
- D. $p = 15,000(1.04)^x$

ID: 8e83d9b7 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that a model predicts the population of Bergen in **2005** was **15,000**. The model also predicts that each year for the next **5** years, the population increased by **4%** of the previous year's population. The predicted population in one of these years can be found by multiplying the predicted population from the previous year by **1.04**. Since the predicted population in **2005** was **15,000**, the predicted population **1** year later is **15,000(1.04)**. The predicted population **2** years later is this value times **1.04**, which is **15,000(1.04)(1.04)**, or **15,000(1.04)²**. The predicted population **3** years later is this value times **1.04**, or **15,000(1.04)³**. More generally, the predicted population, p , x years after **2005** is represented by the equation $p = 15,000(1.04)^x$.

Choice A is incorrect. Substituting **0** for x in this equation indicates the predicted population in **2005** was **0.96** rather than **15,000**.

Choice B is incorrect. Substituting **0** for x in this equation indicates the predicted population in **2005** was **1.04** rather than **15,000**.

Choice C is incorrect. This equation indicates the predicted population is decreasing, rather than increasing, by **4%** each year.

Question Difficulty: Medium

Question ID 74fabd59

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 74fabd59

The product of a positive number x and the number that is 8 more than x is 180. What is the value of x ?

- A. 5
- B. 10
- C. 18
- D. 36

ID: 74fabd59 Answer

Correct Answer: B

Rationale

Choice B is correct. The number that's 8 more than x can be represented by the expression $x + 8$. It's given that the product of x and $x + 8$ is 180, so it follows that $(x)(x + 8) = 180$, or $x^2 + 8x = 180$. Subtracting 180 from each side of this equation yields $x^2 + 8x - 180 = 0$. Factoring the left-hand side of this equation yields $(x - 10)(x + 18) = 0$. Applying the zero product property to this equation yields two solutions: $x = 10$ and $x = -18$. Since x is a positive number, the value of x is 10.

Choice A is incorrect. If $x = 5$, the product of x and the number that's 8 more than x would be $(5)(13)$, or 65, not 180.

Choice C is incorrect. This is the value of the number that's 8 more than x , not the value of x .

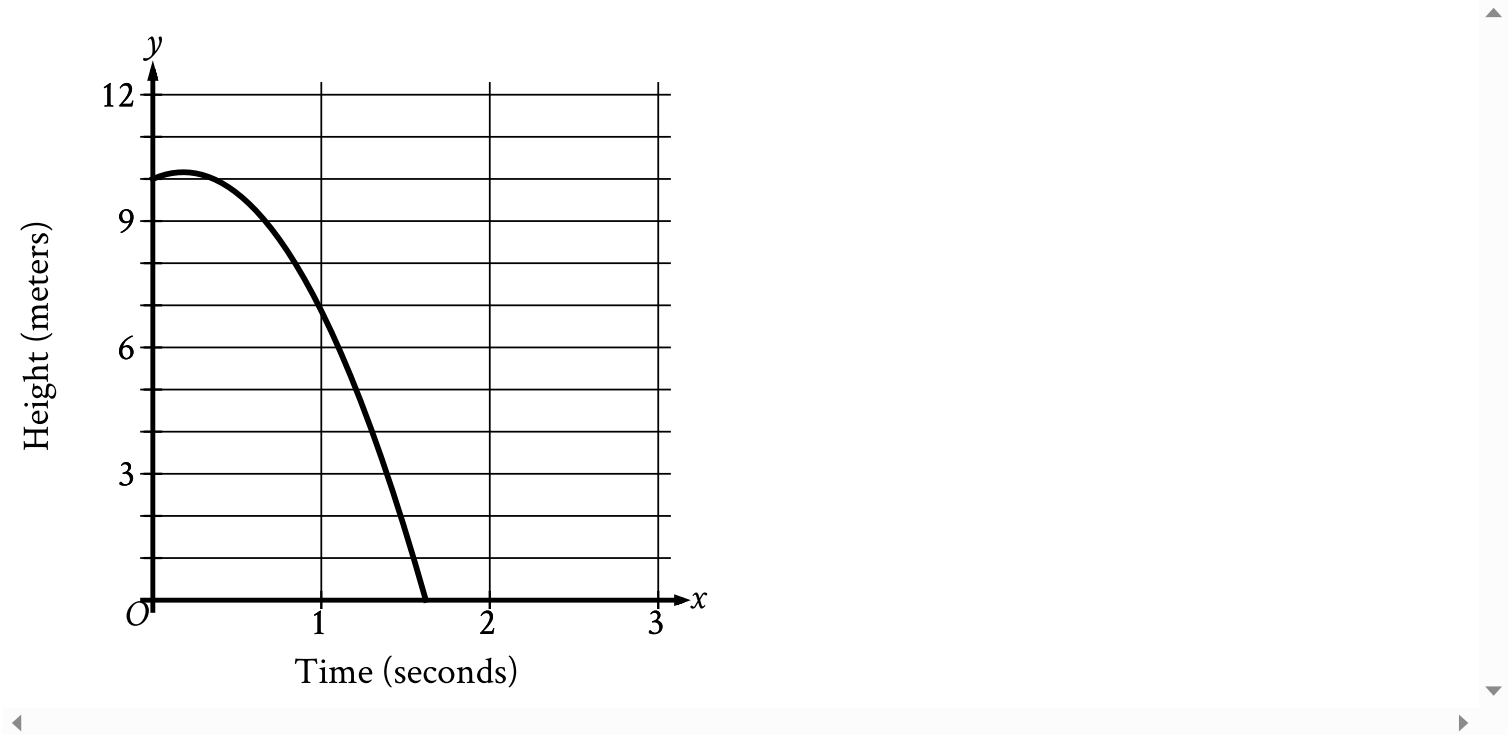
Choice D is incorrect. If $x = 36$, the product of x and the number that's 8 more than x would be $(36)(44)$, or 1,584, not 180.

Question Difficulty: Medium

Question ID 05e193b5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 05e193b5



A competitive diver dives from a platform into the water. The graph shown gives the height above the water y , in meters, of the diver x seconds after diving from the platform. What is the best interpretation of the x -intercept of the graph?

- A. The diver reaches a maximum height above the water at **1.6** seconds.
- B. The diver hits the water at **1.6** seconds.
- C. The diver reaches a maximum height above the water at **0.2** seconds.
- D. The diver hits the water at **0.2** seconds.

ID: 05e193b5 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that the graph shows the height above the water y , in meters, of a diver x seconds after diving from a platform. The x -intercept of a graph is the point at which the graph intersects the x -axis, or when the value of y is **0**. The graph shown intersects the x -axis between $x = 1$ and $x = 2$. In other words, the diver is **0** meters above the water, or hits the water, between **1** and **2** seconds after diving from the platform. Of the given choices, only choice B includes an interpretation where the diver hits the water between **1** and **2** seconds. Therefore, the best interpretation of the x -intercept of the graph is the diver hits the water at **1.6** seconds.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. This is the best interpretation of the maximum value, not the x -intercept, of the graph.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID a85c6c61

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: a85c6c61

$|x + 45| = 48$

What is the positive solution to the given equation?

- A. 3
- B. 48
- C. 93
- D. 96

ID: a85c6c61 Answer

Correct Answer: A

Rationale

Choice A is correct. The given absolute value equation can be rewritten as two linear equations: $x + 45 = 48$ and $-(x + 45) = 48$, or $x + 45 = -48$. Subtracting 45 from both sides of the equation $x + 45 = 48$ yields $x = 3$. Subtracting 45 from both sides of the equation $x + 45 = -48$ yields $x = -93$. Thus, the given equation has two possible solutions, 3 and -93. Therefore, the positive solution to the given equation is 3.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 4bde993d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 4bde993d

$$0.36x^2 + 0.63x + 1.17$$

The given expression can be rewritten as $a(4x^2 + 7x + 13)$, where a is a constant. What is the value of a ?

ID: 4bde993d Answer

Correct Answer: .09, 9/100

Rationale

The correct answer is **.09**. It's given that the expression $0.36x^2 + 0.63x + 1.17$ can be rewritten as $a(4x^2 + 7x + 13)$. Applying the distributive property to the expression $a(4x^2 + 7x + 13)$ yields $4ax^2 + 7ax + 13a$. Therefore, $0.36x^2 + 0.63x + 1.17$ can be rewritten as $4ax^2 + 7ax + 13a$. It follows that in the expressions $0.36x^2 + 0.63x + 1.17$ and $4ax^2 + 7ax + 13a$, the coefficients of x^2 are equivalent, the coefficients of x are equivalent, and the constant terms are equivalent. Therefore, $0.36 = 4a$, $0.63 = 7a$, and $1.17 = 13a$. Solving any of these equations for a yields the value of a . Dividing both sides of the equation $0.36 = 4a$ by 4 yields $0.09 = a$. Therefore, the value of a is **0.09**. Note that .09 and 9/100 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID bc43aed0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: bc43aed0

Which expression is equivalent to $6x + 5x + 4y$?

- A. $15x$
- B. $15y$
- C. $11x + 4y$
- D. $30x + 4y$

ID: bc43aed0 Answer

Correct Answer: C

Rationale

Choice C is correct. In the given expression, $6x$ and $5x$ are like terms. Combining these like terms yields $11x$. It follows that the expression $6x + 5x + 4y$ is equivalent to $11x + 4y$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 629e9228

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 629e9228

$2r = 9s + t$

The given equation relates the variables r , s , and t . Which equation correctly expresses s in terms of r and t ?

- A. $s = \frac{2}{9}r - t$
- B. $s = \frac{2r-t}{9}$
- C. $s = 18r - t$
- D. $s = 9(2r - t)$

ID: 629e9228 Answer

Correct Answer: B

Rationale

Choice B is correct. Subtracting t from each side of the given equation yields $2r - t = 9s$. Dividing each side of this equation by 9 yields $\frac{2r-t}{9} = s$, which can be rewritten as $s = \frac{2r-t}{9}$. Therefore, the equation $s = \frac{2r-t}{9}$ correctly expresses s in terms of r and t .

Choice A is incorrect. This equation is equivalent to $2r = 9(s + t)$, not $2r = 9s + t$.

Choice C is incorrect. This equation is equivalent to $2r = \frac{s+t}{9}$, not $2r = 9s + t$.

Choice D is incorrect. This equation is equivalent to $2r = \frac{s}{9} + t$, not $2r = 9s + t$.

Question Difficulty: Medium

Question ID feba5a37

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: feba5a37

A rectangle has an area of **155** square inches. The length of the rectangle is **4** inches less than **7** times the width of the rectangle. What is the width of the rectangle, in inches?

ID: feba5a37 Answer

Correct Answer: 5

Rationale

The correct answer is **5**. Let x represent the width, in inches, of the rectangle. It's given that the length of the rectangle is **4** inches less than **7** times its width, or $7x - 4$ inches. The area of a rectangle is equal to its width multiplied by its length. Multiplying the width, x inches, by the length, $7x - 4$ inches, yields $x(7x - 4)$ square inches. It's given that the rectangle has an area of **155** square inches, so it follows that $x(7x - 4) = 155$, or $7x^2 - 4x = 155$. Subtracting **155** from both sides of this equation yields $7x^2 - 4x - 155 = 0$. Factoring the left-hand side of this equation yields $(7x + 31)(x - 5) = 0$. Applying the zero product property to this equation yields two solutions: $x = -\frac{31}{7}$ and $x = 5$. Since x is the rectangle's width, in inches, which must be positive, the value of x is **5**. Therefore, the width of the rectangle, in inches, is **5**.

Question Difficulty: Hard

Question ID 78216a4d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 78216a4d

The function g is defined by $g(x) = \sqrt{8x + 1}$. What is the value of $g(3)$?

- A. $\frac{5}{8}$
- B. $\frac{25}{8}$
- C. 5
- D. 25

ID: 78216a4d Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the function g is defined by $g(x) = \sqrt{8x + 1}$. Substituting 3 for x in the given function yields $g(3) = \sqrt{8(3) + 1}$, which is equivalent to $g(3) = \sqrt{25}$, or $g(3) = 5$. Therefore, the value of $g(3)$ is 5.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

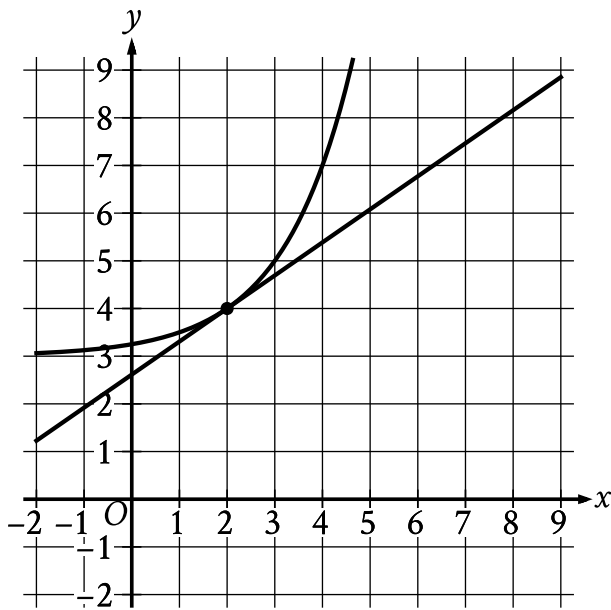
Choice D is incorrect. This is the value of $8(3) + 1$, not $\sqrt{8(3) + 1}$.

Question Difficulty: Medium

Question ID c35b7410

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: c35b7410



The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution (x, y) to this system?

- A. $(0, 0)$
- B. $(0, 2)$
- C. $(2, 4)$
- D. $(4, 0)$

ID: c35b7410 Answer

Correct Answer: C

Rationale

Choice C is correct. The solution to the system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear equation and the nonlinear equation shown intersect at the point $(2, 4)$. Thus, the solution to the system is $(2, 4)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 160f2a35

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 160f2a35

Which quadratic equation has no real solutions?

- A. $x^2 + 14x - 49 = 0$
- B. $x^2 - 14x + 49 = 0$
- C. $5x^2 - 14x - 49 = 0$
- D. $5x^2 - 14x + 49 = 0$

ID: 160f2a35 Answer

Correct Answer: D

Rationale

Choice D is correct. The number of solutions to a quadratic equation in the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. If the value of the discriminant is greater than zero, then the quadratic equation has two distinct real solutions. If the value of the discriminant is equal to zero, then the quadratic equation has exactly one real solution. If the value of the discriminant is less than zero, then the quadratic equation has no real solutions. For the quadratic equation in choice D, $5x^2 - 14x + 49 = 0$, $a = 5$, $b = -14$, and $c = 49$. Substituting 5 for a , -14 for b , and 49 for c in $b^2 - 4ac$ yields $(-14)^2 - 4(5)(49)$, or -784 . Since -784 is less than zero, it follows that the quadratic equation $5x^2 - 14x + 49 = 0$ has no real solutions.

Choice A is incorrect. The value of the discriminant for this quadratic equation is 392 . Since 392 is greater than zero, it follows that this quadratic equation has two real solutions.

Choice B is incorrect. The value of the discriminant for this quadratic equation is 0 . Since zero is equal to zero, it follows that this quadratic equation has exactly one real solution.

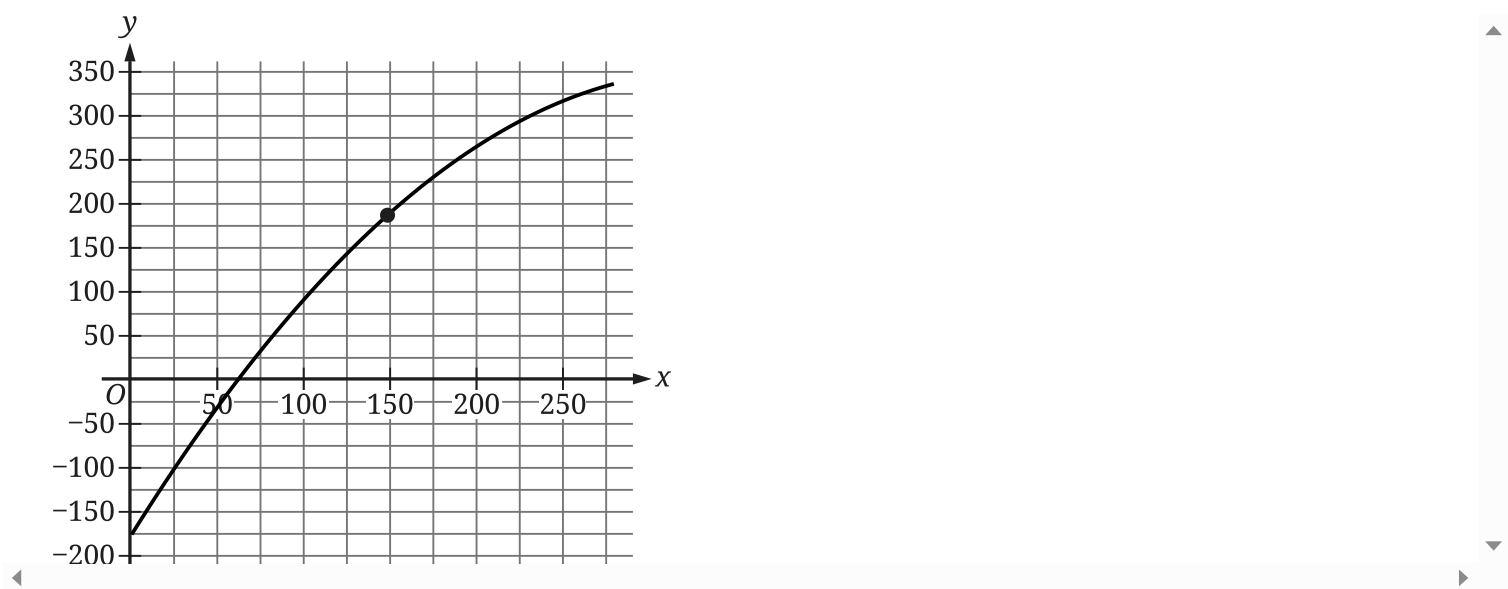
Choice C is incorrect. The value of the discriminant for this quadratic equation is $1,176$. Since $1,176$ is greater than zero, it follows that this quadratic equation has two real solutions.

Question Difficulty: Hard

Question ID 998a3fe3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 998a3fe3



The graph shows the estimated boiling point y , in degrees Celsius, of a normal paraffin with a molecular weight of x grams per mole, where $1 \leq x \leq 280$. Which statement is the best interpretation of the point $(149.02, 186.05)$?

- A. A normal paraffin with a molecular weight of **186.05** grams per mole has an estimated boiling point of **149.02** degrees Celsius.
- B. A normal paraffin with a molecular weight of **149.02** grams per mole has an estimated boiling point of **186.05** degrees Celsius.
- C. The minimum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of **149.02** grams per mole and an estimated boiling point of **186.05** degrees Celsius.
- D. The maximum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of **149.02** grams per mole and an estimated boiling point of **186.05** degrees Celsius.

ID: 998a3fe3 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the graph shows the estimated boiling point y , in degrees Celsius, of a normal paraffin with a molecular weight of x grams per mole. It follows that for the point $(149.02, 186.05)$, **149.02** represents the molecular weight, in grams per mole, of a normal paraffin and **186.05** represents its estimated boiling point, in degrees Celsius. Therefore, the best interpretation of the point $(149.02, 186.05)$ is that a normal paraffin with a molecular weight of **149.02** grams per mole has an estimated boiling point of **186.05** degrees Celsius.

Choice A is incorrect. This statement represents a situation where x , not y , represents the boiling point, in degrees Celsius, and y , not x , represents the molecular weight, in grams, per mole.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID 2fc078ac

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 2fc078ac

$$z^2 + 10z - 24 = 0$$

What is one of the solutions to the given equation?

ID: 2fc078ac Answer

Correct Answer: 2, -12

Rationale

The correct answer is either **2** or **−12**. The left-hand side of the given equation can be rewritten by factoring. The two values that multiply to **−24** and add to **10** are **12** and **−2**. It follows that the given equation can be rewritten as $(z + 12)(z - 2) = 0$. Setting each factor equal to **0** yields two equations: $z + 12 = 0$ and $z - 2 = 0$. Subtracting **12** from both sides of the equation $z + 12 = 0$ results in $z = -12$. Adding **2** to both sides of the equation $z - 2 = 0$ results in $z = 2$. Note that 2 and -12 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID 638544c4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 638544c4

The area of a rectangular banner is **2,661** square inches. The banner's length x , in inches, is **24** inches longer than its width, in inches. Which equation represents this situation?

- A. $0 = x^2 - 24x - 2,661$
- B. $0 = x^2 - 24x + 2,661$
- C. $0 = x^2 + 24x - 2,661$
- D. $0 = x^2 + 24x + 2,661$

ID: 638544c4 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the banner's length x , in inches, is **24** inches longer than its width, in inches. It follows that the banner's width, in inches, can be represented by the expression $x - 24$. The area of a rectangle is the product of its length and its width. It's given that the area of the banner is **2,661** square inches, so it follows that $2,661 = x(x - 24)$, or $2,661 = x^2 - 24x$. Subtracting **2,661** from each side of this equation yields $0 = x^2 - 24x - 2,661$. Therefore, the equation that represents this situation is $0 = x^2 - 24x - 2,661$.

Choice B is incorrect and may result from representing the width, in inches, of the banner as $24 - x$, rather than $x - 24$.

Choice C is incorrect and may result from representing the width, in inches, of the banner as $x + 24$, rather than $x - 24$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 4cc2980e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 4cc2980e

A company opens an account with an initial balance of **\$36,100.00**. The account earns interest, and no additional deposits or withdrawals are made. The account balance is given by an exponential function ***A***, where ***A*(*t*)** is the account balance, in dollars, ***t*** years after the account is opened. The account balance after **13** years is **\$68,071.93**. Which equation could define ***A***?

- A. ***A*(*t*) = 36,100.00(1.05)^{*t*}**
- B. ***A*(*t*) = 31,971.93(1.05)^{*t*}**
- C. ***A*(*t*) = 31,971.93(0.05)^{*t*}**
- D. ***A*(*t*) = 36,100.00(0.05)^{*t*}**

ID: 4cc2980e Answer

Correct Answer: A

Rationale

Choice A is correct. Since it's given that the account balance, ***A*(*t*)**, in dollars, after ***t*** years can be modeled by an exponential function, it follows that function ***A*** can be written in the form ***A*(*t*) = *Nr^t***, where ***N*** is the initial value of the function and ***r*** is a constant related to the growth of the function. It's given that the initial balance of the account is **\$36,100.00**, so it follows that the initial value of the function, or ***N***, must be **36,100.00**. Substituting **36,100.00** for ***N*** in the equation ***A*(*t*) = *Nr^t*** yields ***A*(*t*) = 36,100.00*r^t***. It's given that the account balance after **13** years, or when ***t* = 13**, is **\$68,071.93**. It follows that ***A*(13) = 68,071.93**, or **36,100.00*r¹³* = 68,071.93**. Dividing each side of the equation **36,100.00*r¹³* = 68,071.93** by **36,100.00** yields ***r¹³* = $\frac{68,071.93}{36,100.00}$** . Taking the **13**th root of both sides of this equation yields ***r* = $\sqrt[13]{\frac{68,071.93}{36,100.00}}$** , or ***r*** is approximately equal to **1.05**. Substituting **1.05** for ***r*** in the equation ***A*(*t*) = 36,100.00*r^t*** yields ***A*(*t*) = 36,100.00(1.05)^{*t*}**, so the equation ***A*(*t*) = 36,100.00(1.05)^{*t*}** could define ***A***.

Choice B is incorrect. Substituting **0** for ***t*** in this function indicates an initial balance of **\$31,971.93**, rather than **\$36,100.00**.

Choice C is incorrect. Substituting **0** for ***t*** in this function indicates an initial balance of **\$31,971.93**, rather than **\$36,100.00**. Additionally, this function indicates the account balance is decreasing, rather than increasing, over time.

Choice D is incorrect. This function indicates the account balance is decreasing, rather than increasing, over time.

Question Difficulty: Medium

Question ID 62dd5b77

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 62dd5b77

$$\begin{aligned}f(x) &= x^2 + bx \\ g(x) &= 9x^2 - 27x\end{aligned}$$

Functions f and g are given, and in function f , b is a constant. If $f(x) \cdot g(x) = 9x^4 - 26x^3 - 3x^2$, what is the value of b ?

- A. -26
- B. $-\frac{26}{9}$
- C. $\frac{1}{9}$
- D. 9

ID: 62dd5b77 Answer

Correct Answer: C

Rationale

Choice C is correct. Multiplying the given functions f and g yields $f(x) \cdot g(x) = (x^2 + bx)(9x^2 - 27x)$. Applying the distributive property to the right-hand side of this equation yields $f(x) \cdot g(x) = (x^2)(9x^2 - 27x) + (bx)(9x^2 - 27x)$. Applying the distributive property once again to the right-hand side of this equation yields $f(x) \cdot g(x) = (x^2)(9x^2) - (x^2)(27x) + (bx)(9x^2) - (bx)(27x)$, which is equivalent to $f(x) \cdot g(x) = 9x^4 - 27x^3 + 9bx^3 - 27bx^2$. Factoring out x^3 from the second and third terms yields $f(x) \cdot g(x) = 9x^4 + (-27 + 9b)x^3 - 27bx^2$. Since the left-hand sides of $f(x) \cdot g(x) = 9x^4 + (-27 + 9b)x^3 - 27bx^2$ and $f(x) \cdot g(x) = 9x^4 - 26x^3 - 3x^2$ are equal, it follows that $(-27 + 9b)x^3 = -26x^3$, or $-27 + 9b = -26$, and $-27bx^2 = -3x^2$, or $-27b = -3$. Adding 27 to each side of $-27 + 9b = -26$ yields $9b = 1$. Dividing each side of this equation by 9 yields $b = \frac{1}{9}$. Similarly, dividing each side of $-27b = -3$ by -27 yields $b = \frac{1}{9}$. Therefore, the value of b is $\frac{1}{9}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 628fa44f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 628fa44f

$f(x) = 27(1.20)^{\frac{x}{3}}$

For the given function f , the value of $f(x)$ increases by $p\%$ for every increase of x by 6 . What is the value of p ?

ID: 628fa44f Answer

Correct Answer: 44

Rationale

The correct answer is **44**. For a function defined by an equation of the form $f(x) = a\left(1 + \frac{r}{100}\right)^{\frac{x}{k}}$, where a , r , and k are positive constants, the value of $f(x)$ increases by $r\%$ for every increase of x by k . The given equation can be rewritten as $f(x) = 27(1.20)^{\frac{2x}{6}}$, or $f(x) = 27\left((1.2)^2\right)^{\frac{x}{6}}$, which is equivalent to $f(x) = 27(1.44)^{\frac{x}{6}}$, or $f(x) = 27\left(1 + \frac{44}{100}\right)^{\frac{x}{6}}$. This equation is of the form $f(x) = a\left(1 + \frac{r}{100}\right)^{\frac{x}{k}}$, where $a = 27$, $r = 44$, and $k = 6$. It follows that the value of $f(x)$ increases by **44%** for every increase of x by 6 . Therefore, the value of p is **44**.

Alternate approach: It's given that $f(x) = 27(1.20)^{\frac{x}{3}}$. An increase in the value of x by 6 can be represented by the expression $x + 6$. The value of $f(x + 6)$ can be found by substituting $x + 6$ for x in the equation $f(x) = 27(1.20)^{\frac{x}{3}}$, which yields $f(x + 6) = 27(1.20)^{\frac{x+6}{3}}$. By the product property of exponents, this equation is equivalent to $f(x + 6) = 27(1.20)^{\frac{x}{3}} \cdot (1.20)^{\frac{6}{3}}$, or $f(x + 6) = 27(1.20)^{\frac{x}{3}} \cdot (1.20)^2$, which can be rewritten as $f(x + 6) = 27(1.20)^{\frac{x}{3}} \cdot (1.44)$. Since $f(x) = 27(1.20)^{\frac{x}{3}}$, it follows that $f(x + 6) = f(x) \cdot (1.44)$. Therefore, for any value of x , the value of $f(x + 6)$ is **1.44** times, or **44%** greater than, the value of $f(x)$. It's given that the value of $f(x)$ increases by $p\%$ for every increase in the value of x by 6 . It follows that the value of p is **44**.

Question Difficulty: Hard

Question ID 08ffcaa3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 08ffcaa3

Which expression is equivalent to $16(x + 15)$?

- A. $16x + 31$
- B. $16x + 240$
- C. $16x + 1$
- D. $16x + 15$

ID: 08ffcaa3 Answer

Correct Answer: B

Rationale

Choice B is correct. The expression $16(x + 15)$ can be rewritten as $16(x) + 16(15)$, which is equivalent to $16x + 240$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 339e04b7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 339e04b7

$$\begin{aligned}y &= 5x + 4 \\ y &= 5x^2 + 4\end{aligned}$$

Which ordered pair (x, y) is a solution to the given system of equations?

- A. $(0, 0)$
- B. $(0, 4)$
- C. $(8, 44)$
- D. $(8, 84)$

ID: 339e04b7 Answer

Correct Answer: B

Rationale

Choice B is correct. The second equation in the given system is $y = 5x^2 + 4$. Substituting $5x^2 + 4$ for y in the first equation of the given system yields $5x^2 + 4 = 5x + 4$. Subtracting 4 from both sides of this equation yields $5x^2 = 5x$. Subtracting $5x$ from both sides of this equation yields $5x^2 - 5x = 0$. Factoring out a common factor of $5x$ from the left-hand side of this equation yields $5x(x - 1) = 0$. It follows that $5x = 0$ or $x - 1 = 0$. Dividing both sides of the equation $5x = 0$ by 5 yields $x = 0$. Substituting 0 for x in the equation $y = 5x + 4$ yields $y = 5(0) + 4$, or $y = 4$. Therefore, a solution to the given system of equations is the ordered pair $(0, 4)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID a71e3029

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: a71e3029

$$\frac{20}{p} = \frac{20}{q} - \frac{20}{r} - \frac{20}{s}$$

The given equation relates the positive variables p , q , r , and s . Which of the following is equivalent to q ?

- A. $p + r + s$
- B. $20(p + r + s)$
- C. $\frac{prs}{pr+ps+rs}$
- D. $\frac{prs}{20p+20r+20s}$

ID: a71e3029 Answer

Correct Answer: C

Rationale

Choice C is correct. Multiplying each side of the given equation by $\frac{1}{20}$ yields $\frac{1}{20} \left(\frac{20}{p} \right) = \frac{1}{20} \left(\frac{20}{q} - \frac{20}{r} - \frac{20}{s} \right)$. Distributing $\frac{1}{20}$ on each side of this equation yields $\frac{20}{20p} = \frac{20}{20q} - \frac{20}{20r} - \frac{20}{20s}$, or $\frac{1}{p} = \frac{1}{q} - \frac{1}{r} - \frac{1}{s}$. Adding $\frac{1}{r} + \frac{1}{s}$ to each side of this equation yields $\frac{1}{s} + \frac{1}{r} + \frac{1}{p} = \frac{1}{q}$. Multiplying $\frac{1}{s}$ by $\frac{pr}{pr}$, $\frac{1}{r}$ by $\frac{ps}{ps}$, and $\frac{1}{p}$ by $\frac{rs}{rs}$ yields $\frac{pr}{prs} + \frac{ps}{prs} + \frac{rs}{prs} = \frac{1}{q}$, which is equivalent to $\frac{pr+ps+rs}{prs} = \frac{1}{q}$. Since $\frac{pr+ps+rs}{prs} = \frac{1}{q}$, and it's given that p , q , r , and s are positive, it follows that the reciprocals of each side of this equation are also equal. Thus, $\frac{prs}{pr+ps+rs} = \frac{q}{1}$, or $\frac{prs}{pr+ps+rs} = q$. Therefore, $\frac{prs}{pr+ps+rs}$ is equivalent to q .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 429593eb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 429593eb

<i>x</i>	<i>y</i>
21	−8
23	8
25	−8

The table shows three values of *x* and their corresponding values of *y*, where $y = f(x) + 4$ and *f* is a quadratic function. What is the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane?

ID: 429593eb Answer

Correct Answer: -2112

Rationale

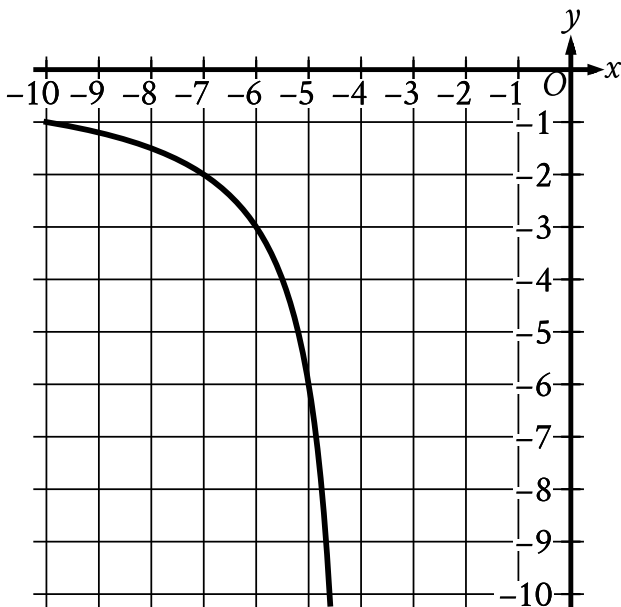
The correct answer is **−2,112**. It's given that *f* is a quadratic function. It follows that *f* can be defined by an equation of the form $f(x) = a(x - h)^2 + k$, where *a*, *h*, and *k* are constants. It's also given that the table shows three values of *x* and their corresponding values of *y*, where $y = f(x) + 4$. Substituting $a(x - h)^2 + k$ for *f*(*x*) in this equation yields $y = a(x - h)^2 + k + 4$. This equation represents a quadratic relationship between *x* and *y*, where *k* + 4 is either the maximum or the minimum value of *y*, which occurs when *x* = *h*. For quadratic relationships between *x* and *y*, the maximum or minimum value of *y* occurs at the value of *x* halfway between any two values of *x* that have the same corresponding value of *y*. The table shows that x-values of **21** and **25** correspond to the same y-value, **−8**. Since **23** is halfway between **21** and **25**, the maximum or minimum value of *y* occurs at an x-value of **23**. The table shows that when *x* = **23**, *y* = **8**. It follows that *h* = **23** and *k* + 4 = **8**. Subtracting **4** from both sides of the equation *k* + 4 = **8** yields *k* = **4**. Substituting **23** for *h* and **4** for *k* in the equation $y = a(x - h)^2 + k + 4$ yields $y = a(x - 23)^2 + 4 + 4$, or $y = a(x - 23)^2 + 8$. The value of *a* can be found by substituting any x-value and its corresponding y-value for *x* and *y*, respectively, in this equation. Substituting **25** for *x* and **−8** for *y* in this equation yields $-8 = a(25 - 23)^2 + 8$, or $-8 = a(2)^2 + 8$. Subtracting **8** from both sides of this equation yields $-16 = a(2)^2$, or $-16 = 4a$. Dividing both sides of this equation by **4** yields $-4 = a$. Substituting -4 for *a*, **23** for *h*, and **4** for *k* in the equation $f(x) = a(x - h)^2 + k$ yields $f(x) = -4(x - 23)^2 + 4$. The y-intercept of the graph of $y = f(x)$ in the xy-plane is the point on the graph where *x* = **0**. Substituting **0** for *x* in the equation $f(x) = -4(x - 23)^2 + 4$ yields $f(0) = -4(0 - 23)^2 + 4$, or $f(0) = -4(-23)^2 + 4$. This is equivalent to $f(0) = -2,112$, so the y-intercept of the graph of $y = f(x)$ in the xy-plane is **(0, −2,112)**. Thus, the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane is **−2,112**.

Question Difficulty: Hard

Question ID cec444a7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: cec444a7



The rational function f is defined by an equation in the form $f(x) = \frac{a}{x+b}$, where a and b are constants. The partial graph of $y = f(x)$ is shown. If $g(x) = f(x + 4)$, which equation could define function g ?

- A. $g(x) = \frac{6}{x}$
- B. $g(x) = \frac{6}{x+4}$
- C. $g(x) = \frac{6}{x+8}$
- D. $g(x) = \frac{6(x+4)}{x+4}$

ID: cec444a7 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $f(x) = \frac{a}{x+b}$ and that the graph shown is a partial graph of $y = f(x)$. Substituting y for $f(x)$ in the equation $f(x) = \frac{a}{x+b}$ yields $y = \frac{a}{x+b}$. The graph passes through the point $(-7, -2)$. Substituting -7 for x and -2 for y in the equation $y = \frac{a}{x+b}$ yields $-2 = \frac{a}{-7+b}$. Multiplying each side of this equation by $-7 + b$ yields $-2(-7 + b) = a$, or $14 - 2b = a$. The graph also passes through the point $(-5, -6)$. Substituting -5 for x and -6 for y in the equation $y = \frac{a}{x+b}$ yields $-6 = \frac{a}{-5+b}$. Multiplying each side of this equation by $-5 + b$ yields $-6(-5 + b) = a$, or $30 - 6b = a$. Substituting $14 - 2b$ for a in this equation yields $30 - 6b = 14 - 2b$. Adding $6b$ to each side of this equation yields $30 = 14 + 4b$. Subtracting 14 from each side of this equation yields $16 = 4b$. Dividing each side of this equation by 4 yields $4 = b$. Substituting 4 for b in the equation $14 - 2b = a$ yields $14 - 2(4) = a$, or $6 = a$. Substituting 6 for a and 4 for b in the equation $f(x) = \frac{a}{x+b}$ yields $f(x) = \frac{6}{x+4}$. It's given that $g(x) = f(x + 4)$. Substituting $x + 4$ for x in the equation $f(x) = \frac{6}{x+4}$ yields $f(x + 4) = \frac{6}{x+4+4}$, which is equivalent to $f(x + 4) = \frac{6}{x+8}$. It follows that $g(x) = \frac{6}{x+8}$.

Choice A is incorrect. This could define function g if $g(x) = f(x - 4)$.

Choice B is incorrect. This could define function g if $g(x) = f(x)$.

Choice D is incorrect. This could define function g if $g(x) = f(x) \cdot (x + 4)$.

Question Difficulty: Hard

Question ID f82cc584

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: f82cc584

A park ranger hung squirrel houses each in the shape of a right rectangular prism for fox squirrels. Each house has a height of **11** inches. The length of each house's base is x inches, which is **1** inch more than the width of the house's base. Which function V gives the volume of each house, in cubic inches, in terms of the length of the house's base?

- A. $V(x) = 11x(x - 1)$
- B. $V(x) = 11x(x + 1)$
- C. $V(x) = x(x + 11)(x - 1)$
- D. $V(x) = x(x + 11)(x + 1)$

ID: f82cc584 Answer

Correct Answer: A

Rationale

Choice A is correct. The volume of a prism is equal to the area of its base times its height. It's given that the length of each house's base is x inches and that this length is **1** inch more than the width, in inches, of the house's base. It follows that the width, in inches, of the house's base is $x - 1$. The area of a rectangle is the product of its length and its width. Therefore, the area of the base of the house is $x(x - 1)$ square inches. It's given that the height of each house is **11** inches. Therefore, the function V that gives the volume of each house, in cubic inches, in terms of the length of the house's base x is $V(x) = x(x - 1)11$, or $V(x) = 11x(x - 1)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 6db620cc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 6db620cc

If $\frac{42}{x} = 7x$, what is the value of $7x^2$?

- A. 6
- B. 7
- C. 42
- D. 294

ID: 6db620cc Answer

Correct Answer: C

Rationale

Choice C is correct. Multiplying both sides of the given equation by x yields $42 = 7x^2$. Therefore, the value of $7x^2$ is 42.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 490e956a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 490e956a

$r(t) = 63t - 2t^2$

The function r is defined by the given equation. The function s is defined by $s(t) = r(t) + 1$. Which expression represents the maximum value of $s(t)$?

- A. $1 + \left(\frac{63}{2}\right)^2$
- B. $1 + 2\left(\frac{63}{4}\right)^2$
- C. $1 - 2\left(\frac{63}{4}\right)^2$
- D. $1 - \left(\frac{63}{2}\right)^2$

ID: 490e956a Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that $s(t) = r(t) + 1$. Since $r(t) = 63t - 2t^2$, it follows that $s(t) = 63t - 2t^2 + 1$, or $s(t) = -2t^2 + 63t + 1$. For a quadratic function s defined by an equation of the form $s(t) = a(t - h)^2 + k$, where a , h , and k are constants and a is negative, the value of $s(t)$ reaches its maximum, k , when the value of t is h . The equation $s(t) = -2t^2 + 63t + 1$ can be rewritten in the form $s(t) = a(t - h)^2 + k$ by completing the square. The equation $s(t) = -2t^2 + 63t + 1$ is equivalent to $s(t) = -2\left(t^2 - \frac{63}{2}t\right) + 1$, or $s(t) = -2\left(t^2 - \frac{63}{2}t + \left(\frac{63}{4}\right)^2 - \left(\frac{63}{4}\right)^2\right) + 1$. This equation can be rewritten as $s(t) = -2\left(\left(t - \frac{63}{4}\right)^2 - \left(\frac{63}{4}\right)^2\right) + 1$, or $s(t) = -2\left(t - \frac{63}{4}\right)^2 + 1 + 2\left(\frac{63}{4}\right)^2$. This equation is in the form $s(t) = a(t - h)^2 + k$, where $a = -2$, $h = \frac{63}{4}$, and $k = 1 + 2\left(\frac{63}{4}\right)^2$. Therefore, the maximum value of $s(t)$ is $1 + 2\left(\frac{63}{4}\right)^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 7820dc15

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 7820dc15

$r^2 + qr = 2r - 55$

In the given equation, q is an integer constant. The given equation has no real solutions. What is the largest possible value of q ?

ID: 7820dc15 Answer

Correct Answer: 16

Rationale

The correct answer is **16**. A quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has no real solution if and only if its discriminant, $b^2 - 4ac$, is negative. It's given that in the equation $r^2 + qr = 2r - 55$, q is an integer constant. Subtracting $2r$ and adding 55 to both sides of this equation yields $r^2 + qr - 2r + 55 = 0$, or $r^2 + (q - 2)r + 55 = 0$. In this equation, $a = 1$, $b = q - 2$, and $c = 55$. Substituting 1 for a , $q - 2$ for b , and 55 for c in the discriminant expression yields $(q - 2)^2 - 4(1)(55)$, or $(q - 2)^2 - 220$. Since the value of the discriminant must be negative, it follows that $(q - 2)^2 - 220 < 0$, or $(q - 2)^2 < 220$. This is equivalent to $|q - 2| < \sqrt{220}$, or $-\sqrt{220} < q - 2 < \sqrt{220}$. Adding 2 to each part of this inequality yields $2 - \sqrt{220} < q < 2 + \sqrt{220}$. Since $\sqrt{220}$ is approximately **14.83**, it follows that $-12.83 < q < 16.83$. Since q is an integer, the largest possible value of q is **16**.

Question Difficulty: Hard

Question ID d1695bb7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: d1695bb7

The function f is defined by $f(x) = 3x^5$. In the xy -plane, the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ up 4 units. Which equation defines the function g ?

- A. $g(x) = 3x^5 + 4$
- B. $g(x) = 3x^5 - 4$
- C. $g(x) = 3x^{20}$
- D. $g(x) = 12x^5$

ID: d1695bb7 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that in the xy -plane, the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ up 4 units. It follows that $g(x) = f(x) + 4$. It's also given that function f is defined by $f(x) = 3x^5$. Substituting $3x^5$ for $f(x)$ into $g(x) = f(x) + 4$ yields $g(x) = 3x^5 + 4$. Therefore, the equation that defines function g is $g(x) = 3x^5 + 4$.

Choice B is incorrect. This equation defines a function g for which the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ down, not up, 4 units.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID ec297cd8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: ec297cd8

$$\begin{aligned}x &= 3 \\ y &= (15 - x)^2\end{aligned}$$

A solution to the given system of equations is (x, y) . What is the value of xy ?

- A. 432
- B. 54
- C. 45
- D. 18

ID: ec297cd8 Answer

Correct Answer: A

Rationale

Choice A is correct. The first equation in the given system of equations is $x = 3$. Substituting **3** for x in the second equation in the given system of equations yields $y = (15 - 3)^2$, or $y = 144$. Substituting **3** for x and **144** for y in the expression xy yields **(3)(144)**, or **432**. Therefore, the value of xy is **432**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 1563ad0d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 1563ad0d

$$\begin{aligned}x &= 49 \\ y &= \sqrt{x} + 9\end{aligned}$$

The graphs of the given equations intersect at the point (x, y) in the xy -plane. What is the value of y ?

- A. 16
- B. 40
- C. 81
- D. 130

ID: 1563ad0d Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the graphs of the given equations intersect at the point (x, y) in the xy -plane. It follows that (x, y) represents a solution to the system consisting of the given equations. The first equation given is $x = 49$. Substituting **49** for x in the second equation given, $y = \sqrt{x} + 9$, yields $y = \sqrt{49} + 9$, which is equivalent to $y = 7 + 9$, or $y = 16$. It follows that the graphs of the given equations intersect at the point **(49, 16)**. Therefore, the value of y is **16**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 3b0ffb71

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 3b0ffb71

Function f is defined by $f(x) = -a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x) - 12$ has a y -intercept at $(0, -\frac{75}{7})$. The product of a and b is $\frac{320}{7}$. What is the value of a ?

ID: 3b0ffb71 Answer

Correct Answer: 20

Rationale

The correct answer is **20**. It's given that $f(x) = -a^x + b$. Substituting $-a^x + b$ for $f(x)$ in the equation $y = f(x) - 12$ yields $y = -a^x + b - 12$. It's given that the y -intercept of the graph of $y = f(x) - 12$ is $(0, -\frac{75}{7})$. Substituting **0** for x and $-\frac{75}{7}$ for y in the equation $y = -a^x + b - 12$ yields $-\frac{75}{7} = -a^0 + b - 12$, which is equivalent to $-\frac{75}{7} = -1 + b - 12$, or $-\frac{75}{7} = b - 13$. Adding **13** to both sides of this equation yields $\frac{16}{7} = b$. It's given that the product of a and b is $\frac{320}{7}$, or $ab = \frac{320}{7}$. Substituting $\frac{16}{7}$ for b in this equation yields $(a)(\frac{16}{7}) = \frac{320}{7}$. Dividing both sides of this equation by $\frac{16}{7}$ yields $a = \mathbf{20}$.

Question Difficulty: Hard

Question ID 46b433ae

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 46b433ae

$x(kx - 56) = -16$

In the given equation, k is an integer constant. If the equation has no real solution, what is the least possible value of k ?

ID: 46b433ae Answer

Correct Answer: 50

Rationale

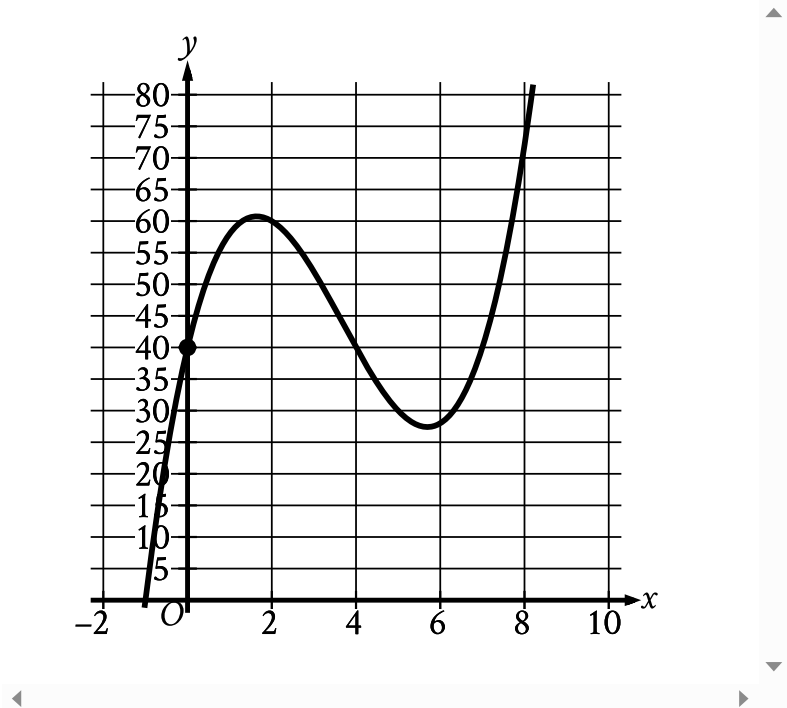
The correct answer is **50**. An equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has no real solutions if and only if its discriminant, $b^2 - 4ac$, is negative. Applying the distributive property to the left-hand side of the equation $x(kx - 56) = -16$ yields $kx^2 - 56x = -16$. Adding **16** to each side of this equation yields $kx^2 - 56x + 16 = 0$. Substituting k for a , -56 for b , and **16** for c in $b^2 - 4ac$ yields a discriminant of $(-56)^2 - 4(k)(16)$, or **3,136** $- 64k$. If the given equation has no real solution, it follows that the value of **3,136** $- 64k$ must be negative. Therefore, **3,136** $- 64k < 0$. Adding **64k** to both sides of this inequality yields **3,136** $< 64k$. Dividing both sides of this inequality by **64** yields **49** $< k$, or $k > 49$. Since it's given that k is an integer, the least possible value of k is **50**.

Question Difficulty: Hard

Question ID 9cc38e5f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 9cc38e5f



The y -intercept of the graph shown is (x, y) . What is the value of y ?

ID: 9cc38e5f Answer

Correct Answer: 40

Rationale

The correct answer is **40**. The y -intercept of a graph in the xy -plane is the point (x, y) on the graph where $x = 0$. The y -intercept of the graph shown is $(0, 40)$. Therefore, the value of y is **40**.

Question Difficulty: Easy

Question ID 285cf49d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 285cf49d

The function f is defined by $f(x) = 5\left(\frac{1}{4} - x\right)^2 + \frac{11}{4}$. What is the value of $f\left(\frac{1}{4}\right)$?

ID: 285cf49d Answer

Correct Answer: 11/4, 2.75

Rationale

The correct answer is $\frac{11}{4}$. It's given that the function f is defined by $f(x) = 5\left(\frac{1}{4} - x\right)^2 + \frac{11}{4}$. Substituting $\frac{1}{4}$ for x in this equation yields $f\left(\frac{1}{4}\right) = 5\left(\frac{1}{4} - \frac{1}{4}\right)^2 + \frac{11}{4}$, which is equivalent $f\left(\frac{1}{4}\right) = 5(0)^2 + \frac{11}{4}$, or $f\left(\frac{1}{4}\right) = \frac{11}{4}$. Therefore, the value of $f\left(\frac{1}{4}\right)$ is $\frac{11}{4}$. Note that 11/4 or 2.75 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID 3784db1a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 3784db1a

The expression $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of $a + c$?

ID: 3784db1a Answer

Correct Answer: 15.5, 31/2

Rationale

The correct answer is **15.5**. It's given that $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers. The expression $(3x^2 + a)(2x^2 + b)$ can be rewritten as $6x^4 + 3bx^2 + 2ax^2 + ab$, or $6x^4 + (3b + 2a)x^2 + ab$. Therefore, the expression $6x^4 + (3b + 2a)x^2 + ab$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3b + 2a = 31$ and $ab = 35$. Since a and b are positive integers and $ab = 35$, the possible values of a and b are 1 and 35, 5 and 7, 7 and 5, and 35 and 1, respectively. Of these options, the only pair that satisfies the equation $3b + 2a = 31$ is $a = 5$ and $b = 7$, since $3(7) + 2(5) = 31$. It's also given that $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. The expression $(3x^2 + c)(2x^2 + d)$ can be rewritten as $6x^4 + 3dx^2 + 2cx^2 + cd$, or $6x^4 + (3d + 2c)x^2 + cd$. Therefore, the expression $6x^4 + (3d + 2c)x^2 + cd$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3d + 2c = 31$ and $cd = 35$. Since it's given that c and d are positive nonintegers, it follows that c and d are not zero. Thus, dividing both sides of the equation $cd = 35$ by c yields $d = \frac{35}{c}$. Substituting $\frac{35}{c}$ for d in the equation $3d + 2c = 31$ yields $3(\frac{35}{c}) + 2c = 31$, or $\frac{105}{c} + 2c = 31$. Multiplying both sides of this equation by c yields $c(\frac{105}{c} + 2c) = c(31)$, or $105 + 2c^2 = 31c$. Subtracting $31c$ from both sides of this equation yields $105 + 2c^2 - 31c = 0$, or $2c^2 - 31c + 105 = 0$. This equation can be rewritten as $2c^2 - 10c - 21c + 105 = 0$. Since the first two terms of the expression $2c^2 - 10c - 21c + 105$ have a common factor of $2c$ and the last two terms of the expression $2c^2 - 10c - 21c + 105$ have a common factor of 21 , the expression $2c^2 - 10c - 21c + 105$ can be rewritten as $2c(c - 5) - 21(c - 5)$. Since each term of this expression has a common factor of $(c - 5)$, the expression $2c(c - 5) - 21(c - 5)$ can be rewritten as $(c - 5)(2c - 21)$. Thus, the equation $2c^2 - 31c + 105 = 0$ can be rewritten as $(c - 5)(2c - 21) = 0$. It follows that $c = 5$ or $c = \frac{21}{2}$. Since c is a positive noninteger, it follows that $c = \frac{21}{2}$, or $c = 10.5$. Substituting 5 for a and 10.5 for c into the expression $a + c$ yields $5 + 10.5$, or **15.5**. Therefore, the value of $a + c$ is **15.5**. Note that 15.5 and 31/2 are examples of ways to enter a correct answer.

Alternate approach: The correct answer is **15.5**. It's given that $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers. The expression $(3x^2 + a)(2x^2 + b)$ can be rewritten as $6x^4 + 3bx^2 + 2ax^2 + ab$, or $6x^4 + (3b + 2a)x^2 + ab$. Therefore, the expression $6x^4 + (3b + 2a)x^2 + ab$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3b + 2a = 31$ and $ab = 35$. Since it's given that a and b are positive integers, it follows that a and b are not zero. Thus, dividing both sides of the equation $ab = 35$ by a yields $b = \frac{35}{a}$. Substituting $\frac{35}{a}$ for b in the equation $3b + 2a = 31$ yields $3(\frac{35}{a}) + 2a = 31$, or $\frac{105}{a} + 2a = 31$. Multiplying both sides of this equation by a yields $105 + 2a^2 = 31a$. Subtracting $31a$ from both sides of this equation yields $2a^2 - 31a + 105 = 0$. This equation can be rewritten as $2a^2 - 10a - 21a + 105 = 0$. Since the first two terms of the expression $2a^2 - 10a - 21a + 105$ have a common factor of $2a$ and the last two terms of the expression $2a^2 - 10a - 21a + 105$ have a common factor of 21 , the expression $2a^2 - 10a - 21a + 105$ can be rewritten as $2a(a - 5) - 21(a - 5)$. Since each term of this expression has a common factor of $(a - 5)$, the expression $2a(a - 5) - 21(a - 5)$ can be rewritten as $(a - 5)(2a - 21)$. Thus, the equation $2a^2 - 31a + 105 = 0$ can be rewritten as $(a - 5)(2a - 21) = 0$. It follows that $a = 5$ or $a = \frac{21}{2}$. Since a is a positive integer, it follows that $a = 5$. It's also given that $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. The expression $(3x^2 + c)(2x^2 + d)$ can be rewritten as $6x^4 + 3dx^2 + 2cx^2 + cd$, or $6x^4 + (3d + 2c)x^2 + cd$. Therefore, the expression $6x^4 + (3d + 2c)x^2 + cd$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3d + 2c = 31$ and $cd = 35$. Since it's given that c and d are positive nonintegers, it follows that c and d are not zero. Thus, dividing both sides of the equation $cd = 35$ by c yields $d = \frac{35}{c}$. Substituting $\frac{35}{c}$ for d in the equation $3d + 2c = 31$ yields $3(\frac{35}{c}) + 2c = 31$, or $\frac{105}{c} + 2c = 31$. Multiplying both

sides of this equation by c yields $105 + 2c^2 = 31c$. Subtracting $31c$ from both sides of this equation yields $2c^2 - 31c + 105 = 0$. Since the equations $2a^2 - 31a + 105 = 0$ and $2c^2 - 31c + 105 = 0$ represent the same equation, it follows that a is equal to c . Therefore, $c = 5$ or $c = \frac{21}{2}$. Since c is a positive noninteger, it follows that $c = \frac{21}{2}$, or $c = 10.5$. Substituting 5 for a and 10.5 for c in the expression $a + c$ yields $5 + 10.5$, or 15.5 . Therefore, the value of $a + c$ is 15.5 . Note that 15.5 and $31/2$ are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 55087e8b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 55087e8b

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. The model indicates the object has an initial height of **10** feet above the ground and reaches its maximum height of **1,034** feet above the ground **8** seconds after being launched. Based on the model, what is the height, in feet, of the object above the ground **10** seconds after being launched?

- A. **234**
- B. **778**
- C. **970**
- D. **1,014**

ID: 55087e8b Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that a quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. This quadratic function can be defined by an equation of the form $f(x) = a(x - h)^2 + k$, where $f(x)$ is the height of the object x seconds after it was launched, and a , h , and k are constants such that the function reaches its maximum value, k , when $x = h$. Since the model indicates the object reaches its maximum height of **1,034** feet above the ground **8** seconds after being launched, $f(x)$ reaches its maximum value, **1,034**, when $x = 8$. Therefore, $k = 1,034$ and $h = 8$. Substituting **8** for h and **1,034** for k in the function $f(x) = a(x - h)^2 + k$ yields $f(x) = a(x - 8)^2 + 1,034$. Since the model indicates the object has an initial height of **10** feet above the ground, the value of $f(x)$ is **10** when $x = 0$. Substituting **0** for x and **10** for $f(x)$ in the equation $f(x) = a(x - 8)^2 + 1,034$ yields $10 = a(0 - 8)^2 + 1,034$, or $10 = 64a + 1,034$. Subtracting **1,034** from both sides of this equation yields $64a = -1,024$. Dividing both sides of this equation by **64** yields $a = -16$. Therefore, the model can be represented by the equation $f(x) = -16(x - 8)^2 + 1,034$. Substituting **10** for x in this equation yields $f(10) = -16(10 - 8)^2 + 1,034$, or $f(10) = 970$. Therefore, based on the model, **10** seconds after being launched, the height of the object above the ground is **970** feet.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 59b7ef3e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 59b7ef3e



The box plots summarize the masses, in kilograms, of two groups of gazelles. Based on the box plots, which of the following statements must be true?

- A. The mean mass of group 1 is greater than the mean mass of group 2.
- B. The mean mass of group 1 is less than the mean mass of group 2.
- C. The median mass of group 1 is greater than the median mass of group 2.
- D. The median mass of group 1 is less than the median mass of group 2.

ID: 59b7ef3e Answer

Correct Answer: C

Rationale

Choice C is correct. The median of a data set represented in a box plot is represented by the vertical line within the box. It follows that the median mass of the gazelles in group 1 is **25** kilograms, and the median mass of the gazelles in group 2 is **24** kilograms. Since **25** kilograms is greater than **24** kilograms, the median mass of group 1 is greater than the median mass of group 2.

Choice A is incorrect. The mean mass of each of the two groups cannot be determined from the box plots.

Choice B is incorrect. The mean mass of each of the two groups cannot be determined from the box plots.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID ef173fc1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: ef173fc1

To estimate the proportion of a population that has a certain characteristic, a random sample was selected from the population. Based on the sample, it is estimated that the proportion of the population that has the characteristic is **0.49**, with an associated margin of error of **0.04**. Based on this estimate and margin of error, which of the following is the most appropriate conclusion about the proportion of the population that has the characteristic?

- A. It is plausible that the proportion is between **0.45** and **0.53**.
- B. It is plausible that the proportion is less than **0.45**.
- C. The proportion is exactly **0.49**.
- D. It is plausible that the proportion is greater than **0.53**.

ID: ef173fc1 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the estimate for the proportion of the population that has the characteristic is **0.49** with an associated margin of error of **0.04**. Subtracting the margin of error from the estimate and adding the margin of error to the estimate gives an interval of plausible values for the true proportion of the population that has the characteristic. Therefore, it's plausible that the proportion of the population that has this characteristic is between **0.45** and **0.53**.

Choice B is incorrect. A value less than **0.45** is outside the interval of plausible values for the proportion of the population that has the characteristic.

Choice C is incorrect. The value **0.49** is an estimate for the proportion based on this sample. However, since the margin of error for this estimate is known, the most appropriate conclusion is not that the proportion is exactly one value but instead lies in an interval of plausible values.

Choice D is incorrect. A value greater than **0.53** is outside the interval of plausible values for the proportion of the population that has the characteristic.

Question Difficulty: Medium

Question ID cf74060f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: cf74060f

How many yards are equivalent to **612 inches**? (**1 yard = 36 inches**)

- A. **0.059**
- B. **17**
- C. **576**
- D. **22,032**

ID: cf74060f Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that **1 yard = 36 inches**. Therefore, **612 inches** is equivalent to **612 inches** $\left(\frac{1 \text{ yard}}{36 \text{ inches}}\right)$, which can be rewritten as $\frac{612 \text{ yards}}{36}$, or **17 yards**.

Choice A is incorrect. This is the number of yards that are equivalent to **2.124 inches**.

Choice C is incorrect. This is the number of yards that are equivalent to **20,736 inches**.

Choice D is incorrect. This is the number of yards that are equivalent to **793,152 inches**.

Question Difficulty: Easy

Question ID 2a2cb9b7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 2a2cb9b7

Task	Time (minutes)
I	5
II	2
III	15
IV	9
V	9

The table shows the amount of time it took Robert to complete each of the 5 tasks in a project. What was the mean time, in minutes, for Robert to complete a task for this project?

ID: 2a2cb9b7 Answer

Correct Answer: 8

Rationale

The correct answer is 8. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. It's given in the table that the times, in minutes, it took Robert to complete each of the tasks are 5, 2, 15, 9, and 9. The sum of these times is $5 + 2 + 15 + 9 + 9$, or 40, minutes. There are 5 tasks in the project. Therefore, the mean time, in minutes, for Robert to complete a task for this project is $40 \div 5$, or 8.

Question Difficulty: Easy

Question ID 37d52288

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 37d52288

210 is $p\%$ greater than 30. What is the value of p ?

ID: 37d52288 Answer

Correct Answer: 600

Rationale

The correct answer is **600**. It's given that **210** is $p\%$ greater than **30**. It follows that $210 = (1 + \frac{p}{100})(30)$. Dividing both sides of this equation by **30** yields $7 = 1 + \frac{p}{100}$. Subtracting **1** from both sides of this equation yields $6 = \frac{p}{100}$. Multiplying both sides of this equation by **100** yields $p = 600$. Therefore, the value of p is **600**.

Question Difficulty: Hard

Question ID 7654a7f7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 7654a7f7

The length of the base of a certain parallelogram is **89%** of the height of the parallelogram. Which expression represents the length of the base of the parallelogram, where ***h*** is the height of the parallelogram?

- A. **$89h$**
- B. **$0.089h$**
- C. **$8.9h$**
- D. **$0.89h$**

ID: 7654a7f7 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the length of the base of the parallelogram is **89%** of the height of the parallelogram. Since ***h*** is the height of the parallelogram, it follows that the length of the base of the parallelogram can be represented by the expression $\frac{89}{100}h$, or **$0.89h$** .

Choice A is incorrect. This expression represents **8,900%**, not **89%**, of the height of the parallelogram.

Choice B is incorrect. This expression represents **8.9%**, not **89%**, of the height of the parallelogram.

Choice C is incorrect. This expression represents **890%**, not **89%**, of the height of the parallelogram.

Question Difficulty: Easy

Question ID df636b8e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: df636b8e

Data set A consists of the heights of **75** objects and has a mean of **25** meters. Data set B consists of the heights of **50** objects and has a mean of **65** meters. Data set C consists of the heights of the **125** objects from data sets A and B. What is the mean, in meters, of data set C?

ID: df636b8e Answer

Correct Answer: 41

Rationale

The correct answer is **41**. The mean of a data set is computed by dividing the sum of the values in the data set by the number of values in the data set. It's given that data set A consists of the heights of **75** objects and has a mean of **25** meters. This can be represented by the equation $\frac{x}{75} = 25$, where x represents the sum of the heights of the objects, in meters, in data set A. Multiplying both sides of this equation by **75** yields $x = 75(25)$, or $x = 1,875$ meters. Therefore, the sum of the heights of the objects in data set A is **1,875** meters. It's also given that data set B consists of the heights of **50** objects and has a mean of **65** meters. This can be represented by the equation $\frac{y}{50} = 65$, where y represents the sum of the heights of the objects, in meters, in data set B. Multiplying both sides of this equation by **50** yields $y = 50(65)$, or $y = 3,250$ meters. Therefore, the sum of the heights of the objects in data set B is **3,250** meters. Since it's given that data set C consists of the heights of the **125** objects from data sets A and B, it follows that the mean of data set C is the sum of the heights of the objects, in meters, in data sets A and B divided by the number of objects represented in data sets A and B, or $\frac{1,875+3,250}{125}$, which is equivalent to **41** meters. Therefore, the mean, in meters, of data set C is **41**.

Question Difficulty: Hard

Question ID ad39571c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: ad39571c

The ratio a to b is equivalent to the ratio 63 to 17 . What is the value of a when $b = 153$?

ID: ad39571c Answer

Correct Answer: 567

Rationale

The correct answer is **567**. It's given that the ratio a to b is equivalent to the ratio 63 to 17 , or $\frac{a}{b} = \frac{63}{17}$. When $b = 153$, the value of a can be calculated by substituting **153** for b in the equation $\frac{a}{b} = \frac{63}{17}$, which yields $\frac{a}{153} = \frac{63}{17}$. Multiplying both sides of this equation by **153** yields $a = 567$. Thus, when $b = 153$, the value of a is **567**.

Question Difficulty: Medium

Question ID 6ef051d6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 6ef051d6

A cherry pitting machine pits **12** pounds of cherries in **3** minutes. At this rate, how many minutes does it take the machine to pit **96** pounds of cherries?

- A. **8**
- B. **15**
- C. **24**
- D. **36**

ID: 6ef051d6 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the cherry pitting machine pits **12** pounds of cherries in **3** minutes. This rate can be written as $\frac{12 \text{ pounds of cherries}}{3 \text{ minutes}}$. If the number of minutes it takes the machine to pit **96** pounds of cherries is represented by x , the value of x can be calculated by solving the equation $\frac{12 \text{ pounds of cherries}}{3 \text{ minutes}} = \frac{96 \text{ pounds of cherries}}{x \text{ minutes}}$, which can be rewritten as $\frac{12}{3} = \frac{96}{x}$, or $4 = \frac{96}{x}$. Multiplying each side of this equation by x yields $4x = 96$. Dividing each side of this equation by **4** yields $x = 24$. Therefore, it takes the machine **24** minutes to pit **96** pounds of cherries.

Choice A is incorrect. This is the number of minutes it takes the machine to pit **32**, not **96**, pounds of cherries.

Choice B is incorrect. This is the number of minutes it takes the machine to pit **60**, not **96**, pounds of cherries.

Choice D is incorrect. This is the number of minutes it takes the machine to pit **144**, not **96**, pounds of cherries.

Question Difficulty: Easy

Question ID fef1f5c9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: fef1f5c9

For a particular machine that produces beads, **29** out of every **100** beads it produces have a defect. A bead produced by the machine will be selected at random. What is the probability of selecting a bead that has a defect?

- A. $\frac{1}{2,900}$
- B. $\frac{1}{29}$
- C. $\frac{29}{100}$
- D. $\frac{29}{10}$

ID: fef1f5c9 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that **29** out of every **100** beads that the machine produces have a defect. It follows that if the machine produces k beads, then the number of beads that have a defect is $\frac{29}{100}k$, for some constant k . If a bead produced by the machine will be selected at random, the probability of selecting a bead that has a defect is given by the number of beads with a defect, $\frac{29}{100}k$, divided by the number of beads produced by the machine, k . Therefore, the probability of selecting a bead that has a defect is $\frac{\frac{29}{100}k}{k}$, or $\frac{29}{100}$.

Choice A is incorrect and may result from conceptual or computational errors.

Choice B is incorrect and may result from conceptual or computational errors.

Choice D is incorrect and may result from conceptual or computational errors.

Question Difficulty: Easy

Question ID 72c5e5dd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 72c5e5dd

4, 4, 4, 4, 8, 8, 8, 13, 13

Which frequency table correctly represents the data listed?

- A.

Number	Frequency
4	4
8	3
13	2
- B.

Number	Frequency
4	4
3	8
2	13
- C.

Number	Frequency
4	16
8	24
13	26
- D.

Number	Frequency
16	4
24	8
26	13

ID: 72c5e5dd Answer

Correct Answer: A

Rationale

Choice A is correct. A frequency table is a table that lists the data value and shows the number of times the data value occurs. In the data listed, the number **4** occurs four times, the number **8** occurs three times, and the number **13** occurs two times. This corresponds to the table in choice A.

Choice B is incorrect. This table has the values for number and frequency reversed.

Choice C is incorrect because the frequency values don't represent the data listed.

Choice D is incorrect. This table represents the listed number values as the frequency values.

Question Difficulty: Easy

Question ID ee20d658

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: ee20d658

0.1152 is 0.08% of the number b . The number b is 25% of the number c . What is the value of c ?

ID: ee20d658 Answer

Correct Answer: 576

Rationale

The correct answer is **576**. It's given that **0.1152** is **0.08%** of the number b . It follows that $0.1152 = \frac{0.08}{100}b$, or $0.1152 = 0.0008b$. Dividing both sides of this equation by **0.0008** yields $144 = b$. It's also given that the number b is **25%** of the number c . It follows that $b = \frac{25}{100}c$, or $b = 0.25c$. Substituting **144** for b in this equation yields $144 = 0.25c$. Dividing both sides of this equation by **0.25** yields $576 = c$. Therefore, the value of c is **576**.

Question Difficulty: Hard

Question ID 1e42e80a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 1e42e80a

Each year, the value of an investment increases by **0.49%** of its value the previous year. Which of the following functions best models how the value of the investment changes over time?

- A. Decreasing exponential
- B. Decreasing linear
- C. Increasing exponential
- D. Increasing linear

ID: 1e42e80a Answer

Correct Answer: C

Rationale

Choice C is correct. Because the value of the investment increases each year, the function that best models how the value of the investment changes over time is an increasing function. It's given that each year, the value of the investment increases by **0.49%** of its value the previous year. Since the value of the investment changes by a fixed percentage each year, the function that best models how the value of the investment changes over time is an exponential function. Therefore, the function that best models how the value of the investment changes over time is an increasing exponential function.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Hard

Question ID 619a9af3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 619a9af3

The number of raccoons in a **131**-square-mile area is estimated to be **2,358**. What is the estimated population density, in raccoons per square mile, of this area?

- A. **18**
- B. **131**
- C. **149**
- D. **2,376**

ID: 619a9af3 Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that there are **2,358** raccoons in a **131**-square-mile area. The estimated population density, in raccoons per square mile, is the estimated number of raccoons divided by the number of square miles. Therefore, the estimated population density of this area is $\frac{2,358 \text{ raccoons}}{131 \text{ square miles}}$, or **18** raccoons per square mile.

Choice B is incorrect. This is the number of square miles in the area, not the estimated number of raccoons per square mile in this area.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 767d5ce1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 767d5ce1

A certain bird species can fly at an average speed of **16** meters per second when in continuous flight. At this rate, how many meters would this bird species fly in **4** seconds?

- A. **64**
- B. **20**
- C. **16**
- D. **12**

ID: 767d5ce1 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that a certain bird species can fly at an average speed of **16** meters per second when in continuous flight. At this rate, in **4** seconds this bird species would fly ($\frac{16 \text{ meters}}{\text{second}}$)(**4 seconds**), or **64** meters.

Choice B is incorrect. This is the value of **16 + 4**, not **16(4)**.

Choice C is incorrect. This is the distance the bird would fly in **1** second, not **4** seconds.

Choice D is incorrect. This is the value of **16 – 4**, not **16(4)**.

Question Difficulty: Easy

Question ID 1c92f5b3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 1c92f5b3

An insect moves at a speed of $\frac{3}{20}$ feet per second. What is this speed, in yards per second? (**3 feet = 1 yard**)

- A. $\frac{1}{20}$
- B. $\frac{9}{20}$
- C. **6**
- D. **20**

ID: 1c92f5b3 Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that **3 feet = 1 yard**. It follows that a speed of $\frac{3}{20}$ feet per second is equivalent to $\left(\frac{\frac{3}{20} \text{ feet}}{1 \text{ second}}\right)\left(\frac{1 \text{ yard}}{3 \text{ feet}}\right)$, which is equivalent to $\left(\frac{3}{20}\right)\left(\frac{1}{3}\right)$, or $\frac{1}{20}$, yards per second.

Choice B is incorrect. This is the speed, in feet per second, that's equivalent to $\frac{3}{20}$ yards per second.

Choice C is incorrect. This is the speed, in yards per second, that's equivalent to **18**, not $\frac{3}{20}$, feet per second.

Choice D is incorrect. This is the speed, in yards per second, that's equivalent to **60**, not $\frac{3}{20}$, feet per second.

Question Difficulty: Medium

Question ID 5dd20f69

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 5dd20f69

37% of the items in a box are green. Of those, **37%** are also rectangular. Of the green rectangular items, **42%** are also metal. Which of the following is closest to the percentage of the items in the box that are not rectangular green metal items?

- A. **1.16%**
- B. **57.50%**
- C. **94.25%**
- D. **98.84%**

ID: 5dd20f69 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that **37%** of the items in a box are green. Let x represent the total number of items in the box. It follows that $\frac{37}{100}x$, or **0.37x**, items in the box are green. It's also given that of those, **37%** are also rectangular. Therefore, $\frac{37}{100}(0.37x)$, or **0.1369x**, items in the box are green rectangular items. It's also given that of the green rectangular items, **42%** are also metal. Therefore, $\frac{42}{100}(0.1369x)$, or **0.057498x**, items in the box are rectangular green metal items. The number of the items in the box that are not rectangular green metal items is the total number of items in the box minus the number of rectangular green metal items in the box. Therefore, the number of items in the box that are not rectangular green metal items is $x - 0.057498x$, or **0.942502x**. The percentage of items in the box that are not rectangular green metal items is the percentage that **0.942502x** is of x . If $p\%$ represents this percentage, the value of p is $100\left(\frac{0.942502x}{x}\right)$, or **94.2502**. Of the given choices, **94.25%** is closest to the percentage of items in the box that are not rectangular green metal items.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 90bcc3ea

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 90bcc3ea

23, 27, 27, 32, 35, 36, 52
What is the range of the 7 scores shown?

ID: 90bcc3ea Answer

Correct Answer: 29

Rationale

The correct answer is **29**. The range of a data set is the difference between its maximum value and its minimum value. For the data set shown, the maximum score is **52** and the minimum score is **23**. The difference between those scores is **52 — 23**, or **29**. Therefore, the range of the 7 scores shown is **29**.

Question Difficulty: Medium

Question ID 304a37da

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 304a37da

Each value in the data set shown represents the height, in centimeters, of a plant.

6, 10, 13, 2, 15, 22, 10, 4, 4, 4

What is the mean height, in centimeters, of these plants?

ID: 304a37da Answer

Correct Answer: 9

Rationale

The correct answer is **9**. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. It follows that the mean height, in centimeters, of these plants is the sum of the heights, in centimeters, of each plant, **6 + 10 + 13 + 2 + 15 + 22 + 10 + 4 + 4 + 4**, or **90**, divided by the number of plants in the data set, **10**. Therefore, the mean height, in centimeters, of these plants is $\frac{90}{10}$, or **9**.

Question Difficulty: Medium

Question ID 5aba7ae0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 5aba7ae0

−13, 4, 23

A data set of three numbers is shown. If a number from this data set is selected at random, what is the probability of selecting a negative number?

- A. 0
- B. $\frac{1}{3}$
- C. $\frac{2}{3}$
- D. 1

ID: 5aba7ae0 Answer

Correct Answer: B

Rationale

Choice B is correct. If a number from the data set is selected at random, the probability of selecting a negative number is the count of negative numbers in the data set divided by the total count of numbers in the data set. It's given that a data set of three numbers is shown. It follows that the total count of numbers in the data set is **3**. In the data set shown, **−13** is the only negative number. It follows that the count of negative numbers in the data set is **1**. Therefore, if a number from the data set is selected at random, the probability of selecting a negative number is $\frac{1}{3}$.

Choice A is incorrect. This is the probability of selecting a negative number from a data set that doesn't contain any negative numbers.

Choice C is incorrect. This is the probability of selecting a positive number, not a negative number, from the data set.

Choice D is incorrect. This is the probability of selecting a negative number from a data set that contains only negative numbers.

Question Difficulty: Easy

Question ID 17afbecc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 17afbecc

A cable provider wanted to know how many of its **30,000** customers would be interested in a new service plan. The provider selected **300** customers at random and asked each customer whether the customer would be interested in the new plan. Of those surveyed, **8** said they would be interested. Which of the following is the best estimate of the total number of customers who would be interested in the new service plan?

- A. 8
- B. 80
- C. 800
- D. 8,000

ID: 17afbecc Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that **8** out of **300** customers in a random sample said they would be interested in the new plan. This means the fraction of sampled customers who said they would be interested is $\frac{8}{300}$. Multiplying this fraction by the total customer population yields $(\frac{8}{300})(30,000)$, or **800**. Therefore, the best estimate of the total number of customers who would be interested in the new service plan is **800**.

Choice A is incorrect. This is the number of customers surveyed who said they would be interested in the new plan, not the best estimate of the total number of customers who would be interested in the new plan.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID e75edd72

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: e75edd72

The table gives the distribution of topping and type of crust for customer orders at a pizza place.

Topping	Type of crust	
	Thin	Thick
Extra cheese	60	30
Pepperoni	20	30
Tomato	80	20

If one of these customer orders is selected at random, what is the probability of selecting an order with thin crust, given the topping is pepperoni? (Express your answer as a decimal or fraction, not as a percent.)

ID: e75edd72 Answer

Correct Answer: 0.4, 2/5

Rationale

The correct answer is $\frac{2}{5}$. Since the order will be selected at random, the probability of selecting an order with thin crust, given the topping is pepperoni, is equal to the number of orders with thin crust where the topping is pepperoni divided by the total number of orders where the topping is pepperoni. According to the table, there are **20** orders with thin crust where the topping is pepperoni. According to the table, there are **30** orders with thick crust where the topping is pepperoni. Therefore, there are a total of **20 + 30**, or **50**, orders where the topping is pepperoni. Thus, the probability of selecting an order with thin crust, given the topping is pepperoni, is $\frac{20}{50}$, or $\frac{2}{5}$. Note that 2/5 and .4 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID b9f4562a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: b9f4562a

Amara grows cherry tomatoes in her backyard. This year, she harvested **750** cherry tomatoes and gave **10%** of them to her neighbor. How many of the harvested cherry tomatoes did Amara give to her neighbor?

ID: b9f4562a Answer

Correct Answer: 75

Rationale

The correct answer is **75**. It’s given that Amara harvested **750** cherry tomatoes and gave **10%** of them to her neighbor. **10%** can be written as $\frac{10}{100}$, or **0.1**. Therefore, the number of cherry tomatoes that Amara gave to her neighbor can be calculated by multiplying the total number of cherry tomatoes by **0.1**, which gives **(0.1)(750)**, or **75**.

Question Difficulty: Easy

Question ID f3c811a6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: f3c811a6

In a sample, 80% of the items are faulty. There are 88 faulty items in the sample. How many total items are in the sample?

ID: f3c811a6 Answer

Correct Answer: 110

Rationale

The correct answer is 110. Let x represent the total number of items in the sample. It's given that 80% of the items are faulty and that there are 88 faulty items in the sample. Therefore, 80% of x is 88. Since 80% can be rewritten as $\frac{80}{100}$, it follows that $\frac{80}{100}x = 88$. Multiplying both sides of this equation by 100 yields $80x = 8,800$. Dividing both sides of this equation by 80 yields $x = 110$. Therefore, there are 110 total items in the sample.

Question Difficulty: Medium

Question ID 354dd385

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 354dd385

A competition consisted of four different events. One participant completed the first event with an average speed of **20.300** miles per hour. What was this average speed, in yards per hour? (**1 mile = 1,760 yards**)

ID: 354dd385 Answer

Correct Answer: 35728

Rationale

The correct answer is **35,728**. It's given that **1 mile = 1,760 yards**. It follows that an average speed of **20.300** miles per hour is equivalent to $\left(\frac{20.300 \text{ miles}}{1 \text{ hour}}\right)\left(\frac{1,760 \text{ yards}}{1 \text{ mile}}\right)$, or **35,728** yards per hour.

Question Difficulty: Medium

Question ID fb8c1a87

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: fb8c1a87

The total mass, in kilograms, of r identical objects is t . Which expression represents the total mass, in kilograms, of $146r$ of these objects?

- A. $146 - t$
- B. $146 + t$
- C. $\frac{t}{146}$
- D. $146t$

ID: fb8c1a87 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the total mass, in kilograms, of r identical objects is t . To obtain the expression $146r$, r is multiplied by 146 . Therefore, to find the total mass, in kilograms, of $146r$ of these objects, t must also be multiplied by 146 . The result of multiplying t by 146 is the expression $146t$. Therefore, the total mass, in kilograms, of $146r$ of these objects, is $146t$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Hard

Question ID a7e8fdec

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: a7e8fdec

21 is 21% of what number?

- A. 0
- B. 1
- C. 42
- D. 100

ID: a7e8fdec Answer

Correct Answer: D

Rationale

Choice D is correct. Let x represent the number that 21 is 21% of. It follows that $\frac{21}{x} = \frac{21}{100}$. Multiplying each side of this equation by x yields $21 = \frac{21x}{100}$. Multiplying each side of this equation by 100 yields $2,100 = 21x$. Dividing each side of this equation by 21 yields $100 = x$. Therefore, 21 is 21% of 100.

Choice A is incorrect. 21% of 0 is 0, not 21.

Choice B is incorrect. 21% of 1 is 0.21, not 21.

Choice C is incorrect. 21% of 42 is 8.82, not 21.

Question Difficulty: Easy

Question ID a958d93a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: a958d93a

An insurance company offers a series of three information sessions. **1,250** people attended the first information session. **72%** of the people who attended the first information session attended the second information session, and **36%** of the people who attended the first and second information sessions attended the third information session. How many people attended all three information sessions?

ID: a958d93a Answer

Correct Answer: 324

Rationale

The correct answer is **324**. It's given that **1,250** people attended the first information session, and that **72%** of the people who attended the first information session attended the second information session. Therefore, the number of people who attended the first and second information sessions can be found by calculating **72%** of **1,250**, which is equal to **1,250($\frac{72}{100}$)**, or **900**. It's also given that **36%** of the people who attended the first and second information sessions attended the third information session. Since **900** people attended the first and second information sessions, the number of people who attended the first, second, and third information sessions can be found by calculating **36%** of **900**, which is equal to **900($\frac{36}{100}$)**, or **324**. Therefore, **324** people attended all three information sessions.

Question Difficulty: Medium

Question ID 410fb541

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 410fb541

For a certain rectangular region, the ratio of its length to its width is **35** to **10**. If the width of the rectangular region increases by **7** units, how must the length change to maintain this ratio?

- A. It must decrease by **24.5** units.
- B. It must increase by **24.5** units.
- C. It must decrease by **7** units.
- D. It must increase by **7** units.

ID: 410fb541 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the ratio of the rectangular region's length to its width is **35** to **10**. This can be written as a proportion: $\frac{\text{length}}{\text{width}} = \frac{35}{10}$, or $\frac{l}{w} = \frac{35}{10}$. This proportion can be rewritten as $10l = 35w$, or $l = 3.5w$. If the width of the rectangular region increases by **7**, then the length will increase by some number x in order to maintain this ratio. The value of x can be found by replacing l with $l + x$ and w with $w + 7$ in the equation, which gives $l + x = 3.5(w + 7)$. This equation can be rewritten using the distributive property as $l + x = 3.5w + 24.5$. Since $l = 3.5w$, the right-hand side of this equation can be rewritten by substituting l for $3.5w$, which gives $l + x = l + 24.5$, or $x = 24.5$. Therefore, if the width of the rectangular region increases by **7** units, the length must increase by **24.5** units in order to maintain the given ratio.

Choice A is incorrect. If the width of the rectangular region increases, the length must also increase, not decrease.

Choice C is incorrect. If the width of the rectangular region increases, the length must also increase, not decrease.

Choice D is incorrect. Since the ratio of the length to the width of the rectangular region is **35** to **10**, if the width of the rectangular region increases by **7** units, the length would have to increase by a proportional amount, which would have to be greater than **7** units.

Question Difficulty: Hard

Question ID 781ebeac

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 781ebeac

To study the moisture content in a group of trees, samples from the trunk of each tree were taken from **25** trees and cut in the shape of a cube. The length of the edge of one of these cubes is **2.00** centimeters. If this cube has a mass of **2.56** grams, what is the density of this cube, in grams per cubic centimeter?

ID: 781ebeac Answer

Correct Answer: 0.32, 8/25

Rationale

The correct answer is **.32**. The volume of a cube is given by the formula $V = s^3$, where s is the length of an edge of the cube. It's given that each edge of the cube has a length of **2.00** centimeters. Substituting **2.00** centimeters for s in the formula $V = s^3$ yields $V = (2.00 \text{ centimeters})^3$, or $V = 8.00$ cubic centimeters. It's given that the cube has a mass of **2.56** grams. Dividing the mass, in grams, of the cube by the volume, in cubic centimeters, of the cube gives its density, in grams per cubic centimeters. Therefore, the density of the cube is $\frac{2.56 \text{ grams}}{8.00 \text{ cubic centimeters}}$, or **.32** grams per cubic centimeter. Note that .32 and 8/25 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 7f5015ca

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 7f5015ca

The table summarizes the distribution of age and assigned group for **90** participants in a study.

	0–9 years	10–19 years	20+ years	Total
Group A	7	14	9	30
Group B	6	4	20	30
Group C	17	12	1	30
Total	30	30	30	90

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least **10** years of age? (Express your answer as a decimal or fraction, not as a percent.)

ID: 7f5015ca Answer

Correct Answer: .3833, 23/60

Rationale

The correct answer is $\frac{23}{60}$. It's given that one of the participants will be selected at random. The probability of selecting a participant from group A given that the participant is at least **10** years of age is the number of participants in group A who are at least **10** years of age divided by the total number of participants who are at least **10** years of age. The table shows that in group A, there are **14** participants who are **10–19** years of age and **9** participants who are **20+** years of age. Therefore, there are **14 + 9**, or **23**, participants in group A who are at least **10** years of age. The table also shows that there are a total of **30** participants who are **10–19** years of age and **30** participants who are **20+** years of age. Therefore, there are a total of **30 + 30**, or **60**, participants who are at least **10** years of age. It follows that the probability of selecting a participant from group A given that the participant is at least **10** years of age is $\frac{23}{60}$. Note that 23/60, .3833, and 0.383 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID e041089e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: e041089e

One side of a flat board has an area of **874** square inches. If a pressure of **19** pounds per square inch of area is exerted on this side of the board, what is the total force, in pounds, exerted on this side of the board?

ID: e041089e Answer

Correct Answer: 16606

Rationale

The correct answer is **16,606**. It's given that one side of a flat board has an area of **874** square inches. If a pressure of **19** pounds per square inch of area is exerted on this side of the board, the total force exerted on this side of the board is **(874 square inches) $\left(\frac{19 \text{ pounds}}{1 \text{ square inch}}\right)$, or 16,606 pounds.**

Question Difficulty: Hard

Question ID bc6b343d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: bc6b343d

A gift shop buys souvenirs at a wholesale price of **7.00** dollars each and resells them each at a retail price that is **290%** of the wholesale price. At the end of the season, any remaining souvenirs are marked at a discounted price that is **80%** off the retail price. What is the discounted price of each remaining souvenir, in dollars?

ID: bc6b343d Answer

Correct Answer: 203/50, 4.06

Rationale

The correct answer is **4.06**. It's given that the retail price is **290%** of the wholesale price of **\$7.00**. Thus, the retail price is **\$7.00($\frac{290}{100}$)**, which is equivalent to **\$7.00(2.9)**, or **\$20.30**. It's also given that the discounted price is **80%** off the retail price. Thus, the discounted price is **\$20.30($1 - \frac{80}{100}$)**, which is equivalent to **\$20.30(0.20)**, or **\$4.06**.

Question Difficulty: Hard

Question ID 30a4f4a3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 30a4f4a3

Data set A consists of the heights of **75** buildings and has a mean of **32** meters. Data set B consists of the heights of **50** buildings and has a mean of **62** meters. Data set C consists of the heights of the **125** buildings from data sets A and B. What is the mean, in meters, of data set C?

ID: 30a4f4a3 Answer

Correct Answer: 44

Rationale

The correct answer is **44**. The mean of a data set is computed by dividing the sum of the values in the data set by the number of values in the data set. It's given that data set A consists of the heights of **75** buildings and has a mean of **32** meters. This can be represented by the equation $\frac{x}{75} = 32$, where x represents the sum of the heights of the buildings, in meters, in data set A. Multiplying both sides of this equation by **75** yields $x = 75(32)$, or $x = 2,400$ meters. Therefore, the sum of the heights of the buildings in data set A is **2,400** meters. It's also given that data set B consists of the heights of **50** buildings and has a mean of **62** meters. This can be represented by the equation $\frac{y}{50} = 62$, where y represents the sum of the heights of the buildings, in meters, in data set B. Multiplying both sides of this equation by **50** yields $y = 50(62)$, or $y = 3,100$ meters. Therefore, the sum of the heights of the buildings in data set B is **3,100** meters. Since it's given that data set C consists of the heights of the **125** buildings from data sets A and B, it follows that the mean of data set C is the sum of the heights of the buildings, in meters, in data sets A and B divided by the number of buildings represented in data sets A and B, or $\frac{2,400+3,100}{125}$, which is equivalent to **44** meters. Therefore, the mean, in meters, of data set C is **44**.

Question Difficulty: Hard

Question ID 0b5424c0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 0b5424c0

To study fluctuations in composition, samples of pumice were taken from **29** locations and cut in the shape of a cube. The length of the edge of one of these cubes is **3.000** centimeters. This cube has a density of **0.230** grams per cubic centimeter. What is the mass of this cube, in grams?

ID: 0b5424c0 Answer

Correct Answer: 6.21

Rationale

The correct answer is **6.21**. It's given that the samples of pumice were cut in the shape of a cube. It's also given that the length of the edge of one of these cubes is **3.000** centimeters. Therefore, the volume of this cube is **(3.000 centimeters)³**, or **27** cubic centimeters. Since the density of this cube is **0.230** grams per cubic centimeter, it follows that the mass of this cube is **($\frac{0.230 \text{ grams}}{1 \text{ cubic centimeter}}$)(27 cubic centimeters)**, or **6.21** grams.

Question Difficulty: Hard

Question ID 859d9971

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 859d9971

The density of a certain type of wood is **353** kilograms per cubic meter. A sample of this type of wood is in the shape of a cube and has a mass of **345** kilograms. To the nearest hundredth of a meter, what is the length of one edge of this sample?

- A. 0.98
- B. 0.99
- C. 1.01
- D. 1.02

ID: 859d9971 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that the density of a certain type of wood is **353** kilograms per cubic meter (kg/m^3), and a sample of this type of wood has a mass of **345 kg**. Let x represent the volume, in m^3 , of the sample. It follows that the relationship between the density, mass, and volume of this sample can be written as $\frac{353 \text{ kg}}{1 \text{ m}^3} = \frac{345 \text{ kg}}{x \text{ m}^3}$, or $353 = \frac{345}{x}$. Multiplying both sides of this equation by x yields $353x = 345$. Dividing both sides of this equation by **353** yields $x = \frac{345}{353}$. Therefore, the volume of this sample is $\frac{345}{353} \text{ m}^3$. Since it’s given that the sample of this type of wood is a cube, it follows that the length of one edge of this sample can be found using the volume formula for a cube, $V = s^3$, where V represents the volume, in m^3 , and s represents the length, in m, of one edge of the cube. Substituting $\frac{345}{353}$ for V in this formula yields $\frac{345}{353} = s^3$. Taking the cube root of both sides of this equation yields $\sqrt[3]{\frac{345}{353}} = s$, or $s \approx 0.99$. Therefore, the length of one edge of this sample to the nearest hundredth of a meter is **0.99**.

Choices A, C, and D are incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 89b3ea02

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 89b3ea02

A band with **45** members has **11** members who play saxophone. If one band member is selected at random, what is the probability of selecting a band member who plays saxophone?

- A. $\frac{1}{45}$
- B. $\frac{11}{45}$
- C. $\frac{34}{45}$
- D. $\frac{45}{45}$

ID: 89b3ea02 Answer

Correct Answer: B

Rationale

Choice B is correct. The probability of an event occurring is the ratio of the number of favorable outcomes to the total number of possible outcomes. It’s given that there are **45** band members, which is the total number of possible outcomes. It's also given that there are **11** band members who play saxophone. Therefore, the number of favorable outcomes is **11**. Thus, the probability of selecting a band member who plays saxophone is $\frac{11}{45}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the probability of selecting a band member who does not play saxophone.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID aae6aebd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: aae6aebd

There are **20** buttons in a bag: **8** white buttons, **2** orange buttons, and **10** brown buttons. If one of these buttons is selected at random, what is the probability of selecting a white button?

- A. $\frac{2}{20}$
- B. $\frac{8}{20}$
- C. $\frac{10}{20}$
- D. $\frac{12}{20}$

ID: aae6aebd Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that there are **20** buttons in a bag and **8** of the buttons are white. If one button from the bag is selected at random, the probability of selecting a white button is the number of white buttons in the bag divided by the total number of buttons in the bag. Therefore, if one button from the bag is selected at random, the probability of selecting a white button is $\frac{8}{20}$.

Choice A is incorrect. This is the probability of selecting an orange button from the bag.

Choice C is incorrect. This is the probability of selecting a brown button from the bag.

Choice D is incorrect. This is the probability of selecting a button that isn't white from the bag.

Question Difficulty: Easy

Question ID e96895bb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: e96895bb

The expression $0.35x$ represents the result of decreasing a positive quantity x by what percent?

- A. 3.5%
- B. 35%
- C. 6.5%
- D. 65%

ID: e96895bb Answer

Correct Answer: D

Rationale

Choice D is correct. Let $n\%$ represent the percent by which the positive quantity x is decreased to result in $0.35x$. The value of n can be found by solving the equation $x - \left(\frac{n}{100}\right)x = 0.35x$. Since x is a common factor of each of the terms on the left-hand side of this equation, the equation can be rewritten as $x\left(1 - \frac{n}{100}\right) = 0.35x$. Dividing each side of this equation by x yields $1 - \frac{n}{100} = 0.35$. Multiplying each side of this equation by 100 yields $100 - n = 35$. Subtracting 100 from each side of this equation yields $-n = -65$. Dividing each side of this equation by -1 yields $n = 65$. Therefore, the expression $0.35x$ represents the result of decreasing the positive quantity x by 65% .

Choice A is incorrect. Decreasing the quantity x by 3.5% yields $x - 0.035x$, or $0.965x$, not $0.35x$.

Choice B is incorrect. Decreasing the quantity x by 35% yields $x - 0.35x$, or $0.65x$, not $0.35x$.

Choice C is incorrect. Decreasing the quantity x by 6.5% yields $x - 0.065x$, or $0.935x$, not $0.35x$.

Question Difficulty: Hard

Question ID 8c80a1f7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 8c80a1f7

A wind turbine completes **900** revolutions in **50** minutes. At this rate, how many revolutions per minute does this turbine complete?

- A. **18**
- B. **850**
- C. **950**
- D. **1,400**

ID: 8c80a1f7 Answer

Correct Answer: A

Rationale

Choice A is correct. Dividing the number of revolutions by the number of minutes gives the number of revolutions the turbine completes per minute. It’s given that the wind turbine completes **900** revolutions in **50** minutes. Therefore, at this rate, this turbine completes $\frac{900}{50}$, or **18**, revolutions per minute.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

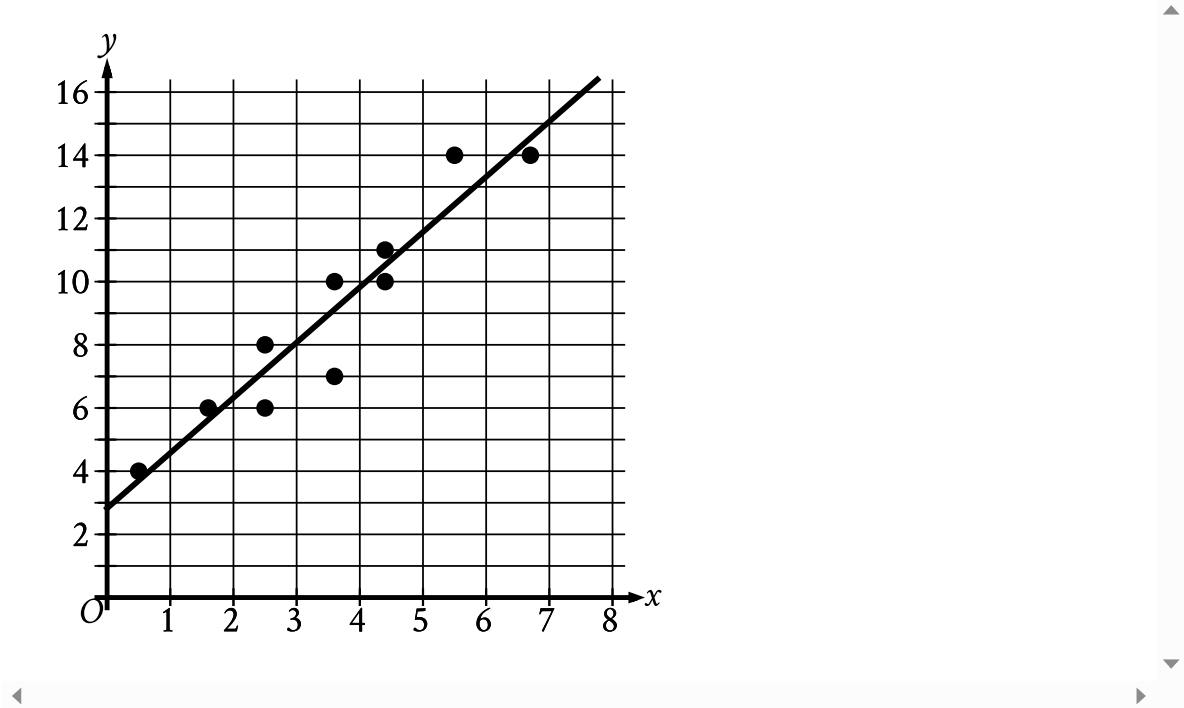
Question Difficulty: Easy

Question ID 9142011e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 9142011e

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

- A. $y = 2.8 + 1.7x$
- B. $y = 2.8 - 1.7x$
- C. $y = -2.8 + 1.7x$
- D. $y = -2.8 - 1.7x$

ID: 9142011e Answer

Correct Answer: A

Rationale

Choice A is correct. The line of best fit shown intersects the y -axis at a positive y -value and has a positive slope. The graph of an equation of the form $y = a + bx$, where a and b are constants, intersects the y -axis at a y -value of a and has a slope of b . Of the given choices, only choice A represents a line that intersects the y -axis at a positive y -value, **2.8**, and has a positive slope, **1.7**.

Choice B is incorrect. This equation represents a line that has a negative slope, not a positive slope.

Choice C is incorrect. This equation represents a line that intersects the y -axis at a negative y -value, not a positive y -value.

Choice D is incorrect. This equation represents a line that intersects the y -axis at a negative y -value, not a positive y -value, and has a negative slope, not a positive slope.

Question Difficulty: Easy

Question ID 5deda8b0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 5deda8b0

For the values j and k , the ratio of j to k is 11 to 12. If j is multiplied by 17, what is k multiplied by in order to maintain the same ratio?

ID: 5deda8b0 Answer

Correct Answer: 17

Rationale

The correct answer is 17. If one value is multiplied by a number, then the other value must be multiplied by the same number in order to maintain the same ratio. It's given that j is multiplied by 17. Therefore, in order to maintain the same ratio, k must also be multiplied by 17.

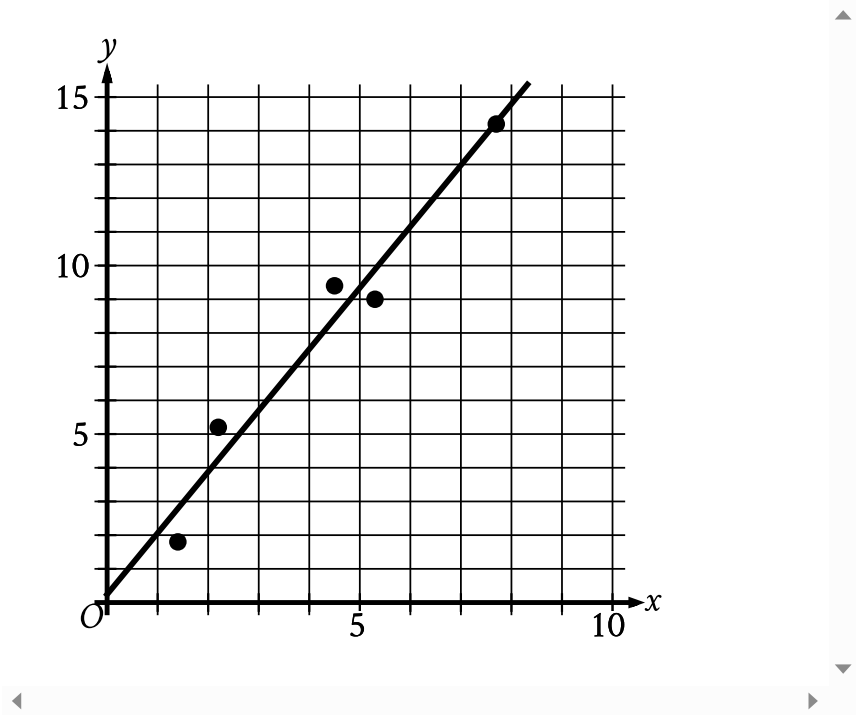
Question Difficulty: Hard

Question ID 150b63a5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 150b63a5

In the given scatterplot, a line of best fit for the data is shown.



Which of the following is closest to the slope of the line of best fit shown?

- A. 0.2
- B. 0.7
- C. 1.8
- D. 2.6

ID: 150b63a5 Answer

Correct Answer: C

Rationale

Choice C is correct. A line in the xy -plane that passes through points (x_1, y_1) and (x_2, y_2) has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points $(0, 0.2)$ and $(5, 9.3)$. It follows that the slope of this line is approximately $\frac{9.3 - 0.2}{5 - 0}$, which is equivalent to $\frac{9.1}{5}$, or **1.82**. Therefore, of the given choices, **1.8** is closest to the slope of the line of best fit shown.

Choice A is incorrect. This value is closest to the y -coordinate of the y -intercept of the line of best fit shown.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID 33ac56b9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 33ac56b9

A fish swam a distance of **5,104** yards. How far did the fish swim, in miles? (**1 mile = 1,760 yards**)

- A. **0.3**
- B. **2.9**
- C. **3,344**
- D. **6,864**

ID: 33ac56b9 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that the fish swam **5,104** yards and that **1** mile is equal to **1,760** yards. Therefore, the fish swam **5,104 yards** $\left(\frac{1 \text{ mile}}{1,760 \text{ yards}}\right)$, which is equivalent to $\frac{5,104}{1,760}$ **miles**, or **2.9** miles.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

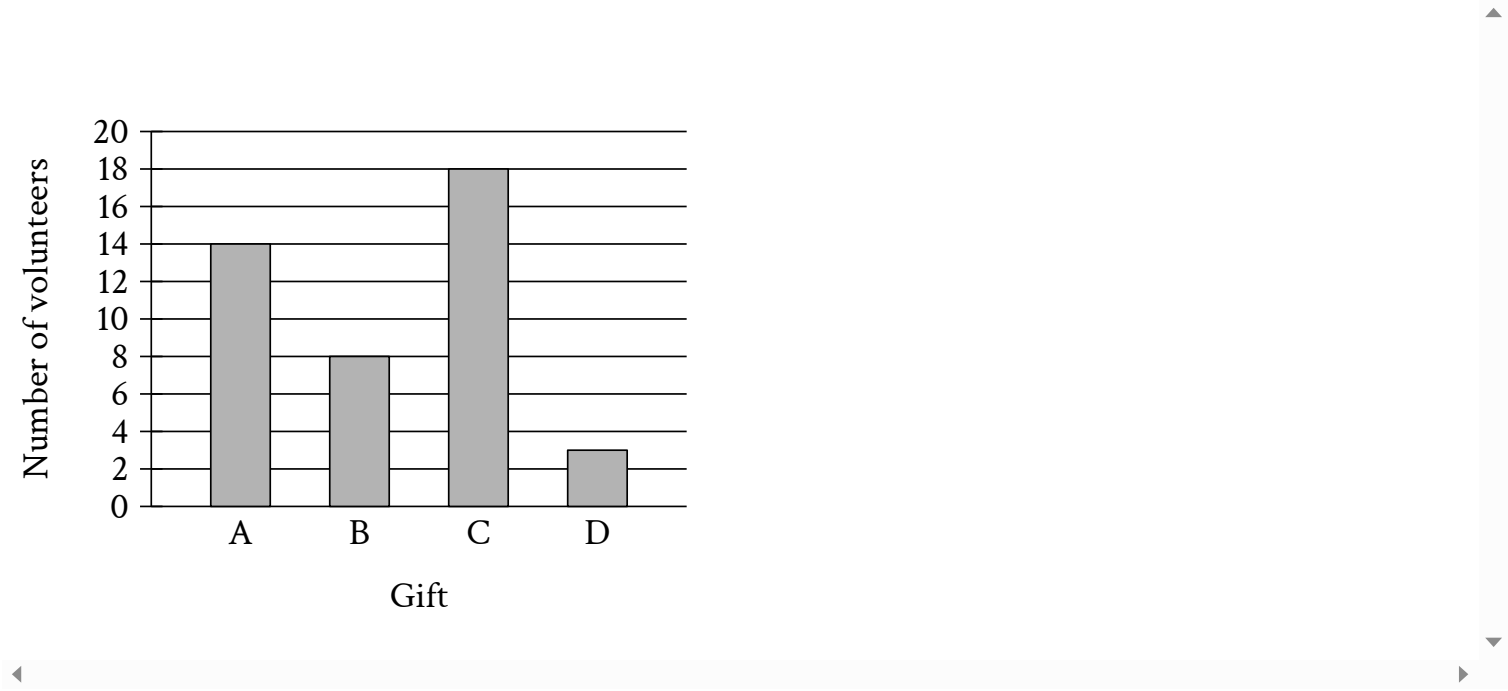
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 09a0ecd6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 09a0ecd6



In April, there were **43** volunteers in a cleanup project. Each volunteer was asked to choose a small gift labeled A, B, C, or D. The bar graph shows the number of volunteers who chose each gift. How many volunteers chose gift C?

- A. **3**
- B. **8**
- C. **14**
- D. **18**

ID: 09a0ecd6 Answer

Correct Answer: D

Rationale

Choice D is correct. The height of each bar in the graph shown represents the number of volunteers who chose the gift labeled with the letter specified at the bottom of the bar. The bar for gift C has a height of **18**. Therefore, **18** volunteers chose gift C.

Choice A is incorrect. This is the number of volunteers who chose gift D, not gift C.

Choice B is incorrect. This is the number of volunteers who chose gift B, not gift C.

Choice C is incorrect. This is the number of volunteers who chose gift A, not gift C.

Question ID 5e5ce50a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 5e5ce50a

The value of z is **1.13** times **100**. The value of z is what percent greater than **100**?

- A. **11.3**
- B. **13**
- C. **130**
- D. **213**

ID: 5e5ce50a Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the value of z is **1.13** times **100**. This can be written as $z = (1.13)(100)$, which is equivalent to $z = (1 + 0.13)(100)$, or $z = (1 + \frac{13}{100})(100)$. It follows that the value of z is **100%** of **100** plus **13%** of **100**. Therefore, the value of z is **13%** greater than **100**.

Choice A is incorrect. This gives a value of z that is **1.113**, not **1.13**, times **100**.

Choice C is incorrect. This gives a value of z that is **2.30**, not **1.13**, times **100**.

Choice D is incorrect. This gives a value of z that is **3.13**, not **1.13**, times **100**.

Question Difficulty: Medium

Question ID 42e6c47c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 42e6c47c

The result of increasing the quantity x by 400% is 60. What is the value of x ?

- A. 12
- B. 15
- C. 240
- D. 340

ID: 42e6c47c Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the result of increasing the quantity x by 400% is 60. This can be written as $x + \left(\frac{400}{100}\right)x = 60$, which is equivalent to $x + 4x = 60$, or $5x = 60$. Dividing each side of this equation by 5 yields $x = 12$. Therefore, the value of x is 12.

Choice B is incorrect. The result of increasing the quantity 15 by 400% is 75, not 60.

Choice C is incorrect. The result of increasing the quantity 240 by 400% is 1,200, not 60.

Choice D is incorrect. The result of increasing the quantity 340 by 400% is 1,700, not 60.

Question Difficulty: Hard

Question ID b0f7f399

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: b0f7f399

A triathlon is a multisport race consisting of three different legs. A triathlon participant completed the cycling leg with an average speed of **19.700** miles per hour. What was the average speed, in yards per hour, of the participant during the cycling leg? (**1 mile = 1,760 yards**)

ID: b0f7f399 Answer

Correct Answer: 34672

Rationale

The correct answer is **34,672**. It's given that **1 mile = 1,760 yards**. It follows that an average speed of **19.700** miles per hour is equivalent to $\left(\frac{19.700 \text{ miles}}{1 \text{ hour}}\right)\left(\frac{1,760 \text{ yards}}{1 \text{ mile}}\right)$, or **34,672** yards per hour.

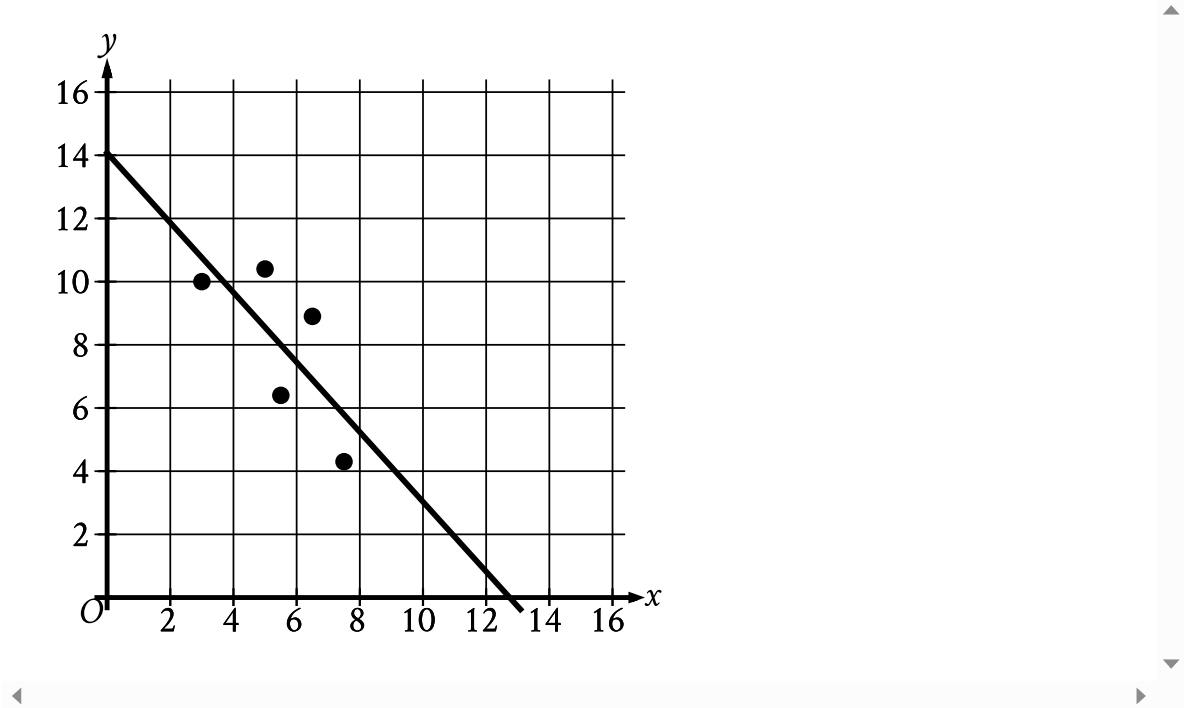
Question Difficulty: Medium

Question ID 00a93f40

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 00a93f40

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



Which of the following is closest to the slope of this line of best fit?

- A. -3.3
- B. -1.1
- C. 1.1
- D. 3.3

ID: 00a93f40 Answer

Correct Answer: B

Rationale

Choice B is correct. A line in the xy -plane that passes through points (x_1, y_1) and (x_2, y_2) has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points $(0, 14)$ and $(13, 0)$. It follows that the slope of this line of best fit is approximately $\frac{0 - 14}{13 - 0}$, or $-\frac{14}{13}$. Of the given choices, -1.1 is closest to $-\frac{14}{13}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 052af884

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 052af884

Data set F consists of **55** integers between **170** and **290**. Data set G consists of all the integers in data set F as well as the integer **10**. Which of the following must be less for data set F than for data set G?

- I. The mean

II. The median
- A. I only

B. II only

C. I and II

D. Neither I nor II

ID: 052af884 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that data set F consists of **55** integers between **170** and **290** and data set G consists of all the integers in data set F as well as the integer **10**. Since the integer **10** is less than all the integers in data set F, the mean of data set G must be less than the mean of data set F. Thus, the mean of data set F isn't less than the mean of data set G. When a data set is in ascending order, the median is between the two middle values when there is an even number of values and the median is the middle value when there is an odd number of values. It follows that the median of data set F is either greater than or equal to the median of data set G. Therefore, the median of data set F isn't less than the median of data set G. Thus, neither the mean nor the median must be less for data set F than for data set G.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID bfb035d6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: bfb035d6

For the last three years, a pig at a farm gave birth to one litter per year. The table shows the number of piglets in each of these three litters.

Litter number	Number of piglets
1	8
2	7
3	9

What is the mean number of piglets per litter for these three litters?

- A. 2
- B. 7
- C. 8
- D. 9

ID: bfb035d6 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the numbers of piglets in litters **1**, **2**, and **3** are **8**, **7**, and **9**, respectively. The mean number of piglets per litter for these three litters is $\frac{8+7+9}{3}$, which is equivalent to $\frac{24}{3}$, or **8**.

Choice A is incorrect. This is the range of the number of piglets per litter for these three litters.

Choice B is incorrect. This is the minimum number of piglets per litter for these three litters.

Choice D is incorrect. This is the maximum number of piglets per litter for these three litters.

Question Difficulty: Medium

Question ID 988ab336

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 988ab336

The population density of Cedar County is **230** people per square mile. The county has a population of **85,100** people. What is the area, in square miles, of Cedar County?

ID: 988ab336 Answer

Correct Answer: 370

Rationale

The correct answer is **370**. It’s given that the population density of Cedar County is **230** people per square mile and the county has a population of **85,100** people. Based on the population density, it follows that the area of Cedar County is $(85,100 \text{ people})\left(\frac{1 \text{ square mile}}{230 \text{ people}}\right)$, or **370** square miles.

Question Difficulty: Medium

Question ID 10ff87f3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 10ff87f3

How many yards are equivalent to **77** rods? (**5.5 yards = 1 rod**)

ID: 10ff87f3 Answer

Correct Answer: 423.5, 847/2

Rationale

The correct answer is **423.5**. It's given that **5.5 yards = 1 rod**. Therefore, **77** rods is equivalent to **(77 rods)** $\left(\frac{5.5 \text{ yards}}{1 \text{ rod}}\right)$, or **423.5** yards. Note that 423.5 and 847/2 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID 90ff6271

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 90ff6271

Tilly earns p dollars for every w hours of work. Which expression represents the amount of money, in dollars, Tilly earns for $39w$ hours of work?

- A. $39p$
- B. $\frac{p}{39}$
- C. $p + 39$
- D. $p - 39$

ID: 90ff6271 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that Tilly earns p dollars for every w hours of work. This can be represented by the proportion $\frac{p}{w}$. The amount of money, x , Tilly earns for $39w$ hours of work can be found by setting up the proportion $\frac{p}{w} = \frac{x}{39w}$. This can be rewritten as $39pw = xw$. Dividing both sides by w results in $x = 39p$.

Choice B is incorrect. This is the amount of money Tilly earns in dollars per hour, not the amount of money Tilly earns for $39w$ hours of work.

Choice C is incorrect. This is the amount of money Tilly earns for w hours of work plus 39 , not the amount of money Tilly earns for $39w$ hours of work.

Choice D is incorrect. This is the amount of money Tilly earns for w hours of work minus 39 , not the amount of money Tilly earns for $39w$ hours of work.

Question Difficulty: Easy

Question ID 2b327d7f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: 2b327d7f

A middle school faculty surveyed students to determine interest in an after-school chess club. Of the **100** students who were selected at random from all students in the school, **32%** expressed interest in the club. Based on the sample mean and the margin of error, the faculty expects that between **24%** and **40%** of all students in the school would express interest in the club. What is the margin of error?

- A. **0%**
- B. **8%**
- C. **16%**
- D. **32%**

ID: 2b327d7f Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that of the **100** students who were selected at random from all students in the school, **32%** expressed interest in the club. It's also given that based on the sample mean and the margin of error, the faculty expects that between **24%** and **40%** of all students in the school would express interest in the club. These values are calculated by subtracting the margin of error from **32%** to get the lower bound, **24%**, and by adding the margin of error to **32%** to get the upper bound, **40%**. It follows that the margin of error can be calculated as the difference between **32%** and **24%** or between **40%** and **32%**. The difference between **32%** and **24%** is **(32 – 24)%**, or **8%**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the difference between **40%** and **24%**, not the difference between **32%** and **24%** or between **40%** and **32%**.

Choice D is incorrect. This is the percentage of the students selected at random who expressed interest in the club, not the margin of error.

Question Difficulty: Hard

Question ID 5ddea380

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 5ddea380

Each vertex of a **14**-sided polygon is labeled with one of the **14** letters ***A*** through ***N***, with a different letter at each vertex. If one vertex is selected at random, what is the probability that the letter ***D*** will be at the selected vertex? (Express your answer as a decimal or fraction, not as a percent.)

ID: 5ddea380 Answer

Correct Answer: .0714, 1/14

Rationale

The correct answer is $\frac{1}{14}$. If one vertex of the polygon is selected at random, the probability that the letter ***D*** will be at the selected vertex is equal to the number of vertices labeled with the letter ***D*** divided by the total number of vertices. It's given that each vertex is labeled with one of the **14** letters ***A*** through ***N***, with a different letter at each vertex. It follows that there is **1** vertex labeled with the letter ***D***. It's also given that the polygon is **14**-sided. It follows that there are a total of **14** vertices. Thus, the probability that the letter ***D*** will be at the selected vertex is $\frac{1}{14}$. Note that 1/14, .0714, and 0.071 are examples of ways to enter a correct answer.

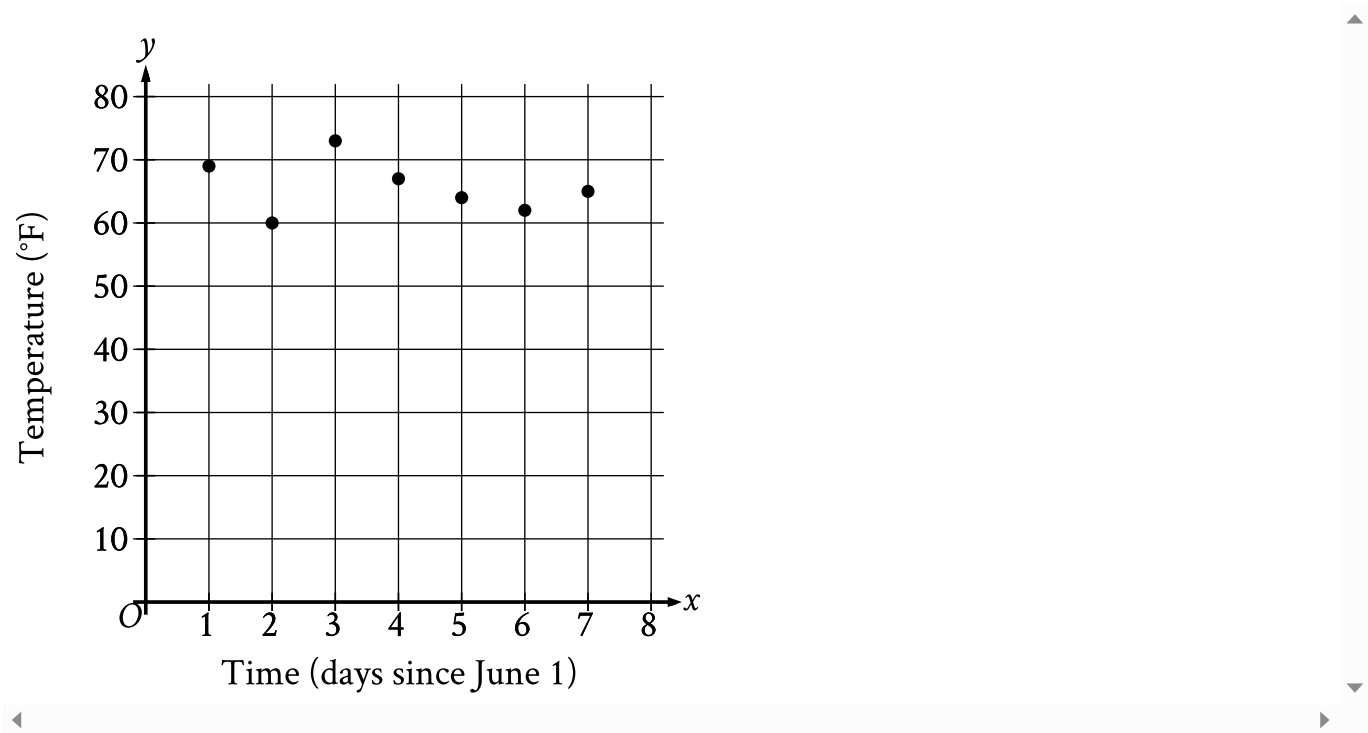
Question Difficulty: Medium

Question ID c387d395

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: c387d395

The scatterplot shows the temperature y , in $^{\circ}\text{F}$, recorded by a meteorologist at various times x , in days since June 1.



During which of the following time periods did the greatest increase in recorded temperature take place?

- A. From $x = 6$ to $x = 7$
- B. From $x = 5$ to $x = 6$
- C. From $x = 2$ to $x = 3$
- D. From $x = 1$ to $x = 2$

ID: c387d395 Answer

Correct Answer: C

Rationale

Choice C is correct. The scatterplot shows that there was an increase in recorded temperature from $x = 2$ to $x = 3$ and from $x = 6$ to $x = 7$. When $x = 2$, the recorded temperature was approximately 60°F and when $x = 3$, the recorded temperature was greater than 70°F . This means that the increase in recorded temperature from $x = 2$ to $x = 3$ was greater than $(70 - 60)^{\circ}\text{F}$, or 10°F . When $x = 6$, the recorded temperature was greater than 60°F and when $x = 7$, the recorded temperature was less than 70°F . This means that the increase in recorded temperature from $x = 6$ to $x = 7$ was less than $(70 - 60)^{\circ}\text{F}$, or 10°F . It follows that the greatest increase in recorded temperature took place from $x = 2$ to $x = 3$.

Choice A is incorrect. The increase in recorded temperature from $x = 6$ to $x = 7$ was less than the increase in recorded temperature from $x = 2$ to $x = 3$.

Choice B is incorrect. From $x = 5$ to $x = 6$, a decrease, not an increase, in recorded temperature took place.

Choice D is incorrect. From $x = 1$ to $x = 2$, a decrease, not an increase, in recorded temperature took place.

Question Difficulty: Easy

Question ID 59b985a2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 59b985a2

Each face of a fair **14**-sided die is labeled with a number from **1** through **14**, with a different number appearing on each face. If the die is rolled one time, what is the probability of rolling a **2**?

- A. $\frac{1}{14}$
- B. $\frac{2}{14}$
- C. $\frac{12}{14}$
- D. $\frac{13}{14}$

ID: 59b985a2 Answer

Correct Answer: A

Rationale

Choice A is correct. The total number of possible outcomes for rolling a fair **14**-sided die is **14**. The number of possible outcomes for rolling a **2** is **1**. The probability of rolling a **2** is the number of possible outcomes for rolling a **2** divided by the total number of possible outcomes, or $\frac{1}{14}$.

Choice B is incorrect. This is the probability of rolling a number no greater than **2**.

Choice C is incorrect. This is the probability of rolling a number greater than **2**.

Choice D is incorrect. This is the probability of rolling a number other than **2**.

Question Difficulty: Easy

Question ID 573d3cbf

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 573d3cbf

71, 72, 73, 76, 77, 79, 83, 87, 93
What is the median of the data shown?

- A. 71
- B. 77
- C. 78
- D. 79

ID: 573d3cbf Answer

Correct Answer: B

Rationale

Choice B is correct. The median of a data set with an odd number of data values is defined as the middle value of the ordered list of values. The data set shown has nine values, so the median is the fifth value in the ordered list, which is **77**.

Choice A is incorrect. This is the minimum value of the data set, not the median.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the mean of the data set, not the median.

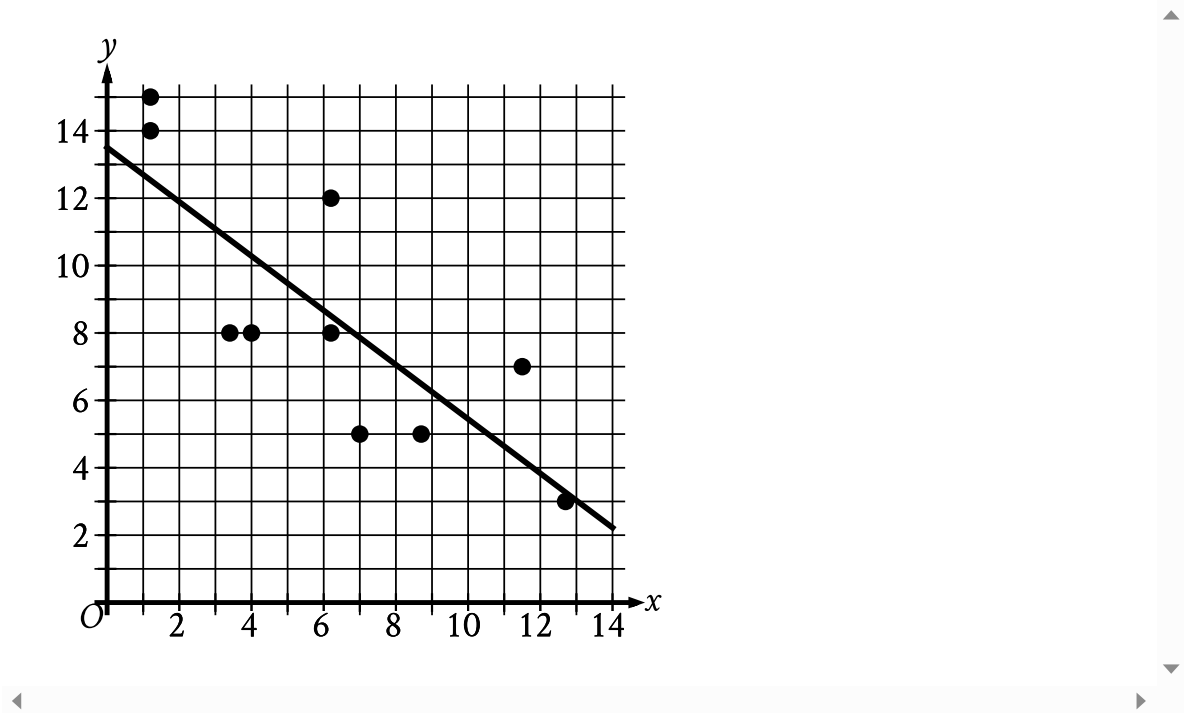
Question Difficulty: Easy

Question ID 839d25cc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 839d25cc

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

- A. $y = 13.5 + 0.8x$
- B. $y = 13.5 - 0.8x$
- C. $y = -13.5 + 0.8x$
- D. $y = -13.5 - 0.8x$

ID: 839d25cc Answer

Correct Answer: B

Rationale

Choice B is correct. The line of best fit shown intersects the y -axis at a positive y -value and has a negative slope. The graph of an equation of the form $y = a + bx$, where a and b are constants, intersects the y -axis at a y -value of a and has a slope of b . Of the given choices, only choice B represents a line that intersects the y -axis at a positive y -value, **13.5**, and has a negative slope, **-0.8**.

Choice A is incorrect. This equation represents a line that has a positive slope, not a negative slope.

Choice C is incorrect. This equation represents a line that intersects the y -axis at a negative y -value, not a positive y -value, and has a positive slope, not a negative slope.

Choice D is incorrect. This equation represents a line that intersects the y -axis at a negative y -value, not a positive y -value.

Question ID aa0af439

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: aa0af439

A kangaroo has a mass of **28** kilograms. What is the kangaroo's mass, in grams? (**1 kilogram = 1,000 grams**)

- A. **28,000**
- B. **1,028**
- C. **972**
- D. **784**

ID: aa0af439 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that a kangaroo has a mass of **28** kilograms and that **1** kilogram is equal to **1,000** grams. Therefore, the kangaroo's mass, in grams, is **28 kilograms** $\left(\frac{1,000 \text{ grams}}{1 \text{ kilogram}}\right)$, which is equivalent to **28,000** grams.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID ecb93995

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: ecb93995

The number a is 55% less than the number b . The number b is 320% greater than 160. What is the value of a ?

ID: ecb93995 Answer

Correct Answer: 1512/5, 302.4

Rationale

The correct answer is 302.4. It's given that the number b is 320% greater than 160. It follows that b is equal to 160 plus 320% of 160, which can be written as $b = 160 + 3.2(160)$, or $b = 672$. It's also given that the number a is 55% less than the number b . It follows that a is equal to b minus 55% of b , which can be written as $a = b - 0.55b$, or $a = 0.45b$. Substituting 672 for b in this equation yields $a = 0.45(672)$, or $a = 302.4$.

Question Difficulty: Hard

Question ID d788fd50

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: d788fd50

Weight (pounds)	Frequency
13	12
14	8
15	5
16	7
17	9
18	10
19	13
20	7

The frequency table summarizes a data set of the weights, rounded to the nearest pound, of **71** tortoises. A weight of **39** pounds is added to the original data set, creating a new data set of the weights, rounded to the nearest pound, of **72** tortoises. Which statement best compares the mean and median of the new data set to the mean and median of the original data set?

- A. The mean of the new data set is greater than the mean of the original data set, and the median of the new data set is greater than the median of the original data set.
- B. The mean of the new data set is greater than the mean of the original data set, and the medians of the two data sets are equal.
- C. The mean of the new data set is less than the mean of the original data set, and the median of the new data set is less than the median of the original data set.
- D. The mean of the new data set is less than the mean of the original data set, and the medians of the two data sets are equal.

ID: d788fd50 Answer

Correct Answer: B

Rationale

Choice B is correct. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. The new data set consists of the weights of the **71** tortoises in the original data set and one additional weight, **39**. Since the additional weight, **39**, is greater than any of the values in the original data set, the mean of the new data set is greater than the mean of the original data set. If a data set contains an odd number of data values, the median is represented by the middle data value in the list when the data values are listed in ascending or descending order. Since the original data set consists of the weights of **71** tortoises and is in ascending order, the median of the original data set is represented by the middle value, or the **36th** value. Based on the frequencies shown in the table, the **36th** value in this data set is **17**. If a data set contains an even number of data values, the median is between the two middle data values when the values are listed in ascending or descending order. Since the new data set consists of the weights of **72** tortoises, the median of the new data set is between the **36th** and

37th data values when the values are arranged in ascending order. To keep the data in ascending order, the additional value of **39** would be placed at the bottom of the frequency table with a frequency of **1**. Therefore, based on the frequencies in the table, the **36th** and **37th** values in the new data set are both **17**. It follows that the median of the new data set is **17**, which is the same as the median of the original data set. Therefore, the mean of the new data set is greater than the mean of the original data set, and the medians of the two data sets are equal.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 6c17c5a3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 6c17c5a3



The histogram summarizes data set A, which represents the number of points per player earned by **50** players of a game. A new player earns **18** points playing the game, and this number of points is added to data set A to create data set B with **51** values. Which of the following must be true?

- I. The median number of points per player for data set B is less than the median number of points per player for data set A.
- II. The mean number of points per player for data set B is less than the mean number of points per player for data set A.

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

ID: 6c17c5a3 Answer

Correct Answer: B

Rationale

Choice B is correct. The mean of a data set is the sum of the values divided by the number of values. It's given that a new player earns **18** points playing the game, and this number of points is added to data set A to create data set B with **51** values. The histogram shows that all the values in data set A are greater than **18**. Since the additional number of points, **18**, is less than any of the values in the original data set, the mean number of points per player for data set B is less than the mean number of points per player for data set A. The median of a data set with **50** values is between the **25th** and **26th** values when the values are listed in ascending order. The median of a data set with **51** values is the **26th** value when the values are listed in ascending order. Therefore, the median of data set B is the **26th** value in data set B, which is the **25th** value in data set A. The histogram shows that the **25th** and **26th** values in data set A are both between **50** and **60** points. Therefore, the **25th** and **26th** values in data set A could be equal, in which case the median of data set A would be equal to both the **25th** and **26th** values in data set A. Then, the median number of points per player for data set A would be equal to the median number of points per player for data set B. Therefore, only statement II must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

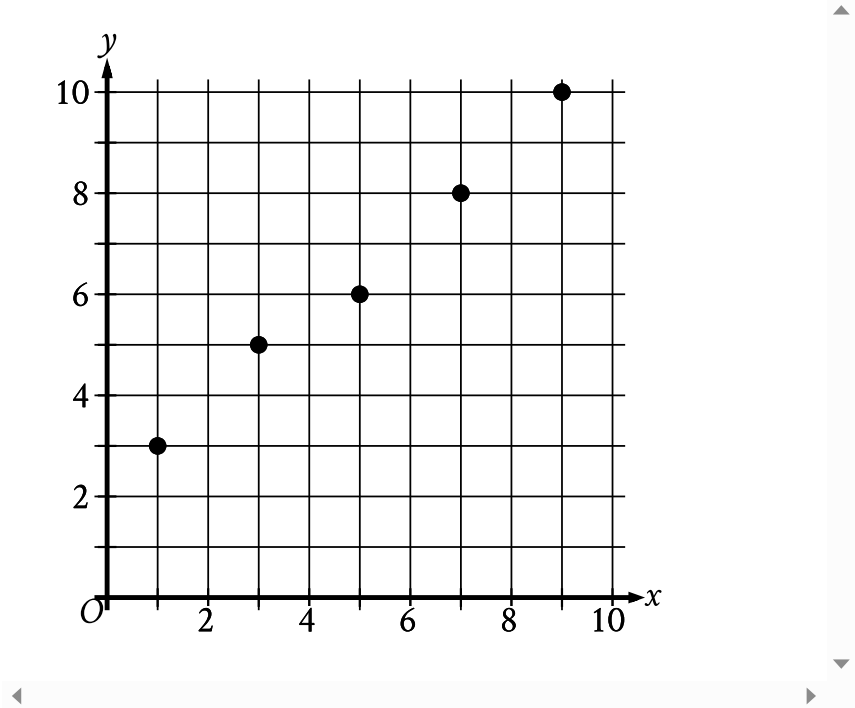
Question Difficulty: Hard

Question ID 68a0f740

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 68a0f740

The scatterplot shows the relationship between two variables, x and y .



Which equation is the most appropriate linear model for this relationship?

- A. $y = -0.9x - 2.2$
- B. $y = -0.9x + 2.2$
- C. $y = -0.9x$
- D. $y = 0.9x + 2.2$

ID: 68a0f740 Answer

Correct Answer: D

Rationale

Choice D is correct. A linear model can be written in the form $y = mx + b$, where m is the slope of the graph of the model in the xy -plane and $(0, b)$ is the y -intercept. The graph of an appropriate linear model for this relationship passes near the points $(1, 3)$ and $(9, 10)$ in the xy -plane. Two points on a line, (x_1, y_1) and (x_2, y_2) , can be used to find the slope of the line using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(1, 3)$ and $(9, 10)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{10 - 3}{9 - 1}$, or $m = 0.875$. Therefore, the value of m for an appropriate linear model is approximately 0.875 . Substituting 0.875 for m in $y = mx + b$ yields $y = 0.875x + b$. Since an appropriate linear model passes near the point $(1, 3)$, the approximate value of b can be found by substituting 1 for x and 3 for y in the equation $y = 0.875x + b$, which yields $3 = (0.875)(1) + b$, or $3 = 0.875 + b$. Subtracting 0.875 from both sides of this equation yields $2.125 = b$. Therefore, the value of b for an appropriate linear model is approximately 2.125 . Thus, of the given choices, $y = 0.9x + 2.2$ is the most appropriate linear model for this relationship.

Alternate approach: A linear model can be written in the form $y = mx + b$, where m is the slope of the graph of the model in the xy -plane and $(0, b)$ is the y -intercept. The scatterplot shows that as the x -values of the data points increase, the y -values of the data points increase, which means the graph of an appropriate linear model has a positive slope. Of the given choices, $y = 0.9x + 2.2$ is the only linear model whose graph has a positive slope.

Choice A is incorrect. The graph of this model has a negative slope, not a positive slope.

Choice B is incorrect. The graph of this model has a negative slope, not a positive slope.

Choice C is incorrect. The graph of this model has a negative slope, not a positive slope.

Question Difficulty: Easy

Question ID 2584cda8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: 2584cda8

An analyst collected data on the price of a carton of grape tomatoes at **30** locations selected at random in Utah. The mean price of a carton of grape tomatoes in Utah was estimated to be **\$4.23**, with an associated margin of error of **\$0.08**. Which of the following is a plausible statement about the mean price of a carton of grape tomatoes for all locations that sell this product in Utah?

- A. It is between **\$4.15** and **\$4.31**.
- B. It is either less than **\$4.15** or greater than **\$4.31**.
- C. It is less than **\$4.15**.
- D. It is greater than **\$4.31**.

ID: 2584cda8 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the mean price of a carton of grape tomatoes in Utah was estimated to be **\$4.23**, with an associated margin of error of **\$0.08**. It follows that plausible values for this mean price are between **\$4.23 – \$0.08** and **\$4.23 + \$0.08**. Therefore, it's plausible that the mean price of a carton of grape tomatoes for all locations that sell this product in Utah is between **\$4.15** and **\$4.31**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

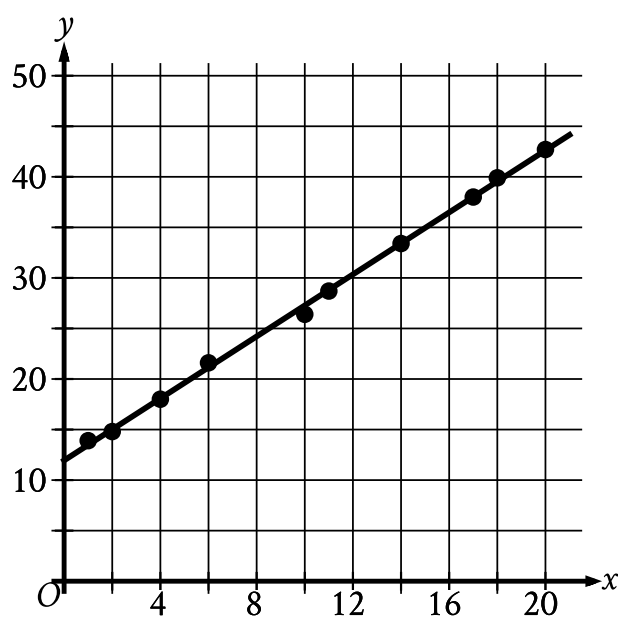
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 14baeb51

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 14baeb51



The scatterplot shows the relationship between two variables, x and y , for data set E. A line of best fit is shown. Data set F is created by multiplying the y -coordinate of each data point from data set E by **3.9**. Which of the following could be an equation of a line of best fit for data set F?

- A. $y = 46.8 + 5.9x$
- B. $y = 46.8 + 1.5x$
- C. $y = 12 + 5.9x$
- D. $y = 12 + 1.5x$

ID: 14baeb51 Answer

Correct Answer: A

Rationale

Choice A is correct. An equation of a line of best fit for data set F can be written in the form $y = a + bx$, where a is the y -coordinate of the y -intercept of the line of best fit and b is the slope. The line of best fit shown for data set E has a y -intercept at approximately $(0, 12)$. It's given that data set F is created by multiplying the y -coordinate of each data point from data set E by **3.9**. It follows that a line of best fit for data set F has a y -intercept at approximately $(0, 12(3.9))$, or $(0, 46.8)$. Therefore, the value of a is approximately **46.8**. The slope of a line that passes through points (x_1, y_1) and (x_2, y_2) can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. Since the line of best fit shown for data set E passes approximately through the point $(12, 30)$, it follows that a line of best fit for data set F passes approximately through the point $(12, 30(3.9))$, or $(12, 117)$. Substituting $(0, 46.8)$ and $(12, 117)$ for (x_1, y_1) and (x_2, y_2) , respectively, in $\frac{y_2 - y_1}{x_2 - x_1}$ yields $\frac{117 - 46.8}{12 - 0}$, which is equivalent to $\frac{70.2}{12}$, or **5.85**. Therefore, the value of b is approximately **5.85**, or approximately **5.9**. Thus, $y = 46.8 + 5.9x$ could be an equation of a line of best fit for data set F.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This could be an equation of a line of best fit for data set E, not data set F.

Question Difficulty: Hard

Question ID c6574ec0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: c6574ec0

In March, the price of a collectible card was **\$15.50**. In April, the price of the collectible card was **\$17.36**. The price of the collectible card in April was $p\%$ of the price of the collectible card in March. What is the value of p ?

- A. 12
- B. 88
- C. 112
- D. 188

ID: c6574ec0 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the price of the collectible card was **\$15.50** in March and **\$17.36** in April. It's also given that the price of the collectible card in April was $p\%$ of the price in March. It follows that **\$17.36** is $p\%$ of **\$15.50**. Therefore, the value of p can be calculated by solving the equation $17.36 = (\frac{p}{100})(15.50)$, or $17.36 = \frac{15.50p}{100}$. Multiplying each side of this equation by 100 yields $1,736 = 15.50p$. Dividing each side of this equation by 15.50 yields $112 = p$. Therefore, the value of p is 112.

Choice A is incorrect. 12% is the percent increase in the price of the collectible card from March to April.

Choice B is incorrect and may result from conceptual or calculation errors.

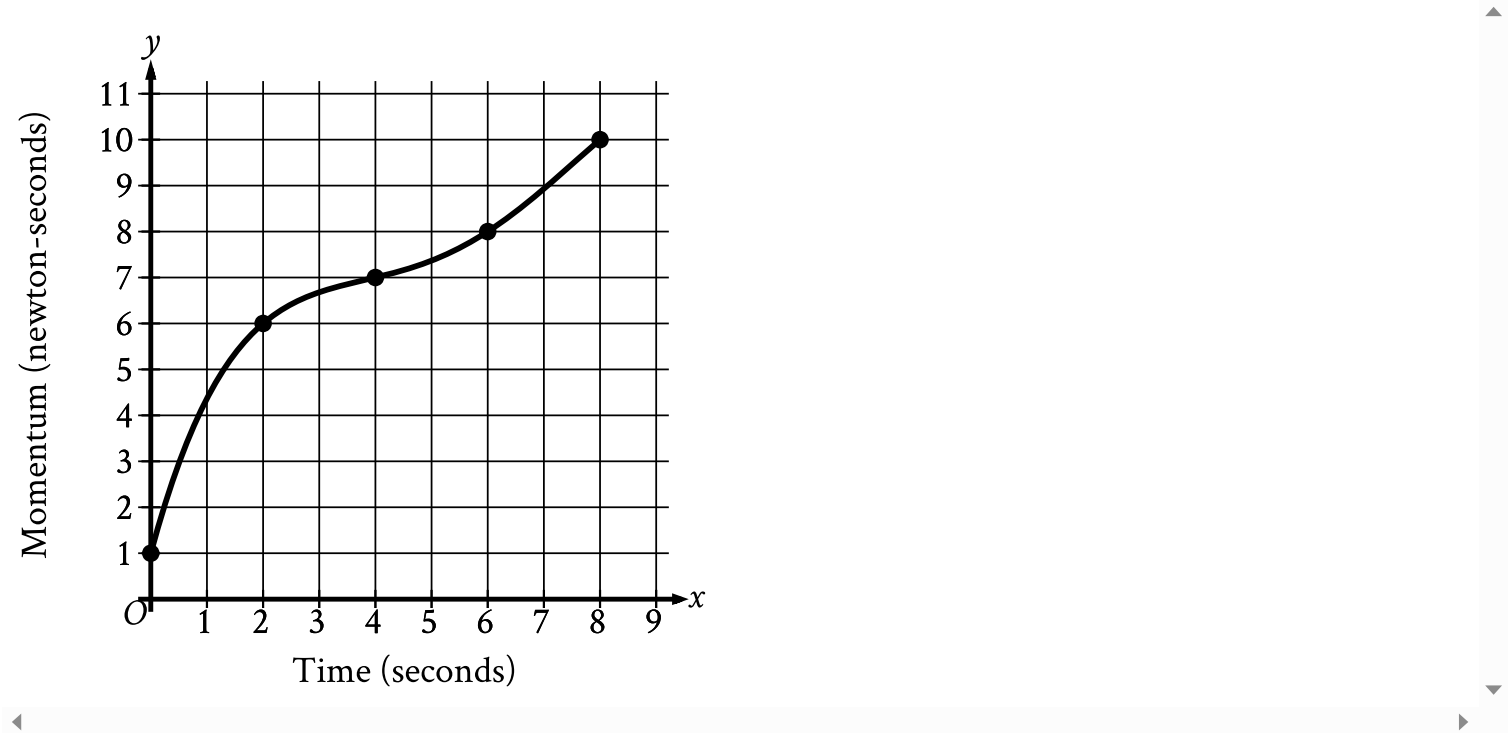
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 8f250149

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 8f250149



The graph shows the momentum y , in newton-seconds, of an object x seconds after the object started moving, for $0 \leq x \leq 8$. What is the average rate of change, in newton-seconds per second, in the momentum of the object from $x = 2$ to $x = 6$?

ID: 8f250149 Answer

Correct Answer: 0.5, 1/2

Rationale

The correct answer is $\frac{1}{2}$. For the graph shown, x represents time, in seconds, and y represents momentum, in newton-seconds. Therefore, the average rate of change, in newton-seconds per second, in the momentum of the object between two x -values is the difference in the corresponding y -values divided by the difference in the x -values. The graph shows that at $x = 2$, the corresponding y -value is 6. The graph also shows that at $x = 6$, the corresponding y -value is 8. It follows that the average rate of change, in newton-seconds per second, from $x = 2$ to $x = 6$ is $\frac{8-6}{6-2}$, which is equivalent to $\frac{2}{4}$, or $\frac{1}{2}$. Note that 1/2 and .5 are examples of ways to enter a correct answer.

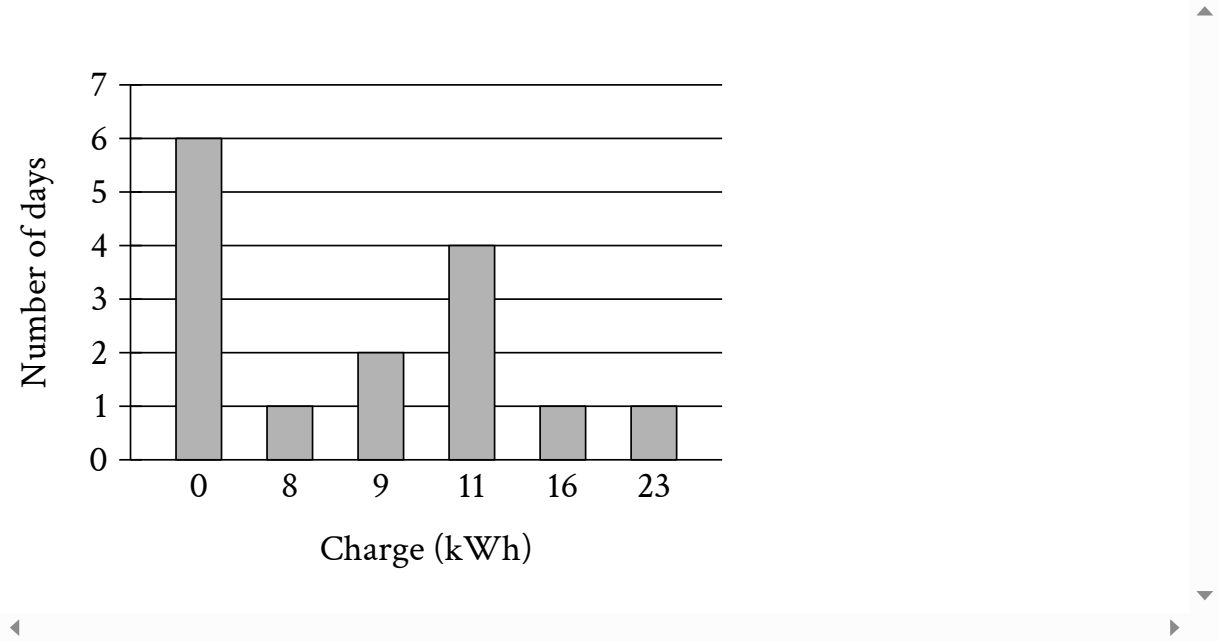
Question Difficulty: Hard

Question ID 625bd2bb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 625bd2bb

The bar graph summarizes the charge, in kilowatt-hours (**kWh**), a battery received each day for **15** days.



For how many of these **15** days did the battery receive a charge of **0 kWh**?

- A. 0
- B. 1
- C. 4
- D. 6

ID: 625bd2bb Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the bar graph summarizes the charge, in kilowatt-hours (**kWh**), a battery received each day for **15** days. The height of each bar in the bar graph shown represents the number of days the battery received the charge, in **kWh**, specified at the bottom of the bar. The bar for a charge of **0 kWh** reaches a height of **6**. Therefore, the battery received a charge of **0 kWh** for **6** of these days.

Choice A is incorrect. This is the charge, in **kWh**, that the battery received, not the number of days the battery received this charge.

Choice B is incorrect. This is the number of days the battery received a charge of either **8**, **16**, or **23 kWh**.

Choice C is incorrect. This is the number of days the battery received a charge of **11 kWh**.

Question Difficulty: Easy

Question ID c0f5ceb0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: c0f5ceb0

What length, in centimeters, is equivalent to a length of **51** meters? (**1 meter = 100 centimeters**)

- A. **0.051**
- B. **0.51**
- C. **5,100**
- D. **51,000**

ID: c0f5ceb0 Answer

Correct Answer: C

Rationale

Choice C is correct. Since **1** meter is equal to **100** centimeters, **51** meters is equal to **51 meters**($\frac{100 \text{ centimeters}}{1 \text{ meter}}$), or **5,100** centimeters.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from dividing, rather than multiplying, **51** by **100**.

Choice D is incorrect. This is the length, in millimeters rather than centimeters, that is equivalent to a length of **51** meters.

Question Difficulty: Easy

Question ID f18032ac

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: f18032ac



A data set consists of **173** colors. The bar graph shows the number of times each color appears in the data set. Which color appears **70** times?

- A. Blue
- B. Green
- C. Red
- D. Yellow

ID: f18032ac Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that a data set consists of **173** colors and the bar graph shows the number of times each color appears in the data set. Therefore, for each color specified at the bottom of the bar, the frequency corresponds to the number of times that color appears in the data set. The color that appears **70** times in the data set has a frequency of **70** on the bar graph. Since the bar with a frequency of **70** corresponds to green, green is the color that appears **70** times.

Choice A is incorrect. The color blue appears about **27** times, not **70** times.

Choice C is incorrect. The color red appears about **33** times, not **70** times.

Choice D is incorrect. The color yellow appears about **43** times, not **70** times.

Question Difficulty: Easy

Question ID dc41a752

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: dc41a752

	1 visit	2 or more visits	Total
Less than 40 years old	15	15	30
At least 40 years old	20	85	105
Total	35	100	135

The table summarizes customers who visited a car dealership in the last month by age and number of visits they made to the dealership. If a customer from the last month is selected at random, what is the probability that the selected customer is at least 40 years old?

- A. $\frac{30}{135}$
- B. $\frac{35}{135}$
- C. $\frac{100}{135}$
- D. $\frac{105}{135}$

ID: dc41a752 Answer

Correct Answer: D

Rationale

Choice D is correct. Based on the table, there are a total of 135 customers who visited the car dealership in the last month, and 105 of these customers are at least 40 years old. If a customer from the last month is selected at random, the probability that the selected customer is at least 40 years old is equal to the number of customers who are at least 40 years old divided by the total number of customers. Therefore, the probability that the selected customer is at least 40 years old is $\frac{105}{135}$.

Choice A is incorrect. This is the probability that the selected customer is less than 40 years old.

Choice B is incorrect. This is the probability that the selected customer visited the dealership 1 time in the last month.

Choice C is incorrect. This is the probability that the selected customer visited the dealership 2 or more times in the last month.

Question Difficulty: Easy

Question ID 338eaa47

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 338eaa47

A certain park has an area of **11,863,808** square yards. What is the area, in square miles, of this park? (**1 mile = 1,760 yards**)

- A. **1.96**
- B. **3.83**
- C. **3,444.39**
- D. **6,740.8**

ID: 338eaa47 Answer

Correct Answer: B

Rationale

Choice B is correct. Since **1** mile is equal to **1,760** yards, **1** square mile is equal to **1,760²**, or **3,097,600**, square yards. It’s given that the park has an area of **11,863,808** square yards. Therefore, the park has an area of **(11,863,808 square yards) $\left(\frac{1 \text{ square mile}}{3,097,600 \text{ square yards}}\right)$** , or $\frac{11,863,808}{3,097,600}$ square miles. Thus, the area, in square miles, of the park is **3.83**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the square root of the area of the park in square yards, not the area of the park in square miles.

Choice D is incorrect and may result from converting **11,863,808** yards to miles, rather than converting **11,863,808** square yards to square miles.

Question Difficulty: Hard

Question ID cc5923de

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: cc5923de

A landscaper uses a hose that puts $88x$ ounces of water in a bucket in $5y$ minutes. Which expression represents the number of ounces of water the hose puts in the bucket in $9y$ minutes at this rate?

- A. $\frac{9x}{440}$
- B. $\frac{440x}{9}$
- C. $\frac{5x}{792}$
- D. $\frac{792x}{5}$

ID: cc5923de Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that a hose puts $88x$ ounces of water in a bucket in $5y$ minutes. Therefore, the rate at which the hose puts water in the bucket, in ounces per minute, can be represented by the expression $\frac{88x}{5y}$. Let w represent the number of ounces of water the hose puts in the bucket in $9y$ minutes at this rate. It follows that the rate at which the hose puts water in the bucket, in ounces per minute, can be represented by the expression $\frac{w}{9y}$. The expressions $\frac{88x}{5y}$ and $\frac{w}{9y}$ represent the same rate, so it follows that $\frac{88x}{5y} = \frac{w}{9y}$. Multiplying both sides of this equation by $9y$ yields $\frac{792xy}{5y} = w$, or $\frac{792x}{5} = w$. Therefore, the number of ounces of water the hose puts in the bucket in $9y$ minutes can be represented by the expression $\frac{792x}{5}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 76ae9e64

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 76ae9e64

What is 10% of 470?

- A. 37
- B. 47
- C. 423
- D. 460

ID: 76ae9e64 Answer

Correct Answer: B

Rationale

Choice B is correct. 10% of a quantity means $\frac{10}{100}$ times the quantity. Therefore, 10% of 470 can be represented as $\frac{10}{100}(470)$, which is equivalent to $0.10(470)$, or 47. Therefore, 10% of 470 is 47.

Choice A is incorrect. This is 10% of 370, not 10% of 470.

Choice C is incorrect. This is 90% of 470, not 10% of 470.

Choice D is incorrect. This is $470 - 10$, not 10% of 470.

Question Difficulty: Easy

Question ID 53b13005

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 53b13005

Last year, Cedric had **35** plants in his garden. This year, the number of plants in Cedric’s garden is **60%** greater than the number of plants in his garden last year. How many plants does Cedric have in his garden this year?

ID: 53b13005 Answer

Correct Answer: 56

Rationale

The correct answer is **56**. It’s given that Cedric had **35** plants in his garden last year and that the number of plants in Cedric's garden this year is **60%** greater than the number of plants in his garden last year. It follows that the number of plants in Cedric’s garden this year is **35** plus **60%** of **35**, which is equal to **$35 + 35\left(\frac{60}{100}\right)$** , or **$35 + 35(0.6)$** . This expression is equivalent to **$35 + 21$** , or **56**. Therefore, Cedric has **56** plants in his garden this year.

Question Difficulty: Medium

Question ID 95bf6ccc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 95bf6ccc

There are **320** marbles in a container. Of these marbles, **10%** are red. How many marbles in the container are red?

- A. **32**
- B. **288**
- C. **320**
- D. **352**

ID: 95bf6ccc Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that **10%** of the **320** marbles in a container are red. Therefore, the number of red marbles can be calculated by multiplying the number of marbles in the container by $\frac{10}{100}$, which gives **320** $\left(\frac{10}{100}\right)$, or **32**.

Choice B is incorrect. This is the number of marbles in the container that aren't red.

Choice C is incorrect. This is the total number of marbles in the container.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 74a66620

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 74a66620

The ratio of the length of line segment XY to the length of line segment ZV is **6** to **1**. If the length of line segment XY is **102** inches, what is the length, in inches, of line segment ZV ?

- A. **17**
- B. **96**
- C. **102**
- D. **612**

ID: 74a66620 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the ratio of the length of line segment XY to the length of line segment ZV is **6** to **1**, which means $\frac{XY}{ZV} = \frac{6}{1}$. It's given that the length of line segment XY is **102** inches. If the length, in inches, of line segment ZV is represented by x , the value of x can be calculated by solving the equation $\frac{102}{x} = \frac{6}{1}$, or $\frac{102}{x} = 6$. Multiplying each side of this equation by x yields $102 = 6x$. Dividing each side of this equation by **6** yields **17** = x . Therefore, the length of line segment ZV is **17** inches.

Choice B is incorrect. This is the length, in inches, of line segment ZV if the length of line segment XY is **576**, not **102**, inches.

Choice C is incorrect. This is the length, in inches, of line segment XY , not line segment ZV .

Choice D is incorrect. This is the length, in inches, of line segment ZV if the ratio of the length of line segment XY to the length of line segment ZV is **1** to **6**, not **6** to **1**.

Question Difficulty: Medium

Question ID 1f99c16c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 1f99c16c

A square scale drawing has a side length of **55** inches, and **1** inch on the drawing represents an actual distance of **19** miles. A larger version of the same drawing is printed as a square with the side length **65%** longer than the side length of the previous drawing. On the larger drawing, which of the following is closest to the actual distance, in miles, represented by **1** inch?

- A. **6.65**
- B. **11.52**
- C. **31.35**
- D. **35.75**

ID: 1f99c16c Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that **1** inch on the drawing represents an actual distance of **19** miles and that the larger drawing has a side length that is **65%** longer than the side length of the previous drawing. It follows that **(1.65)(1 inch)**, or **1.65** inches, on the larger drawing also represents an actual distance of **19** miles. Therefore, the scale of the larger drawing can be represented as $\frac{19 \text{ miles}}{1.65 \text{ inches}}$, which is approximately **11.52** miles per inch. Therefore, **11.52** is closest to the actual distance, in miles, represented by **1** inch on the larger drawing.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

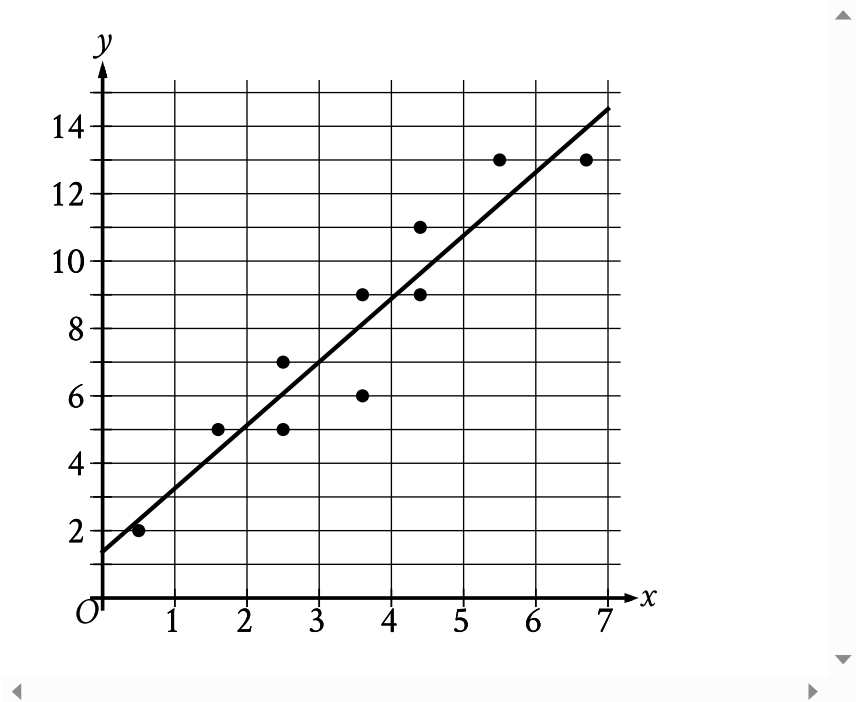
Question Difficulty: Hard

Question ID 8f6842b8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 8f6842b8

In the given scatterplot, a line of best fit for the data is shown.



Which of the following is closest to the slope of the line of best fit shown?

- A. 0
- B. $\frac{1}{2}$
- C. 1
- D. 2

ID: 8f6842b8 Answer

Correct Answer: D

Rationale

Choice D is correct. A line in the xy -plane that passes through the points (x_1, y_1) and (x_2, y_2) has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points $(1, 3.3)$ and $(7, 14.5)$. It follows that the slope of this best fit line is approximately $\frac{14.5 - 3.3}{7 - 1}$, which is equivalent to $\frac{11.2}{6}$, or approximately **1.87**. Therefore, of the given choices, **2** is closest to the slope of the line of best fit shown.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

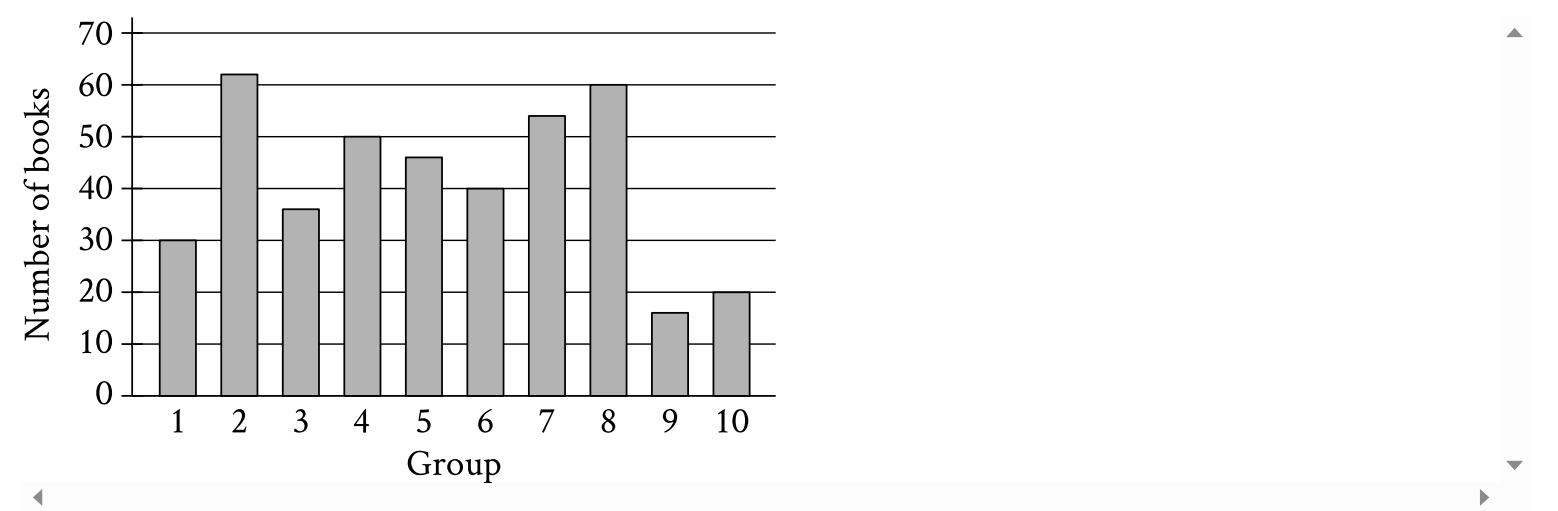
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 88de9ee2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 88de9ee2



The bar graph shows the distribution of **414** books collected by **10** different groups for a book drive. How many books were collected by group **1**?

ID: 88de9ee2 Answer

Correct Answer: 30

Rationale

The correct answer is **30**. The height of each bar in the bar graph shown represents the number of books collected by the group specified at the bottom of the bar. The bar for group **1** reaches a height of **30**. Therefore, group **1** collected **30** books.

Question Difficulty: Easy

Question ID d60b2133

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: d60b2133

The speeds of objects A, B, and C are a meters per second, b meters per second, and c meters per second, respectively. If the speed of object A is 5,900% of the speed of object C and the speed of object C is 0.008% of the speed of object B, which expression represents the value of $a + b$ in terms of c ?

- A. 1,840c
- B. 5,908c
- C. 6,025c
- D. 12,559c

ID: d60b2133 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the speeds of objects A, B, and C are a meters per second, b meters per second, and c meters per second, respectively. If the speed of object A is 5,900% of the speed of object C, this means that $a = \frac{5,900}{100}c$, or $a = 59c$. If the speed of object C is 0.008% of the speed of object B, this means that $c = \frac{0.008}{100}b$, or $c = 0.00008b$. Dividing both sides of this equation by 0.00008 yields $\frac{1}{0.00008}c = b$, or $12,500c = b$. Since $a = 59c$ and $b = 12,500c$, it follows that $a + b = 59c + 12,500c$, or $a + b = 12,559c$. Therefore, the expression 12,559c represents the value of $a + b$ in terms of c .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 4cf4c124

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 4cf4c124

Shape	Number of cards
Square	5
Circle	9
Triangle	14
Star	12

The table summarizes the distribution of shape printed on one side for cards in a deck of cards. How many cards in the deck have a triangle printed on one side?

ID: 4cf4c124 Answer

Correct Answer: 14

Rationale

The correct answer is **14**. The table summarizes the distribution of shape printed on one side for cards in a deck of cards. The table shows that the number of cards when the shape is a triangle is **14**. Therefore, there are **14** cards in the deck that have a triangle printed on one side.

Question Difficulty: Easy

Question ID d0e32a44

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: d0e32a44

In a box of pens, the ratio of black pens to red pens is **8** to **1**. There are **40** black pens in the box. How many red pens are in the box?

- A. **5**
- B. **8**
- C. **40**
- D. **320**

ID: d0e32a44 Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that the ratio of black pens to red pens is **8** to **1**. Therefore, there are $\frac{1}{8}$ as many red pens as black pens in the box. It’s also given that there are **40** black pens in the box. Therefore, the number of red pens is $\frac{1}{8}$ of the **40** black pens. Thus, the number of red pens is $40\left(\frac{1}{8}\right)$, or **5**.

Choice B is incorrect. This is the number of black pens in the box for every red pen.

Choice C is incorrect. This is the number of black pens in the box.

Choice D is incorrect. This is the number of red pens in the box if the ratio of black pens to red pens is **1** to **8**.

Question Difficulty: Easy

Question ID 2dc0f863

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 2dc0f863

Set K consists of all the positive integers that are less than or equal to **160**, where none of the integers are repeated. If a number is selected at random from set K, what is the probability of selecting a number that is even and less than or equal to **50**?

- A. $\frac{5}{32}$
- B. $\frac{3}{16}$
- C. $\frac{5}{21}$
- D. $\frac{5}{16}$

ID: 2dc0f863 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that set K consists of all the positive integers that are less than or equal to **160**, where none of the integers are repeated. Therefore, set K consists of the integers **1** through **160**, which is **160** numbers. Of these, the integers **1** through **50** are less than or equal to **50**. Therefore, there are **50** numbers that belong to set K that are less than or equal to **50**. Of these, half are even. Therefore, $\frac{50}{2}$, or **25**, numbers that belong to set K are even and less than or equal to **50**. The probability of selecting at random a number from set K that is even and less than or equal to **50** is equal to the number of numbers that belong to set K that are even and less than or equal to **50**, which is **25**, divided by the total number of numbers that belong to set K, which is **160**. It follows that the probability of selecting at random from set K a number that is even and less than or equal to **50** is $\frac{25}{160}$, or $\frac{5}{32}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the probability of selecting at random a number from set K that is less than or equal to **50**, not a number that is even and less than or equal to **50**.

Question Difficulty: Hard

Question ID 5d09fea2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 5d09fea2

Of **900,000** beads, **828,000** are silver. What percentage of the beads are silver?

- A. **8%**
- B. **36%**
- C. **72%**
- D. **92%**

ID: 5d09fea2 Answer

Correct Answer: D

Rationale

Choice D is correct. The proportion of the beads that are silver can be written as $\frac{828,000}{900,000}$, or **0.92**. Therefore, the percentage of the beads that are silver is **0.92(100)**, or **92%**.

Choice A is incorrect. This is the percentage of the beads that are not silver.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 4192efd7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 4192efd7

140 is $p\%$ greater than 10. What is the value of p ?

- A. 1,400
- B. 1,300
- C. 140
- D. 130

ID: 4192efd7 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that 140 is $p\%$ greater than 10. It follows that $140 = 10 + \left(\frac{p}{100}\right)10$, which is equivalent to $140 = 10 + \frac{10}{100}p$, or $140 = 10 + 0.1p$. Subtracting 10 from each side of this equation yields $130 = 0.1p$. Dividing each side of this equation by 0.1 yields $1,300 = p$, or $p = 1,300$.

Choice A is incorrect. This would be the value of p if 140 were $p\%$ of 10, not $p\%$ greater than 10.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID adcab929

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: adcab929

Data value	Frequency
6	3
7	3
8	8
9	8
10	9
11	11
12	9
13	0
14	6

The frequency table summarizes the **57** data values in a data set. What is the maximum data value in the data set?

ID: adcab929 Answer

Correct Answer: 14

Rationale

The correct answer is **14**. The maximum value is the largest value in the data set. The frequency refers to the number of times a data value occurs. The given frequency table shows that for this data set, the data value **6** occurs three times, the data value **7** occurs three times, the data value **8** occurs eight times, the data value **9** occurs eight times, the data value **10** occurs nine times, the data value **11** occurs eleven times, the data value **12** occurs nine times, the data value **13** occurs zero times, and the data value **14** occurs six times. Therefore, the maximum data value in the data set is **14**.

Question Difficulty: Hard

Question ID ce18cfcc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: ce18cfcc

The result of increasing the quantity x by 1,800% is 684. What is the value of x ?

- A. 12,996
- B. 12,312
- C. 38
- D. 36

ID: ce18cfcc Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the result of increasing the quantity x by 1,800% is 684. It follows that $x + \left(\frac{1,800}{100}\right)x = 684$, which is equivalent to $x + 18x = 684$, or $19x = 684$. Dividing each side of this equation by 19 yields $x = 36$. Therefore, the value of x is 36.

Choice A is incorrect. The result of increasing the quantity 12,996 by 1,800% is 246,924, not 684.

Choice B is incorrect. The result of increasing the quantity 12,312 by 1,800% is 233,928, not 684.

Choice C is incorrect. The result of increasing the quantity 38 by 1,800% is 722, not 684.

Question Difficulty: Hard

Question ID 614c1b5f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 614c1b5f

There are a total of **840** seats in a school auditorium. During an assembly, students occupied **50%** of the seats in the auditorium. How many seats did the students occupy during this assembly?

- A. **25**
- B. **50**
- C. **420**
- D. **790**

ID: 614c1b5f Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that during an assembly, students occupied **50%** of the **840** seats in the school auditorium. Therefore, the number of seats that the students occupied during this assembly can be calculated by multiplying the number of seats in the school auditorium by $\frac{50}{100}$. Thus, the students occupied **$840\left(\frac{50}{100}\right)$** , or **420**, seats during this assembly.

Choice A is incorrect. This is approximately **3%**, not **50%**, of **840**.

Choice B is incorrect. This is approximately **6%**, not **50%**, of **840**.

Choice D is incorrect. This is approximately **94%**, not **50%**, of **840**.

Question Difficulty: Easy

Question ID c17aedbe

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: c17aedbe

Each of **157** gemstones can be classified as one of three classifications, as shown in the frequency table.

Classification	Frequency
color X	119
color Y	3
color Z	35

If one of the gemstones is selected at random, what is the probability of selecting a gemstone of color Y?

- A. $\frac{3}{157}$
- B. $\frac{35}{157}$
- C. $\frac{119}{157}$
- D. $\frac{154}{157}$

ID: c17aedbe Answer

Correct Answer: A

Rationale

Choice A is correct. If one of the gemstones is selected at random, the probability of selecting a gemstone of color Y is equal to the number of gemstones of color Y divided by the total number of gemstones. According to the table, there are **3** gemstones of color Y, and it's given that the total number of gemstones is **157**. Therefore, if one of the gemstones is selected at random, the probability of selecting a gemstone of color Y is $\frac{3}{157}$.

Choice B is incorrect. This is the probability of selecting a gemstone of color Z.

Choice C is incorrect. This is the probability of selecting a gemstone of color X.

Choice D is incorrect. This is the probability of selecting a gemstone that's not of color Y.

Question Difficulty: Easy

Question ID 6e3e4278

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: 6e3e4278

Scott selected **20** employees at random from all **400** employees at a company. He found that **16** of the employees in this sample are enrolled in exactly three professional development courses this year. Based on Scott’s findings, which of the following is the best estimate of the number of employees at the company who are enrolled in exactly three professional development courses this year?

- A. **4**
- B. **320**
- C. **380**
- D. **384**

ID: 6e3e4278 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that from the sample of **20** employees at the company, **16** of the employees are enrolled in exactly three professional development courses this year. Since $\left(\frac{16}{20}\right)$ is equal to **0.80**, or $\frac{80}{100}$, it follows that **80%** of the employees in the sample are enrolled in exactly three professional development courses this year. Therefore, the best estimate for the percentage of employees at the company who are enrolled in exactly three professional development courses this year is **80%**. It’s given that there are a total of **400** employees at the company. Therefore, the best estimate of the number of employees at the company who are enrolled in exactly three professional development courses this year is $\left(\frac{80}{100}\right)(400)$, or **320**.

Choice A is incorrect. This is the number of employees from the sample who aren't enrolled in exactly three professional development courses this year.

Choice C is incorrect. This is the number of employees who weren't selected for the sample.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 3c4d9ad7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 3c4d9ad7

The table shown summarizes the number of employees at each of the **17** restaurants in a town.

Number of employees	Number of restaurants
2 to 7	2
8 to 13	4
14 to 19	2
20 to 25	7
26 to 31	2

Which of the following could be the median number of employees for the restaurants in this town?

- A. **2**
- B. **9**
- C. **15**
- D. **21**

ID: 3c4d9ad7 Answer

Correct Answer: D

Rationale

Choice D is correct. If a data set contains an odd number of data values, the median is represented by the middle data value in the list when the data values are listed in ascending or descending order. Since the numbers of employees are given as ranges of values rather than specific values, it's only possible to determine the range in which the median falls, rather than the exact median. Since there are **17** restaurants included in the data set and the numbers of employees are listed in ascending order, it follows that the median number of employees will be represented by the ninth restaurant in the list. Since the first **2** restaurants each have **2 to 7** employees, numbers of employees in the **2 to 7** range would be represented by the first and second restaurants in the list. The next **4** restaurants each have **8 to 13** employees. Therefore, numbers of employees in the **8 to 13** range will be represented by the third through sixth restaurants in the list. The next **2** restaurants each have **14 to 19** employees. Therefore, numbers of employees in the **14 to 19** range will be represented by the seventh and eighth restaurants in the list. Since the next **7** restaurants each have **20 to 25** employees, numbers of employees in the **20 to 25** range will be represented by the ninth through fifteenth restaurants in the list. This means that the ninth restaurant in the list, which has the median number of employees for the restaurants in this town, has a number of employees in the **20 to 25** range. Of the given choices, the only number of employees in the **20 to 25** range is **21**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the position of the median in the list, not the value of the median.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID 8f13878c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: 8f13878c

A company fills boxes with approximately **23** pounds of oranges. To test the accuracy of the filling process, **344** boxes of oranges were selected at random and weighed. Based on the sample, it is estimated that the average weight of all boxes of oranges filled by the company in an **8**-hour period is **23.1** pounds, with an associated margin of error of **0.19** pounds. Which of the following is the best interpretation of this estimate?

- A. Plausible values for the average weight of all boxes of oranges filled by the company are between **22.91** pounds and **23.29** pounds.
- B. Plausible values for the average weight of all boxes of oranges filled by the company are less than **22.91** pounds or greater than **23.29** pounds.
- C. The average weight of all boxes of oranges filled by the company is less than **23.01** pounds.
- D. The average weight of all boxes of oranges filled by the company is greater than **23.01** pounds.

ID: 8f13878c Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the estimate for the average weight of all boxes of oranges filled by the company in an **8**-hour period is **23.1** pounds, with an associated margin of error of **0.19** pounds. It follows that plausible values for this average weight are between **23.1 — 0.19** pounds and **23.1 + 0.19** pounds. Therefore, plausible values for the average weight of all boxes of oranges filled by the company are between **22.91** pounds and **23.29** pounds.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID a5bc3f67

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: a5bc3f67

A participant in a bicycle race completes the race with an average speed of **24,816** yards per hour. What is this average speed, in miles per hour? (**1 mile = 1,760 yards**)

ID: a5bc3f67 Answer

Correct Answer: 14.1

Rationale

The correct answer is **14.1**. It's given that a participant completes the bicycle race with an average speed of **24,816** yards per hour and **1 mile = 1,760 yards**. It follows that this average speed is equivalent to $\left(\frac{24,816 \text{ yards}}{1 \text{ hour}}\right)\left(\frac{1 \text{ mile}}{1,760 \text{ yards}}\right)$, which yields $\frac{14.1 \text{ miles}}{1 \text{ hour}}$, or **14.1** miles per hour.

Question Difficulty: Medium

Question ID 9cd1e599

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 9cd1e599

For $x > 0$, the function f is defined as follows:
 $f(x)$ equals 201% of x
Which of the following could describe this function?

- A. Decreasing exponential
- B. Decreasing linear
- C. Increasing exponential
- D. Increasing linear

ID: 9cd1e599 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that for $x > 0$, $f(x)$ is equal to 201% of x . This is equivalent to $f(x) = \frac{201}{100}x$, or $f(x) = 2.01x$, for $x > 0$. This function indicates that as x increases, $f(x)$ also increases, which means f is an increasing function. Furthermore, $f(x)$ increases at a constant rate of 2.01 for each increase of x by 1. A function with a constant rate of change is linear. Thus, the function f can be described as an increasing linear function.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This could describe the function $f(x) = (2.01)^x$, where $f(x)$ is equal to 201% of $f(x - 1)$, not x , for $x > 0$.

Question Difficulty: Hard

Question ID 6a85d2ca

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 6a85d2ca

Each of the following frequency tables represents a data set. Which data set has the greatest mean?

A.

Value	Frequency
70	4
80	5
90	6
100	7

B.

Value	Frequency
70	6
80	6
90	6
100	6

C.

Value	Frequency
70	7
80	6
90	6
100	7

D.

Value	Frequency
70	8
80	5
90	5
100	8

ID: 6a85d2ca Answer

Correct Answer: A

Rationale

Choice A is correct. The tables in choices B, C, and D each represent a data set where the values **80** and **90** have the same frequency and the values **70** and **100** have the same frequency. It follows that each of these data sets is symmetric around the value halfway between **80** and **90**, or **85**. When a data set is symmetric around a value, that value is the mean of the data set. Therefore, the data sets represented by the tables in choices B, C, and D each have a mean of **85**. The table in choice A represents a data set where the value **90** has a greater frequency than the value **80** and the value **100** has a greater frequency than the value **70**. It follows that this data set has a mean greater than **85**. Therefore, of the given choices, choice A represents the data set with the greatest mean.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

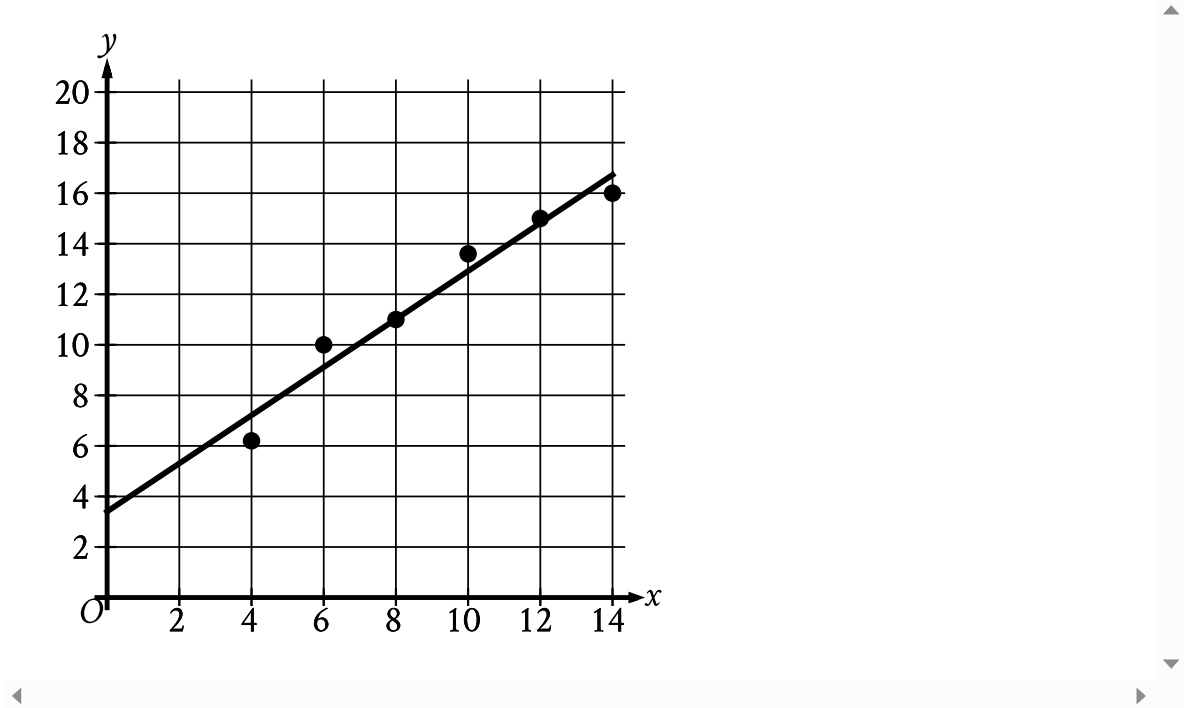
Question Difficulty: Hard

Question ID a61857af

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: a61857af

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

- A. $y = x + 3.4$
- B. $y = x - 3.4$
- C. $y = -x + 3.4$
- D. $y = -x - 3.4$

ID: a61857af Answer

Correct Answer: A

Rationale

Choice A is correct. The line of best fit shown has a positive slope and intersects the y -axis at a positive y -value. The graph of an equation of the form $y = mx + b$, where m and b are constants, has a slope of m and intersects the y -axis at a y -value of b . Of the given choices, only $y = x + 3.4$ represents a line that has a positive slope, 1, and intersects the y -axis at a positive y -value, 3.4.

Choice B is incorrect. This equation represents a line that intersects the y -axis at a negative y -value, not a positive y -value.

Choice C is incorrect. This equation represents a line that has a negative slope, not a positive slope.

Choice D is incorrect. This equation represents a line that has a negative slope, not a positive slope, and intersects the y -axis at a negative y -value, not a positive y -value.

Question ID f7ad05cf

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: f7ad05cf

How many feet are equivalent to **34** yards? (**1 yard = 3 feet**)

ID: f7ad05cf Answer

Correct Answer: 102

Rationale

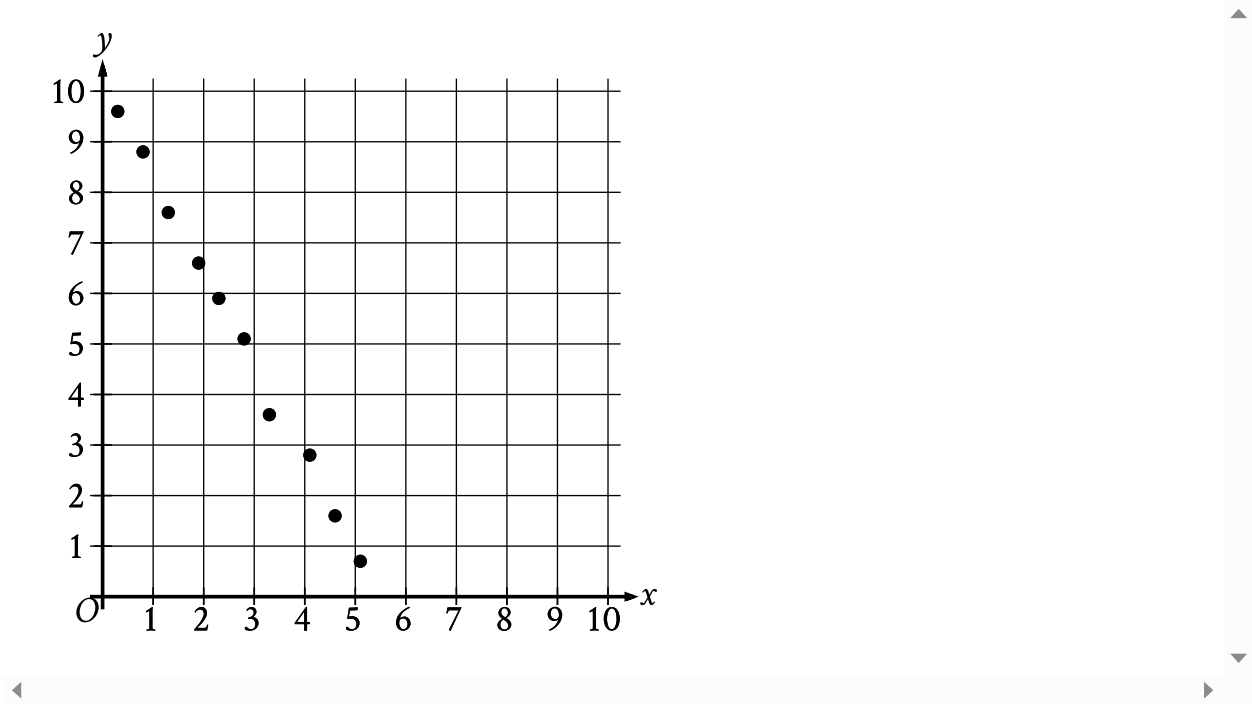
The correct answer is **102**. It's given that **1** yard is equivalent to **3** feet. Therefore, **34** yards is equivalent to **(34 yards)** $\left(\frac{3 \text{ feet}}{1 \text{ yard}}\right)$, or **102** feet.

Question Difficulty: Easy

Question ID 2e6e5ee3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 2e6e5ee3



Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

- A. $y = -1.9x - 10.1$
- B. $y = -1.9x + 10.1$
- C. $y = 1.9x - 10.1$
- D. $y = 1.9x + 10.1$

ID: 2e6e5ee3 Answer

Correct Answer: B

Rationale

Choice B is correct. The equation representing a linear model can be written in the form $y = a + bx$, or $y = bx + a$, where b is the slope of the graph of the model and $(0, a)$ is the y-intercept of the graph of the model. The scatterplot shows that as the x-values of the data points increase, the y-values of the data points decrease, which means the graph of an appropriate linear model has a negative slope. Therefore, $b < 0$. The scatterplot also shows that the data points are close to the y-axis at a positive value of y . Therefore, the y-intercept of the graph of an appropriate linear model has a positive y-coordinate, which means $a > 0$. Of the given choices, only choice B, $y = -1.9x + 10.1$, has a negative value for b , the slope, and a positive value for a , the y-coordinate of the y-intercept.

Choice A is incorrect. The graph of this model has a y-intercept with a negative y-coordinate, not a positive y-coordinate.

Choice C is incorrect. The graph of this model has a positive slope, not a negative slope, and a y-intercept with a negative y-coordinate, not a positive y-coordinate.

Choice D is incorrect. The graph of this model has a positive slope, not a negative slope.

Question Difficulty: Medium

Question ID 99801f3f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 99801f3f

Data Set Q	
Value	Frequency
a	110
b	30
c	0

Data Set R	
Value	Frequency
a	0
b	3
c	11

Data sets Q and R are summarized in the tables shown, where a , b , and c are positive, consecutive integers and $a < b < c$. Which statement comparing the means of the data sets is true?

- A. The mean of data set R is $\frac{2(11)}{14}$ less than the mean of data set Q.
- B. The mean of data set R is $\frac{2(11)}{14}$ greater than the mean of data set Q.
- C. The mean of data set Q is 10 times the mean of data set R.
- D. The means of the two data sets are equal.

ID: 99801f3f Answer

Correct Answer: B

Rationale

Choice B is correct. The mean of a data set is the sum of the values of the data set divided by the total number of values in the data set. When a data set is represented in a frequency table, the sum of the values in the data set is the sum of the products of each value and its frequency. For data set Q, the sum of the products of each value and its frequency is $110a + 30b + 0c$, or $110a + 30b$, and the number of values is $110 + 30 + 0$, or 140. Therefore, the mean of data set Q is $\frac{110a+30b}{140}$. For data set R, the sum of the products of each value and its frequency is $0a + 3b + 11c$, or $3b + 11c$, and the number of values is $0 + 3 + 11$, or 14. Therefore, the mean of data set R is $\frac{3b+11c}{14}$. Since a , b , and c are positive, consecutive integers and $a < b < c$, it follows that $b = a + 1$ and $c = a + 2$. Substituting $a + 1$ for b in the mean of data set Q yields $\frac{110a+30(a+1)}{140}$, or $\frac{110a+30a+30}{140}$, which is equivalent to $\frac{140a+30}{140}$, or $a + \frac{3}{14}$. Substituting $a + 1$ for b and $a + 2$ for c in the mean of data set R yields $\frac{3(a+1)+11(a+2)}{14}$, or $\frac{3a+3+11a+22}{14}$, which is equivalent to $\frac{14a+25}{14}$, or $a + \frac{25}{14}$. Subtracting the mean of data set Q from the mean of

data set R yields $\left(a + \frac{3}{14} + \frac{2(11)}{14}\right) - \left(a + \frac{3}{14}\right)$, or $\frac{2(11)}{14}$. Therefore, the mean of data set R is $\frac{2(11)}{14}$ greater than the mean of data set Q.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 4e5389dc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 4e5389dc

A total of **50** children attended a summer camp and were offered **4** types of sandwiches. The table shows the number of children who chose each type of sandwich.

Type of sandwich	Number of children
Turkey	15
Chicken	23
Ham	3
Vegetarian	9
Total	50

If one of these children is selected at random, what is the probability of selecting a child who chose a vegetarian sandwich?

- A. $\frac{9}{100}$
- B. $\frac{9}{50}$
- C. $\frac{1}{4}$
- D. $\frac{9}{10}$

ID: 4e5389dc Answer

Correct Answer: B

Rationale

Choice B is correct. If one of these children is selected at random, the probability of selecting a child who chose a vegetarian sandwich is equal to the number of children who chose a vegetarian sandwich divided by the total number of children. According to the table, there are a total of **50** children, and **9** of these children chose a vegetarian sandwich. Therefore, the probability of selecting a child who chose a vegetarian sandwich is $\frac{9}{50}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 738856d4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 738856d4

How many fluid ounces are equivalent to **76** quarts? (**8 fluid ounces = 1 cup and 4 cups = 1 quart**)

ID: 738856d4 Answer

Correct Answer: 2432

Rationale

The correct answer is **2,432**. It's given that **4 cups = 1 quart**. It follows that **76** quarts is equivalent to $(76 \text{ quarts})\left(\frac{4 \text{ cups}}{1 \text{ quart}}\right)$, or **304** cups. It's also given that **8 fluid ounces = 1 cup**. It follows that **304** cups is equivalent to $(304 \text{ cups})\left(\frac{8 \text{ fluid ounces}}{1 \text{ cup}}\right)$, or **2,432** fluid ounces.

Question Difficulty: Medium

Question ID 206e4916

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: 206e4916

A sample consisting of **720** adults who own televisions was selected at random for a study. Based on the sample, it is estimated that **32%** of all adults who own televisions use their televisions to watch nature shows, with an associated margin of error of **3.41%**. Which of the following is the most plausible conclusion about all adults who own televisions?

- A. More than **35.41%** of all adults who own televisions use their televisions to watch nature shows.
- B. Between **28.59%** and **35.41%** of all adults who own televisions use their televisions to watch nature shows.
- C. Since the sample included adults who own televisions and not just those who use their televisions to watch nature shows, no conclusion can be made.
- D. Since the sample did not include all the people who watch nature shows, no conclusion can be made.

ID: 206e4916 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that based on a sample selected at random, it's estimated that **32%** of all adults who own televisions use their televisions to watch nature shows, with an associated margin of error of **3.41%**. Subtracting the margin of error from the estimate and adding the margin of error to the estimate gives an interval of plausible values for the true percentage of adults who own televisions who use their televisions to watch nature shows. This means it's plausible that between **32% – 3.41%**, or **28.59%**, and **32% + 3.41%**, or **35.41%**, of all adults who own televisions use their televisions to watch nature shows. Therefore, of the given choices, the most plausible conclusion is that between **28.59%** and **35.41%** of all adults who own televisions use their televisions to watch nature shows.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. To make a plausible conclusion about all adults who own televisions, the sample must be selected at random from all adults who own televisions, not just those who use their televisions to watch nature shows.

Choice D is incorrect. Since the sample was selected at random from all adults who own televisions, a plausible conclusion can be made about all adults who own televisions.

Question Difficulty: Medium

Question ID d515af91

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: d515af91

	Site A	Site B	Total
Tulip	35	15	50
Daffodil	31	21	52
Total	66	36	102

The table shows the distribution of two types of flowers at two different sites. If a flower represented in the table is selected at random, what is the probability of selecting a flower from site A, given that the flower is a tulip? (Express your answer as a decimal or fraction, not as a percent.)

ID: d515af91 Answer

Correct Answer: 0.7, 7/10

Rationale

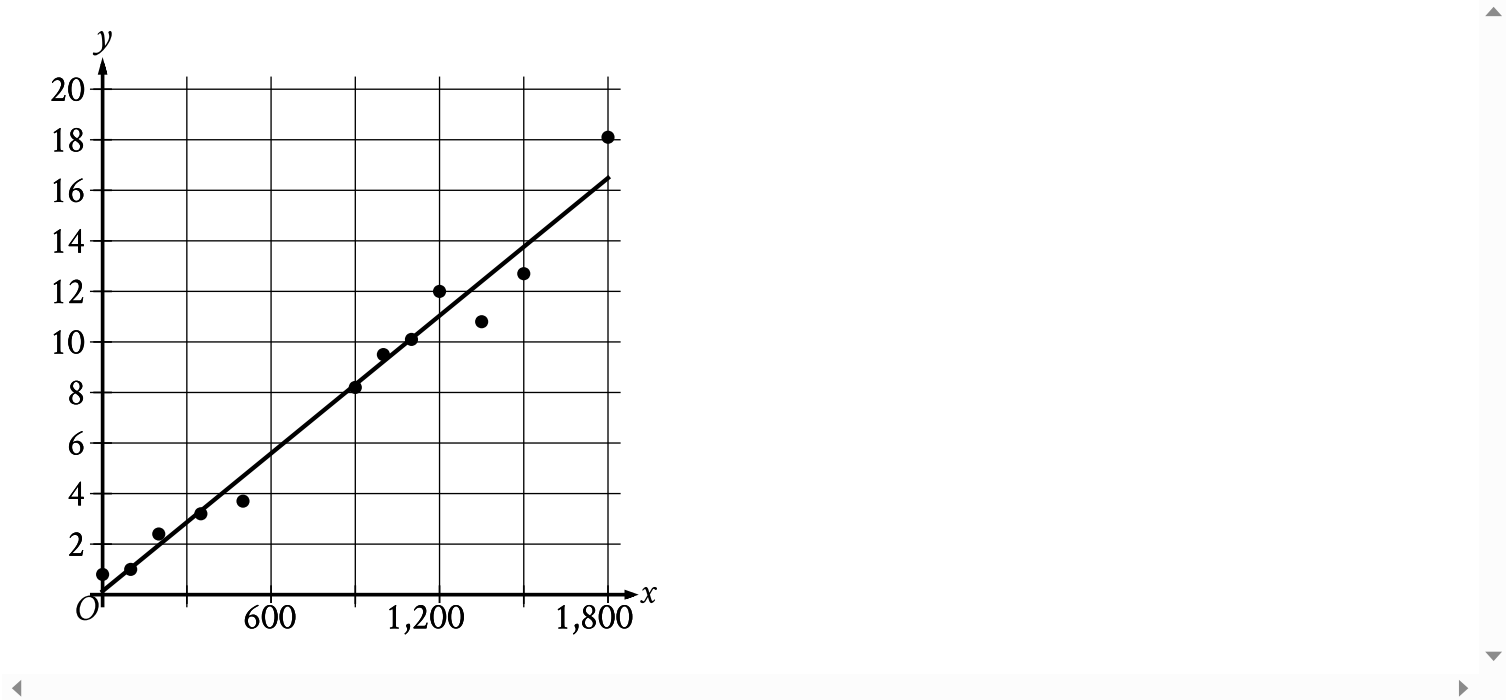
The correct answer is $\frac{35}{50}$. Based on the table, there are a total of **50** tulips, and **35** of these tulips are from site A. The probability of selecting at random a flower from site A, given that the flower is a tulip, is equal to the number of tulips from site A divided by the total number of tulips, which can be written as $\frac{35}{50}$, or $\frac{7}{10}$. Note that 35/50, 7/10, and .7 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID f8bfe491

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: f8bfe491



Twelve data points are shown in the scatterplot. A line of best fit for the data is also shown. At $x = 1,200$, which of the following is closest to the y -value predicted by the line of best fit?

- A. 16
- B. 14
- C. 11
- D. 6

ID: f8bfe491 Answer

Correct Answer: C

Rationale

Choice C is correct. On the line of best fit, an x -value of **1,200** corresponds to a y -value between **10** and **12**. Therefore, of the given choices, **11** is closest to the y -value predicted by the line of best fit at $x = 1,200$.

Choice A is incorrect. This is the integer value closest to the y -value predicted by the line of best fit at $x = 1,800$.

Choice B is incorrect. This is the integer value closest to the y -value predicted by the line of best fit at $x = 1,500$.

Choice D is incorrect. This is the integer value closest to the y -value predicted by the line of best fit at $x = 600$.

Question Difficulty: Easy

Question ID 4e15a974

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 4e15a974

An object travels at a constant speed of **6** centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel **24** centimeters?

ID: 4e15a974 Answer

Correct Answer: 4

Rationale

The correct answer is **4**. It's given that the object travels at a constant speed of **6** centimeters per second. The speed of the object can be written as $\frac{6 \text{ centimeters}}{1 \text{ second}}$. Let x represent the time, in seconds, it would take for the object to travel **24** centimeters. The value of x can be calculated by solving the equation $\frac{6 \text{ centimeters}}{1 \text{ second}} = \frac{24 \text{ centimeters}}{x \text{ seconds}}$, which can be written as $\frac{6}{1} = \frac{24}{x}$, or $6 = \frac{24}{x}$. Multiplying each side of this equation by x yields $6x = 24$. Dividing each side of this equation by **6** yields $x = 4$. Therefore, it would take the object **4** seconds to travel **24** centimeters.

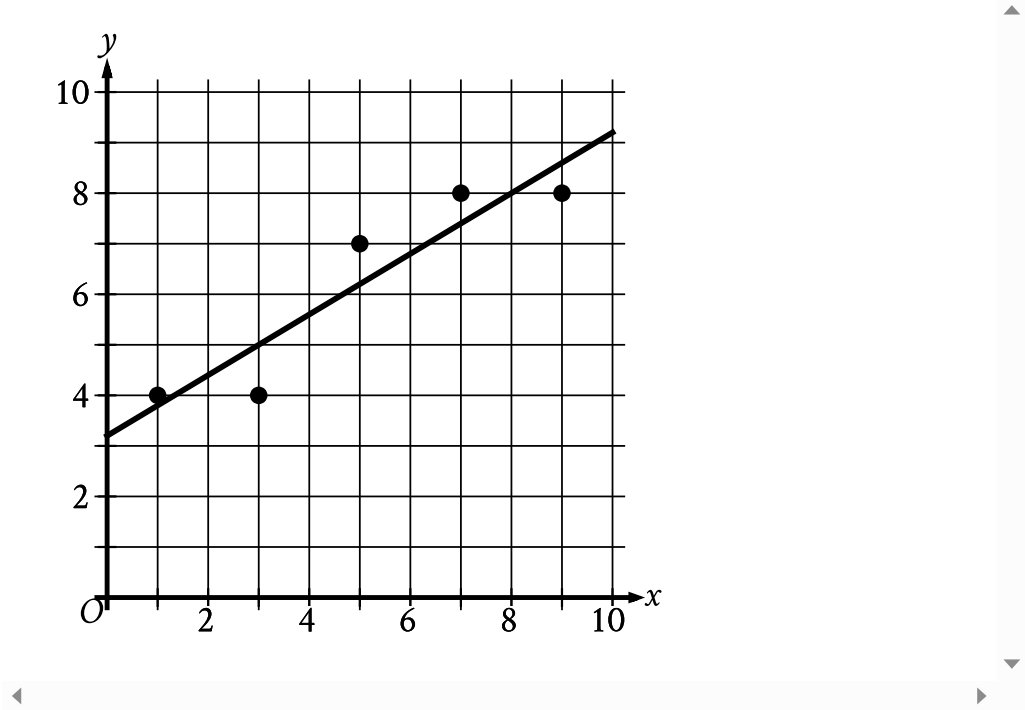
Question Difficulty: Easy

Question ID 289b65e7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 289b65e7

The scatterplot shows the relationship between x and y . A line of best fit is also shown.



Which of the following is closest to the slope of this line of best fit?

- A. 0.60
- B. 2.50
- C. 7.80
- D. 8.00

ID: 289b65e7 Answer

Correct Answer: A

Rationale

Choice A is correct. A line in the xy -plane that passes through the points (x_1, y_1) and (x_2, y_2) has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes through the points with approximate coordinates $(0, 3.2)$ and $(2, 4.4)$. It follows that the slope of the line of best fit is approximately $\frac{4.4 - 3.2}{2 - 0}$, which is equivalent to $\frac{1.2}{2}$, or 0.6 . Therefore, of the given choices, **0.6** is closest to the slope of the line of best fit shown.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID f73da1c2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: f73da1c2

The amount of Hanna's bill for a food order was **\$50**. Hanna gave a tip of **20%** of the amount of the bill. What is the amount, in dollars, of the tip Hanna gave?

ID: f73da1c2 Answer

Correct Answer: 10

Rationale

The correct answer is **10**. It's given that the amount of Hanna's food order was **\$50** and that Hanna gave a tip of **20%** of the amount of the bill. **20%** of **50** can be calculated as $(\frac{20}{100})(50)$, which yields $\frac{1000}{100}$, or **10**. Therefore, the amount, in dollars, of the tip Hanna gave is **10**.

Question Difficulty: Medium

Question ID d760b4fe

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: d760b4fe

There are **170** blocks in a container. Of these blocks, **10%** are green. How many blocks in the container are green?

ID: d760b4fe Answer

Correct Answer: 17

Rationale

The correct answer is **17**. It's given that there are **170** blocks in a container, and of these blocks, **10%** are green. Since **10%** can be rewritten as $\frac{10}{100}$, or **0.1**, the number of green blocks in the container is **0.1(170)**, or **17**.

Question Difficulty: Easy

Question ID 21f398ee

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 21f398ee



The box plot summarizes 15 data values. What is the median of this data set?

- A. 2
- B. 3
- C. 5
- D. 8

ID: 21f398ee Answer

Correct Answer: C

Rationale

Choice C is correct. The median of a data set represented in a box plot is given by the vertical line within the box. In the given box plot, the vertical line within the box occurs at 5. Therefore, the median of this data set is 5.

Choice A is incorrect. This is the minimum value of the data set.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. This is the maximum value of the data set.

Question Difficulty: Easy

Question ID 70763935

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 70763935

Last year, **200** students enrolled in an interior design program. This year, the number of students enrolled is **147%** of last year’s number. How many students are enrolled in the interior design program this year?

- A. **247**
- B. **294**
- C. **347**
- D. **394**

ID: 70763935 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the number of students enrolled in an interior design program this year is **147%** of last year's number, which is **200**. **147%** of **200** can be expressed as $(\frac{147}{100})(200)$, or $(1.47)(200)$, which is equivalent to **294**. Therefore, **294** students are enrolled in the interior design program this year.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 877d34aa

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 877d34aa



The dot plot represents the **15** values in data set A. Data set B is created by adding **56** to each of the values in data set A. Which of the following correctly compares the medians and the ranges of data sets A and B?

- A. The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.
- B. The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.
- C. The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.
- D. The median of data set B is greater than the median of data set A, and the range of data set B is greater than the range of data set A.

ID: 877d34aa Answer

Correct Answer: C

Rationale

Choice C is correct. The median of a data set with an odd number of values, in ascending or descending order, is the middle value of the data set, and the range of a data set is the positive difference between the maximum and minimum values in the data set. Since the dot plot shown gives the values in data set A in ascending order and there are **15** values in the data set, the eighth value in data set A, **23**, is the median. The maximum value in data set A is **26** and the minimum value is **22**, so the range of data set A is **26 – 22**, or **4**. It’s given that data set B is created by adding **56** to each of the values in data set A. Increasing each of the **15** values in data set A by **56** will also increase its median value by **56** making the median of data set B **79**. Increasing each value of data set A by **56** does not change the range, since the maximum value of data set B is **26 + 56**, or **82**, and the minimum value is **22 + 56**, or **78**, making the range of data set B **82 – 78**, or **4**. Therefore, the median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

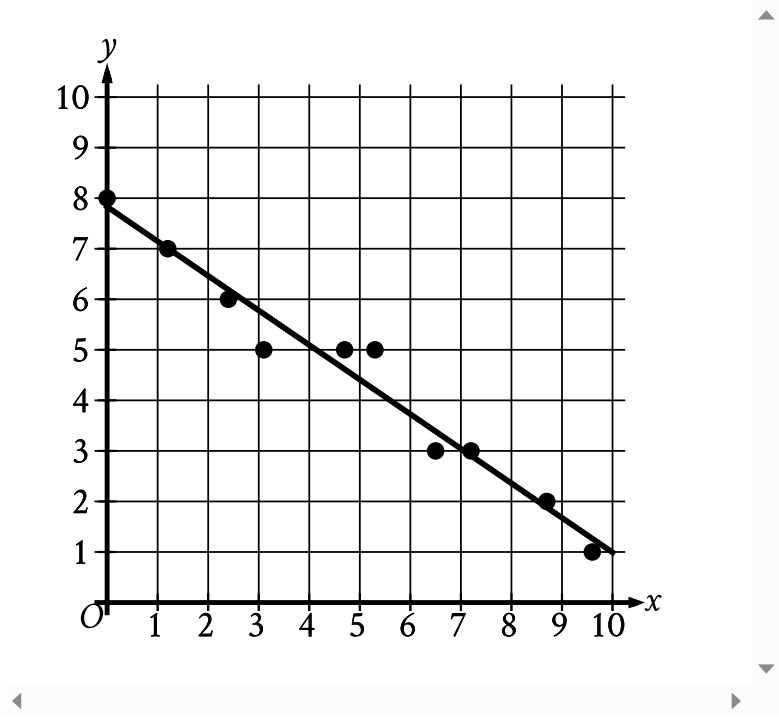
Question Difficulty: Hard

Question ID 9d2afaa0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 9d2afaa0

In the given scatterplot, a line of best fit for the data is shown.



Which of the following is closest to the slope of this line of best fit?

- A. 7
- B. 0.7
- C. -0.7
- D. -7

ID: 9d2afaa0 Answer

Correct Answer: C

Rationale

Choice C is correct. A line of best fit is shown in the scatterplot such that as the value of x increases, the value of y decreases. It follows that the slope of the line of best fit shown is negative. The slope of a line in the xy -plane that passes through the points (x_1, y_1) and (x_2, y_2) can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points $(0, 8)$ and $(10, 1)$. Substituting $(0, 8)$ for (x_1, y_1) and $(10, 1)$ for (x_2, y_2) in $\frac{y_2 - y_1}{x_2 - x_1}$ yields the slope of the line being approximately $\frac{1 - 8}{10 - 0}$, which is equivalent to $\frac{-7}{10}$, or -0.7 . Therefore, of the given choices, -0.7 is the closest to the slope of this line of best fit.

Choice A is incorrect. The line of best fit shown has a negative slope, not a positive slope.

Choice B is incorrect. The line of best fit shown has a negative slope, not a positive slope.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID b1d65b2d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: b1d65b2d

The number a is **70%** less than the positive number b . The number c is **60%** greater than a . The number c is how many times b ?

ID: b1d65b2d Answer

Correct Answer: .48, 12/25

Rationale

The correct answer is **.48**. It's given that the number a is **70%** less than the positive number b . Therefore, $a = (1 - \frac{70}{100})b$, which is equivalent to $a = (1 - 0.70)b$, or $a = 0.30b$. It's also given that the number c is **60%** greater than a . Therefore, $c = (1 + \frac{60}{100})a$, which is equivalent to $c = (1 + 0.60)a$, or $c = 1.60a$. Since $a = 0.30b$, substituting $0.30b$ for a in the equation $c = 1.60a$ yields $c = 1.60(0.30b)$, or $c = 0.48b$. Thus, c is **0.48** times b . Note that .48 and 12/25 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID d295f5ed

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: d295f5ed

A box contains **13** red pens and **37** blue pens. If one of these pens is selected at random, what is the probability of selecting a red pen? (Express your answer as a decimal or fraction, not as a percent.)

ID: d295f5ed Answer

Correct Answer: .26, 13/50

Rationale

The correct answer is $\frac{13}{50}$. It's given that a box contains **13** red pens and **37** blue pens. If one of these pens is selected at random, the probability of selecting a red pen is the number of red pens in the box divided by the number of red and blue pens in the box. The number of red and blue pens in the box is **13 + 37**, or **50**. Since there are **13** red pens in the box, it follows that the probability of selecting a red pen is $\frac{13}{50}$. Note that 13/50 and .26 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID 8ec5cb70

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 8ec5cb70

A list of **10** data values is shown.
6, 8, 16, 4, 17, 26, 8, 5, 5, 5
What is the mean of these data?

ID: 8ec5cb70 Answer

Correct Answer: 10

Rationale

The correct answer is **10**. The mean of a data set is calculated by dividing the sum of the data values by the number of data values in the data set. For this data set, the mean can be calculated as $\frac{6+8+16+4+17+26+8+5+5+5}{10}$, which is equivalent to $\frac{100}{10}$, or **10**.

Question Difficulty: Medium

Question ID d16f452e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: d16f452e

The table summarizes the distribution of age and assigned group for **90** participants in a study.

	0–9 years	10–19 years	20+ years	Total
Group A	5	17	8	30
Group B	6	8	16	30
Group C	19	5	6	30
Total	30	30	30	90

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least 10 years of age?

- A. $\frac{5}{18}$
- B. $\frac{5}{12}$
- C. $\frac{17}{30}$
- D. $\frac{5}{6}$

ID: d16f452e Answer

Correct Answer: B

Rationale

Choice B is correct. Since the participant will be selected at random, the probability of selecting a participant from group A, given that the participant is at least **10** years of age, is equal to the number of participants from group A who are at least **10** years of age divided by the total number of participants who are at least **10** years of age. Based on the table, in group A, there are **17** participants who are **10–19** years of age and **8** participants who are **20+** years of age. Therefore, there are a total of **17 + 8**, or **25**, participants in group A who are at least **10** years of age. Based on the table, of the total number of participants, there are **30** participants who are **10–19** years of age and **30** participants who are **20+** years of age. Therefore, a total of **30 + 30**, or **60**, of the participants are at least **10** years of age. Thus, the probability of selecting a participant from group A, given that the participant is at least **10** years of age, is $\frac{25}{60}$, or $\frac{5}{12}$.

Choice A is incorrect. This is the number of participants from group A who are at least **10** years of age divided by the total number of participants, rather than divided by the number of participants who are at least **10** years of age.

Choice C is incorrect. This is the probability of randomly selecting a participant from group A, given that the participant is **10–19** years of age, rather than given that the participant is at least **10** years of age.

Choice D is incorrect. This is the probability of randomly selecting a participant who is at least **10** years of age, given that the participant is in group A.

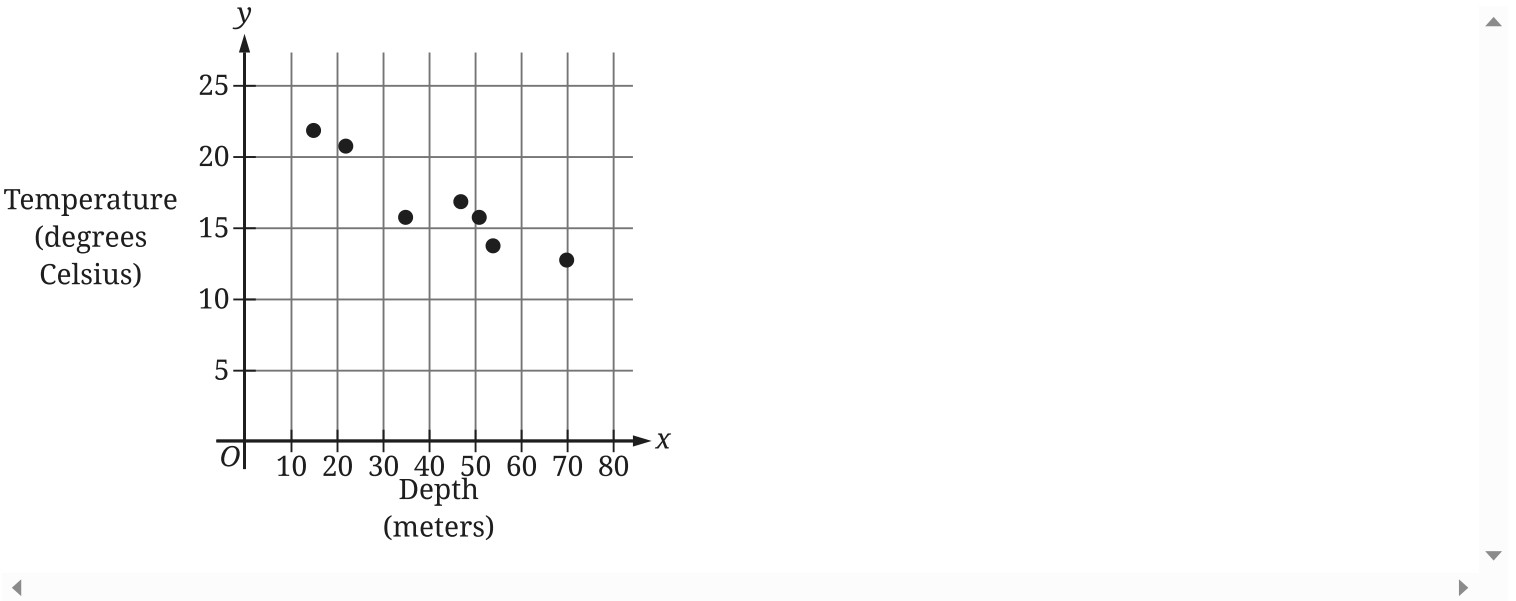
Question Difficulty: Hard

Question ID cf1cdbff

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: cf1cdbff

As part of a study of lake conditions, the water temperature was recorded at different depths below the surface of a certain lake. The data shown in the scatterplot give the recorded temperature, in degrees Celsius, for 7 depths, in meters, below the surface of the lake.



Which of the following is closest to the slope of a line of best fit for the data shown?

- A. **−0.17**
- B. **−2.17**
- C. **−6.02**
- D. **−8.02**

ID: cf1cdbff Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the data shown in the scatterplot give the recorded temperature, in degrees Celsius, for 7 depths, in meters, below the surface of a lake. The graph of an appropriate line of best fit for the data shown passes approximately through the points (15, 21.8) and (70, 12.7). The slope of a line in the xy-plane that passes through the points (x₁, y₁) and (x₂, y₂) can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. Substituting (15, 21.8) and (70, 12.7) for (x₁, y₁) and (x₂, y₂), respectively, in the expression $\frac{y_2 - y_1}{x_2 - x_1}$ yields $\frac{12.7 - 21.8}{70 - 15}$, or $\frac{-9.1}{55}$, which is approximately **−0.165**. Therefore, of the given choices, **−0.17** is closest to the slope of a line of best fit for the data shown.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID 588427e5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 588427e5

A special camera is used for underwater ocean research. The camera is at a depth of **39** fathoms. What is the camera's depth in feet? (**1 fathom = 6 feet**)

- A. **234**
- B. **117**
- C. **45**
- D. **7**

ID: 588427e5 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that a special camera is used for underwater ocean research, and this camera is at a depth of **39** fathoms. It's also given that **1** fathom is equal to **6** feet. Thus, **39** fathoms is equivalent to **(39 fathoms) ($\frac{6 \text{ feet}}{1 \text{ fathom}}$)**, or **234** feet. Therefore, the camera's depth, in feet, is **234**.

Choice B is incorrect. This is the camera's depth, in feet, if the camera is at a depth of **19.5** fathoms.

Choice C is incorrect. This is the camera's depth, in feet, if the camera is at a depth of **7.5** fathoms.

Choice D is incorrect and may result from conceptual or calculation errors.

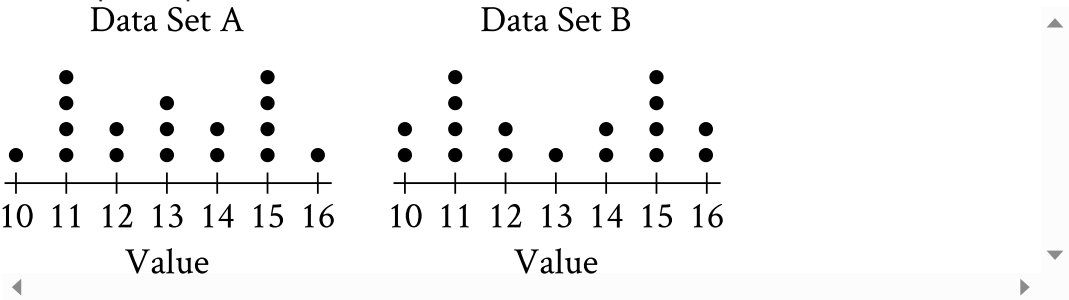
Question Difficulty: Easy

Question ID a24e4c19

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: a24e4c19

The dot plots represent the distributions of values in data sets A and B.



Which of the following statements must be true?

- I. The median of data set A is equal to the median of data set B.

II. The standard deviation of data set A is equal to the standard deviation of data set B.
- A. I only

B. II only

C. I and II

D. Neither I nor II

ID: a24e4c19 Answer

Correct Answer: A

Rationale

Choice A is correct. The median of a data set with an odd number of values that are in ascending or descending order is the middle value of the data set. Since the distribution of the values of both data set A and data set B form symmetric dot plots, and each data set has an odd number of values, it follows that the median is given by the middle value in each of the dot plots. Thus, the median of data set A is **13**, and the median of data set B is **13**. Therefore, statement I is true. Data set A and data set B have the same frequency for each of the values **11**, **12**, **14**, and **15**. Data set A has a frequency of **1** for values **10** and **16**, whereas data set B has a frequency of **2** for values **10** and **16**. Standard deviation is a measure of the spread of a data set; it is larger when there are more values further from the mean, and smaller when there are more values closer to the mean. Since both distributions are symmetric with an odd number of values, the mean of each data set is equal to its median. Thus, each data set has a mean of **13**. Since more of the values in data set A are closer to **13** than data set B, it follows that data set A has a smaller standard deviation than data set B. Thus, statement II is false. Therefore, only statement I must be true.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID a9fae561

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: a9fae561

Response	Frequency
Once a week or more	3
Two or three times a month	16
About once a month	26
A few times a year	73
Almost never	53
Never	29
Total	200

The table gives the results of a survey of **200** people who were asked how often they see a movie in a theater. How many people responded either “never” or “almost never”?

- A. **24**
- B. **53**
- C. **82**
- D. **118**

ID: a9fae561 Answer

Correct Answer: C

Rationale

Choice C is correct. The table gives the results of **200** people who were asked how often they see a movie in a theater. The table shows that **29** people responded “never” and **53** people responded “almost never.” Therefore, **29 + 53**, or **82**, people responded either “never” or “almost never.”

Choice A is incorrect. This is the difference between the number of people who responded “almost never” and the number of people who responded “never.”

Choice B is incorrect. This is the number of people who responded “almost never” but doesn't include those who responded “never.”

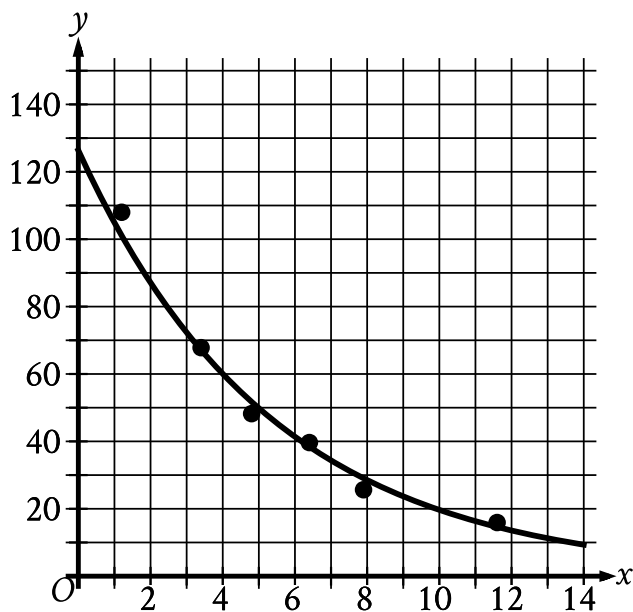
Choice D is incorrect. This is the number of people who responded something other than “never” or “almost never,” rather than the number of people who responded either “never” or “almost never.”

Question Difficulty: Easy

Question ID a900a374

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: a900a374



The scatterplot shows the relationship between two variables, x and y . An equation for the exponential model shown can be written as $y = a(b)^x$, where a and b are positive constants. Which of the following is closest to the value of b ?

- A. 0.83
- B. 1.83
- C. 18.36
- D. 126.35

ID: a900a374 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that an equation for the exponential model shown can be written as $y = a(b)^x$, where a and b are positive constants. For an exponential model written in this form, if the value of b is greater than 0 but less than 1, the model is decreasing. If the value of b is greater than 1, the model is increasing. The exponential model shown is decreasing. Therefore, the value of b is greater than 0 but less than 1. Of the given choices, only 0.83 is a value greater than 0 but less than 1. Thus, 0.83 is closest to the value of b .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID 12ee6d83

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 12ee6d83

A certain town has an area of **4.36** square miles. What is the area, in square yards, of this town? (**1 mile = 1,760 yards**)

- A. **404**
- B. **7,674**
- C. **710,459**
- D. **13,505,536**

ID: 12ee6d83 Answer

Correct Answer: D

Rationale

Choice D is correct. Since the number of yards in **1 mile** is **1,760**, the number of square yards in **1 square mile** is **(1,760)(1,760) = 3,097,600**. Therefore, if the area of the town is **4.36** square miles, it is **4.36(3,097,600) = 13,505,536**, in square yards.

Choice A is incorrect and may result from dividing the number of yards in a mile by the square mileage of the town.

Choice B is incorrect and may result from multiplying the number of yards in a mile by the square mileage of the town.

Choice C is incorrect and may result from dividing the number of square yards in a square mile by the square mileage of the town.

Question Difficulty: Hard

Question ID f15866f4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: f15866f4

−11, −9, 26

A data set of three numbers is shown. If a number from this data set is selected at random, what is the probability of selecting a positive number?

- A. 0
- B. $\frac{1}{3}$
- C. $\frac{2}{3}$
- D. 1

ID: f15866f4 Answer

Correct Answer: B

Rationale

Choice B is correct. The probability of selecting a positive number is the number of positive numbers in the data set divided by the total number of numbers in the data set. There is **1** positive number in this data set. There are **3** total numbers in this data set. Thus, if a number from this data set is selected at random, the probability of selecting a positive number is $\frac{1}{3}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the probability of selecting a negative number from this data set.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 900ac3a3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 900ac3a3



- Data set A has **15** values and is represented by the dot plot shown. Data set B is created by adding **46** to each of the values in data set A. Which of the following correctly compares the medians and the ranges of data sets A and B?
- A. The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.
 - B. The median of data set B is greater than the median of data set A, and the range of data set B is greater than the range of data set A.
 - C. The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.
 - D. The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.

ID: 900ac3a3 Answer

Correct Answer: A

Rationale

Choice A is correct. The median is the middle value in a data set when the data are arranged in order from least to greatest. Since there are **15** values in data set A, the median is the **8th** value. The **8th** value is **23**, so the median of data set A is **23**. The range is found by subtracting the minimum value in a data set from the maximum value. The minimum value in data set A is **22** and the maximum value is **26**. Therefore, the range of data set A is **26 – 22**, or **4**. It’s given that data set B is created by adding **46** to each of the values in data set A. Therefore, the **8th** value in data set B is **23 + 46**, or **69**, so the median of data set B is **69**. The minimum value in data set B is **22 + 46**, or **68**, and the maximum value is **26 + 46**, or **72**. Therefore, the range of data set B is **72 – 68**, or **4**. Since the median of data set A is **23**, and the median of data set B is **69**, the median of data set B is greater than the median of data set A. Since the ranges of data sets A and B are both **4**, the range of data set B is equal to the range of data set A.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 66db54cd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 66db54cd

How many tablespoons are equivalent to **14** teaspoons? (**3 teaspoons = 1 tablespoon**)

ID: 66db54cd Answer

Correct Answer: 14/3, 4.666, 4.667

Rationale

The correct answer is $\frac{14}{3}$. It's given that **3** teaspoons is equivalent to **1** tablespoon. Therefore, **14** teaspoons is equivalent to **(14 teaspoons)** $\left(\frac{1 \text{ tablespoon}}{3 \text{ teaspoons}}\right)$, or $\frac{14}{3}$ tablespoons. Note that 14/3, 4.666, and 4.667 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 459ec83b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 459ec83b

The population density of Worthington is **290** people per square mile. Worthington has a population of **92,800** people. What is the area, in square miles, of Worthington?

- A. **102,400**
- B. **93,090**
- C. **320**
- D. **32**

ID: 459ec83b Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that the population density of Worthington is **290** people per square mile and Worthington has a population of **92,800** people. Therefore, the area of Worthington is **92,800 people** $\left(\frac{1 \text{ square mile}}{290 \text{ people}}\right)$, which is equivalent to $\frac{92,800 \text{ square miles}}{290}$, or **320** square miles.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 88120df7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 88120df7

13 is $p\%$ of 25. What is the value of p ?

ID: 88120df7 Answer

Correct Answer: 52

Rationale

The correct answer is 52. It's given that 13 is $p\%$ of 25. It follows that $\frac{13}{25} = \frac{p}{100}$. Multiplying both sides of this equation by 100 gives $52 = p$. Therefore, the value of p is 52.

Question Difficulty: Hard

Question ID 82bc092c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 82bc092c

The number of coins in a collection increased from **9** to **90**. What was the percent increase in the number of coins in this collection?

- A. **10%**
- B. **81%**
- C. **90%**
- D. **900%**

ID: 82bc092c Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the number of coins in the collection increased from **9** to **90**. It follows that the number of coins in the collection increased by **90 — 9**, or **81**. Let ***x*%** represent the percentage that **81** is of **9**. The value of ***x*** can be found using the proportion $\frac{81}{9} = \frac{x}{100}$, or **9 = $\frac{x}{100}$** . Multiplying both sides of this equation by **100** yields **900 = *x***. Thus, when the number of coins in the collection increased from **9** to **90**, the percent increase was **900%**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 48a06fda

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 48a06fda

Objects R and S each travel at a constant speed. The speed of object R is half the speed of object S. Object R travels a distance of $4x$ inches in y seconds. Which expression represents the time, in seconds, it takes object S to travel a distance of $24x$ inches?

- A. $12y$
- B. $3y$
- C. $16y$
- D. $6y$

ID: 48a06fda Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that object R travels a distance of $4x$ inches in y seconds. This speed can be written as $\frac{4x \text{ inches}}{y \text{ seconds}}$. It's given that the speed of object R is half the speed of object S. It follows that the speed of object S is twice the speed of object R, which is $2\left(\frac{4x \text{ inches}}{y \text{ seconds}}\right)$, or $\frac{8x \text{ inches}}{y \text{ seconds}}$. Let n represent the time, in seconds, it takes object S to travel a distance of $24x$ inches. The value of n can be found by solving the equation $\frac{8x \text{ inches}}{y \text{ seconds}} = \frac{24x \text{ inches}}{n \text{ seconds}}$, which can be written as $\frac{8x}{y} = \frac{24x}{n}$. Multiplying each side of this equation by ny yields $8xn = 24xy$. Dividing each side of this equation by $8x$ yields $n = 3y$. Therefore, the expression $3y$ represents the time, in seconds, it takes object S to travel a distance of $24x$ inches.

Choice A is incorrect. This expression represents the time, in seconds, it would take object S to travel a distance of $24x$ inches if the speed of object R were twice, not half, the speed of object S.

Choice C is incorrect. This expression represents the time, in seconds, it takes object S to travel a distance of $128x$ inches, not $24x$ inches.

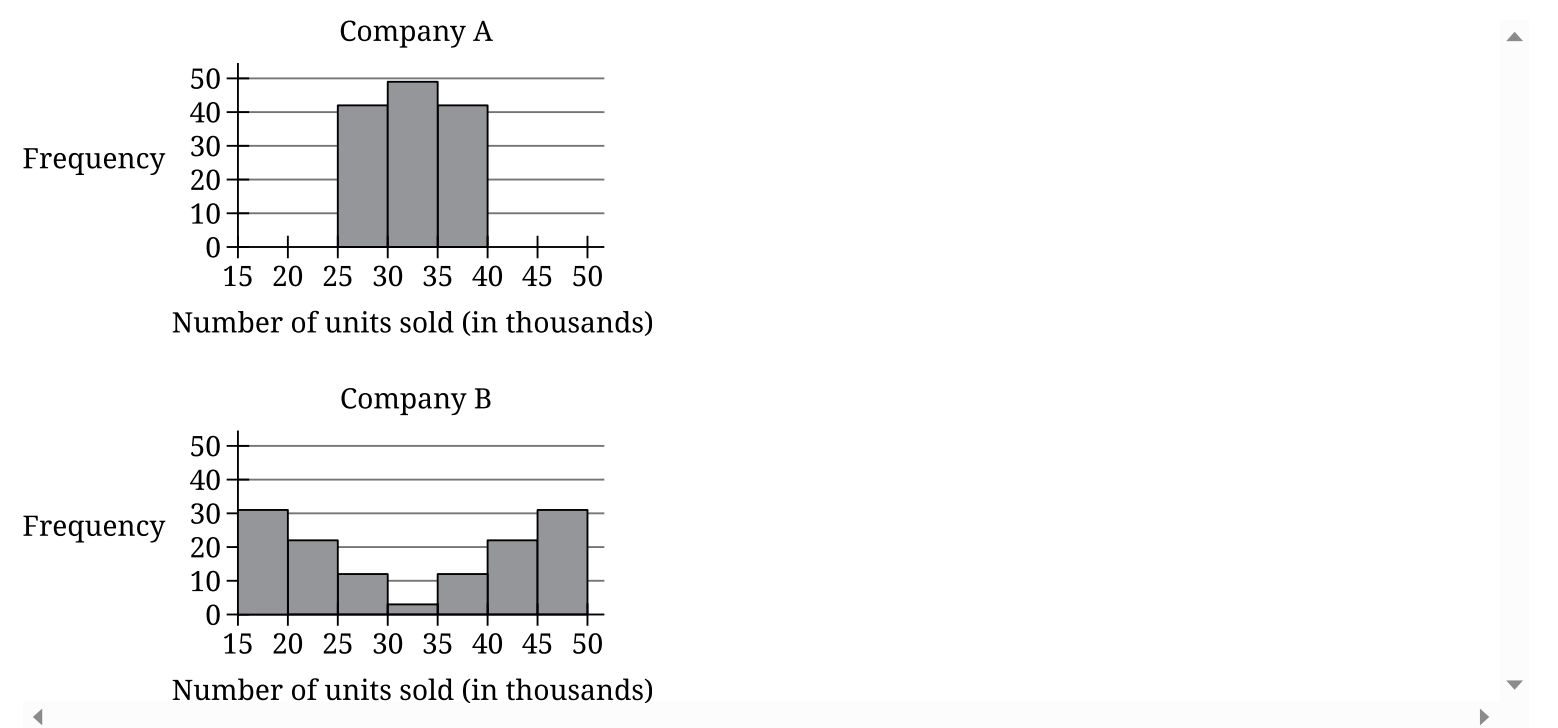
Choice D is incorrect. This expression represents the time, in seconds, it takes object R, not object S, to travel a distance of $24x$ inches.

Question Difficulty: Hard

Question ID 0df0ed4d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 0df0ed4d



The histograms summarize the distributions of number of units sold, in thousands, for company A and company B. Which statement best compares the standard deviations of number of units sold for these companies?

- A. The standard deviation of number of units sold for company A is less than the standard deviation of number of units sold for company B.
- B. The standard deviation of number of units sold for company A is greater than the standard deviation of number of units sold for company B.
- C. The standard deviation of number of units sold for company A is equal to the standard deviation of number of units sold for company B.
- D. There is not enough information to compare the standard deviations.

ID: 0df0ed4d Answer

Correct Answer: A

Rationale

Choice A is correct. Standard deviation measures the spread of a given data set from its mean. In a data set with a smaller standard deviation, there are more values close to the mean. In a data set with a greater standard deviation, there are more values farther from the mean. The two histograms shown have the same scale on the horizontal axis. Therefore, their standard deviations can be compared by visually comparing the spreads of their histograms. The distribution summarized by each histogram is symmetric. Therefore, the mean of the data set for each histogram is a value in the middle bar of that histogram. The middle bar of each histogram has a value of at least **30** thousand units sold but less than **35** thousand units sold. Therefore, the mean of the data set for each histogram is at least **30** thousand and less than **35** thousand. The histogram for company A shows

all the values in that data set are close to the mean. For company B, the histogram shows there are fewer values close to the mean and more values farther from the mean. Therefore, the standard deviation of number of units sold for company A is less than the standard deviation of number of units sold for company B.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

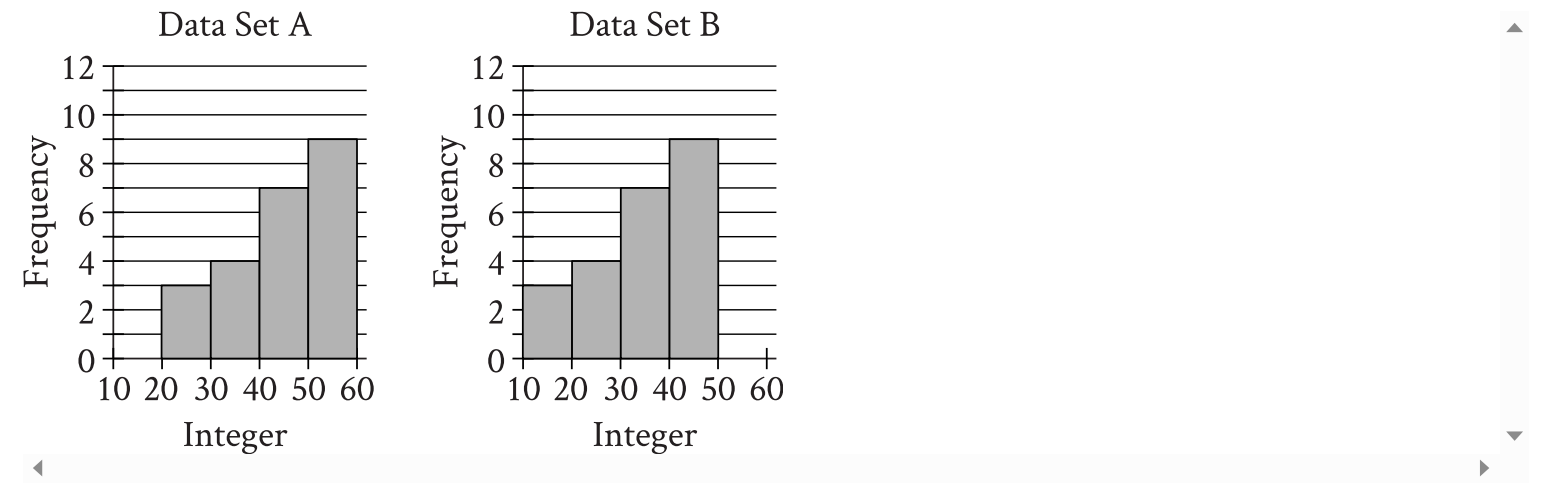
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 8d82c32b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 8d82c32b



Two data sets of **23** integers each are summarized in the histograms shown. For each of the histograms, the first interval represents the frequency of integers greater than or equal to **10**, but less than **20**. The second interval represents the frequency of integers greater than or equal to **20**, but less than **30**, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?

- A. 0
- B. 1
- C. 10
- D. 23

ID: 8d82c32b Answer

Correct Answer: B

Rationale

Choice B is correct. The histograms shown have the same shape, but data set A contains values between **20** and **60** and data set B contains values between **10** and **50**. Thus, the mean of data set A is greater than the mean of data set B. Therefore, the smallest possible difference between the mean of data set A and the mean of data set B is the difference between the smallest possible mean of data set A and the greatest possible mean of data set B. In data set A, since there are **3** integers in the interval greater than or equal to **20** but less than **30**, **4** integers greater than or equal to **30** but less than **40**, **7** integers greater than or equal to **40** but less than **50**, and **9** integers greater than or equal to **50** but less than **60**, the smallest possible mean for data set A is $\frac{(3 \cdot 20) + (4 \cdot 30) + (7 \cdot 40) + (9 \cdot 50)}{23}$. In data set B, since there are **3** integers greater than or equal to **10** but less than **20**, **4** integers greater than or equal to **20** but less than **30**, **7** integers greater than or equal to **30** but less than **40**, and **9** integers greater than or equal to **40** but less than **50**, the largest possible mean for data set B is $\frac{(3 \cdot 19) + (4 \cdot 29) + (7 \cdot 39) + (9 \cdot 49)}{23}$. Therefore, the smallest possible difference between the mean of data set A and the mean of data set B is $\frac{(3 \cdot 20) + (4 \cdot 30) + (7 \cdot 40) + (9 \cdot 50)}{23} - \frac{(3 \cdot 19) + (4 \cdot 29) + (7 \cdot 39) + (9 \cdot 49)}{23}$, which is equivalent to $\frac{(3 \cdot 20) - (3 \cdot 19) + (4 \cdot 30) - (4 \cdot 29) + (7 \cdot 40) - (7 \cdot 39) + (9 \cdot 50) - (9 \cdot 49)}{23}$. This expression can be rewritten as $\frac{3(20 - 19) + 4(30 - 29) + 7(40 - 39) + 9(50 - 49)}{23}$, or $\frac{23}{23}$, which is equal to **1**. Therefore, the smallest possible difference between the mean of data set A and the mean of data set B is **1**.

Choice A is incorrect. This is the smallest possible difference between the ranges, not the means, of the data sets.

Choice C is incorrect. This is the difference between the greatest possible mean, not the smallest possible mean, of data set A and the greatest possible mean of data set B.

Choice D is incorrect. This is the smallest possible difference between the sum of the values in data set A and the sum of the values in data set B, not the smallest possible difference between the means.

Question Difficulty: Hard

Question ID 0563ce88

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 0563ce88

If $\frac{4a}{b} = 6.7$ and $\frac{a}{bn} = 26.8$, what is the value of n ?

ID: 0563ce88 Answer

Correct Answer: .0625, 1/16

Rationale

The correct answer is **.0625**. It's given that $\frac{4a}{b} = 6.7$ and $\frac{a}{bn} = 26.8$. The equation $\frac{4a}{b} = 6.7$ can be rewritten as $(4)(\frac{a}{b}) = 6.7$. Dividing both sides of this equation by 4 yields $\frac{a}{b} = 1.675$. The equation $\frac{a}{bn} = 26.8$ can be rewritten as $(\frac{a}{b})(\frac{1}{n}) = 26.8$. Substituting **1.675** for $\frac{a}{b}$ in this equation yields $(1.675)(\frac{1}{n}) = 26.8$, or $\frac{1.675}{n} = 26.8$. Multiplying both sides of this equation by n yields **1.675 = 26.8n**. Dividing both sides of this equation by **26.8** yields $n = 0.0625$. Therefore, the value of n is **0.0625**. Note that .0625, 0.062, 0.063, and 1/16 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 7aefc67c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 7aefc67c

There are **250** trees in a park. Of these trees, **6%** are birch trees. How many birch trees are in the park?

- A. **6**
- B. **15**
- C. **75**
- D. **244**

ID: 7aefc67c Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that there are **250** trees in a park and of these trees, **6%** are birch trees. The number of birch trees in the park can be calculated by multiplying the number of trees in the park by $\frac{6}{100}$. Therefore, the number of birch trees in the park is **250**($\frac{6}{100}$), or **15**.

Choice A is incorrect. This is the percentage of trees in the park that are birch trees, not the number of birch trees in the park.

Choice C is incorrect. This is **30%**, not **6%**, of **250**.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 5d625c74

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 5d625c74

The table shows the distribution of rooms in a certain facility by seating capacity.

Seating capacity	Proportion
Less than 18 seats	26%
18–40 seats	21%
41–65 seats	29%
Greater than 65 seats	24%

If a room in this facility is selected at random, which of the following is closest to the probability of selecting a room that has a seating capacity greater than **65** seats, given that the room has a seating capacity of at least **18** seats?

- A. **0.24**
- B. **0.32**
- C. **0.50**
- D. **0.92**

ID: 5d625c74 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that the proportion of rooms with less than **18** seats is **26%**, which is equivalent to **0.26**. It’s also given that the proportion of rooms with **18–40** seats is **21%**, or **0.21**; that the proportion of rooms with **41–65** seats is **29%**, or **0.29**; and that the proportion of rooms with greater than **65** seats is **24%**, or **0.24**. The probability of selecting a room that has a seating capacity greater than **65** seats given that the room has a seating capacity of at least **18** seats is equal to the proportion of rooms that have a seating capacity of greater than **65** seats divided by the proportion of rooms that have a seating capacity of at least **18** seats. Rooms with a seating capacity of at least **18** seats include the categories **18–40** seats, **41–65** seats, and greater than **65** seats, so the total proportion for this condition can be written as **0.21 + 0.29 + 0.24**, or **0.74**. Therefore, the probability of selecting a room that has a seating capacity greater than **65** seats given that it has a seating capacity of at least **18** seats is $\frac{0.24}{0.74}$, or approximately **0.324**. Of the given choices, **0.32** is closest to **0.324**.

Choice A is incorrect. This is the probability of selecting a room that has a seating capacity greater than **65** seats from all rooms in the facility, not just rooms that meet the condition of having a seating capacity of at least **18** seats.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 5b384052

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 5b384052

The list gives the mass, in grams, of **5** alpine marmots.
4,010; 4,010; 3,030; 4,050; 3,050
What is the mean mass, in grams, of these **5** alpine marmots?

ID: 5b384052 Answer

Correct Answer: 3630

Rationale

The correct answer is **3,630**. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. The sum of the masses, in grams, of these alpine marmots is **4,010 + 4,010 + 3,030 + 4,050 + 3,050**, or **18,150** grams. The number of alpine marmots in the data set is **5**. Therefore, the mean mass, in grams, of these **5** alpine marmots is $\frac{18,150}{5}$, or **3,630**.

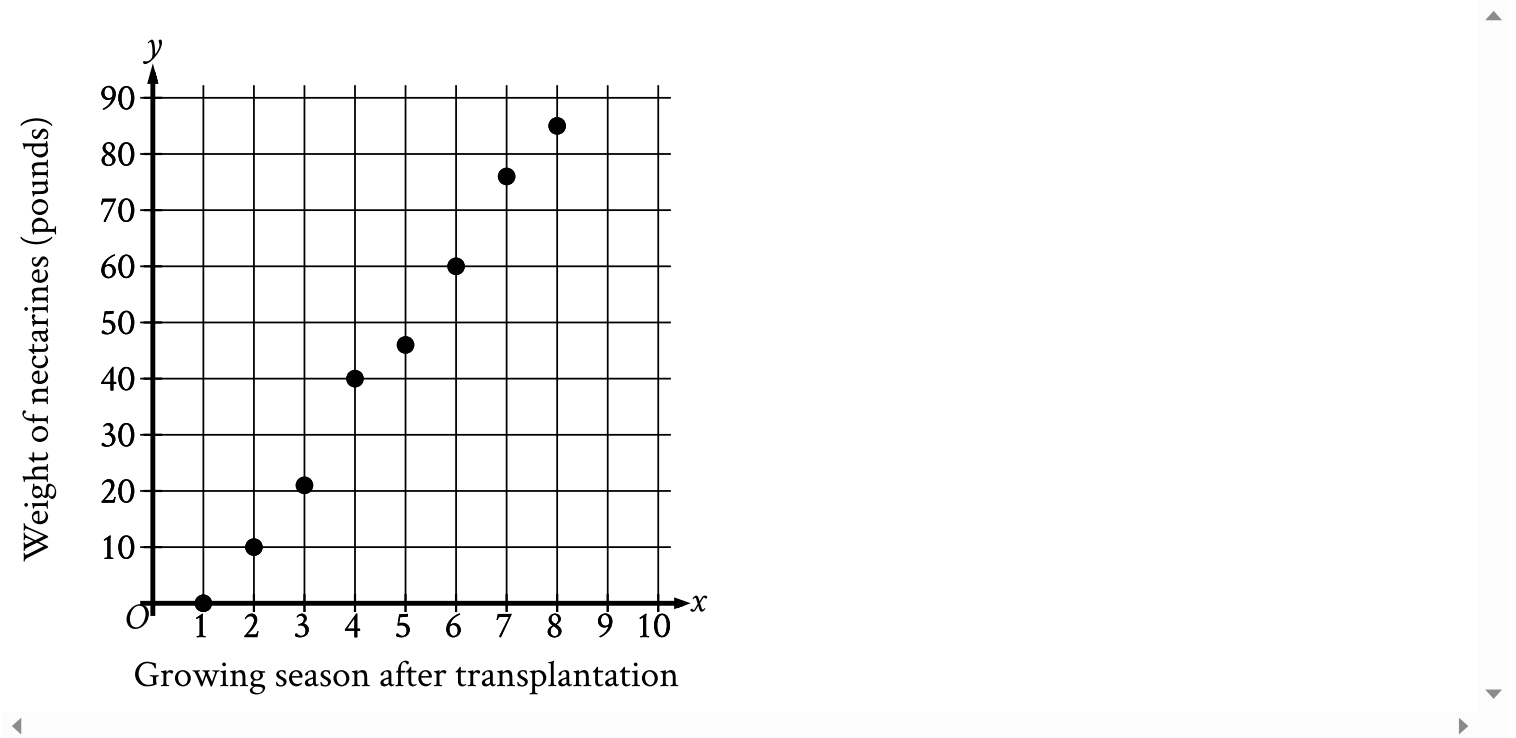
Question Difficulty: Medium

Question ID f6571d2b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: f6571d2b

An orchard owner recorded the weight, in pounds, of all nectarines that grew on a dwarf nectarine tree during each growing season after the tree's transplantation. The scatterplot shows this weight, in pounds, for each growing season after the tree's transplantation.



What was the weight, to the nearest pound, of all nectarines that grew on the tree during the **4th** growing season after the tree's transplantation?

ID: f6571d2b Answer

Correct Answer: 40

Rationale

The correct answer is **40**. For each data point on the scatterplot, the x-value represents the growing season after transplantation and the y-value represents the weight, in pounds, of all nectarines that grew on the tree during the season. The scatterplot shows a data point at **(4, 40)**. It follows that during the **4th** growing season after the tree's transplantation, **40** pounds of nectarines grew on the tree.

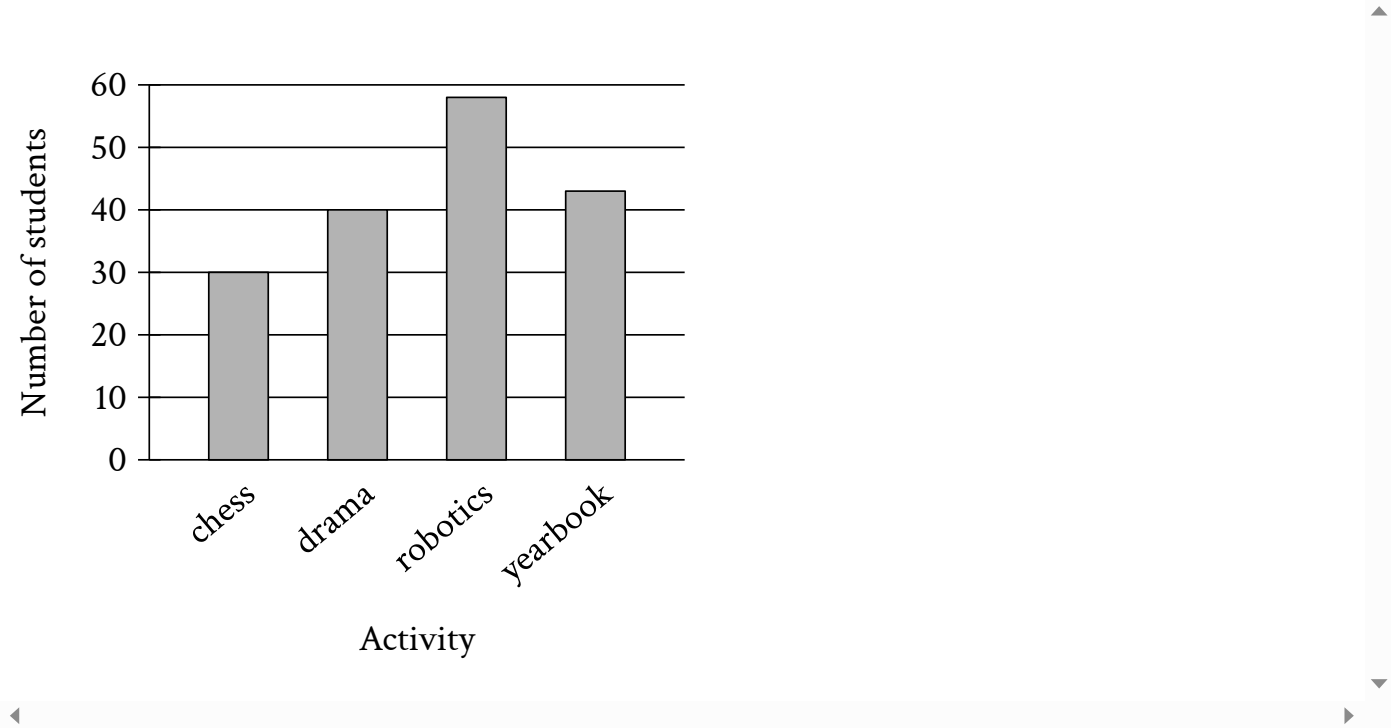
Question Difficulty: Easy

Question ID 7495dbde

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 7495dbde

The bar graph shows the distribution of the number of students in each of four extracurricular activities at a high school.



How many more students are in drama than in chess?

- A. 10
- B. 30
- C. 40
- D. 70

ID: 7495dbde Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the bar graph shows the distribution of the number of students in each of four extracurricular activities at a high school. The bar representing drama has a height of **40**; therefore, there are **40** students in drama. The bar representing chess has a height of **30**; therefore, there are **30** students in chess. Thus, there are **40 — 30**, or **10** more students in drama than in chess.

Choice B is incorrect. This is the number of students in chess.

Choice C is incorrect. This is the number of students in drama.

Choice D is incorrect. This is the sum of the number of students in drama and in chess.

Question ID b8d2c4c9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: b8d2c4c9

For data set A, the table summarizes the distribution of the number of deliveries received by an office each day during a period of **11** days.

Deliveries	Days
0	2
3	2
4	2
5	2
6	1
7	1
13	1

The data value **13** was recorded in error and is removed from data set A to create data set B, which consists of the remaining **10** data values. Which statement best compares the median of data set A and the median of data set B?

- A. The median of data set B is less than the median of data set A.
- B. The median of data set B is greater than the median of data set A.
- C. The median of data set B is equal to the median of data set A.
- D. There is not enough information to compare the medians of the two data sets.

ID: b8d2c4c9 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that for data set A, the table summarizes the number of deliveries received by an office each day over a period of **11** days. If a data set contains an odd number of data values, the median is represented by the middle data value in the list when the data values are listed in ascending or descending order. Since data set A contains **11** data values in ascending order, the median of data set A is the **6th** value. Based on the number of days for each value shown in the table, the **6th** value is **4**, so the median of data set A is **4**. It's also given that the data value **13** was recorded in error and is removed from data set A to create data set B. It follows that data set B contains **10** data values in ascending order. If a data set contains an even number of data values, the median is between the two middle data values when the values are listed in ascending or descending order. It follows that the median of data set B is between the **5th** and **6th** values. Based on the number of days for each value shown in the table, the **5th** and **6th** values are both **4**, so the median of data set B is **4**. Therefore, the median of data set B is equal to the median of data set A.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 8e083159

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 8e083159

How many cups are equivalent to **83** quarts? (**4 cups = 1 quart**)

- A. **21**
- B. **79**
- C. **87**
- D. **332**

ID: 8e083159 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that **4 cups** is equal to **1 quart**. Multiplying $\frac{4 \text{ cups}}{1 \text{ quart}}$ by **83** quarts yields **4 • 83** cups, or **332** cups.

Choice A is incorrect. This is the number of quarts in **83** cups, rounded to the nearest whole number.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID f99fee2a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: f99fee2a

73, 74, 75, 77, 79, 82, 84, 85, 91
What is the median of the data shown?

ID: f99fee2a Answer

Correct Answer: 79

Rationale

The correct answer is **79**. The median of a data set with an odd number of values is the middle value of the set when the values are ordered from least to greatest. Because the given data set consists of nine values that are ordered from least to greatest, the median is the fifth value in the data set. Therefore, the median of the data shown is **79**.

Question Difficulty: Easy

Question ID b1f33dbf

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: b1f33dbf

One of a planet's moons orbits the planet every **252** days. A second moon orbits the planet every **287** days. How many more days does it take the second moon to orbit the planet **29** times than it takes the first moon to orbit the planet **29** times?

ID: b1f33dbf Answer

Correct Answer: 1015

Rationale

The correct answer is **1,015**. It's given that the first moon orbits the planet every **252** days. Therefore, it takes the first moon **252(29)**, or **7,308**, days to orbit the planet **29** times. It's also given that the second moon orbits the planet every **287** days. Therefore, it takes the second moon **287(29)**, or **8,323**, days to orbit the planet **29** times. Since it takes the first moon **7,308** days and the second moon **8,323** days, it takes the second moon **8,323 — 7,308**, or **1,015**, more days than it takes the first moon to orbit the planet **29** times.

Question Difficulty: Hard

Question ID 07f62f7b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 07f62f7b

There is a proportional relationship between p and t . If $p = 5,130$ when $t = 6,840$, then what is the value of p when $t = 5,472$?

- A. 7,296
- B. 7,182
- C. 4,104
- D. 3,762

ID: 07f62f7b Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that there is a proportional relationship between p and t . This relationship can be written as $\frac{p}{t}$. If $p = 5,130$ when $t = 6,840$, the value of p when $t = 5,472$ can be calculated by solving the equation $\frac{5,130}{6,840} = \frac{p}{5,472}$. Multiplying each side of this equation by $5,472$ yields $(5,472) \frac{5,130}{6,840} = \frac{p}{5,472} (5,472)$, or $p = 4,104$. Therefore, when $t = 5,472$, the value of p is $4,104$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 65d0d66e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 65d0d66e

A scientist studying the life cycle of dragonflies counted the number of dragonflies in a certain habitat each day for **46** days. On February **15**, there were **99** dragonflies in the habitat. The percent increase in the number of dragonflies in the habitat from January **1** to February **15** was **12.50%**. How many dragonflies were in the habitat on January **1**?

- A. **88**
- B. **87**
- C. **12**
- D. **8**

ID: 65d0d66e Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that a scientist studying the life cycle of dragonflies counted the number of dragonflies in a certain habitat each day for **46** days. It’s also given that on February **15**, there were **99** dragonflies in the habitat and that the percent increase in the number of dragonflies in the habitat from January **1** to February **15** was **12.50%**. This can be represented by the equation $99 = (1 + \frac{12.50}{100})x$, where x represents the number of dragonflies in the habitat on January **1**. This equation can be rewritten as $99 = 1.125x$. Dividing both sides of this equation by **1.125** yields $88 = x$. Therefore, there were **88** dragonflies in the habitat on January **1**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 167f9a80

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 167f9a80

According to a set of standards, a certain type of substance can contain a maximum of **0.001%** phosphorus by mass. If a sample of this substance has a mass of **140** grams, what is the maximum mass, in grams, of phosphorus the sample can contain to meet these standards?

ID: 167f9a80 Answer

Correct Answer: .0014

Rationale

The correct answer is **.0014**. It's given that a certain type of substance can contain a maximum of **0.001%** phosphorus by mass to meet a set of standards. If a sample of the substance has a mass of **140** grams, it follows that the maximum mass, in grams, of phosphorus the sample can contain to meet the standards is **0.001%** of **140**, or $\frac{0.001}{100} (140)$, which is equivalent to **(0.00001)(140)**, or **0.0014**. Note that .0014 and 0.001 are examples of ways to enter a correct answer.

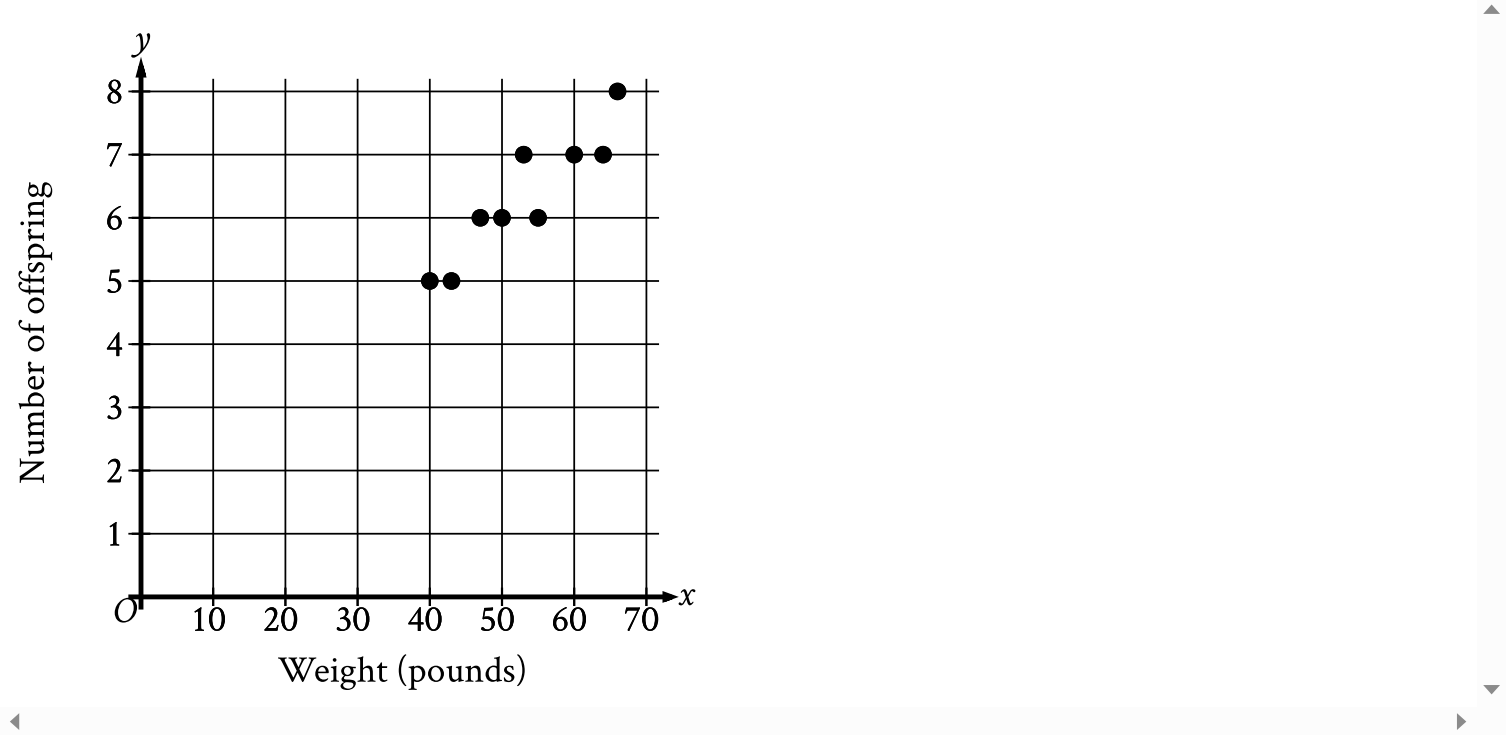
Question Difficulty: Hard

Question ID 1075bb9c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 1075bb9c

The scatterplot shows the relationship between the weight, in pounds, of each of **9** female gray wolves on April **30** and the number of offspring each gray wolf produced.



How many offspring did the **50**-pound gray wolf produce?

- A. **8**
- B. **7**
- C. **6**
- D. **5**

ID: 1075bb9c Answer

Correct Answer: C

Rationale

Choice C is correct. For each point on the scatterplot shown, the x-value represents the weight, in pounds, of a female gray wolf and the y-value represents the number of offspring that wolf produced. The point on the graph with an x-value of **50** has a y-value of **6**. Therefore, the **50**-pound gray wolf produced **6** offspring.

Choice A is incorrect. One of the wolves produced **8** offspring, but its weight was greater than **50** pounds.

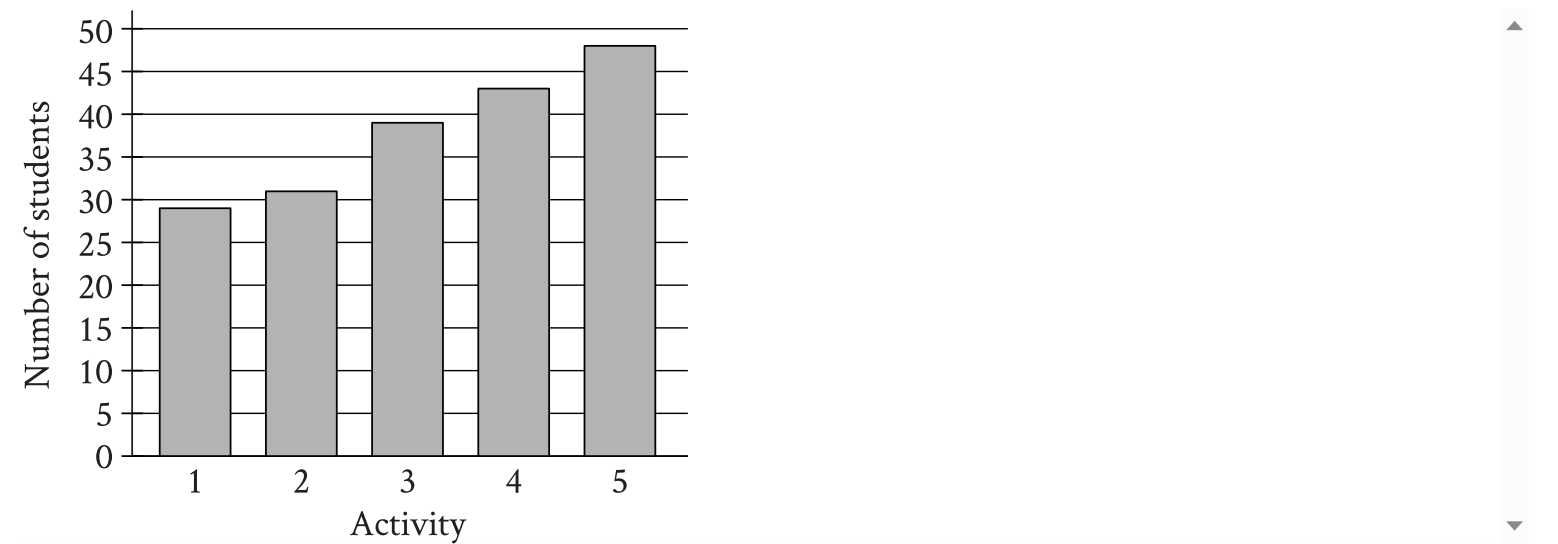
Choice B is incorrect. Three of the wolves produced **7** offspring each, but their weights were each greater than **50** pounds.

Choice D is incorrect. Two of the wolves produced **5** offspring each, but their weights were each less than **50** pounds.

Question ID 73f42b7a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 73f42b7a



A group of students voted on five after-school activities. The bar graph shows the number of students who voted for each of the five activities. How many students chose activity **3**?

- A. **25**
- B. **39**
- C. **48**
- D. **50**

ID: 73f42b7a Answer

Correct Answer: B

Rationale

Choice B is correct. The height of each bar in the bar graph given represents the number of students that voted for the activity specified at the bottom of the bar. The bar for activity **3** has a height that is between **35** and **40**. In other words, the number of students that chose activity **3** is between **35** students and **40** students. Of the given choices, **39** is the only value between **35** and **40**. Therefore, **39** students chose activity **3**.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. This is the number of students that chose activity **5**, not activity **3**.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID f317ce0b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: f317ce0b

The number a is 110% greater than the number b . The number b is 90% less than 47. What is the value of a ?

ID: f317ce0b Answer

Correct Answer: 9.87, 987/100

Rationale

The correct answer is **9.87**. It's given that the number a is 110% greater than the number b . It follows that $a = (1 + \frac{110}{100})b$, or $a = 2.1b$. It's also given that the number b is 90% less than 47. It follows that $b = (1 - \frac{90}{100})(47)$, or $b = 0.1(47)$, which yields $b = 4.7$. Substituting 4.7 for b in the equation $a = 2.1b$ yields $a = 2.1(4.7)$, which is equivalent to $a = 9.87$. Therefore, the value of a is **9.87**.

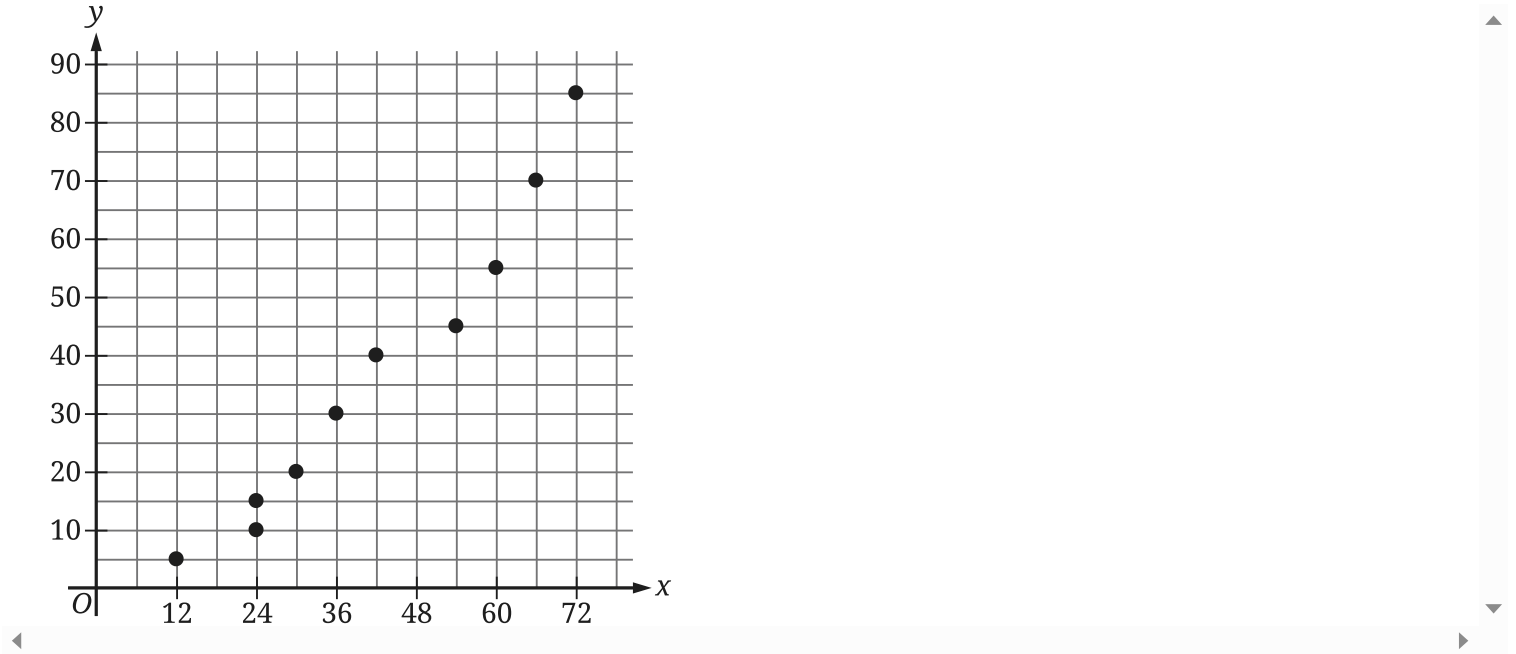
Question Difficulty: Hard

Question ID aa482e9e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: aa482e9e

The owner of a restaurant supply store selects **10** cutting boards to put on sale. The graph shows the width x , in centimeters, of each cutting board and its corresponding sale price y , in dollars, where each value of y is an integer.



According to the graph, which statement is true?

- A. Cutting boards that have a width greater than **48** centimeters have an average sale price of **\$59.00**.
- B. Cutting boards that have a width greater than **48** centimeters have an average sale price of **\$40.00**.
- C. Cutting boards that have a width between **18** and **48** centimeters have an average sale price of **\$31.20**.
- D. Cutting boards that have a width between **18** and **48** centimeters have an average sale price of **\$23.00**.

ID: aa482e9e Answer

Correct Answer: D

Rationale

Choice D is correct. Each point (x, y) on the scatterplot gives the width x , in centimeters, and the sale price y , in dollars, of one of the **10** cutting boards put on sale at this store. According to the graph, **5** cutting boards have widths that are between **18** and **48** centimeters. These widths, in centimeters, are **24, 24, 30, 36, and 42**. The sale prices, in dollars, corresponding to these widths are **10, 15, 20, 30, and 40**, respectively. The average of these sale prices, in dollars, is $\frac{10+15+20+30+40}{5}$, or $\frac{115}{5}$, which is equivalent to **23**. Therefore, cutting boards that have a width between **18** and **48** centimeters have an average sale price of **\$23.00**.

Choice A is incorrect. Cutting boards that have a width greater than **48** centimeters have an average sale price of **\$63.75**, not **\$59.00**.

Choice B is incorrect. Cutting boards that have a width greater than **48** centimeters have an average sale price of **\$63.75**, not **\$40.00**.

Choice C is incorrect. Cutting boards that have a width between **18** and **48** centimeters have an average sale price of **\$23.00**, not **\$31.20**.

Question Difficulty: Medium

Question ID 06adcfe0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 06adcfe0

A box holds **104** tokens, of which **13** are brown. If a token is selected at random from the box, what is the probability of selecting a token that is brown?

- A. $\frac{1}{13}$
- B. $\frac{1}{10}$
- C. $\frac{1}{8}$
- D. $\frac{1}{4}$

ID: 06adcfe0 Answer

Correct Answer: C

Rationale

Choice C is correct. If a token is selected at random from the box, the probability of selecting a token that is brown is equal to the number of tokens that are brown divided by the total number of tokens in the box. Since there are **13** brown tokens and **104** total tokens in the box, the probability of selecting a token that is brown is $\frac{13}{104}$, or $\frac{1}{8}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 297e7895

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 297e7895

There are **350** objects in a box. Of these objects, **4%** are balls. How many balls are in the box?

- A. **4**
- B. **14**
- C. **70**
- D. **346**

ID: 297e7895 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that **4%** of the **350** objects in the box are balls. Therefore, the number of balls in the box can be found by calculating **4%** of **350**, which is equal to **$350\left(\frac{4}{100}\right)$** , or **14**.

Choice A is incorrect. This is the percentage of objects in the box that are balls, not the number of balls in the box.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID b3d9da38

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: b3d9da38

The area of a rectangular region is increasing at a rate of **250** square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use **1 meter = 3.28 feet**.)

- A. **0.39**
- B. **1.27**
- C. **13.67**
- D. **23.24**

ID: b3d9da38 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that **1 meter = 3.28 feet**. It follows that **1² square meter = 3.28² square feet**, or **1 square meter = 10.7584 square feet**. Since **1 hour = 60 minutes**, it follows that **250 square feet per hour** is equivalent to $\left(\frac{250 \text{ square feet}}{1 \text{ hour}}\right)\left(\frac{1 \text{ square meter}}{10.7584 \text{ square feet}}\right)\left(\frac{1 \text{ hour}}{60 \text{ minutes}}\right)$, or $\frac{250 \text{ square meters}}{645.504 \text{ minutes}}$, which is approximately **0.3873** square meters per minute. Of the given choices, **0.39** is closest to **0.3873**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID a645a9e2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: a645a9e2

The number of zebras in a population in **2018** was **1.27** times the number of zebras in this population in **2014**. If the number of zebras in this population in **2014** is $p\%$ of the number of zebras in this population in **2018**, what is the value of p , to the nearest whole number?

ID: a645a9e2 Answer

Correct Answer: 79

Rationale

The correct answer is **79**. Let x represent the number of zebras in the population in **2014** and let y represent the number of zebras in the population in **2018**. It's given that the number of zebras in this population in **2018** was **1.27** times the number of zebras in this population in **2014**. It follows that the equation $y = 1.27x$ represents this situation. Dividing both sides of this equation by **1.27** yields $\frac{y}{1.27} = x$, or $(\frac{1}{1.27})y = x$. Therefore, the number of zebras in this population in **2014** is $\frac{1}{1.27}$ times the number of zebras in this population in **2018**. If the number of zebras in this population in **2014** is $p\%$ of the number of zebras in this population in **2018**, then $x = \frac{p}{100}y$. It follows that $\frac{1}{1.27} = \frac{p}{100}$, or $\frac{100}{1.27} = p$, which means p is approximately equal to **78.74**. Therefore, the value of p , to the nearest whole number, is **79**.

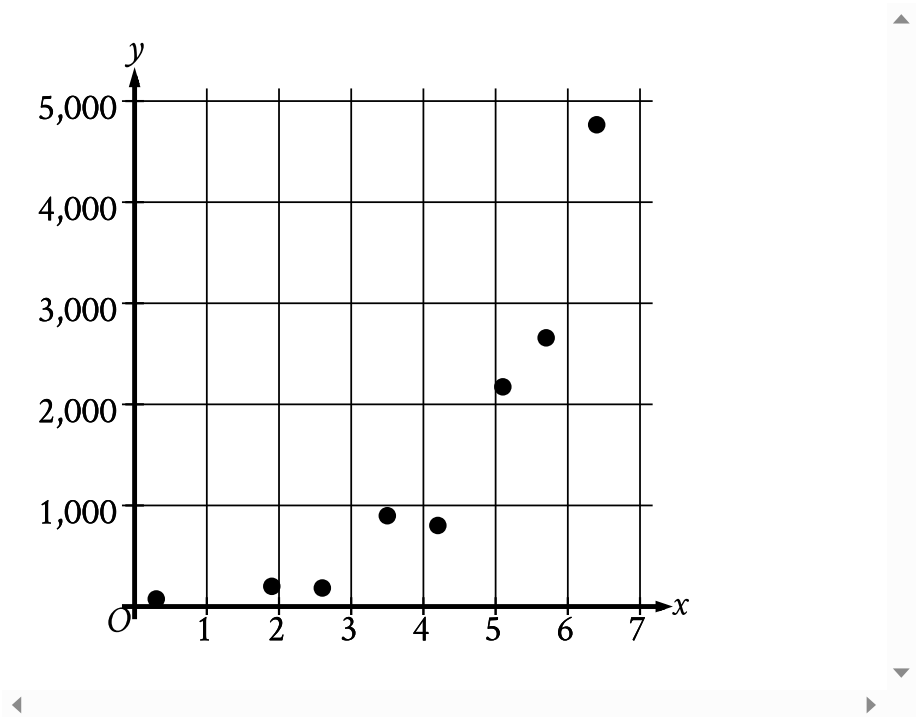
Question Difficulty: Hard

Question ID 634bde56

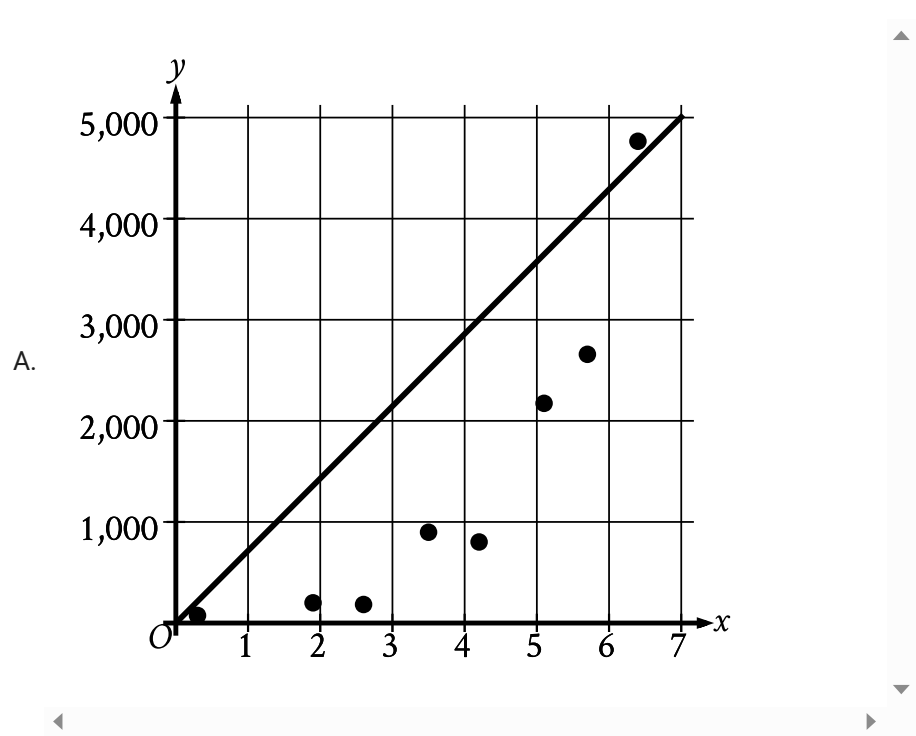
Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 634bde56

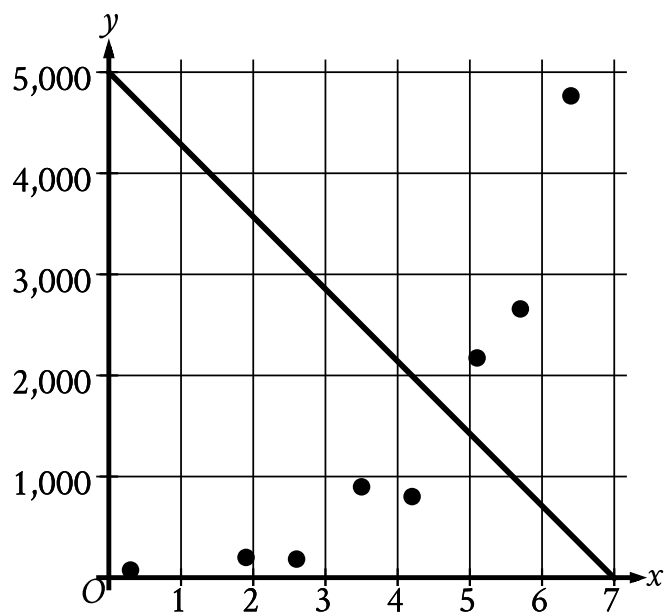
The scatterplot shows the relationship between two variables, x and y .



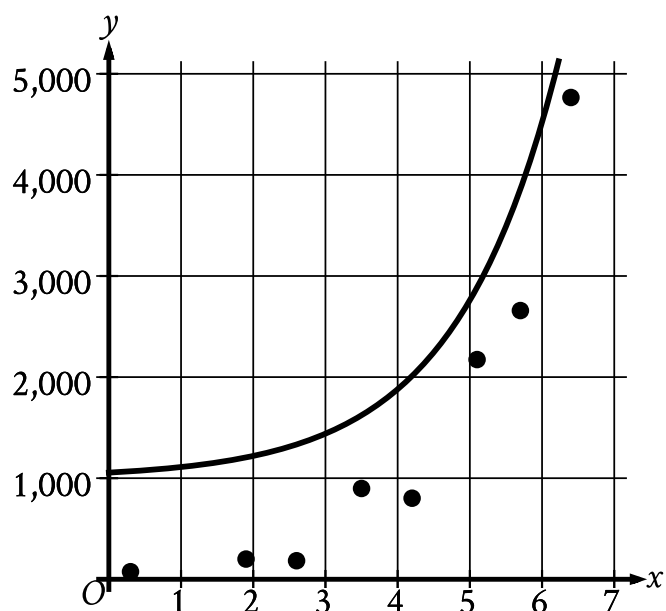
Which of the following graphs shows the most appropriate model for the data?



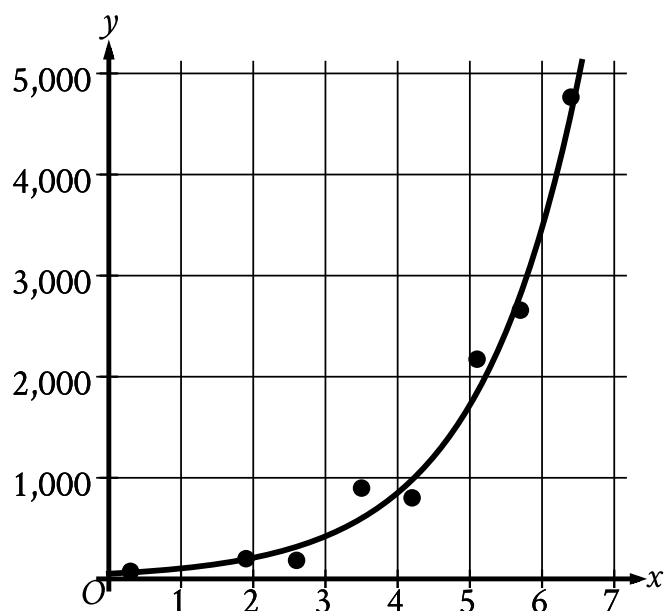
B.



C.



D.



Correct Answer: D

Rationale

Choice D is correct. An appropriate model should follow the trend of the data points and should have data points both above and below the model. The scatterplot shows that the data points have an increasing trend that is curved. Therefore, an appropriate model should be an increasing curve with data points both above and below the model. Of the given choices, only the model in choice D is an increasing curve with data points both above and below the model.

Choice A is incorrect. Since the trend of the data points isn't linear, a line isn't the most appropriate model for the data.

Choice B is incorrect. Since the trend of the data points is increasing and isn't linear, a decreasing line isn't the most appropriate model for the data.

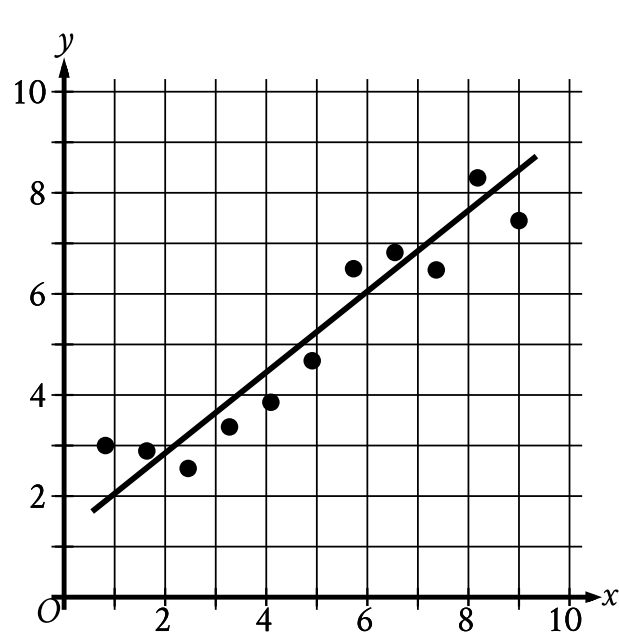
Choice C is incorrect. All the data points are below the model shown in this graph.

Question Difficulty: Easy

Question ID b9f9a101

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: b9f9a101



The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown. For how many of the 11 data points does the line of best fit predict a greater y -value than the actual y -value?

ID: b9f9a101 Answer

Correct Answer: 6

Rationale

The correct answer is **6**. The line of best fit predicts a greater y -value than the actual y -value for any data point that's located below the line of best fit. For the scatterplot shown, **6** of the data points are below the line of best fit. Therefore, the line of best fit predicts a greater y -value than the actual y -value for **6** of the data points.

Question Difficulty: Hard

Question ID 696c8935

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 696c8935

An object's speed is **64 yards** per second. What is the object's speed, in feet per second? (**1 yard = 3 feet**)

- A. **61**
- B. **67**
- C. **94**
- D. **192**

ID: 696c8935 Answer

Correct Answer: D

Rationale

Choice D is correct. Since **1** yard is equal to **3** feet, **64** yards is equal to **64 yards** $\left(\frac{3 \text{ feet}}{1 \text{ yard}}\right)$, or **192** feet. It follows that **64** yards per second is equivalent to **192** feet per second. Therefore, the object's speed is **192** feet per second.

Choice A is incorrect. A speed of **61** feet per second is equivalent to $\frac{61}{3}$, not **64**, yards per second.

Choice B is incorrect. A speed of **67** feet per second is equivalent to $\frac{67}{3}$, not **64**, yards per second.

Choice C is incorrect. A speed of **94** feet per second is equivalent to $\frac{94}{3}$, not **64**, yards per second.

Question Difficulty: Easy

Question ID 87bcd883

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 87bcd883

The population of City A increased by 7% from 2015 to 2016. If the 2016 population is k times the 2015 population, what is the value of k ?

- A. 0.07
- B. 0.7
- C. 1.07
- D. 1.7

ID: 87bcd883 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the population of City A increased by 7% from 2015 to 2016. Therefore, the population of City A in 2016 includes 100% of the population of City A in 2015 plus an additional 7% of the population of City A in 2015. This means that the population of City A in 2016 is 107% of the population in 2015. Thus, the population of City A in 2016 is $\frac{107}{100}$, or 1.07, times the 2015 population. Therefore, the value of k is 1.07.

Choice A is incorrect. This would be the value of k if the population in 2016 was 7% of the population in 2015.

Choice B is incorrect. This would be the value of k if the population in 2016 was 70% of the population in 2015.

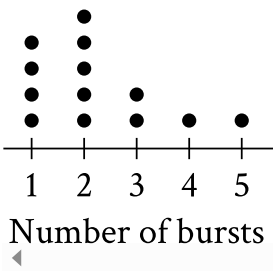
Choice D is incorrect. This would be the value of k if the population increased by 70%, not 7%, from 2015 to 2016.

Question Difficulty: Hard

Question ID 8521e464

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 8521e464



The dot plot represents a data set of the number of bursts for **13** eruptions of a steam vent. If an additional eruption with **11** bursts is added to this data set to create a new data set of **14** eruptions, which of the following measures will be greater for the new data set than for the original data set?

- I. The median number of bursts

II. The mean number of bursts
- A. I and II

B. I only

C. II only

D. Neither I nor II

ID: 8521e464 Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that the dot plot represents a data set of the number of bursts for **13** eruptions of a steam vent. The median of a data set with an odd number of elements is the middle element when the elements are in numerical order. For **13** elements in numerical order, this is the **7th** element. For this data set, the first **4** elements have a value of **1**, and the next **5** elements have a value of **2**. Thus, the **7th** element in the ordered data set is **2** and the median number of bursts for the original data set is **2**. If an additional eruption with **11** bursts is added to this data set to create a new data set of **14** eruptions, the median of the new data set will be between the **7th** and **8th** elements in the ordered set, which will also be **2**. Therefore, the median number of bursts for the new data set will be the same as the median number of bursts for the original data set. The mean number of bursts for the original data set is found by adding the values of all **13** elements and dividing that sum by the number of elements, **13**. Since the data is shown in a dot plot, the sum of the values of the elements can be found by multiplying each element's value by its frequency: $1(4) + 2(5) + 3(2) + 4(1) + 5(1)$, or **29**. Therefore, the mean number of bursts for the original data set is $\frac{29}{13}$. If an additional eruption with **11** bursts is added to this data set to create a new data set of **14** bursts, the mean number of bursts for the new data set is $\frac{29+11}{14}$, or $\frac{40}{14}$. Since $\frac{40}{14} > \frac{29}{13}$, the mean number of bursts for the new data set is greater than the mean number of bursts for the original data set. Therefore, of the median number of bursts and the mean number of bursts, only the mean number of bursts is greater for the new data set than for the original data set.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

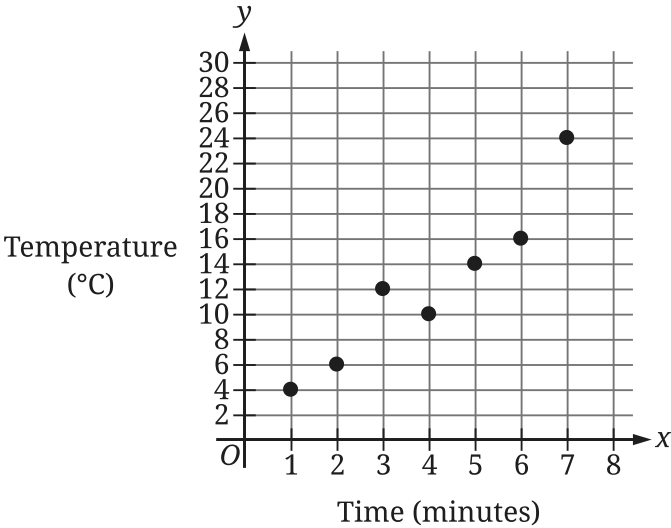
Question Difficulty: Hard

Question ID 7196a6b2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 7196a6b2

During a study, the temperature, **in degrees Celsius ($^{\circ}\text{C}$)**, of the air in a chamber was recorded to the nearest integer at certain times. The scatterplot shows the recorded temperature **y , in $^{\circ}\text{C}$** , of the air in the chamber **x minutes** after the start of the study.



What was the average rate of change, **in $^{\circ}\text{C}$ per minute**, of the recorded temperature of the air in the chamber from **$x = 5$ to $x = 7$** ?

ID: 7196a6b2 Answer

Correct Answer: 5

Rationale

The correct answer is **5**. For the graph shown, **x** represents time, in minutes, and **y** represents temperature, in degrees Celsius ($^{\circ}\text{C}$). Therefore, the average rate of change, in $^{\circ}\text{C}$ per minute, of the recorded temperature of the air in the chamber between two x -values is the difference in the corresponding y -values divided by the difference in the x -values. The graph shows that at **$x = 5$** , the corresponding y -value is **14**. The graph also shows that at **$x = 7$** , the corresponding y -value is **24**. It follows that the average rate of change, in $^{\circ}\text{C}$ per minute, from **$x = 5$ to $x = 7$** is $\frac{24-14}{7-5}$, which is equivalent to $\frac{10}{2}$, or **5**.

Question Difficulty: Hard

Question ID 51136782

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 51136782

The number a is 190% greater than the number b . The number b is 80% less than 24 . What is the value of a ?

- A. 9.12
- B. 13.92
- C. 26.40
- D. 36.48

ID: 51136782 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the number b is 80% less than 24 . It follows that b is equal to 24 minus 80% of 24 , which can be written as $b = 24 - (\frac{80}{100})24$. This is equivalent to $b = 24 - 0.8(24)$, or $b = 4.8$. It's also given that the number a is 190% greater than the number b . It follows that a is equal to b plus 190% of b . Since $b = 4.8$, this can be written as $a = 4.8 + (\frac{190}{100})4.8$. This is equivalent to $a = 4.8 + 1.9(4.8)$, or $a = 13.92$.

Choice A is incorrect. This would be the value of a if a were 190% of b , not 190% greater than b .

Choice C is incorrect. This is $(190 - 80)\%$ of 24 .

Choice D is incorrect. This would be the value of a if b were 80% of 24 , not 80% less than 24 , and a were 190% of b , not 190% greater than b .

Question Difficulty: Hard

Question ID 5fe97124

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 5fe97124

A rectangle has an area of **3,168** square inches. What is the area, in **square feet**, of this rectangle? (**1 foot = 12 inches**)

- A. **22**
- B. **56.3**
- C. **264**
- D. **675.4**

ID: 5fe97124 Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that **1 foot = 12 inches**. It follows that **1² square foot = 12² square inches**, or **1 square foot = 144 square inches**. Thus, **3,168** square inches is equivalent to **$\left(3,168 \text{ square inches}\right)\left(\frac{1 \text{ square foot}}{144 \text{ square inches}}\right)$** , or **22** square feet.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $\frac{3,168}{12}$, not $\frac{3,168}{144}$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID b85e0fb8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: b85e0fb8

The ratio x to y is equivalent to the ratio 12 to t . When $x = 156$, what is the value of y in terms of t ?

- A. $13t$
- B. $12t$
- C. $144t$
- D. $168t$

ID: b85e0fb8 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the ratio x to y is equivalent to the ratio 12 to t . This can be represented by $\frac{x}{y} = \frac{12}{t}$. Substituting 156 for x in this equation yields $\frac{156}{y} = \frac{12}{t}$. This can be rewritten as $12y = 156t$. Dividing both sides of this equation by 12 yields $y = 13t$. Therefore, when $x = 156$, the value of y in terms of t is $13t$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 0c9a623e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 0c9a623e

Data set X: **5, 9, 9, 13**
Data set Y: **5, 9, 9, 13, 27**

The lists give the values in data sets X and Y. Which statement correctly compares the mean of data set X and the mean of data set Y?

- A. The mean of data set X is greater than the mean of data set Y.
- B. The mean of data set X is less than the mean of data set Y.
- C. The means of data set X and data set Y are equal.
- D. There is not enough information to compare the means.

ID: 0c9a623e Answer

Correct Answer: B

Rationale

Choice B is correct. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. It follows that the mean of data set X is $\frac{5+9+9+13}{4}$, or **9**, and the mean of data set Y is $\frac{5+9+9+13+27}{5}$, or **12.6**. Since **9** is less than **12.6**, the mean of data set X is less than the mean of data set Y.

Alternate approach: Data set Y consists of the **4** values in data set X and one additional value, **27**. Since the additional value, **27**, is larger than any value in data set X, the mean of data set X is less than the mean of data set Y.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID ced14ecd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: ced14ecd

6, 6, 8, 8, 8, 10, 21

Which of the following lists represents a data set that has the same median as the data set shown?

- A. 4, 6, 6, 6, 8, 8
- B. 6, 6, 8, 8, 10, 10
- C. 6, 8, 10, 10, 10, 12
- D. 8, 8, 10, 10, 21, 21

ID: ced14ecd Answer

Correct Answer: B

Rationale

Choice B is correct. If a data set contains an odd number of data values, the median is represented by the middle data value in the list when the data values are listed in ascending or descending order. Since the data set shown has **7** data values and is in ascending order, it follows that the median is the fourth data value in the list, or **8**. If a data set contains an even number of data values, the median is between the two middle data values when the values are listed in ascending or descending order. Since each of the choices consists of a data set with **6** data values in ascending order, it follows that the median is between the third and fourth data value. The third and fourth data values in choice B are **8** and **8**. Thus, choice B represents a data set with a median of **8**. Since the median of the data set shown is **8** and choice B represents a data set with a median of **8**, it follows that choice B represents a data set that has the same median as the data set shown.

Choice A is incorrect. This list represents a data set with a median of **6**, not **8**.

Choice C is incorrect. This list represents a data set with a median of **10**, not **8**.

Choice D is incorrect. This list represents a data set with a median of **10**, not **8**.

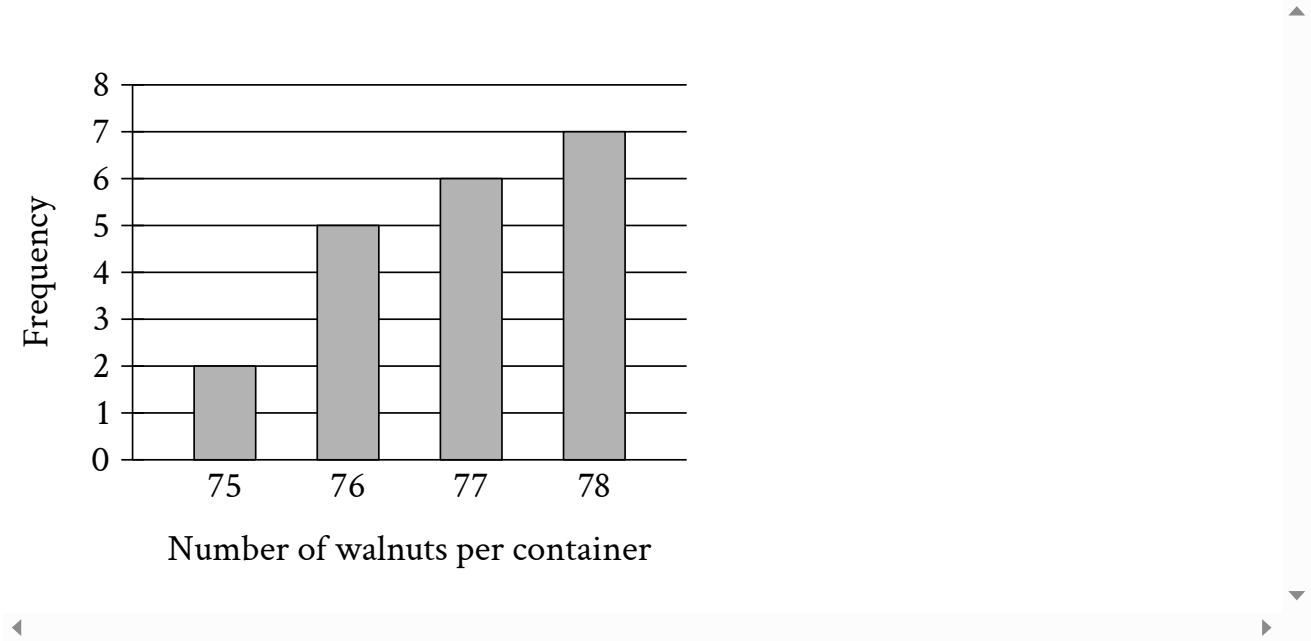
Question Difficulty: Easy

Question ID 4bb1ac93

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 4bb1ac93

The bar graph shows the distribution of the number of walnuts per container for **20** containers at a grocery store.



How many of these containers of walnuts contain exactly **78** walnuts?

- A. **2**
- B. **7**
- C. **20**
- D. **78**

ID: 4bb1ac93 Answer

Correct Answer: B

Rationale

Choice B is correct. The height of each bar in the graph shown represents the number of containers that contain the number of walnuts specified at the bottom of the bar. The bar for **78** walnuts has a height of **7**. Therefore, **7** of these containers of walnuts contain exactly **78** walnuts.

Choice A is incorrect. This is the number of containers that contain exactly **75** walnuts, not **78** walnuts.

Choice C is incorrect. This is the total number of containers of walnuts represented in the bar graph, not the number that contain exactly **78** walnuts.

Choice D is incorrect. This is the number of walnuts in a container that contains exactly **78** walnuts, not the number of containers that contain exactly **78** walnuts.

Question ID ce4fa697

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: ce4fa697

Each rock in a collection of **70** rocks was classified as either igneous, metamorphic, or sedimentary, as shown in the frequency table.

Classification	Frequency
igneous	10
metamorphic	33
sedimentary	27



If one of these rocks is selected at random, what is the probability of selecting a rock that is igneous?

- A. $\frac{10}{27}$
- B. $\frac{10}{33}$
- C. $\frac{10}{60}$
- D. $\frac{10}{70}$

ID: ce4fa697 Answer

Correct Answer: D

Rationale

Choice D is correct. If one of the rocks in the collection is selected at random, the probability of selecting a rock that is igneous is equal to the number of igneous rocks in the collection divided by the total number of rocks in the collection. According to the table, there are **10** igneous rocks in the collection, and it's given that there's a total of **70** rocks in the collection. Therefore, if one of the rocks in the collection is selected at random, the probability of selecting a rock that is igneous is $\frac{10}{70}$.

Choice A is incorrect. This is the number of igneous rocks in the collection divided by the number of sedimentary rocks in the collection, not divided by the total number of rocks in the collection.

Choice B is incorrect. This is the number of igneous rocks in the collection divided by the number of metamorphic rocks in the collection, not divided by the total number of rocks in the collection.

Choice C is incorrect. This is the number of igneous rocks in the collection divided by the number of rocks in the collection that aren't igneous, not divided by the total number of rocks in the collection.

Question Difficulty: Easy

Question ID a283471c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: a283471c

Isabel grows potatoes in her garden. This year, she harvested **760** potatoes and saved **10%** of them to plant next year. How many of the harvested potatoes did Isabel save to plant next year?

- A. **66**
- B. **76**
- C. **84**
- D. **86**

ID: a283471c Answer

Correct Answer: B

Rationale

Choice B is correct. The number of harvested potatoes Isabel saved to plant next year can be calculated by multiplying the total number of potatoes Isabel harvested, **760**, by the proportion of potatoes she saved. Since she saved **10%** of the potatoes she harvested, the proportion of potatoes she saved is $\frac{10}{100}$, or **0.1**. Multiplying **760** by this proportion gives **760(0.1)**, or **76**, potatoes that she saved to plant next year.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 00f86169

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 00f86169

At a particular track meet, the ratio of coaches to athletes is **1** to **26**. If there are x coaches at the track meet, which of the following expressions represents the number of athletes at the track meet?

- A. $\frac{x}{26}$
- B. $26x$
- C. $x + 26$
- D. $\frac{26}{x}$

ID: 00f86169 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that at a particular track meet, the ratio of coaches to athletes is **1** to **26**. If one number in a ratio is multiplied by a value, the other number must be multiplied by the same value in order to maintain the same ratio. If there are x coaches at the track meet, multiplying both numbers in the ratio by x yields **1(x)** to **26(x)**, or x to **26 x** . Therefore, the expression **26 x** represents the number of athletes at the track meet.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 6d5919de

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 6d5919de

In a bag, there are **7** red, **4** white, **33** blue, and **33** yellow cubes. If one of these cubes is selected at random, what is the probability of selecting a cube that is neither blue nor yellow?

- A. $\frac{6}{7}$
- B. $\frac{7}{11}$
- C. $\frac{1}{3}$
- D. $\frac{1}{7}$

ID: 6d5919de Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that there are **7** red, **4** white, **33** blue, and **33** yellow cubes in the bag. Therefore, there are a total of **7 + 4 + 33 + 33**, or **77**, cubes in the bag. If the cube is neither blue nor yellow, then it must be either red or white. Therefore, the probability of selecting a cube that is neither blue nor yellow is equivalent to the probability of selecting a cube that is either red or white. If one of these cubes is selected at random, the probability of selecting a cube that is either red or white is equal to the sum of the number of red cubes and white cubes divided by the total number of cubes in the bag. There are **7** red cubes, **4** white cubes, and **77** total cubes in the bag. Therefore, the probability of selecting a red or white cube is $\frac{7+4}{77}$, which is equivalent to $\frac{11}{77}$, or $\frac{1}{7}$. Thus, if one cube is selected at random, the probability of selecting a cube that is neither blue nor yellow is $\frac{1}{7}$.

Choice A is incorrect. This is the probability of selecting a cube that is either blue or yellow, rather than the probability of selecting a cube that is neither blue nor yellow.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 6da71b2d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: 6da71b2d

Poll Results	
Angel Cruz	483
Terry Smith	320

The table shows the results of a poll. A total of **803** voters selected at random were asked which candidate they would vote for in the upcoming election. According to the poll, if **6,424** people vote in the election, by how many votes would Angel Cruz be expected to win?

- A. **163**
- B. **1,304**
- C. **3,864**
- D. **5,621**

ID: 6da71b2d Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that **483** out of **803** voters responded that they would vote for Angel Cruz. Therefore, the proportion of voters from the poll who responded they would vote for Angel Cruz is $\frac{483}{803}$. It's also given that there are a total of **6,424** voters in the election. Therefore, the total number of people who would be expected to vote for Angel Cruz is $6,424(\frac{483}{803})$, or **3,864**. Since **3,864** of the **6,424** total voters would be expected to vote for Angel Cruz, it follows that $6,424 - 3,864$, or **2,560** voters would be expected not to vote for Angel Cruz. The difference in the number of votes for and against Angel Cruz is $3,864 - 2,560$, or **1,304** votes. Therefore, if **6,424** people vote in the election, Angel Cruz would be expected to win by **1,304** votes.

Choice A is incorrect. This is the difference in the number of voters from the poll who responded that they would vote for and against Angel Cruz.

Choice C is incorrect. This is the total number of people who would be expected to vote for Angel Cruz.

Choice D is incorrect. This is the difference between the total number of people who vote in the election and the number of voters from the poll.

Question Difficulty: Hard

Question ID 8ba43cc1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 8ba43cc1

How many meters are equivalent to **2,300** centimeters? (**100 centimeters = 1 meter**)

- A. **0.043**
- B. **23**
- C. **2,400**
- D. **230,000**

ID: 8ba43cc1 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that **100** centimeters is equal to **1** meter. Therefore, **2,300** centimeters is equivalent to **(2,300 centimeters)** $(\frac{1 \text{ meter}}{100 \text{ centimeters}})$, or **23** meters.

Choice A is incorrect. **0.043** meters is equivalent to **4.3**, not **2,300**, centimeters.

Choice C is incorrect. **2,400** meters is equivalent to **240,000**, not **2,300**, centimeters.

Choice D is incorrect. **230,000** meters is equivalent to **23,000,000**, not **2,300**, centimeters.

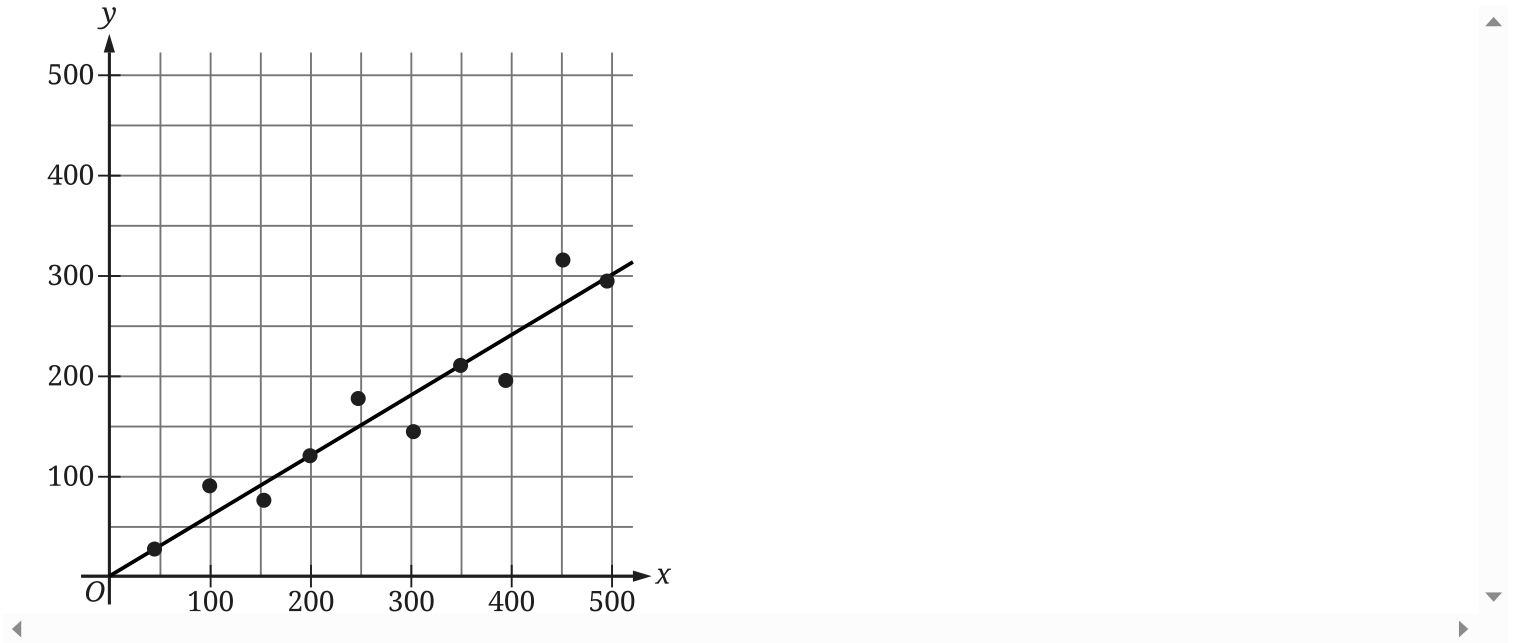
Question Difficulty: Easy

Question ID cd563f0d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: cd563f0d

A group of **10** botanists recorded data on the germination rates of a certain type of seed for one growing season. The scatterplot shows the relationship between the number of seeds planted, x , and the number of seeds that germinated, y , for each of the botanists. A line of best fit is also shown.



Which of the following is the best interpretation of the slope of the line of best fit in this context?

- A. The number of seeds planted is predicted to increase by **60** seeds every **100** days.
- B. The number of seeds planted is predicted to increase by **300** seeds every **100** days.
- C. The number of seeds that germinate is predicted to increase by **60** seeds for every additional **100** seeds that are planted.
- D. The number of seeds that germinate is predicted to increase by **300** seeds for every additional **100** seeds that are planted.

ID: cd563f0d Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the scatterplot shows the relationship between the number of seeds planted, x , and the number of seeds that germinated, y . The slope of a line of best fit represents the change in the y -value for a given change in the x -value. According to the graph, the y -value of the line of best fit increases by about **60** when the x -value increases by about **100**, which represents an increase of about **60** germinated seeds for every additional **100** seeds planted. Therefore, the best interpretation of the slope of the line of best fit is that the number of seeds that germinate is predicted to increase by **60** seeds for every additional **100** seeds that are planted.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 1e261383

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 1e261383

A product costs **11.00** dollars per pound. What is the cost, in dollars, for **6** pounds of the product?

ID: 1e261383 Answer

Correct Answer: 66

Rationale

The correct answer is **66**. It’s given that a product costs **11.00** dollars per pound. Therefore, the cost for **6** pounds of the product is $\left(\frac{11.00 \text{ dollars}}{1 \text{ pound}}\right)(6 \text{ pounds})$, which is equivalent to **66.00**, or **66**, dollars.

Question Difficulty: Easy

Question ID e76d325a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: e76d325a

A company that produces socks wants to estimate the percent of the socks produced in a typical week that are defective. A random sample of **310** socks produced in a certain week were inspected. Based on the sample, it is estimated that **12%** of all socks produced by the company in this week are defective, with an associated margin of error of **3.62%**. Based on the estimate and associated margin of error, which of the following is the most appropriate conclusion about all socks produced by the company during this week?

- A. **3.62%** of the socks are defective.
- B. It is plausible that between **8.38%** and **15.62%** of the socks are defective.
- C. **12%** of the socks are defective.
- D. It is plausible that more than **15.62%** of the socks are defective.

ID: e76d325a Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that, based on the sample, an estimate of **12%** of all socks produced by the company in a certain week are defective, with an associated margin of error of **3.62%**. This estimate, plus or minus the margin of error, gives an interval of plausible values for the actual percent of all socks produced by the company that week that are defective. Subtracting **3.62%** from **12%** yields **8.38%**. Adding **3.62%** to **12%** yields **15.62%**. Therefore, it is plausible that between **8.38%** and **15.62%** of all socks produced by the company are defective.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. **12%** is the estimated percent of defective socks based on the sample. However, since the margin of error for this estimate is known, the most appropriate conclusion is not that the percent of defective socks is exactly **12%** but instead that it lies in an interval of plausible percents.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 3626b397

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 3626b397

The density of a certain solid substance is **813** kilograms per cubic meter. A sample of this substance is in the shape of a cube, where each edge has a length of **0.60** meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

- A. **176**
- B. **488**
- C. **1,355**
- D. **3,764**

ID: 3626b397 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the sample is in the shape of a cube with edge lengths of **0.60** meters. Therefore, the volume of the sample is **0.60³**, or **0.216**, cubic meters. It's also given that the sample has a density of **813** kilograms per **1** cubic meter. Therefore, the mass of this sample is **(0.216 cubic meters) ($\frac{813 \text{ kilograms}}{1 \text{ cubic meter}}$)**, or **175.608** kilograms. Rounding this mass to the nearest whole number gives **176** kilograms. Therefore, to the nearest whole number, the mass, in kilograms, of this sample is **176**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 2fc22f88

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 2fc22f88

Which expression represents the result of increasing a positive quantity w by **43%**?

- A. **1.43** w
- B. **0.57** w
- C. **43** w
- D. **0.43** w

ID: 2fc22f88 Answer

Correct Answer: A

Rationale

Choice A is correct. The result of increasing a positive quantity w by $x\%$ can be represented by the expression $(1 + \frac{x}{100})w$. Therefore, the result of increasing a positive quantity w by **43%** can be found by substituting **43** for x in the expression $(1 + \frac{x}{100})w$, which gives $(1 + \frac{43}{100})w$, or **1.43** w . Thus, the expression **1.43** w represents the result of increasing a positive quantity w by **43%**.

Choice B is incorrect. This is the result of decreasing a positive quantity w by **43%**.

Choice C is incorrect. This is the result of increasing a positive quantity w by **4,200%**.

Choice D is incorrect. This is the result of decreasing a positive quantity w by **57%**.

Question Difficulty: Medium

Question ID 1047fccf

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 1047fccf

The number a is 60% greater than the positive number b . The number c is 45% less than a . The number c is how many times b ?

ID: 1047fccf Answer

Correct Answer: .88, 22/25

Rationale

The correct answer is .88. It's given that the number a is 60% greater than the positive number b . Therefore, $a = (1 + \frac{60}{100})b$, which is equivalent to $a = (1 + 0.60)b$, or $a = 1.60b$. It's also given that the number c is 45% less than a . Therefore, $c = (1 - \frac{45}{100})a$, which is equivalent to $c = (1 - 0.45)a$, or $c = 0.55a$. Since $a = 1.60b$, substituting $1.60b$ for a in the equation $c = 0.55a$ yields $c = 0.55(1.60b)$, or $c = 0.88b$. Thus, the number c is 0.88 times the number b . Note that .88 and 22/25 are examples of ways to enter a correct answer.

Question Difficulty: Hard